BORIS user guide

v. 9.15.0

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1. User Guide for BORIS, the Behavioral Observation Research Interactive Software

BORIS is a user-friendly application designed for event logging during video/audio coding and live observations. It is free and open source, and it runs on GNU/Linux, Microsoft-Windows and macOS with some limitations.

The official BORIS website is **https://www.boris.unito.it**.

This user guide applies to BORIS version **9.15.0**.

A PDF version of this guide is also available.

2. User guide

2.1 Installation

BORIS can be installed by following the instructions in the download section of the BORIS website.

All previous versions of BORIS are available in the Releases section of the GitHub repository.

2.1.1 GNU / Linux

BORIS runs on several Linux distributions, including Ubuntu, Debian, Raspberry Pi OS, and ChromeOS.

See the BORIS for Linux page to install BORIS for Linux.

2.1.2 Microsoft-Windows

See the BORIS for Microsoft-Windows page to install BORIS for Windows.

Two versions are available: BORIS Portable and BORIS Setup.

2.1.3 macOS

BORIS for macOS is currently experimental. Please report any issues on GitHub.

See the BORIS for macOS page to install BORIS for macOS.

2.2 Starting BORIS

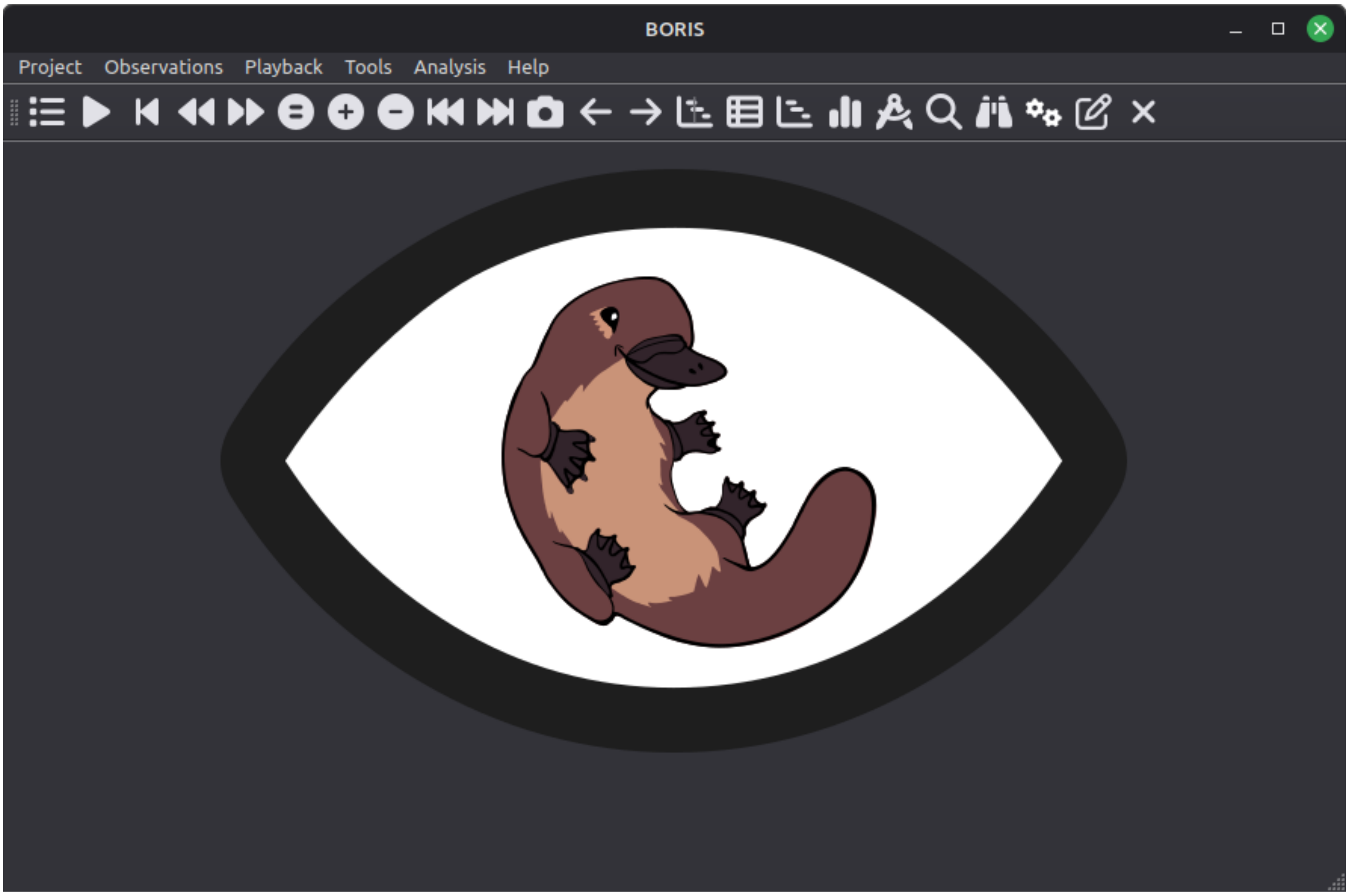
Once BORIS is installed, it can be launched by clicking on its icon.

> **Warning for Windows users**
>
> BORIS does not yet use signed binaries, so you will need to allow the downloaded executable to run. If the option is not immediately visible, click "More info" in the warning dialog and then click "Run anyway".

> **First launch**
>
> BORIS may take a little time to open the first time. Please be patient.

The BORIS main window will appear. At this stage, all toolbar commands are disabled except the Preferences button.

The BORIS main window



The BORIS main window - Dark theme

To launch BORIS from source code, refer to the Run BORIS from source code section.

2.3 Create a project

The BORIS project file serves as a container for all project-related information, excluding media files. It includes the **ethogram**, **independent variables**, **subject definitions**, **behavioral coding maps**, **converters**, and **observation** data. To save the project to your local file system, use **File** > **Save Project** or **Save Project As...**.

Additionally, you can enable the automatic backup feature in the Preferences section.

> ⚠ **Very important**
>
> It is **EXTREMELY IMPORTANT** to perform regular backups of your project files to prevent the loss of data. While software can be reinstalled, your data might be irretrievably lost. Consider using an external drive and/or a cloud service for secure backup.

BORIS lets you create an unlimited number of projects, but only one project can be open at a time.

A video tutorial about creating a project is available at this link.

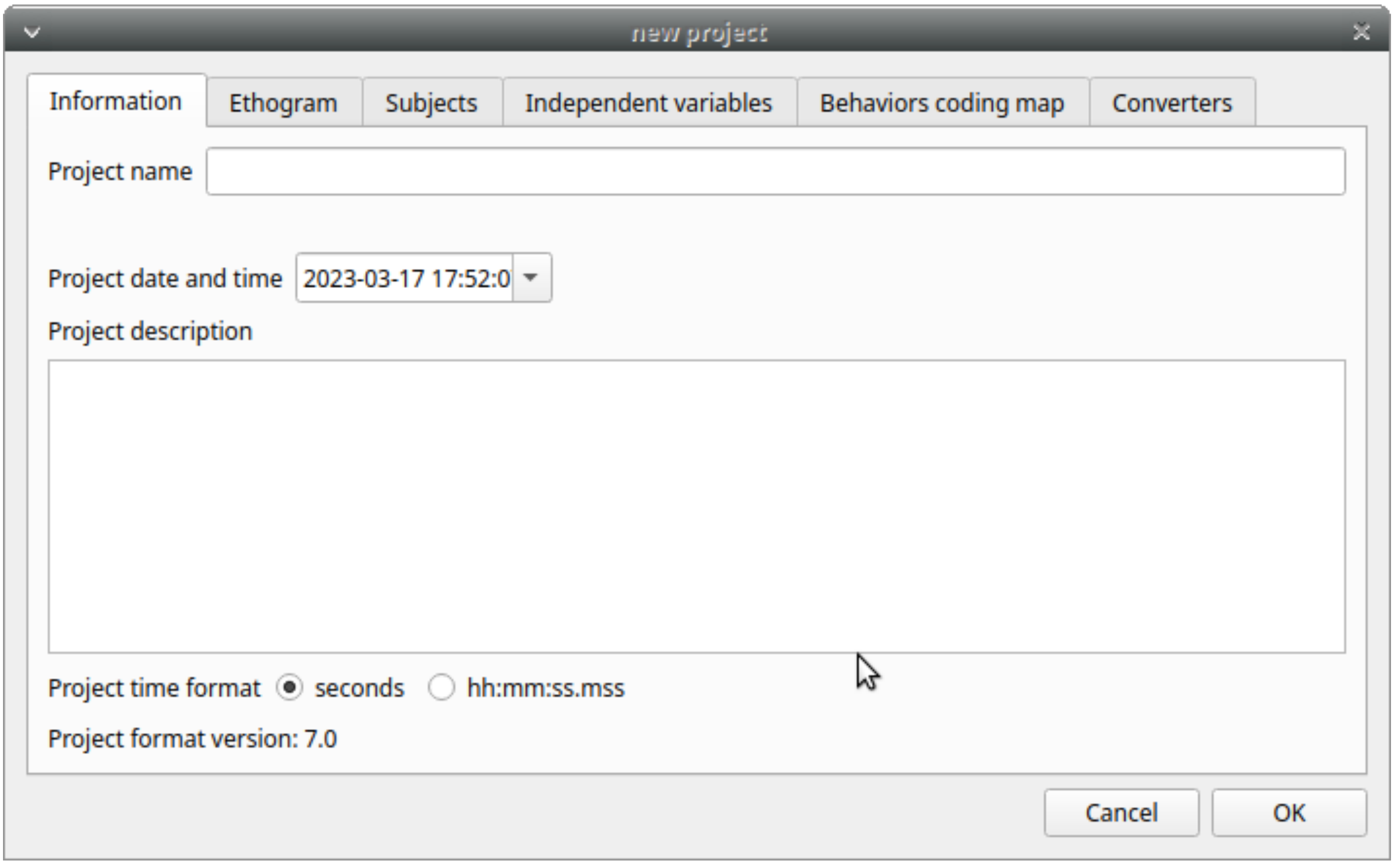
To create a new project, select **File** > **New project**.

Enter the project name in the **Project name** field in the **Information** tab. Once the project has been saved, **Project file path** will show the full path to the project file.

Date is automatically set to the current date and time, but you can change it to match your media date and time, or any other value you prefer.

Description can contain any relevant information about your project, or it can be left empty.

Time format can be set to either **seconds** or **hh🇲🇲ss.mss**. This setting can be changed at any time under **File** > **Preferences** or by using the **Preferences** button in the toolbar.



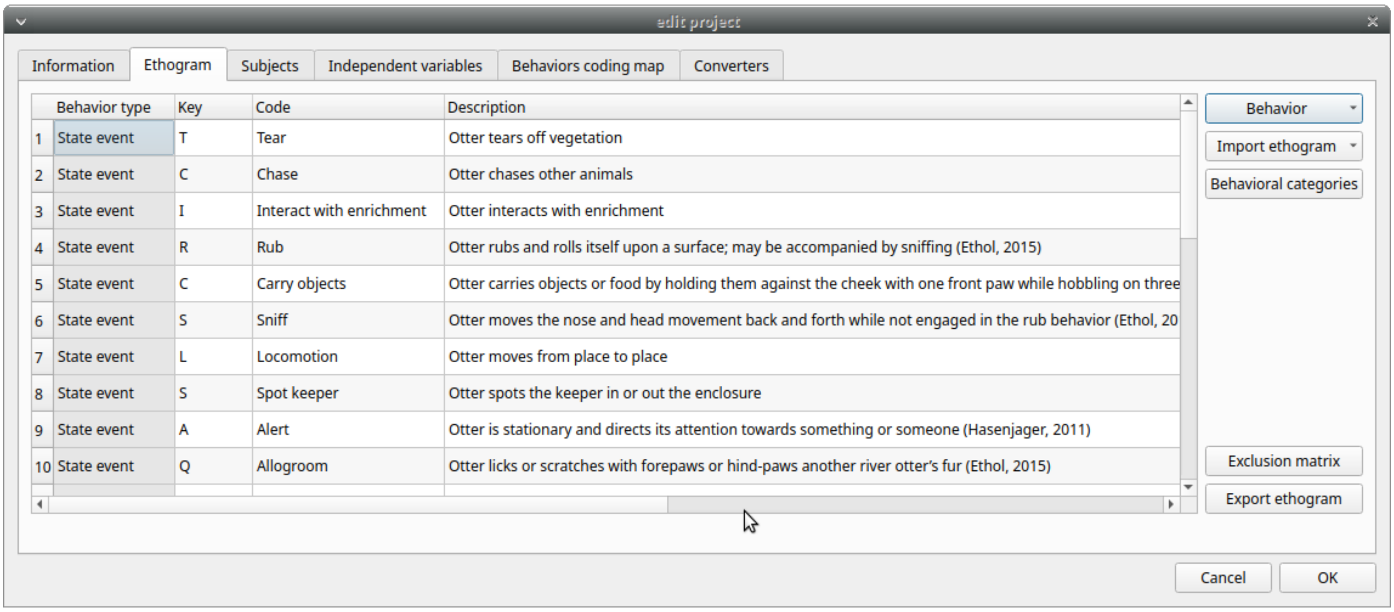
BORIS main window

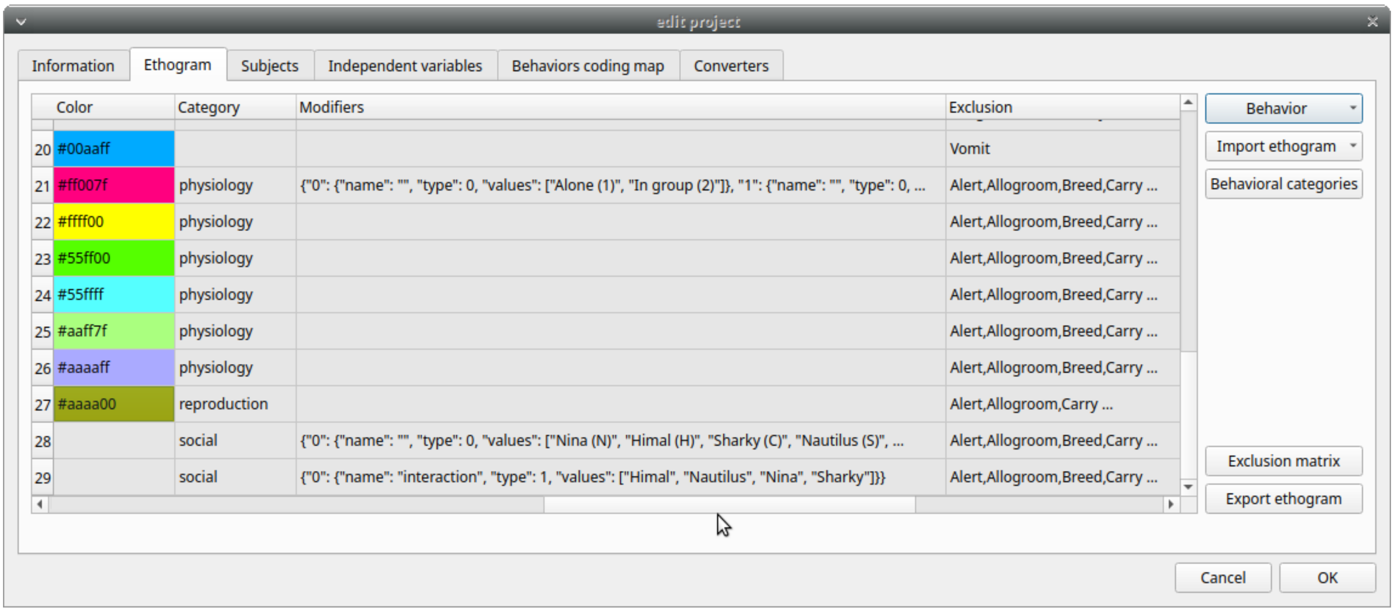
2.3.1 Set an ethogram

See the Wikipedia ethogram definition.

In the **Ethogram** tab, you can:

- Create an ethogram from scratch;
- Import an existing ethogram from another BORIS project;
- Import an ethogram from a JWatcher global definition file (`.gdf`);
- Import an ethogram from a plain text file or a spreadsheet file (`.xlsx` or `.ods`).





Set your ethogram from scratch

ADD A BEHAVIOR

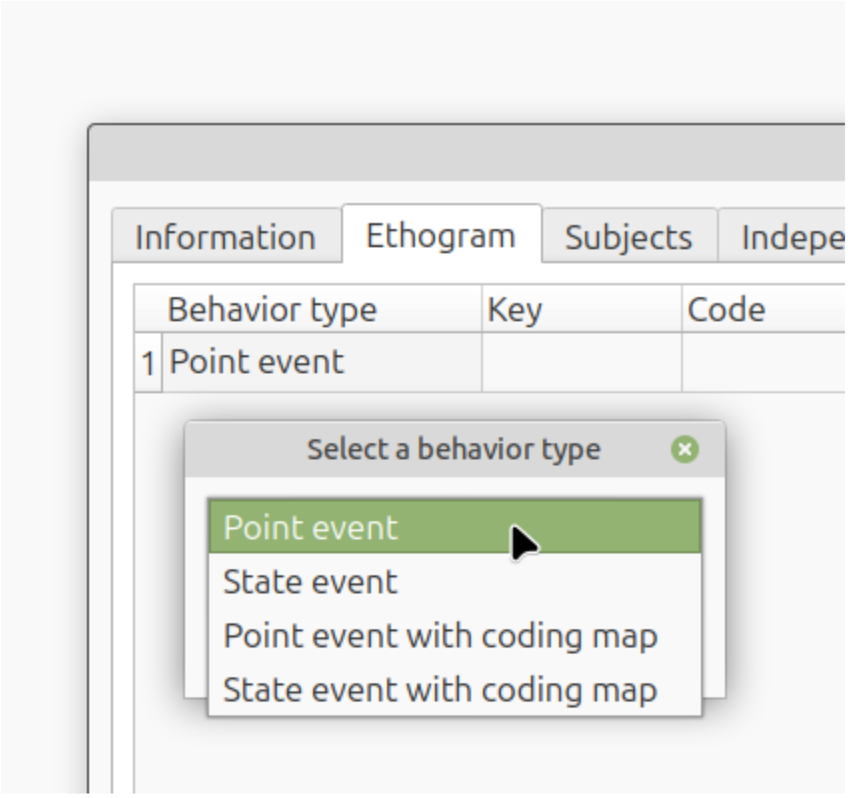
Click **Behavior > Add behavior** to add a new row to the **Ethogram** table. The behavior type is automatically set to **Point event**.

Cells with a gray background cannot be edited directly. You must double-click on them and then select a value.

The order of behaviors in the table can be changed by right-clicking on the row of the behavior you want to move up or down.

SET THE TYPE OF BEHAVIOR

You can define **two types** of behavior. Double-click the cell and select the behavior type:



- **Point event** behavior when the behavior has **no duration**.

  The behavior is coded by pressing the assigned keyboard key (see below) or by double-clicking the corresponding row in the Ethogram table.

- **State event** behavior when the behavior has a **duration**.

  The start and stop of the behavior are coded by pressing the assigned keyboard key (see below) or by double-clicking the corresponding row in the Ethogram table. These behaviors **must** have both a start event and a stop event; otherwise, an **UNPAIRED events** warning will appear when you close the observation or run an analysis.

- **Point event with a coding map**

  A **Point event** that can be coded using a **coding map**.

- **State event with a coding map**

  A **State event** that can be coded using a **coding map**.

You can switch between behavior types at any time by double-clicking the **Behavior type** cell. You can also add a **Coding map** to either a **State event** (**State event with coding map**) or a **Point event** (**Point event with coding map**). See the **Coding map** section for details.

An existing behavior can be duplicated using the **Clone behavior** button. Its code must then be changed. To remove a selected behavior, click **Remove behavior**. The **Remove all behaviors** button clears the **Ethogram** table. Both operations require confirmation.

Behaviors can be sorted by clicking the Ethogram table headers. They cannot be sorted manually.

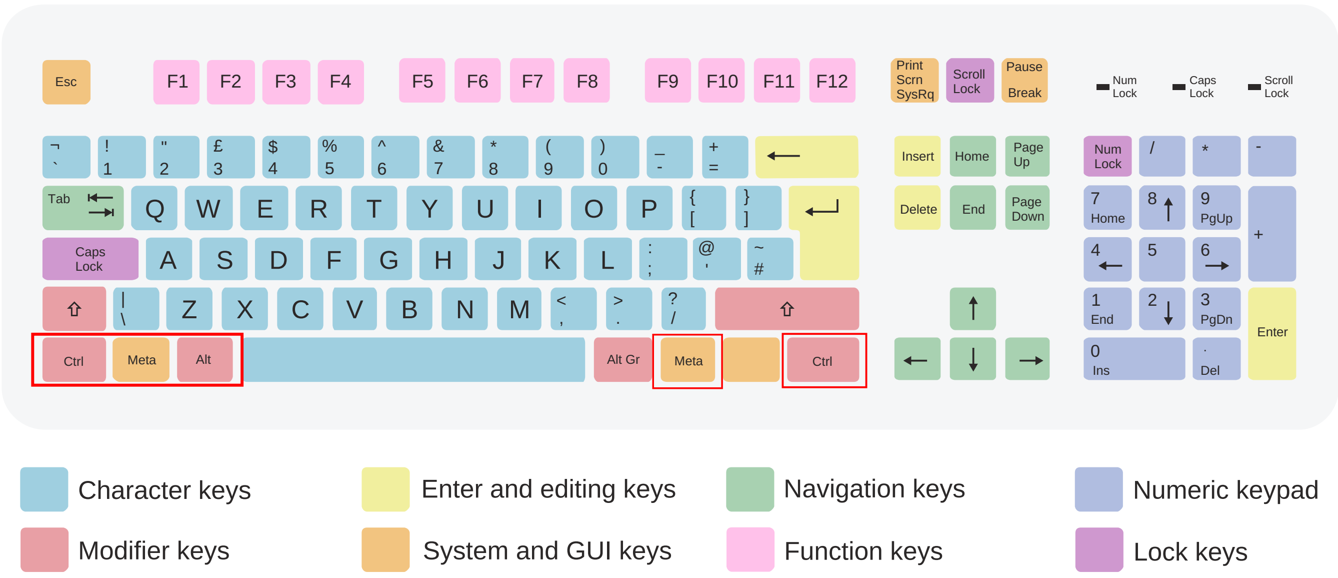
SET A KEY FOR THE BEHAVIOR (OPTIONAL)

For each behavior, you can assign a keyboard key in the **Key** column; this key will be used to code the corresponding events. You can choose whether to set a unique key for each behavior or use the same key for more than one behavior. If you assign the same key to more than one behavior, BORIS will pause coding and ask which behavior you want to record. The keys are **case-sensitive**.

> **Note**
>
> If your project was created with an older version of BORIS (< v.7), you can use **Convert keys to lower case** to convert all keys to lowercase. Otherwise, you will need to code the observation using uppercase keys.
>
> If you open a project file created with a version earlier than v.7, BORIS will ask you to convert uppercase behavior and subject keys to lowercase.

> **Important**
>
> **Do not use the / and * keys! They are reserved for the frame-by-frame mode.**

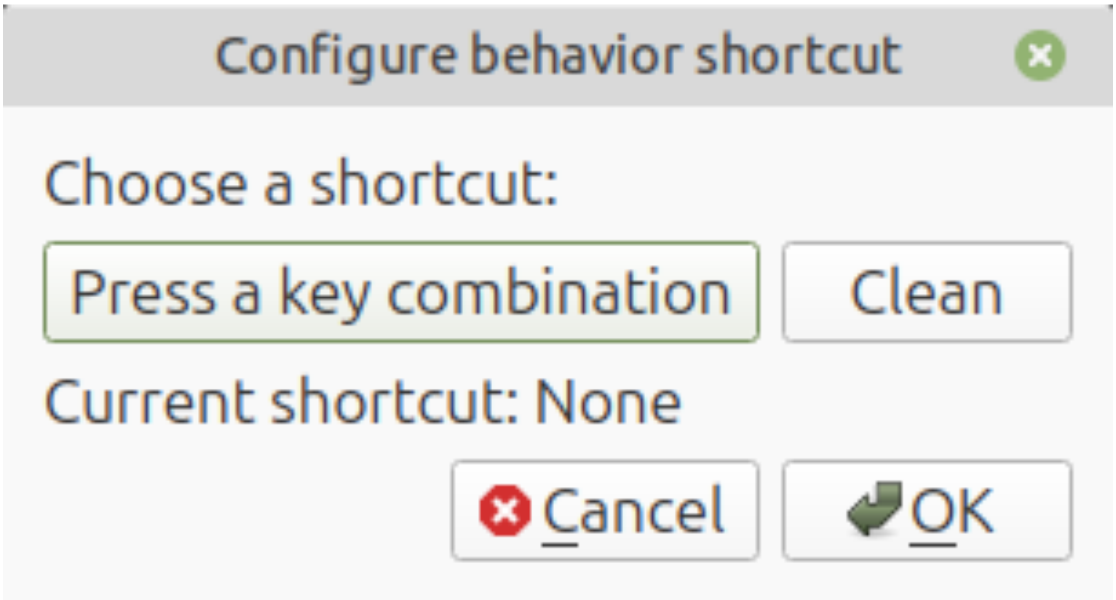
From **version 9.9** onward, you can assign key combinations to trigger a behavior, including **Ctrl**, **Alt**, and **Meta** (also known as the "Windows key") combined with a single key.



ISO keyboard with 105 keys

For the **Meta** key (orange in the keyboard diagram), see https://en.wikipedia.org/wiki/Windows_key.

To set a key combination, double-click the **Key** cell. The following window will appear:



Press the key combination you want to assign. A single modifier key, such as ⌃ Ctrl, ⇧ Shift, ◆ Meta, or ⎇ Alt, is not a valid key combination.

ctrl	alt	Meta
ctrl shift	ctrl shift meta	shift meta

You can also use the shortcut tool to select a single key.

Examples of combinations:

⌃ Ctrl + K, ◆ Meta + K, ⎇ Alt + K, ⌃ Ctrl + ⎇ Alt + K, ⌃ Ctrl + ◆ Meta + K

⌃ Ctrl + 1

Function keys (pink in the keyboard diagram above) can also be assigned to a behavior, from F1 to F12.

Examples of combinations including function keys:

⇧ Shift + F1, ⌃ Ctrl + ⇧ Shift + F12, ◆ Meta + F3

Note

BORIS uses keyboard shortcuts for several functions, such as:

- Saving a project (⌃ Ctrl + S)
- Creating a new observation (⌃ Ctrl + N)
- Starting an observation (⌃ Ctrl + O)
- Closing the current observation (++ctrl+q+)
- Editing an observation (⌃ Ctrl + E)
- Opening the list of observations (⌃ Ctrl + L)
- Adding an event (⌃ Ctrl + A)
- Jumping forward in playback (⌃ Ctrl + F)
- Jumping backward (⌃ Ctrl + B)
- Jumping to a specific time (⌃ Ctrl + T)
- ...

None of these shortcuts can be used to code behaviors.

Important

Consider that assigning a key combination to a behavior makes the project unusable in BORIS versions earlier than **9.9**.

Note

On MacOS on the ethogram table:

- The **control** key will be indicated as **Meta**
- The **option** key will be indicated as **Alt**
- The **command** key will be indicated as **Ctrl**

ISO keyboard with 105 keys

SET A CODE FOR THE BEHAVIOR (MANDATORY)

In the **Code** column, you must add a unique code for each behavior. Duplicate codes are not accepted, and BORIS will display a red warning about duplicates at the bottom left of the **Ethogram** tab. The code can be an alphanumeric string (which must not include the pipe character |).

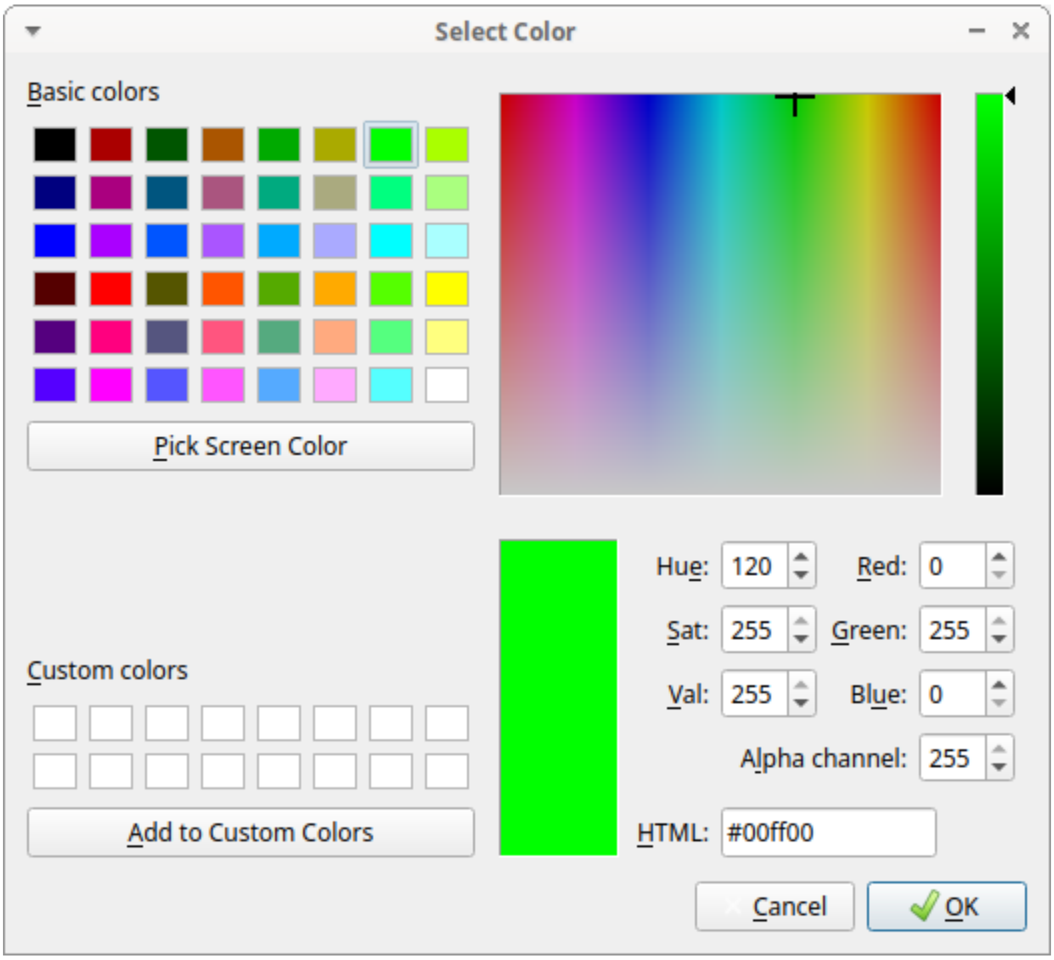
GIVE A DESCRIPTION FOR THE BEHAVIOR (OPTIONAL)

The **Description** of your behavior is optional. The **Description** column can be useful for adding information about a specific behavior, its characteristics (e.g., to standardize observations between different users), or to refer to external information (e.g., a reference to a previous ethogram).

The columns with a grey background (**Behavior type**, **Color**, **Category**, **Modifiers**, **Exclusion**, **Modifiers coding map**) cannot be edited directly.

SELECT A COLOR FOR THE BEHAVIOR (OPTIONAL)

The **Color** column allows you to select a color for the behavior. This color will be used for plotting events. Double-click on the cell and select the color you want to associate with the behavior.



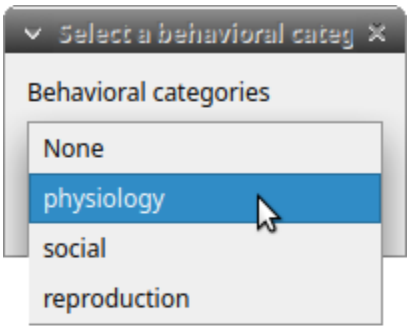
Select the color to associate to the behavior

CATEGORIES OF BEHAVIORS

Defining categories of behaviors can be useful for the analysis of coded events (for example, the time budget analysis).

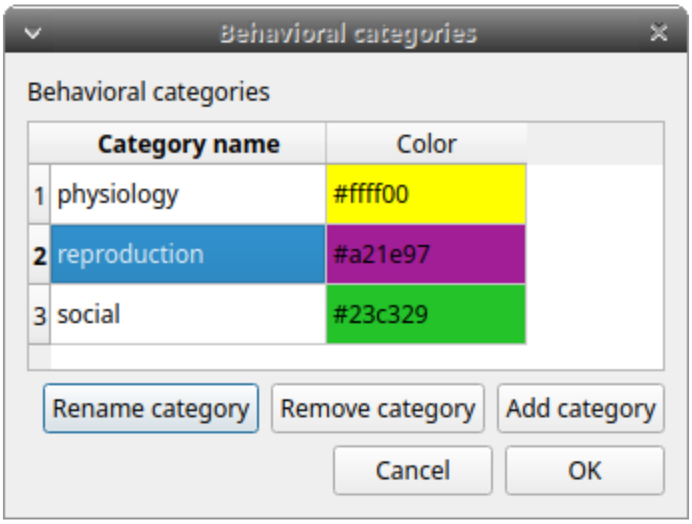
The **Category** column allows you to assign the behavior to a predefined behavioral category.

Double-click on the cell and select the behavioral category for the behavior.



Choose a behavioral category for the behavior

To add, remove, or rename a behavioral category, click the **Behavioral categories** button. A color can also be assigned to a behavioral category.



Behavioral categories manager

SET THE MODIFIERS

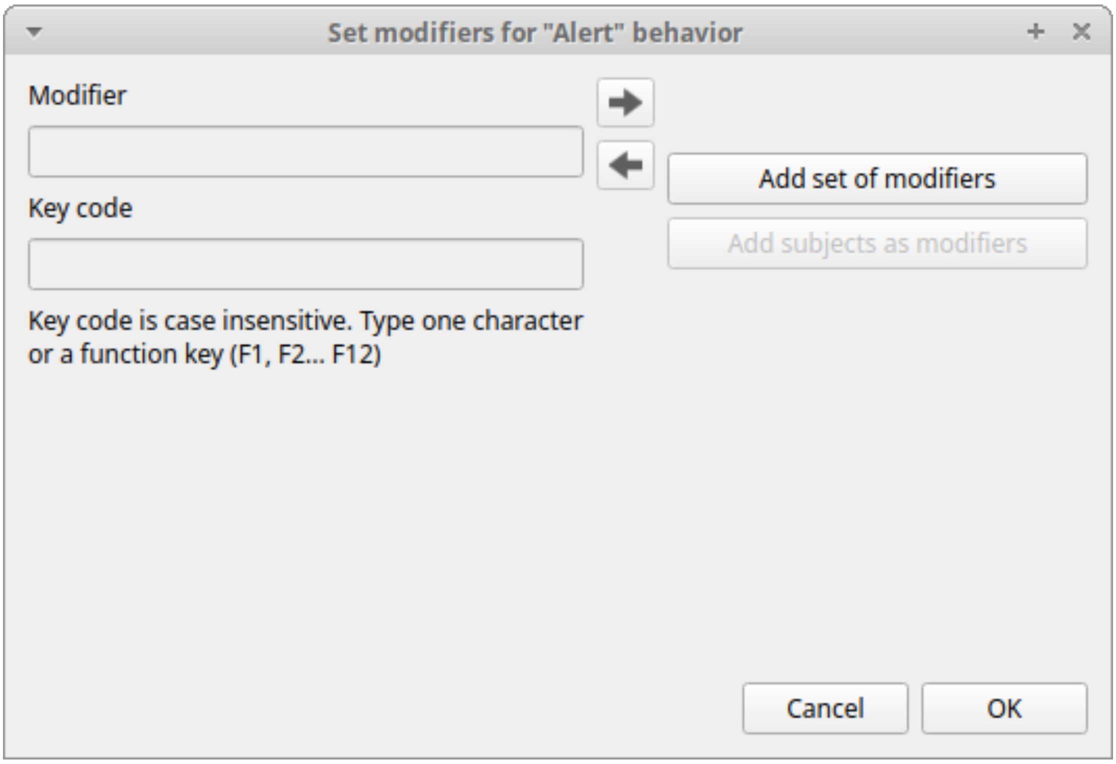
Modifiers can be used to add attributes to a behavior. A single behavior can have two or more modifiers attached (e.g., the behavior **play** may have **solitary** or **social** as modifiers). Using modifiers can significantly reduce the number of keys and simplify behavioral coding.

Four types of modifiers are available: **Single selection**, **Multiple selection**, **Numeric**, and **Value from external data file**:

- The **Single selection** type allows the observer to select only **one** modifier for the current behavior.
- The **Multiple selection** type allows the observer to select one or more modifiers for the current behavior.
- The **Numeric** type allows the observer to input a number, for example, an interaction distance.
- The **Value from external data file** type saves the value of a variable from an external data file.

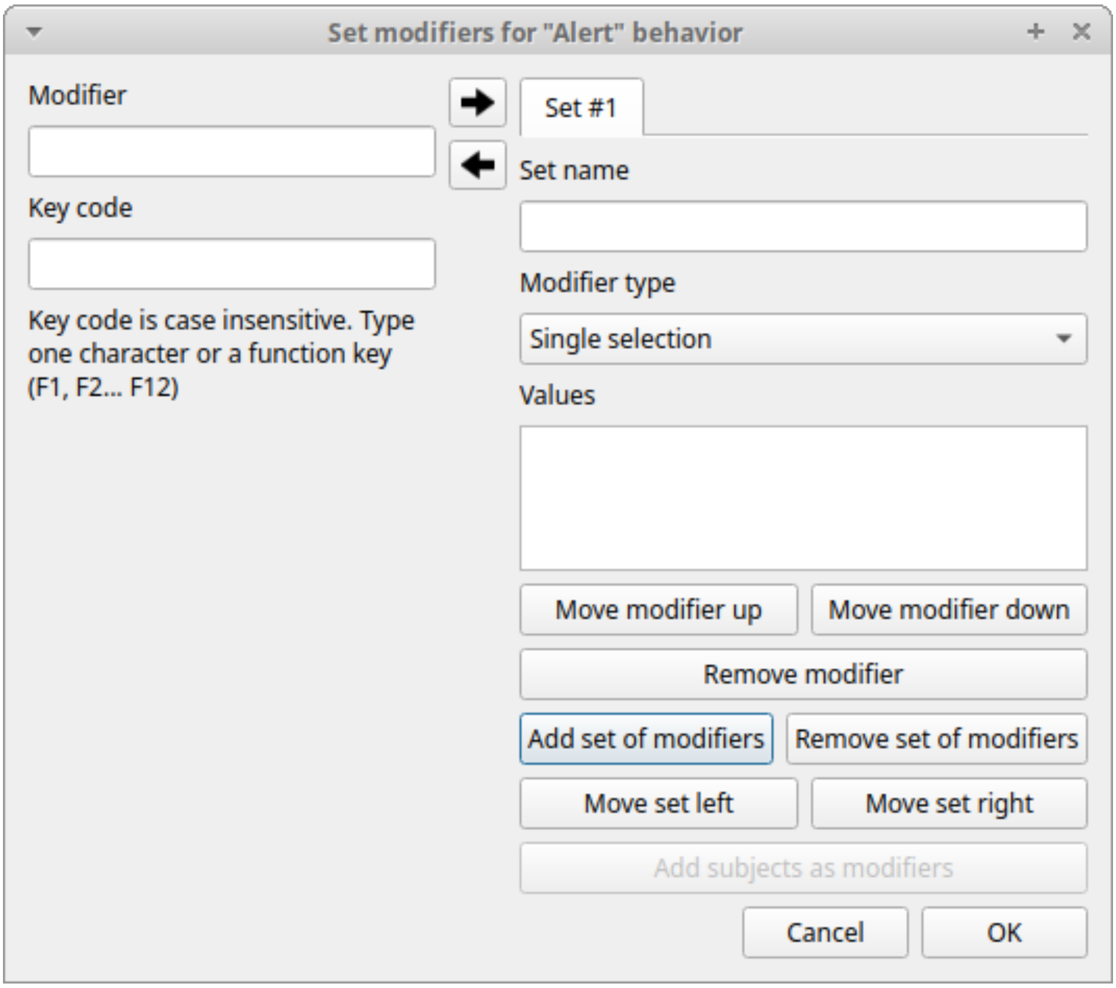
In BORIS, modifiers can also be organized into different modifier sets (e.g., **play social** may have a modifier set (#1) for **brothers** and another (#2) for **sisters**). When using sets of modifiers, you can select one or more modifiers for each set.

To add modifiers to a behavior, double-click the **Modifiers** cell corresponding to the behavior you want to add modifiers to. The following window will appear:



Modifiers configuration

Click the **Add a set of modifiers** button:



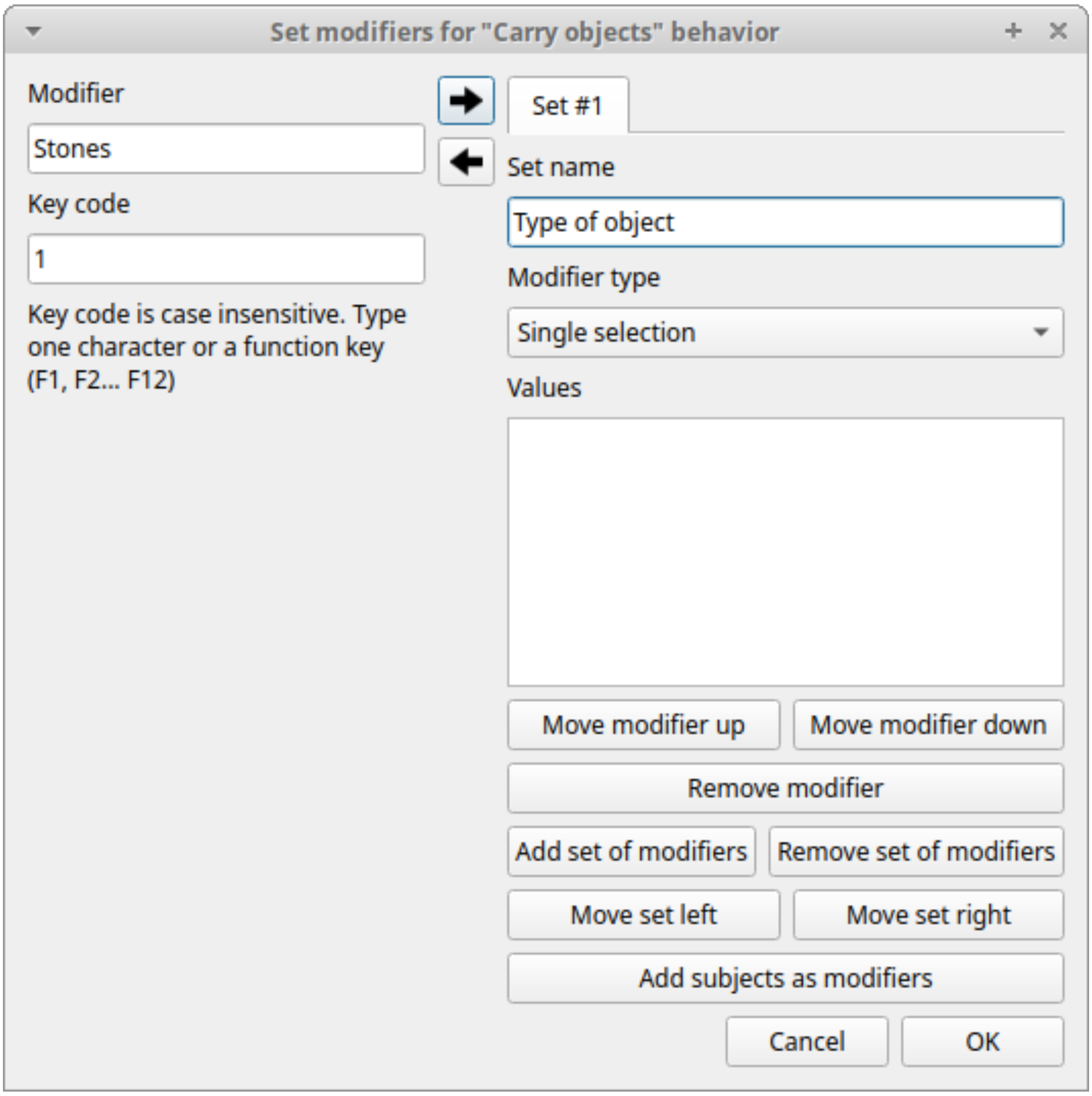
Modifiers configuration

Select the modifier type using the **Modifier type** combo box. You must choose between **Single selection**, **Multiple selection**, **Numeric**, and **Value from external data file**.

Single selection and Multiple selection modifiers

Set a name for the new modifier set by typing it in the **Set name** edit box. Setting a modifier set name is not mandatory.

Within a set of modifiers, you can add a modifier by typing the modifier in the **Modifier** edit box. You can choose a shortcut (one character, case-sensitive) for this modifier (optional). Then press the **right-arrow** button to add the new modifier to the set.

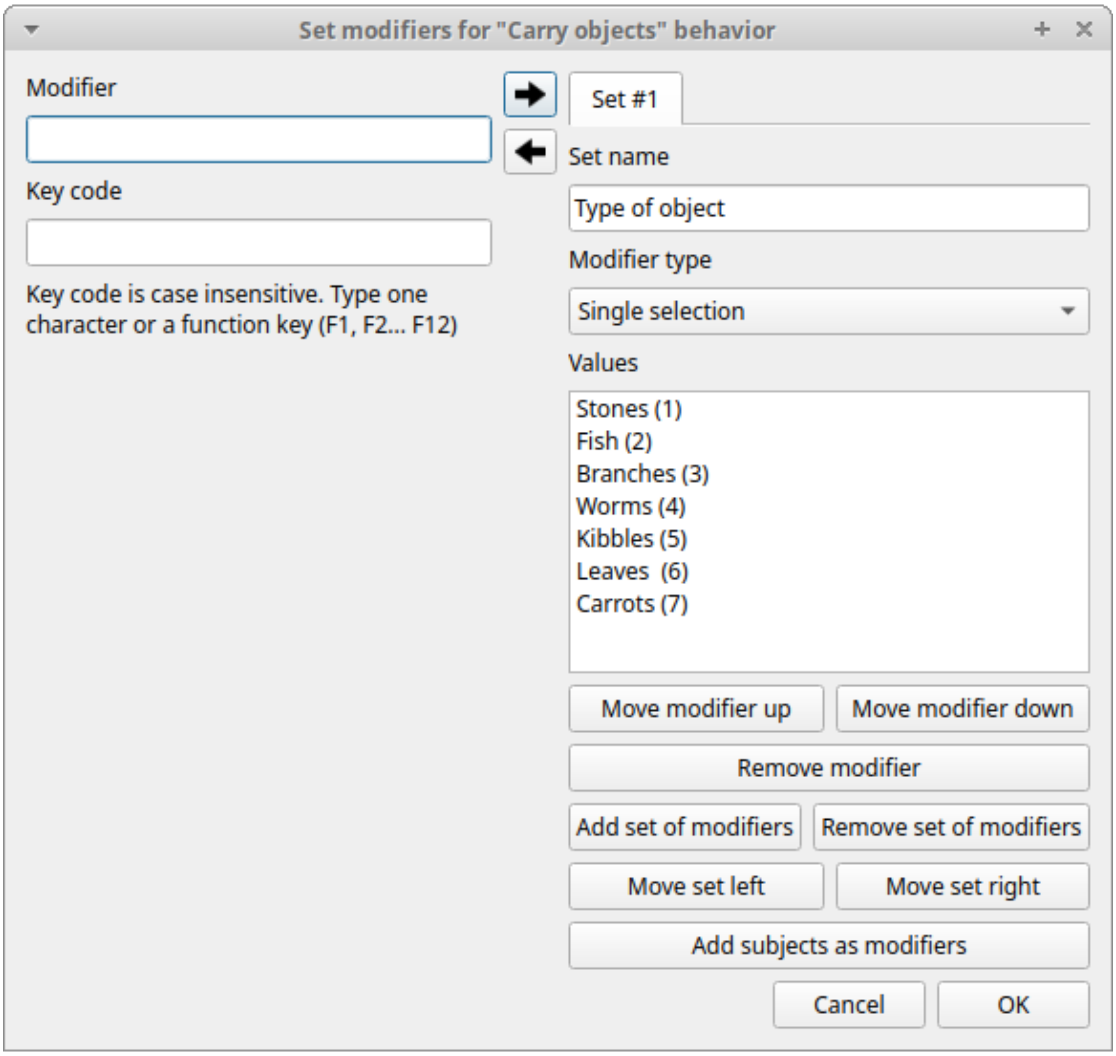


Modifiers configuration

To modify a modifier, select it and press the **left-arrow** button, edit the modifier, and press the **right-arrow** button.

A modifier can be removed by pressing the **Remove modifier** button.

After adding all modifiers, the window will appear like this:



Modifiers configuration

All defined subjects can be added as modifiers using the **Add subjects as modifiers** button. This can be helpful, for example, when coding interactions between subjects.

The modifiers can be loaded from a plain text file. Use the **Load modifiers from file** button.

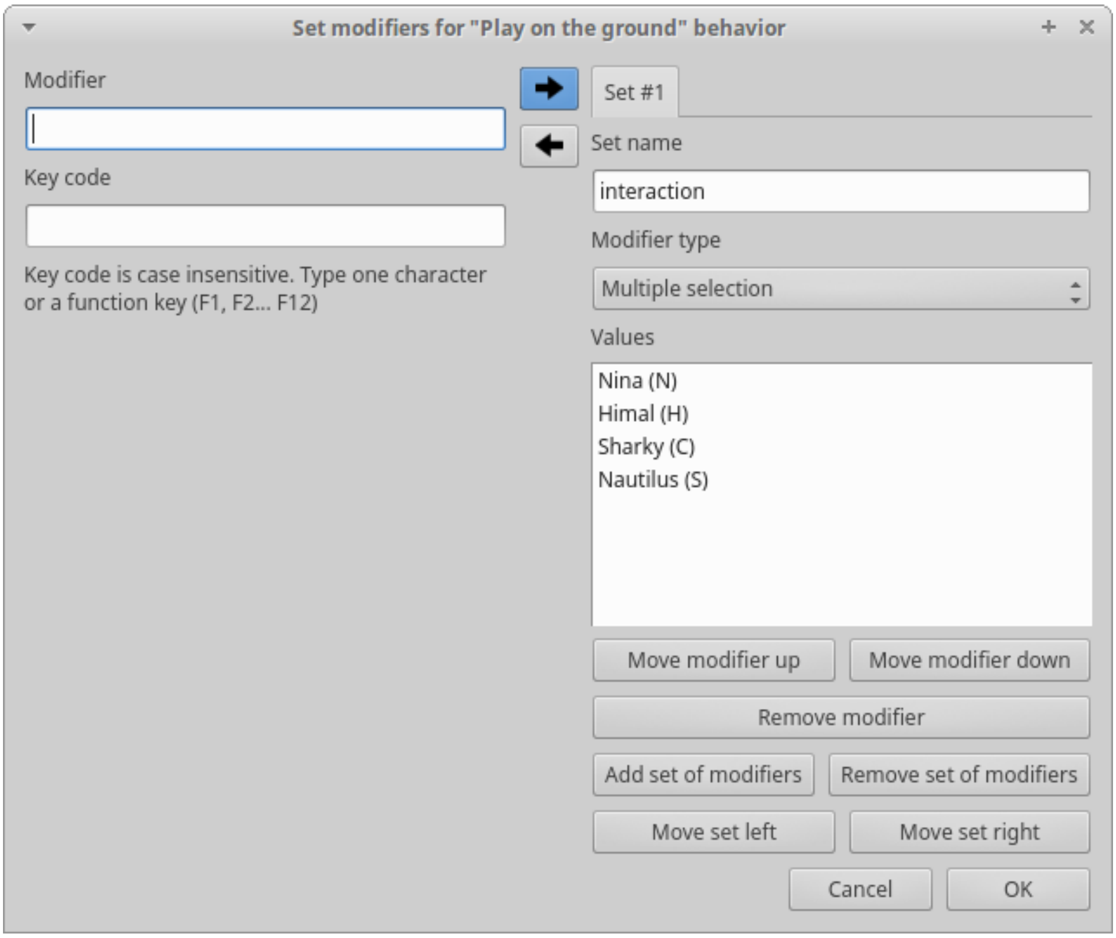
The modifier position within the modifier set can be manually set using the **Move modifier up** and **Move modifier down** buttons. The modifiers can be sorted alphabetically (use the **Sort modifiers** button).

You can add and/or remove sets using the **Add set of modifiers** and **Remove set of modifiers** buttons.

The position of a modifier set can be customized (using the **Move set left** and **Move set right** buttons).

Modifiers cannot contain the following characters: **(|) , ` ~ !**

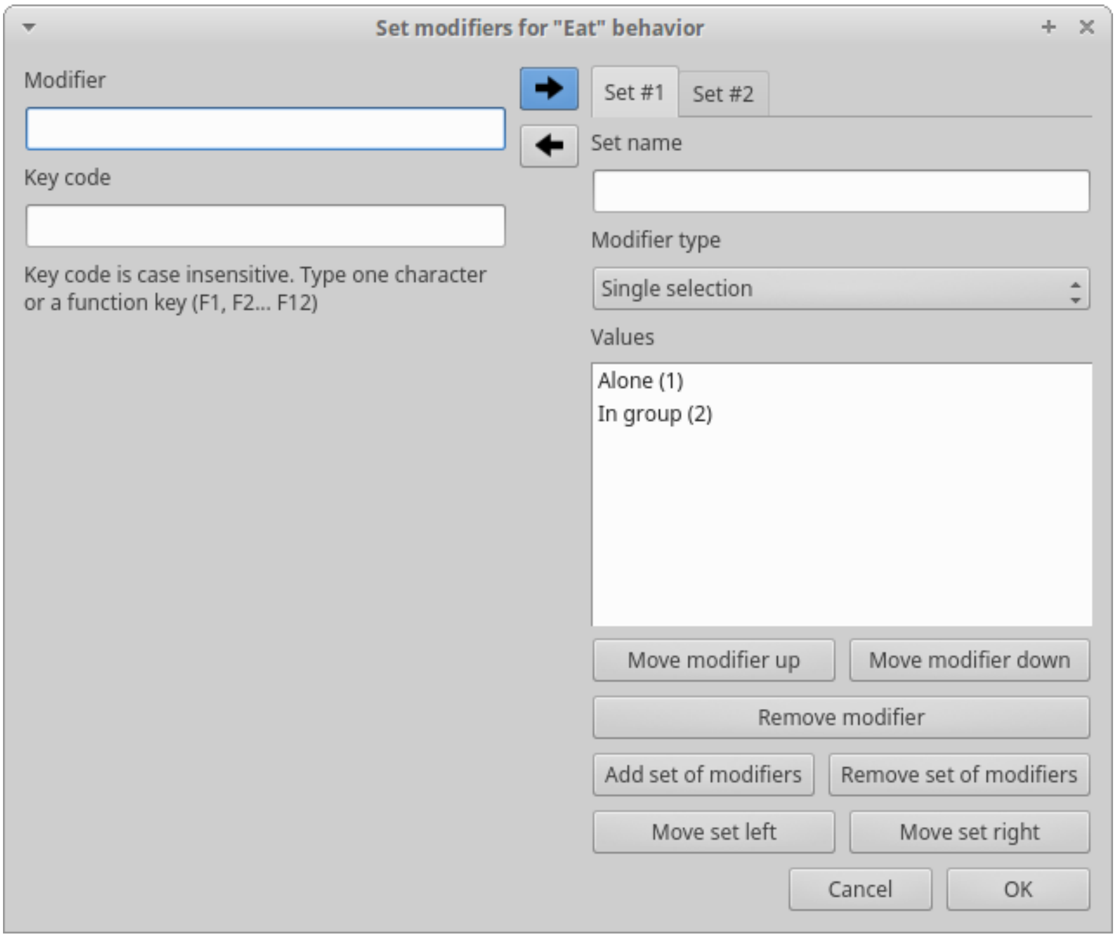
Example of a **multiple selection** modifiers set:



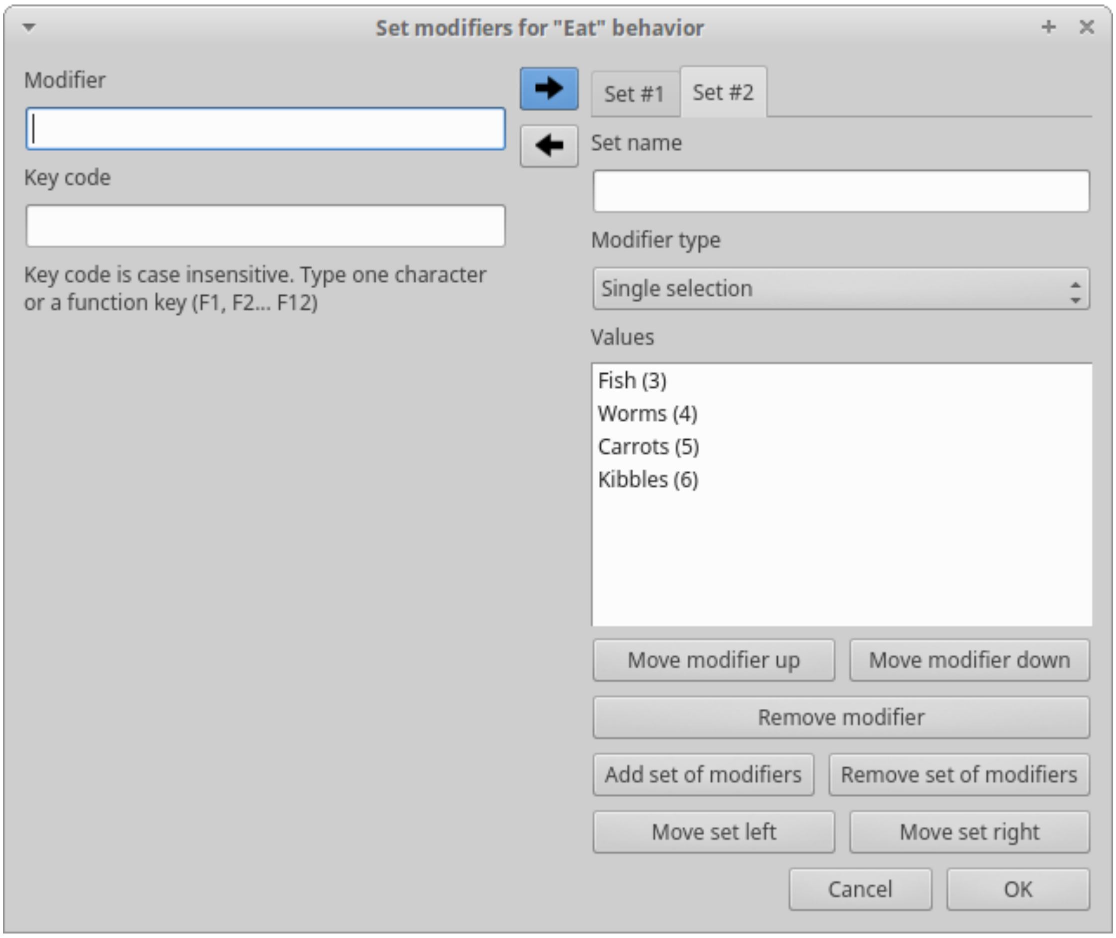
Modifiers configuration

Multiple values can be selected together.

Example of 2 sets of modifiers:



Modifiers configuration



Modifiers configuration

Numeric modifier

Set a name for the new set by typing it in the **Set name** edit box. Setting a modifier set name is not mandatory.

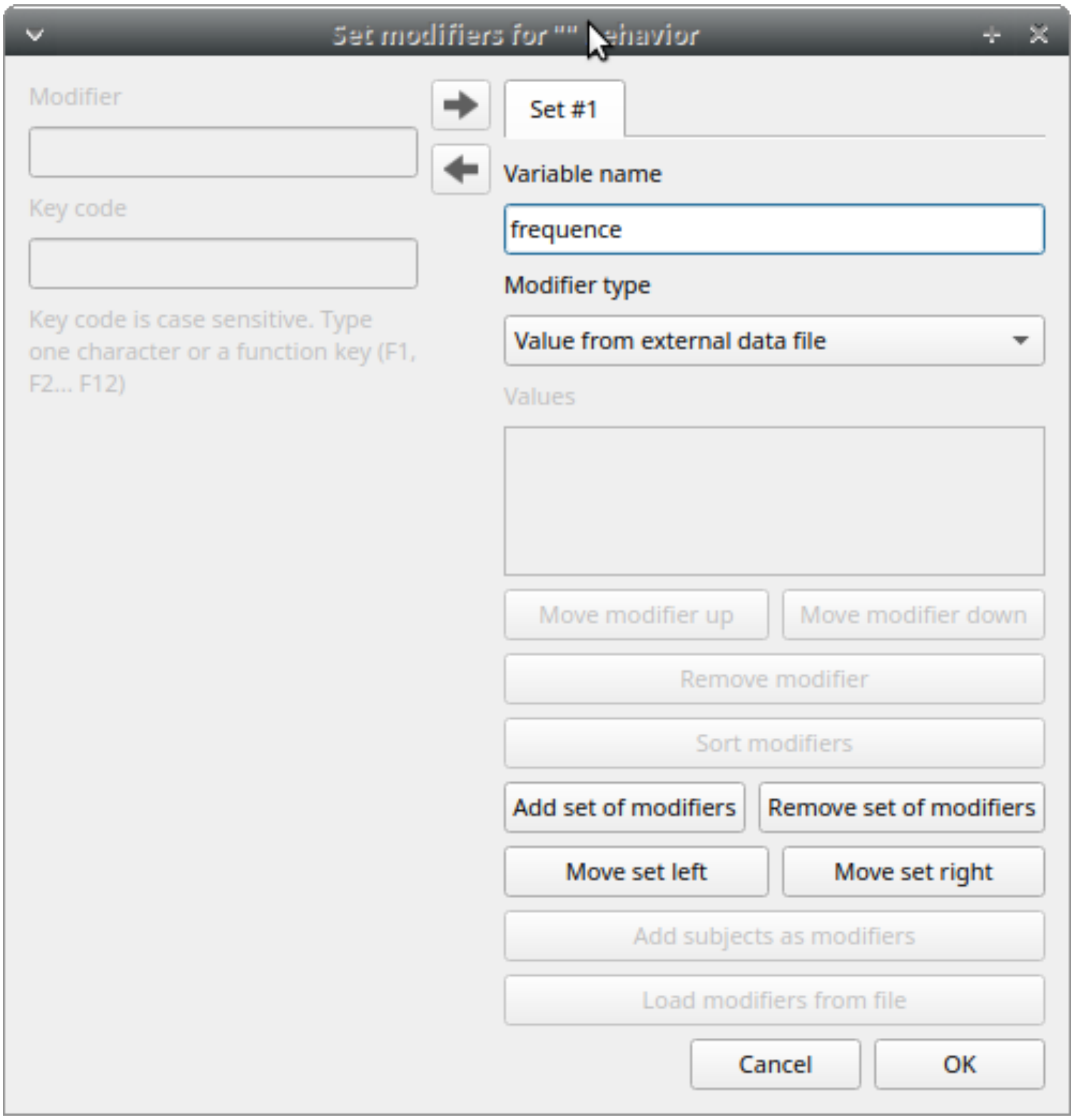
When a **Numeric** modifier is triggered, BORIS will ask the observer for a numeric value.

Value from external data file modifier

This modifier can be used to record the value of a variable from an external data file (defined during the creation of the observation).

You must define the variable name in the **Variable name** edit box. This is mandatory, and the variable name **must** match the variable defined in the observation.

See External data files



modifier value from external data file

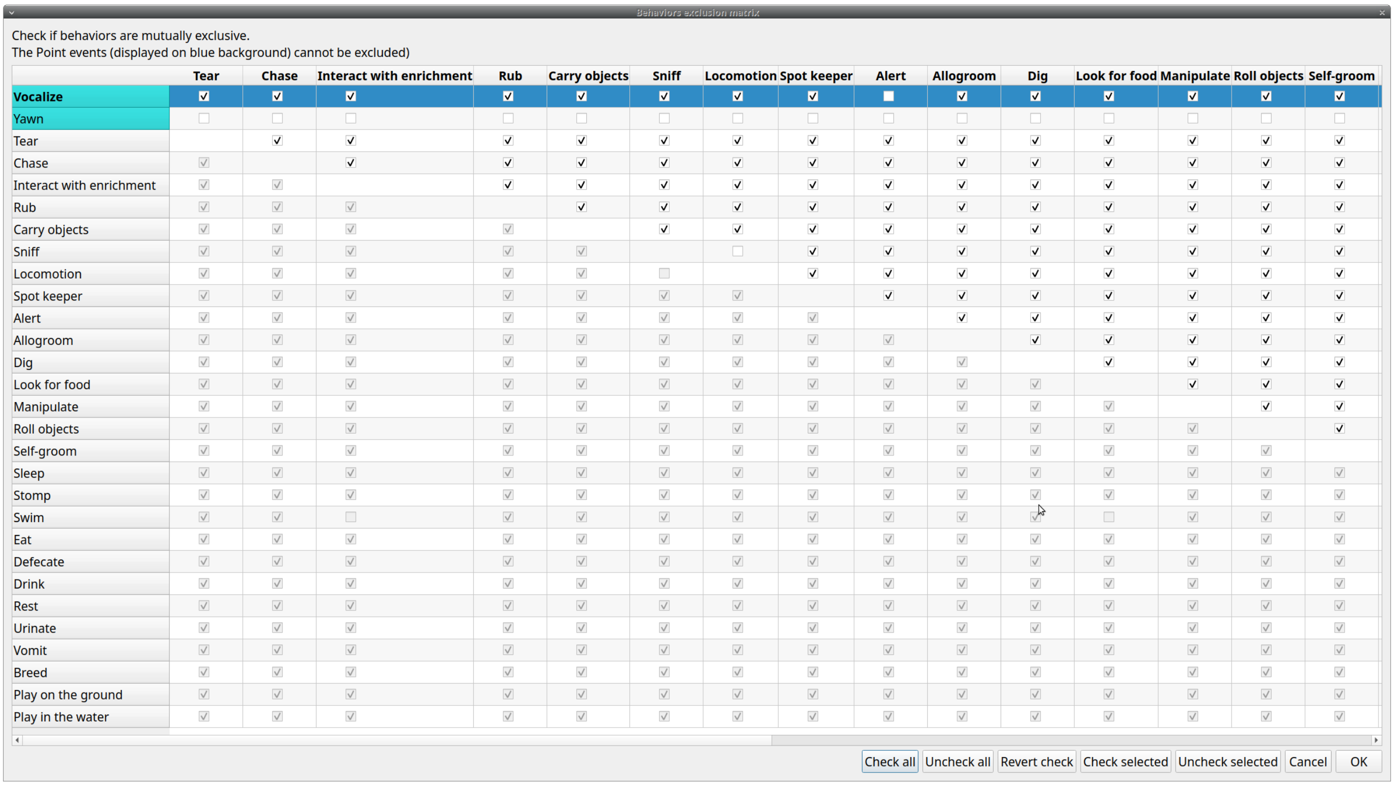
Click **OK** to save modifiers in the **Ethogram** table.

Set the exclusion matrix

The occurrence of an event (State or Point) can exclude the occurrence of a state event. This can be set using the **Exclusion matrix** window, which can be opened by clicking the **Exclusion matrix** button. BORIS will ask whether to include **Point events** or not, and a new **Exclusion matrix** window will open.

Exclusive behaviors may be selected by checking the corresponding checkbox in the automatically-generated matrix. We suggest working on the **Exclusion matrix** after all behaviors have been added to your ethogram.

All behaviors can be excluded by a particular behavior by selecting the corresponding entire row (click on the row header of the behavior) and clicking the **Check selected** button. You can also uncheck all behaviors by clicking the **Uncheck selected** button.



Example of an exclusion matrix

For example in the previous figure, the **Alert** behavior will exclude the following behaviors: **Allogroom**, **Breed**, **Carry objects**, **Chase** ...

During the observation, the excluding event will stop all the current excluded state events one millisecond before the occurrence of the event.

Set the Modifiers coding map

If the behavior is defined as a **Point event with coding map** or a **State event with coding map**, you can associate a **Modifiers coding map** to select the modifiers from a map.

Import an ethogram from an existing project

Behaviors within an ethogram can be imported from an existing BORIS project (.boris) using the **Import ethogram > from a BORIS project** button. BORIS will ask you to select a BORIS project file and whether imported behaviors should replace or be appended to the **Ethogram** table. Imported behaviors will retain all previously defined behavior parameters (namely Behavior type, Key, Code, Description, Modifiers, and Exclusion information).

Import an ethogram from a spreadsheet file

Behaviors can be imported from a spreadsheet file using the **Import ethogram > from spreadsheet file (XLSX/ODS)** button.

The first row of your spreadsheet (header) must contain the following labels. The order is not mandatory:

- Behavior code
- Behavior type
- Description
- Key
- Behavioral category
- Excluded behaviors

Behavior code is mandatory; the other fields can be empty.

Optional fields can be added:

- Color
- Modifiers (JSON)

BORIS will ask you to select a spreadsheet file (by default: .*xlsx or* .ods) and whether imported behaviors should replace or be appended to the **Ethogram** table. The missing information for the imported behaviors must be redefined.

Import an ethogram from a plain text file

Behaviors can be imported from a plain text file using the **Import ethogram > from text file** button. The fields must be separated by TAB, comma (,), or semicolon (;). All rows must contain the same number of fields.

The first row of your plain text file must contain the following labels. The order is not mandatory, but case must be respected:

- Behavior code
- Behavior type
- Description
- Key
- Behavioral category
- Excluded behaviors

Behavior code is mandatory; the other fields can be empty.

Example of a plain text ethogram definition:

```
Behavior type,Behavior code,Key,Behavioral category,Description,Excluded behaviors
state event,Play,p,,Play on the garden,s
point event,Sleep,s,,Subject is sleeping,p
```

BORIS will ask you to select a plain text file (by default: *.txt *.csv *.tsv) and whether imported behaviors should replace or be appended to the **Ethogram** table. The missing information for the behaviors imported from the text file must be redefined.

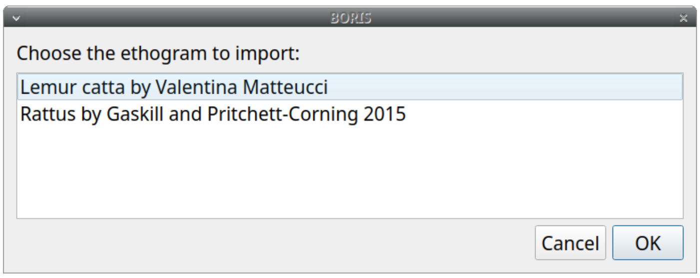
Import an ethogram from a JWatcher global definition file (.gdf)

Behaviors can be imported from a JWatcher global definition file (.gdf) using the **Import ethogram > from JWatcher** button. BORIS will ask you to select a JWatcher file (.gdf) and whether imported behaviors should replace or be appended to the **Ethogram** table. Behavior type and exclusion information for the behaviors imported from JWatcher must be redefined.

Access to the BORIS ethogram repository

This function can be activated by clicking the **Import ethogram > from the BORIS repository** button.

A list of available ethograms will open, and an ethogram can be loaded into the current project.

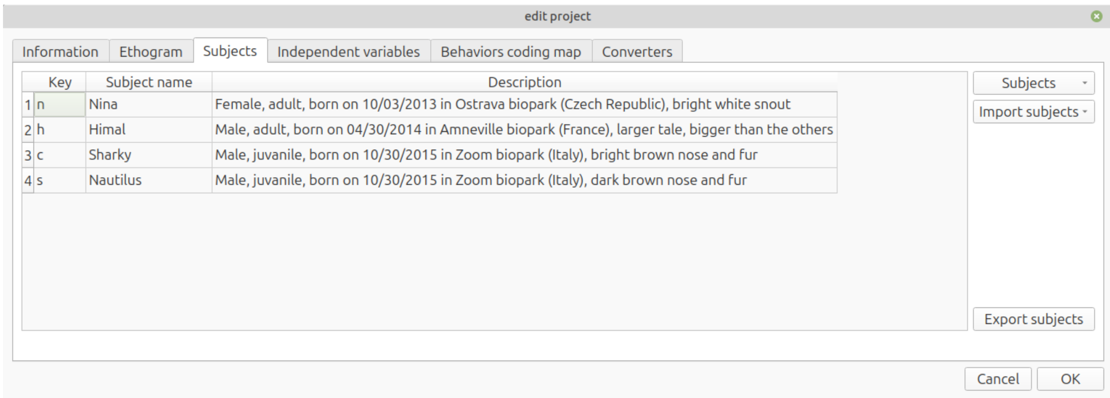


BORIS ethogram repository

Export the ethogram

The entire ethogram can be exported in various formats (TSV, CSV, XLSX, ODS, HTML). See **File** > **Edit project** > **Ethogram tab** > **Export ethogram**.

2.3.2 Define the subjects



configuration of subjects

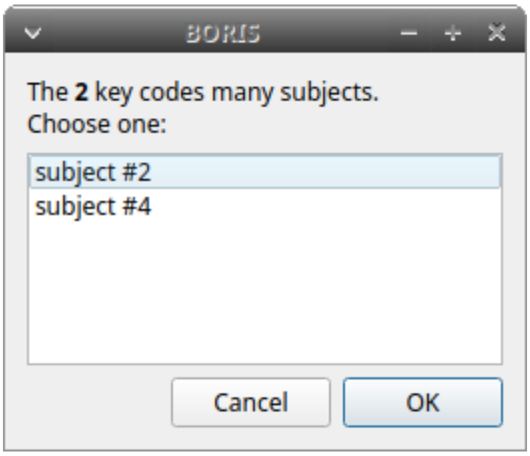
BORIS allows coding behaviors for different subjects within a single observation. The **Subject** table allows you to select subjects using a keyboard **Key**, **Subject name** (e.g., **Kanzi**), and **Description** (e.g., male, born on October 28, 1980).

Set a key for the subject (optional)

With the subjects defined in the previous figure, pressing **n** will set **Nina** as the focal subject for behavioral coding. Pressing **n** again will deselect **Nina** and set the focal subject to **No focal subject**.

Defining a key is not mandatory. In this case, you will have to select the current subject from the subjects list with a double-click.

The keys are **case-sensitive**, and the same key can be used to select more than one subject. In this case, a dialog will appear allowing you to select.



Choose a subject

Defining one or more subjects is not mandatory. Adding, removing, and sorting subjects follows the same logic as the **Ethogram** table (see Set your ethogram from scratch for details).

From **version 9.9** onward, it is possible to assign a combination of keyboard keys to select a subject, allowing the use of **Ctrl**, **Alt**, and **Meta** (also known as "Windows key") combined with a single key.

Refer to Set a key for the behavior for details.

> **Important**
>
> Consider that assigning a key combination to a subject makes the project unusable in BORIS versions earlier than **9.9**.

> **Note**
>
> If your project was created with a version of BORIS earlier than 7, you can use **Convert keys to lower case** to convert all keys to lowercase. Otherwise, you will need to code your observations using uppercase keys.

The subjects can also be imported from an existing BORIS project: use the **Import Subjects from a BORIS project** button.

Import subject from a spreadsheet

The subjects can be imported from a spreadsheet (Google Sheets, Microsoft Excel, LibreOffice Calc).

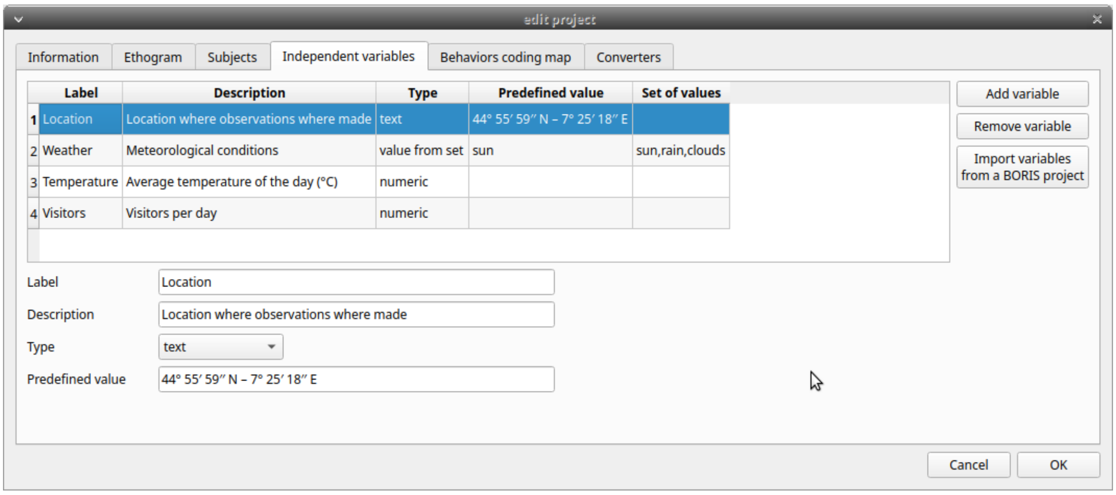
The spreadsheet must contain one subject per row and be organized as above:

- 1st column: Subject key (One character, case-sensitive, optional)
- 2nd column: Subject name (mandatory)
- 3rd column: Description of subject (optional)

Select all cells of your spreadsheet (`^ Ctrl` + `A`), copy to clipboard (`^ Ctrl` + `C`). Click the **Import from clipboard** button.

> **Note**
>
> If you open a project file created with a version older than v.7 BORIS will ask you to convert the upper case behavior and subject keys to lower case.

2.3.3 Define the Independent variables

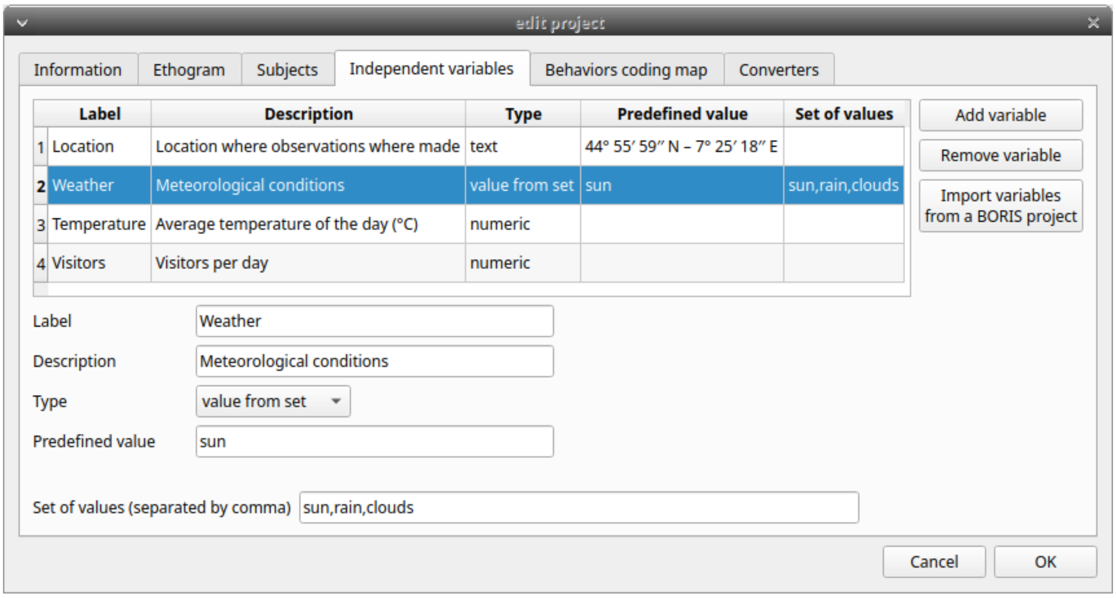


Independent variables

BORIS allows adding information about the observation using **Independent variables**. This can be used to specify factors that may influence behaviors (e.g., group composition, temperature, weather conditions) but will not change during a single observation within a project. Each independent variable can be defined by a **Label** (e.g., weather), a **Description** (e.g., weather conditions), a **Type** (*text*, *numeric*, *value from set*, or *timestamp*).

The values of a set are defined in the **Set of values** column, separating the available values with a comma (,). Please note that the first value of the set will be selected by default. It is useful to define a NA value as the first value of every set.

The values for the independent variables will be requested when creating a new observation. Adding, removing, and sorting independent variables follows the same logic as the **Ethogram** table (see **Set your ethogram from scratch** for details). Independent variables can also be imported from an existing BORIS project using the **Import Variables from a BORIS project**.

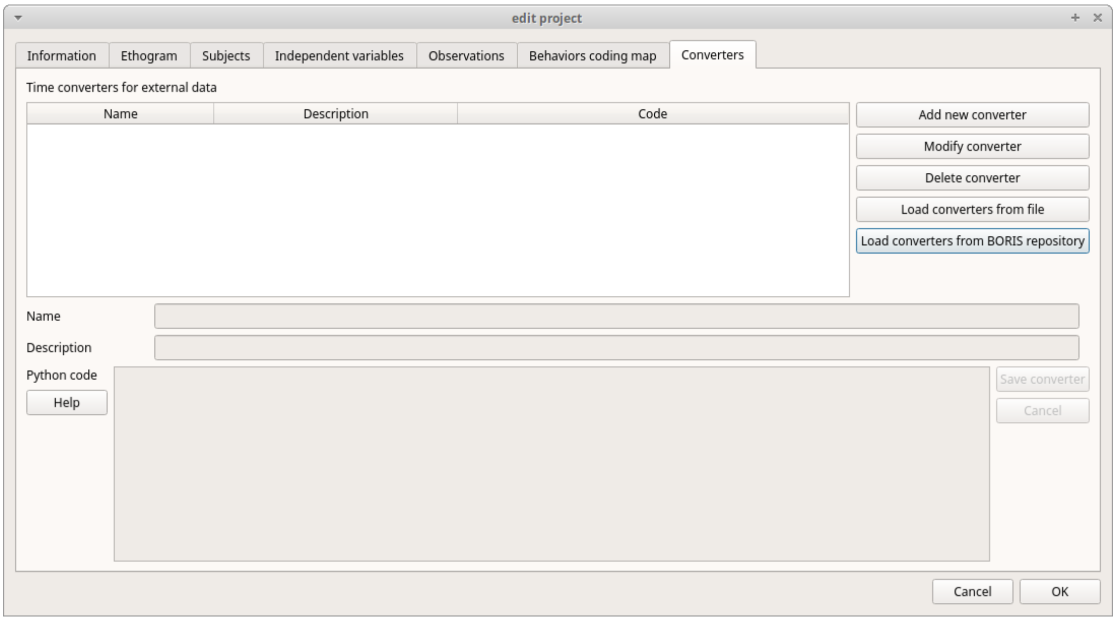


Example of an independent variable (Weather) defined as "set of values"

The predefined value must be contained in the set of values.

2.3.4 Converters' table

Converters are used for plotting external data when the timestamp values are not expressed in seconds. Converters can be written by the user, loaded from a file, or loaded from the BORIS website repository (http://www.boris.unito.it/static/converters.json).



Converters tab

Add new converter

Converters can be written using the Python 3 programming language.

The **INPUT** variable will be loaded with the original value from the external data file (for example, 01:22:32).

The **OUTPUT** variable must contain the converted value in seconds (a dot must be used as the decimal separator).

Example code to convert `HH:MM:SS` format to seconds:

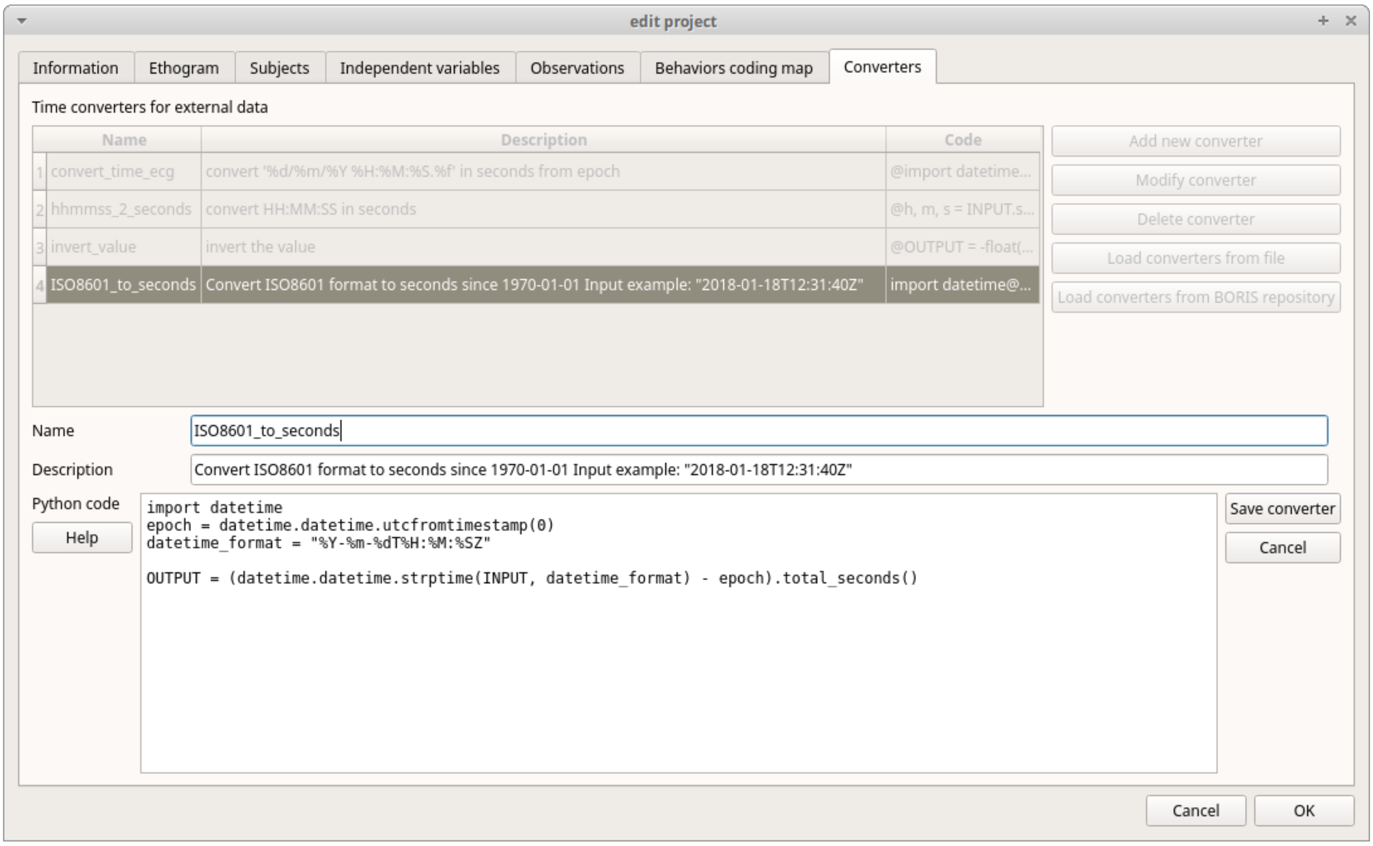
```
h, m, s = INPUT.split(':')
OUTPUT = int(h) * 3600 + int(m) * 60 + int(s)
```

The Python function **strptime()** from the **datetime** module can be useful for converting time values: https://docs.python.org/3/library/datetime.html#strftime-strptime-behavior

Example code to convert a date in ISO 8601 format to seconds using the `strptime()` function:

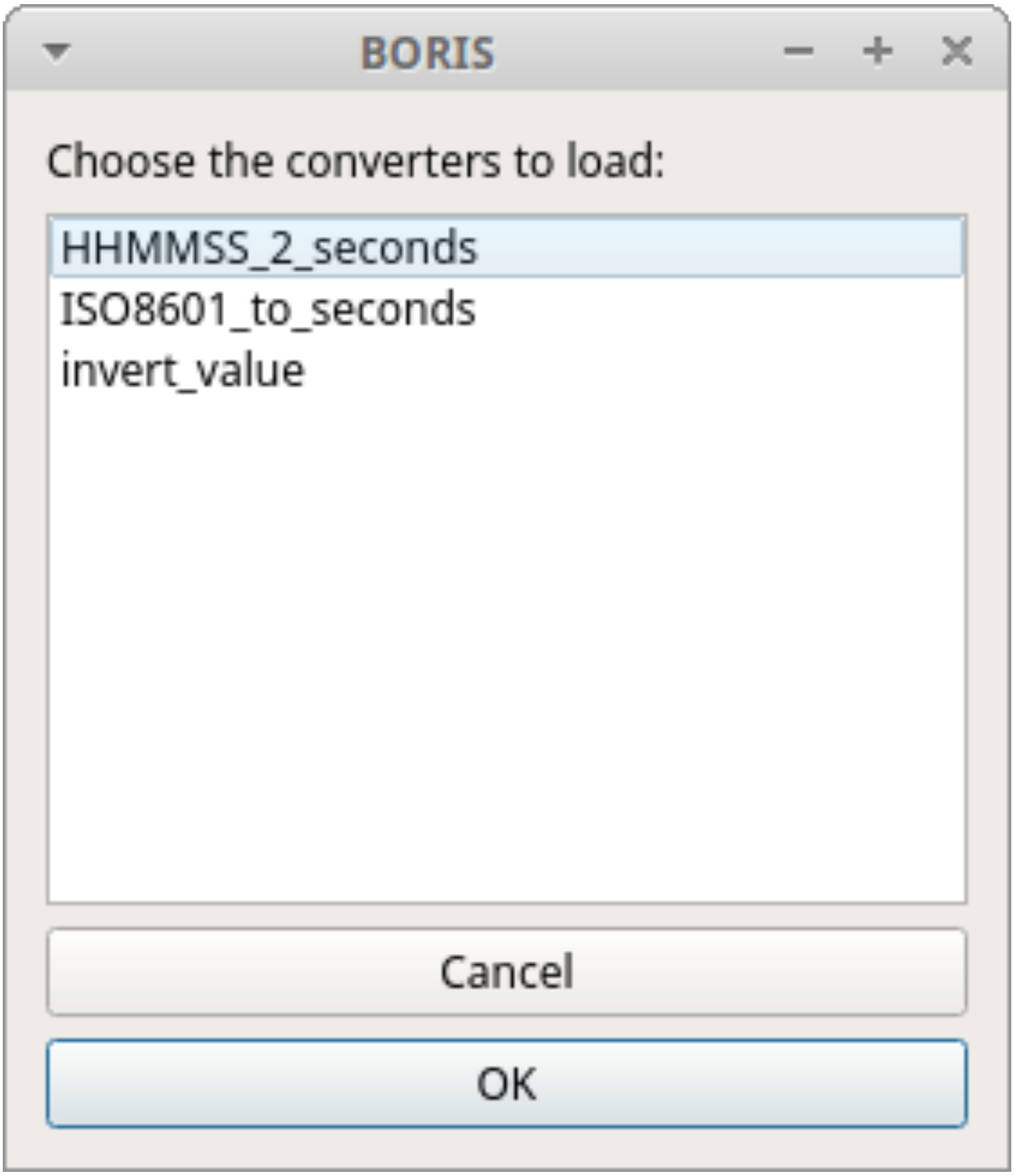
```
import datetime
epoch = datetime.datetime.utcfromtimestamp(0)
datetime_format = "%Y-%m-%dT%H:%M:%SZ"

OUTPUT = (datetime.datetime.strptime(INPUT, datetime_format) - epoch).total_seconds()
```

File > **Edit project** > **Converters**

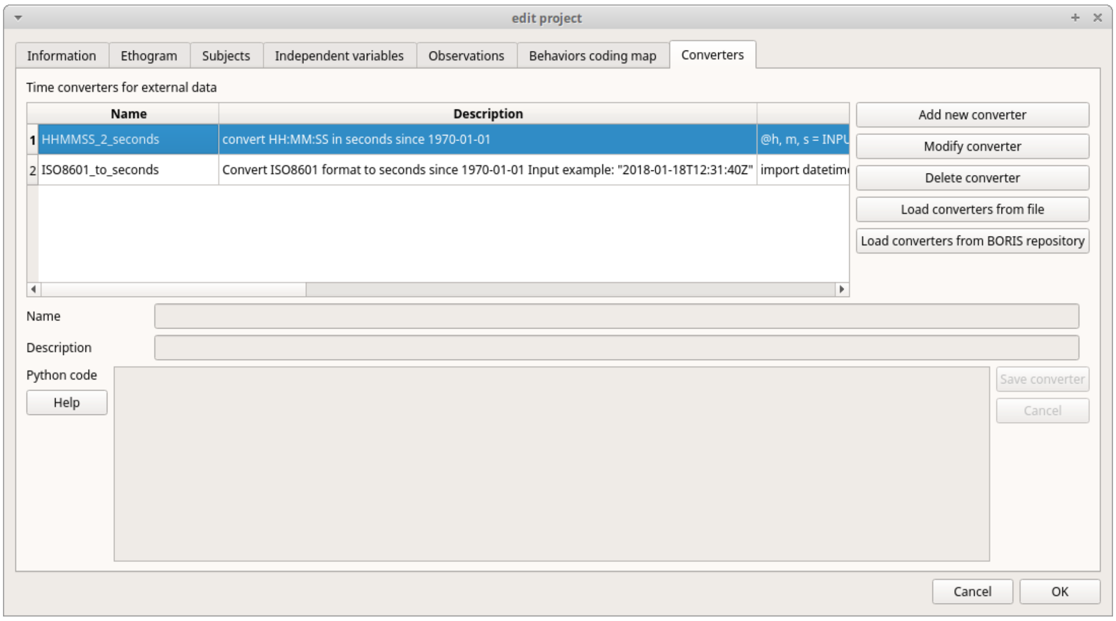


Load converters from BORIS web site

Click **Load converters from BORIS repository** and select the converters to be added to your project.



Converters selection from repository



Converters tab with 2 converters defined

Writing a converter

See Converters for external data values.

The converters loaded in your project can then be selected for converting timestamps (or other values) in external data files.

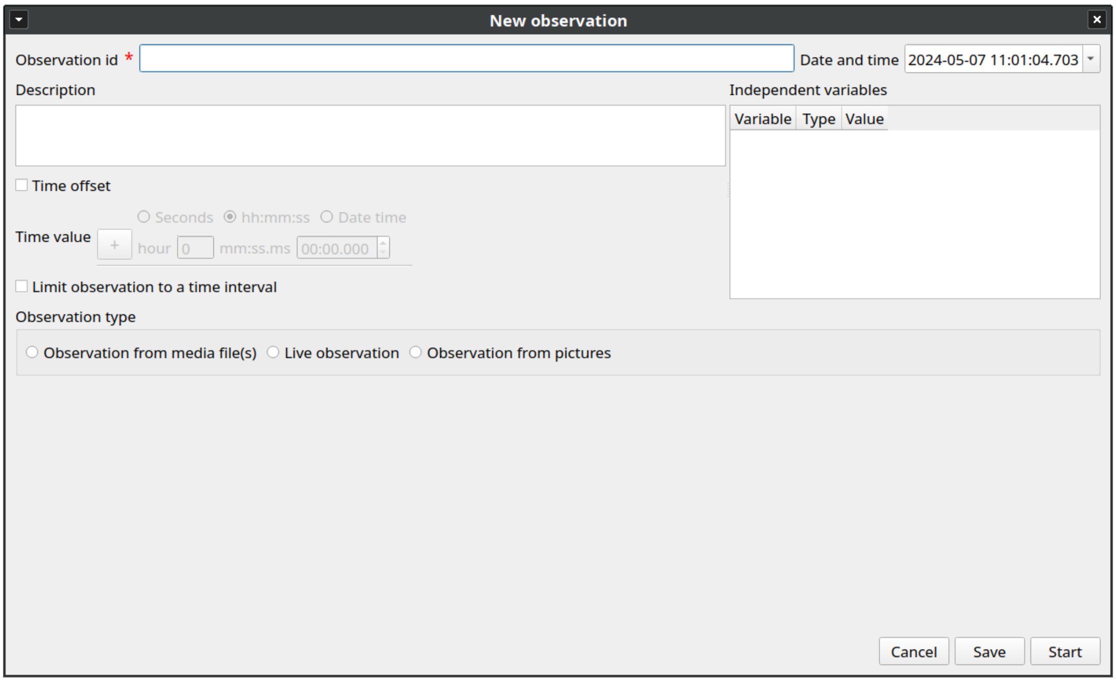
See Converters

2.4 Create a new observation

A video tutorial about making an observation is available at this link.

To create a new observation, you must first create a new BORIS project or open an existing one.

Click **Observations** > **New observation** to open the **New observation** window.



New observation window

This window allows you to enter several observation details:

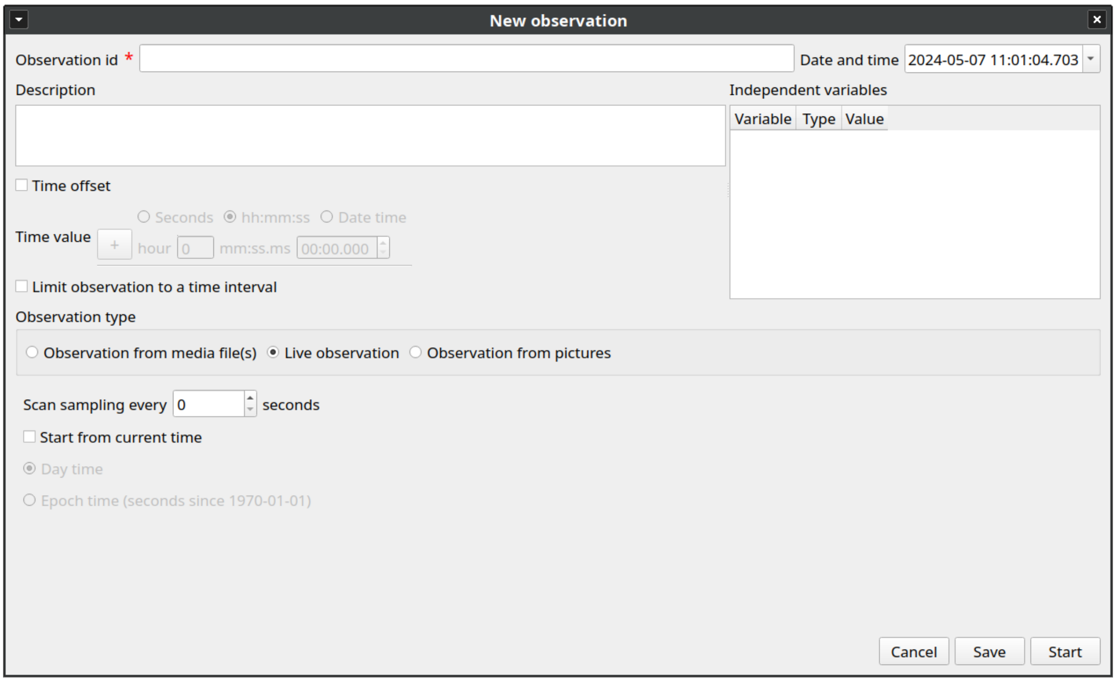
- a mandatory **Observation id** (must be unique across all observations in the open project);
- **Date**, which is automatically set to the current date and time, but can be changed to match your media date and time or any other value you prefer.
- **Description**, which can contain any relevant information about your observation, or it can be left empty.
- **Independent variables** (e.g., to specify factors that may influence behaviors but will not change during the observation within a project). See the independent variables section for details.
- **Time offset**. BORIS allows you to specify a time offset that can be added to or subtracted from the media timecode.
- The **Limit observation to a time interval** option can be used to limit the observation to an arbitrary time interval.

You must then indicate whether you want to create an observation based on **pre-recorded media (audio/video)** or a **live observation**.

2.4.1 Live observation

During a live observation, BORIS displays a timer used to record event times.

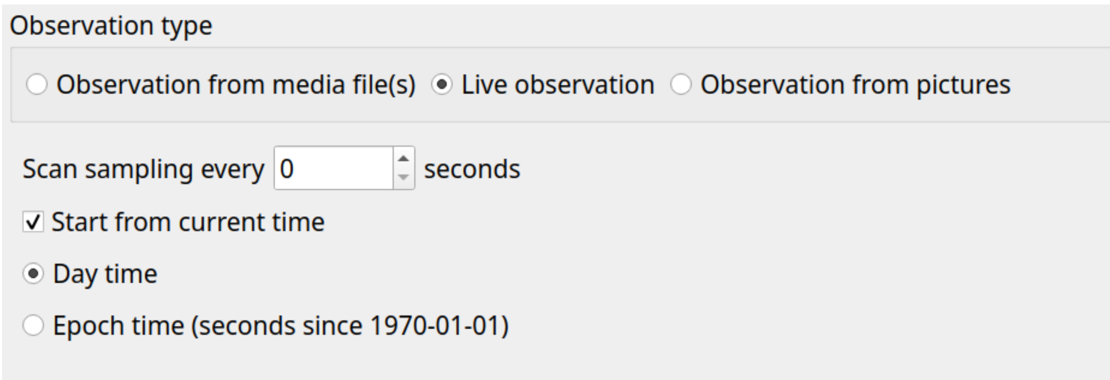
Click on the **Live observation** radio button to create a live observation.



New live observation

Start from current time

If you want the timer to start from the current time, select the **Start from current time** checkbox.



Set a live observation to start from current time

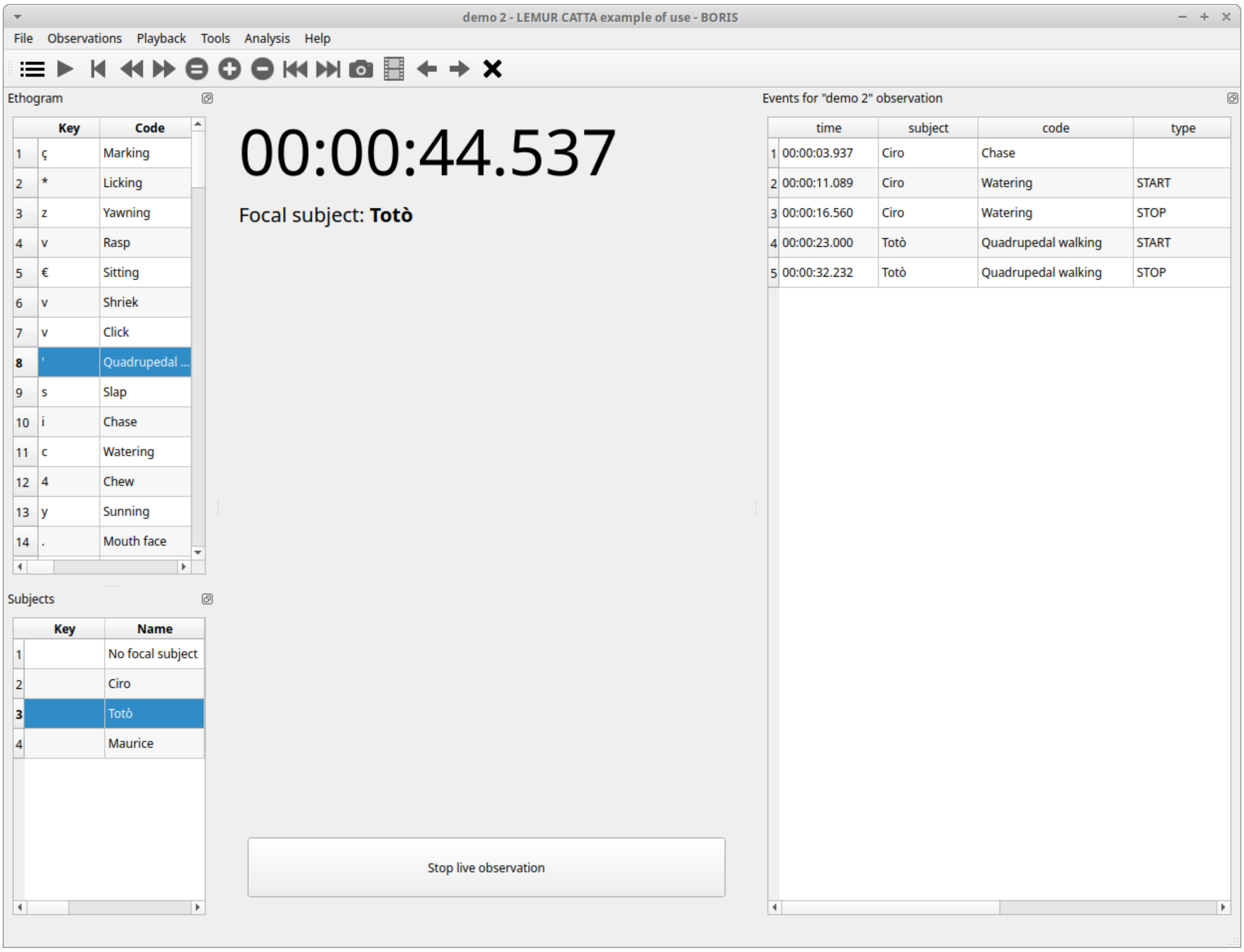
If the **Day time** option is checked, the start time will be the computer's current time when you press the **Start** button.

If the **Epoch time** option is checked, the start time will be the number of seconds since January 1, 1970 (`1970-01-01`). See Unix time for details. This option is useful for long observations lasting several days or for observations that start before midnight and end after midnight.

Start the observation

Click the **Start** button to begin the live observation, or **Save** to save it in the Observations list.

The main window during a live observation will look like this:

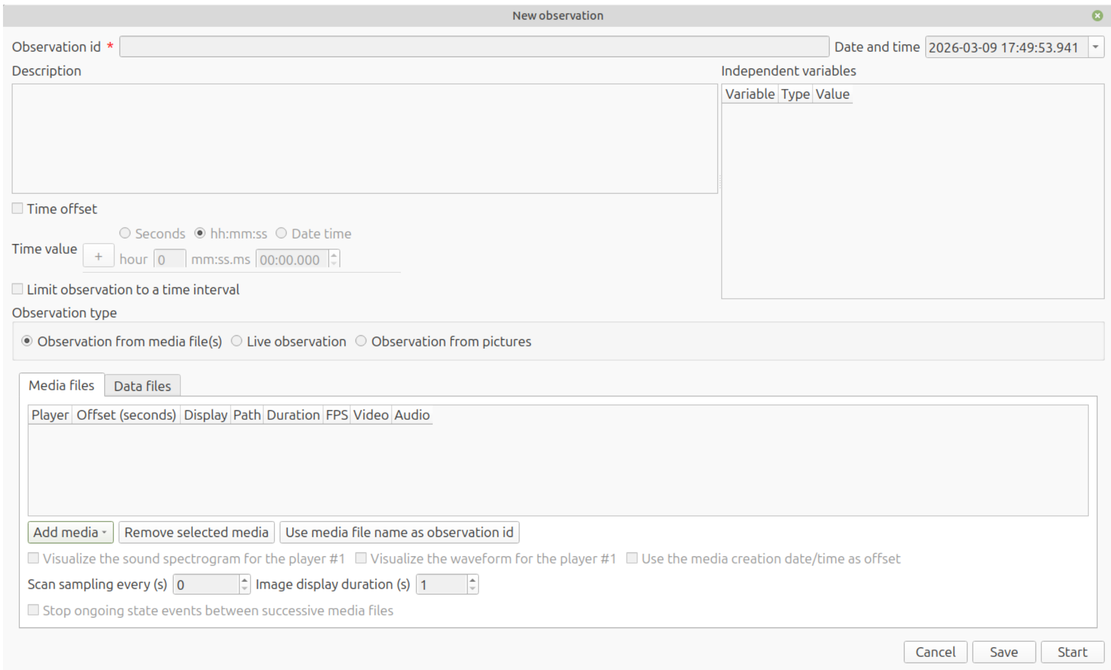


The main window during a live observation

See the Live observations section to start coding.

2.4.2 Observation from media file(s)

Select **Observation from media file(s)** to create an observation based on one or more media files.



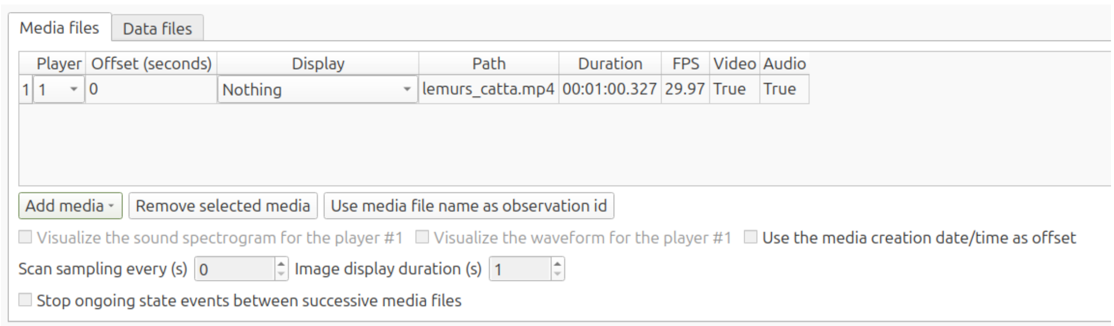
Observation from media files

The **Observation from media file(s)** section contains 2 tabs: **Media files** and **Data files**.

Click the **Media files** tab and add one or more media files using the **Add media** button. You have 4 options:

- **with absolute path**: the full media file path will be recorded in the project
- **with relative path**: the media file path will be recorded relative to the BORIS project file location. This option is useful if you need to move your BORIS project file to another computer.
- **from directory with absolute path**: all media files found in the directory will be added to the playlist (the full media file path will be recorded in the project)
- **from directory with relative path**: all media files found in the directory will be added to the playlist (the relative media file path will be recorded in the project)

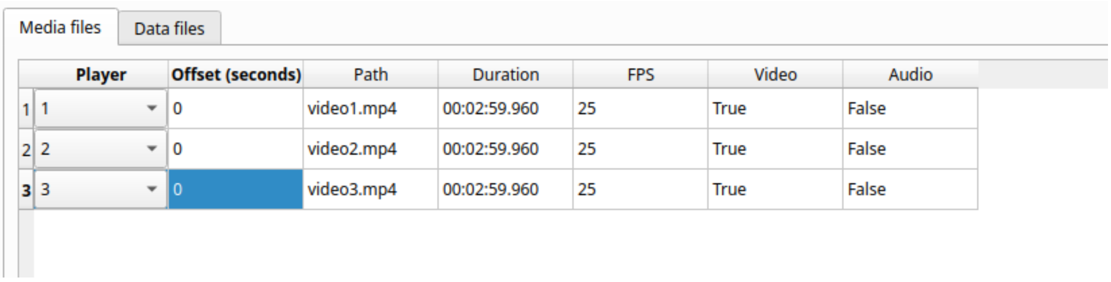
Information about the selected media file will be extracted and displayed in the media list: media file path, media duration, number of frames per second (FPS), presence of a video stream, and presence of an audio stream.



Media files

You can use the media file name as the **Observation id** by clicking the **Use media file name as observation id** button.

The drop-down list in the first column lets you choose a player (up to a maximum of 8). If you want to observe more media files simultaneously, you must use consecutive players starting from 1. See the example below:



	Player	Offset (seconds)	Path	Duration	FPS	Video	Audio
1	1	0	video1.mp4	00:02:59.960	25	True	False
2	2	0	video2.mp4	00:02:59.960	25	True	False
3	3	0	video3.mp4	00:02:59.960	25	True	False

Media files tab

If you need to synchronize two or more videos, you can use the **Offset** column to indicate when the second player should start. For example, if the video loaded in the second player starts 15 seconds after the first video, enter **15** in the **Offset** cell. If the second video starts before the first player, you can enter a negative value in the **Offset** cell.

If you want to play several videos sequentially, select the same player (#1) for all loaded videos. This means that an event occurring at time t~x~ in the media file queued second (e.g., second_video.mp4) in the playlist will be recorded at time t~1~ + t~x~ (where t~1~ is the duration of the first media file, e.g., first_video.mp4).

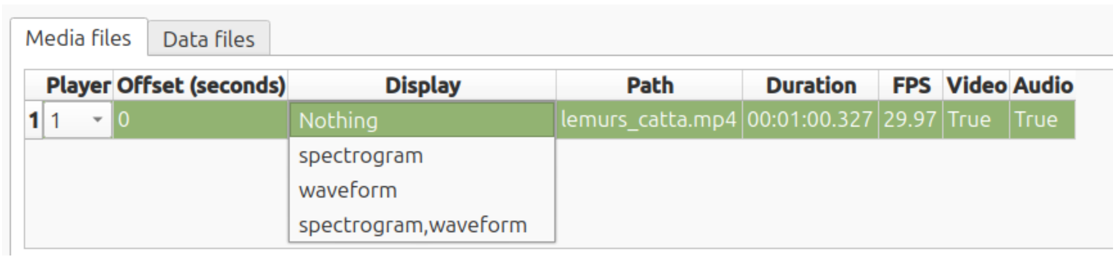
The **Remove selected media** button can be used to remove all selected media files.

BORIS can play all media types supported by the MPV player.

The **Use media file name as observation id** button sets the first media file name as the **observation id**.

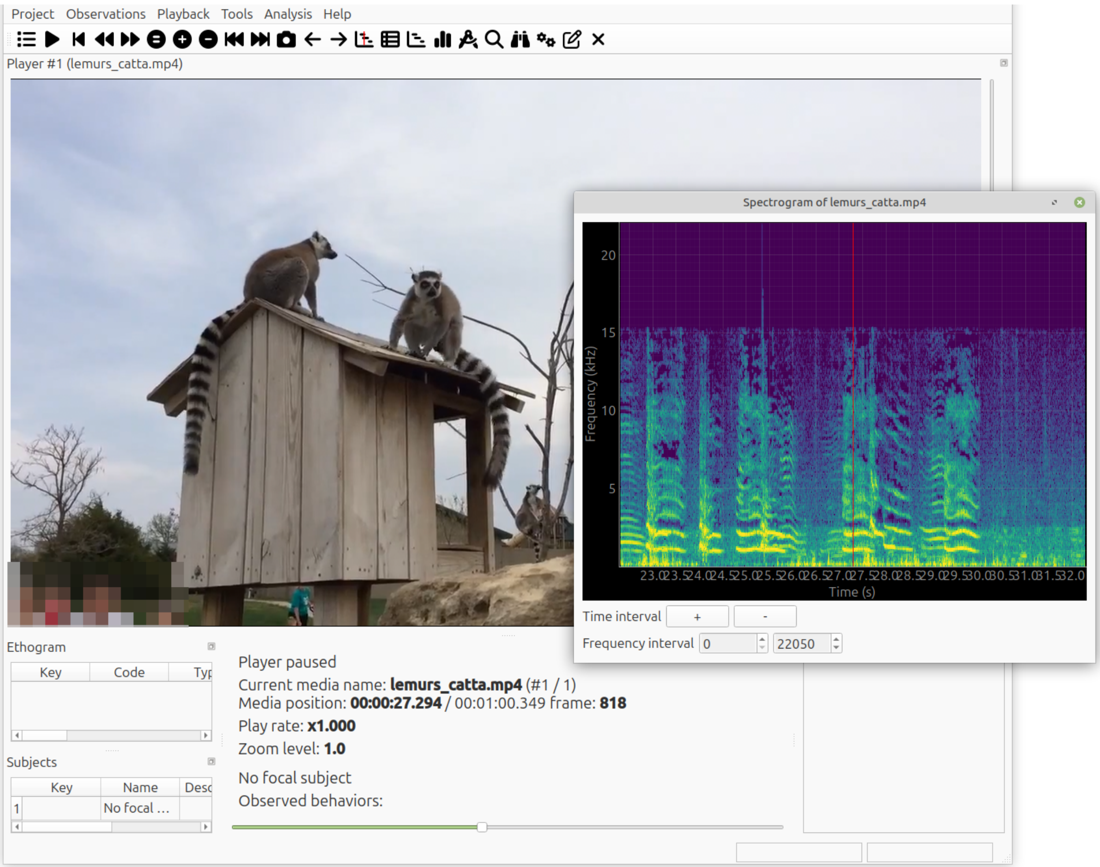
Spectrogram and waveform visualization

BORIS can display audio information, such as the sound spectrogram and/or waveform, during media coding. In the **Display** column, choose the type of plot you want to visualize.

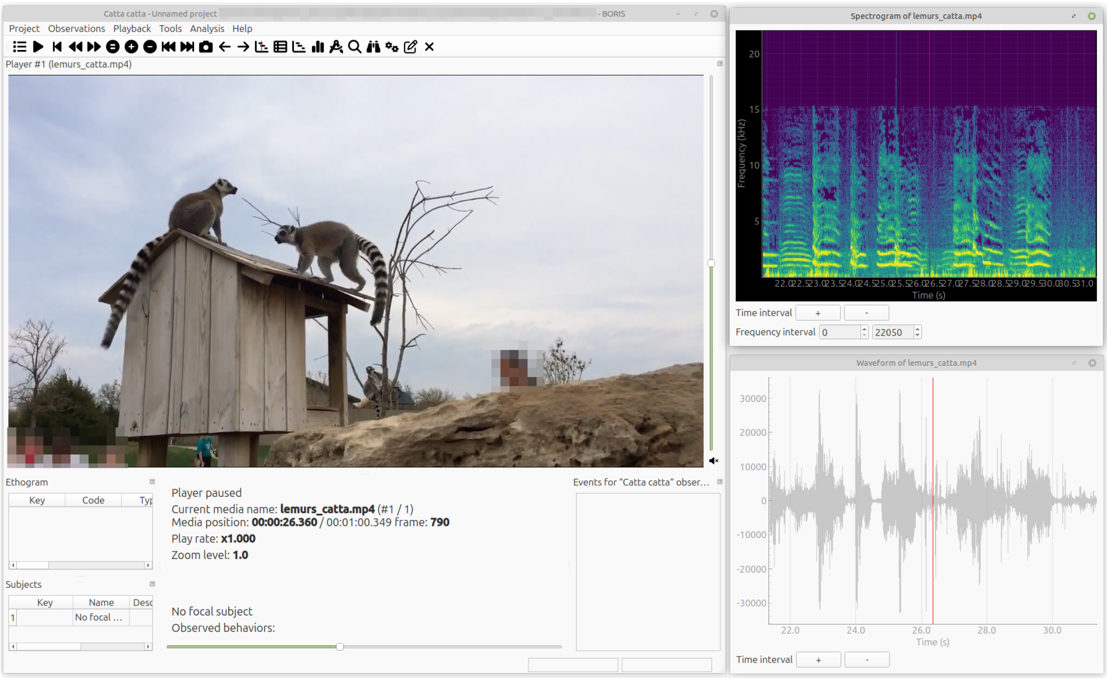


Select display

The spectrogram visualization will be synchronized to the media position during the observation.



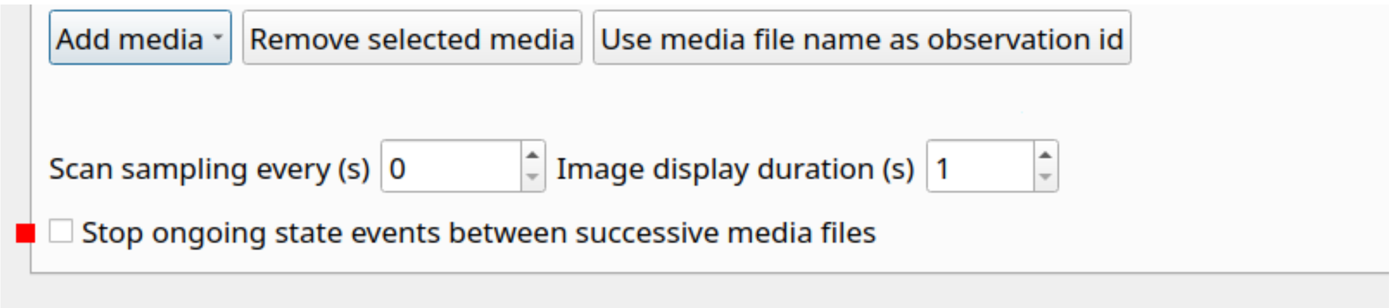
Spectrogram visualization



Spectrogram and waveform visualization

Stop all state events between media files

If your media files are not contiguous, you can enable the **Stop ongoing events between successive media files** option to automatically stop state events between media files. This option can be useful, for example, when coding videos from camera-trap systems.

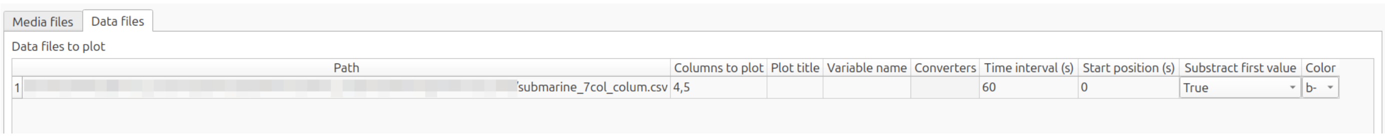


Stop state events between media files

External data files

> **Note**
>
> **At this time, only 2 external data series can be plotted with your media file**

You can select one or more external data files to be plotted synchronously with your media. Click the **Data files** tab and use the **Add data file** button to select a data file.



External data file table

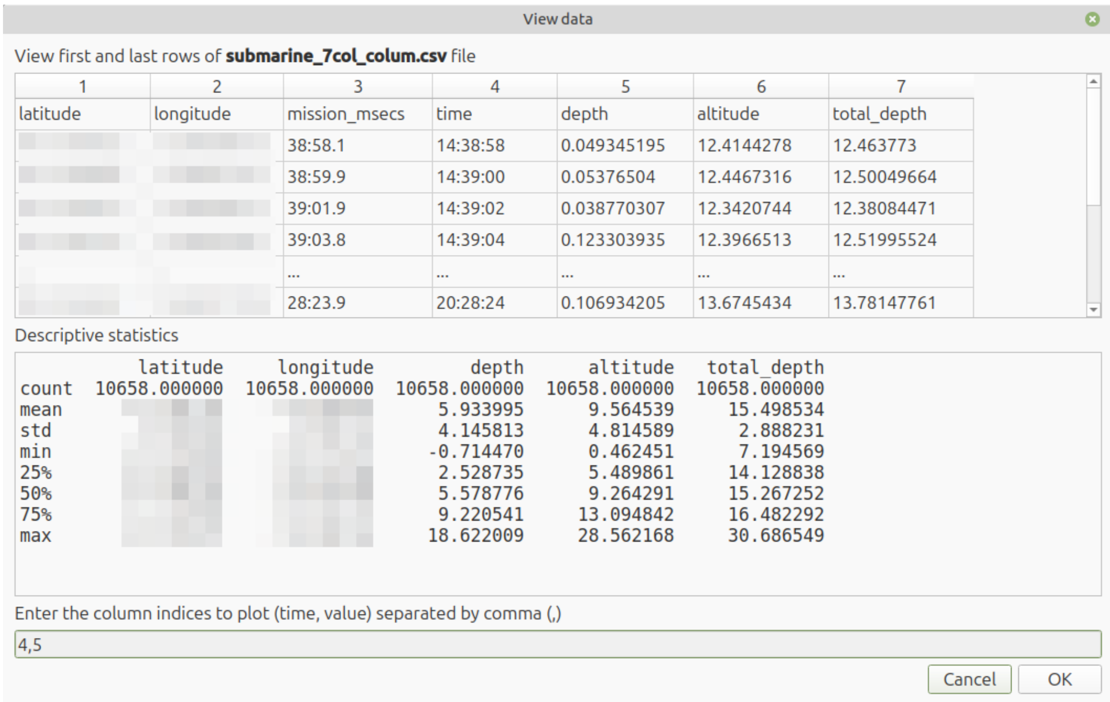
The data files must be plain text files with at least **2 columns** separated by a comma or a TAB character. One column must contain a timestamp that will be used to synchronize the plot with the media. The sampling rate can be variable.

Example of a plain text data file with 5 columns separated by comma (,):

```
Display,X Pos,Y Pos,Start Time (secs),Pupil Diameter
1,864,509,549.233,0.00295773451216519
1,863,505,549.25,0.00281810853630304
1,863,503,549.266,0.00287826382555068
1,861,502,549.283,0.0030536837875843
1,858,501,549.3,0.00308083021081984
1,856,499,549.316,0.00306266942061484
1,854,499,549.333,0.00305776367895305
[...]
```

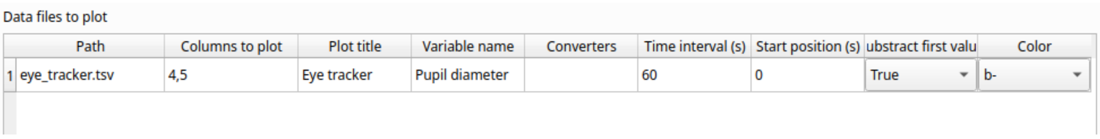
In the above example, the 4th column contains the timestamp and the 5th contains the value to be plotted.

Enter the index of the column containing the timestamp and the index of the column containing the value to be plotted. The two indices must be separated by a comma (,). Click **OK** to close the window.



Selection of columns (time, value)

A new row will be added in the data files table.



Data file table

You can modify or complete the following parameters by typing directly in the table cells:

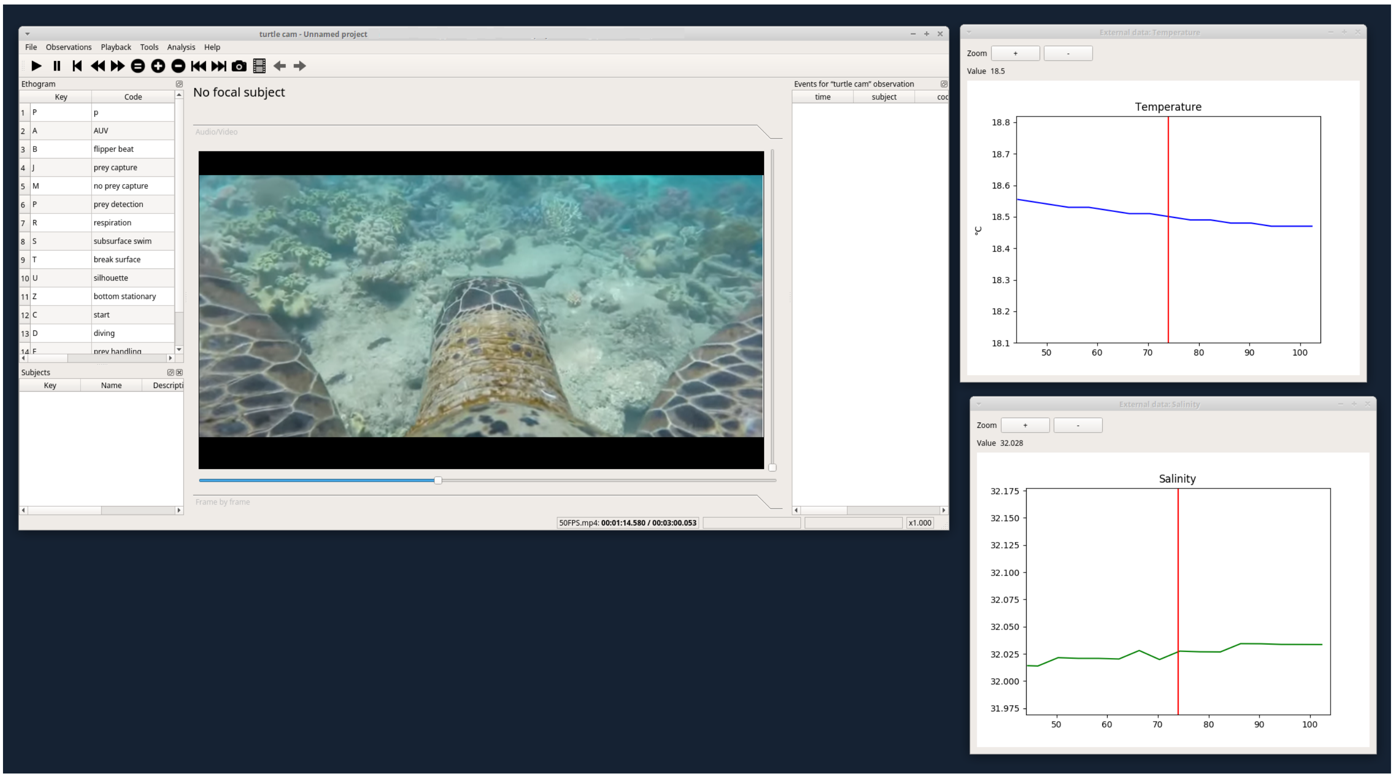
- Columns to plot
- Plot title: the title of the plot
- Variable name
- Converters: Used if the timestamp is not expressed in seconds (see below for details)
- Time interval: The time interval that will be plotted (in seconds)
- Start position: the start position of data for synchronization with the media (in seconds)
- Subtract first value: if the timestamp does not start with a zero value, you can choose to subtract the first value from all timestamp values.
- Color: the color of the plot

NOTE: If you want to record the value of the plotted variable in a modifier of a behavior (see Value from external data file modifier), the modifier must have the same **variable name**.

You can check if the data from the file can be correctly plotted by using the **Show plot** button. If the data are compatible, you will see a plot; otherwise, you will receive a message with an explanation.

Currently, only two values can be plotted synchronously with your media file. The values can come from the same file or from two different files.

During the observation, the values you selected in the external data files will be plotted synchronously with the media file.



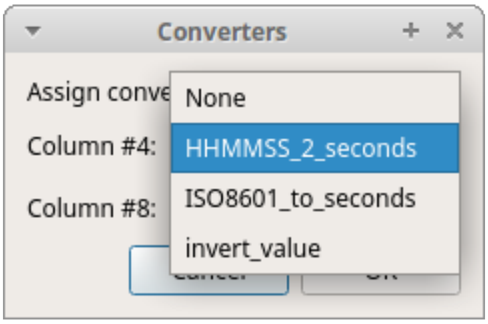
Observation with 2 values plotted from external data files: Temperature and salinity

CONVERTERS

If the values in the timestamp column are not expressed in seconds (e.g., 12.45) but in another format (such as HH:MM:SS, MM:SS, or ISO 8601 like 2018-01-18T12:31:40Z), you must use a converter to transform the current format into seconds.

See the Converters' table in the project configuration.

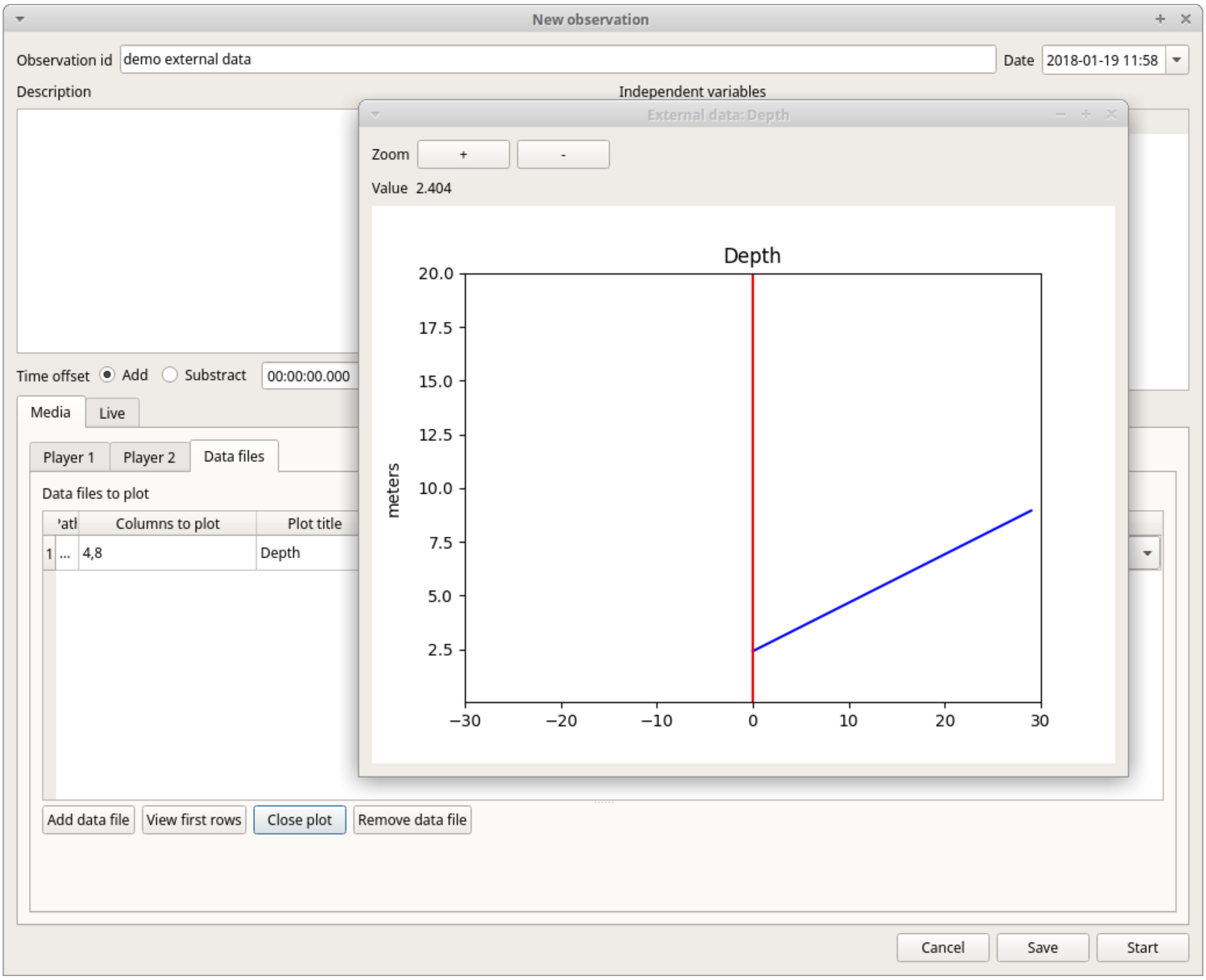
A **double-click** on the converters cell will allow you to select a converter for each column to be plotted.



Data files to plot

	Path	Columns to plot	Plot title	Variable name	Converters	Time interval (s)	Start position (s)
1	/home/user/external_data2.csv	4,8	Depth	meters	{'4': 'HHMMSS_2_seconds'}	60	0

Use the **Show plot** button to verify if your external data can be plotted without issues. The **Close plot** button will close the plot window.



Converters can also be used to convert values that are not time values.

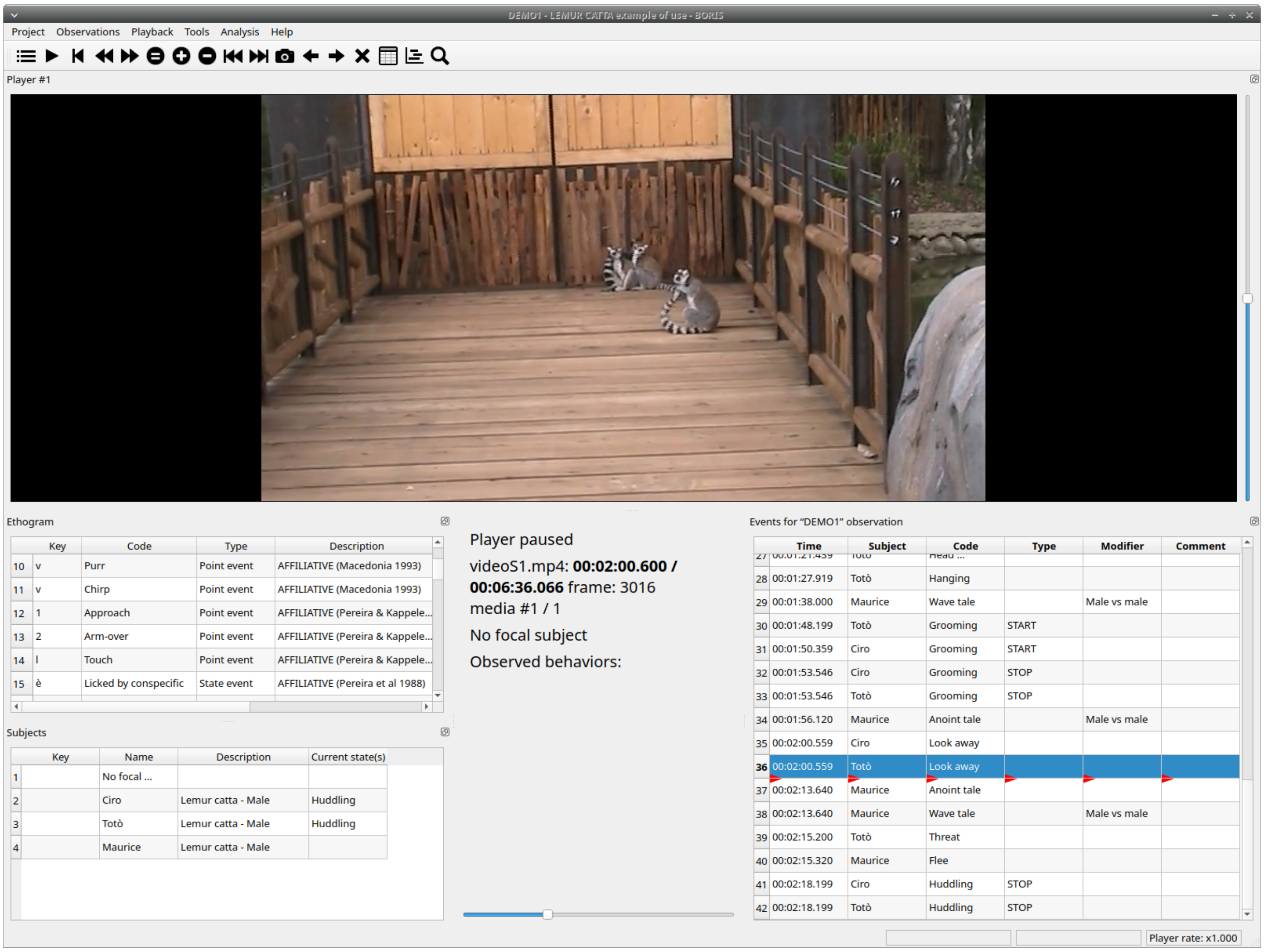
Example of a converter for inverting a value:

```
OUTPUT = - float(INPUT)
```

Start the observation

Click the **Start** button to start coding. The **Observation** window will close, and you'll be transferred to the main **BORIS** window. If you do not want to start the observation, click the **Save** button. The observation will be saved in the observations list.

The main window during the observation of a single media file will look like this:

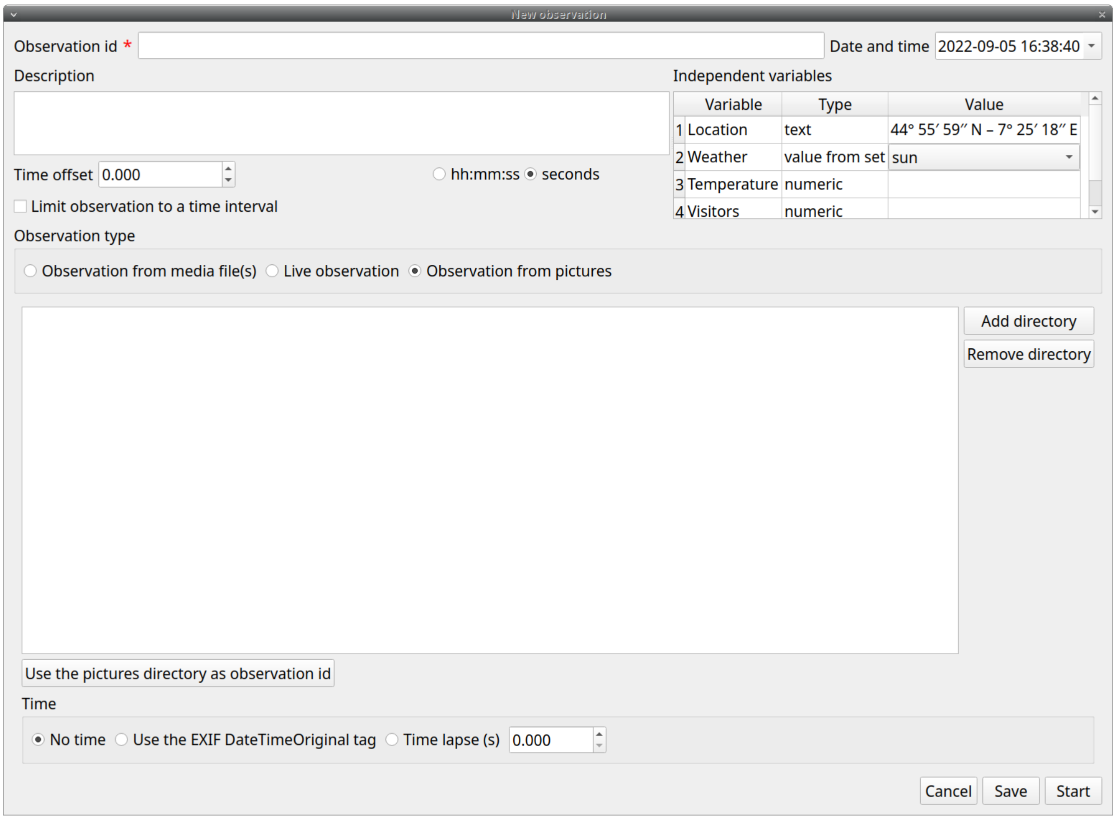


The main window during the observation of one video

See the Media based coding section to start coding.

2.4.3 Observation from pictures

Click on the **Observation from pictures** radio button to create an observation based on pictures.



Observation from pictures tab

Use the **Add directory** button to select a directory containing the pictures you want to code. You can select multiple directories; in this case, the pictures will be browsed in the order the directories were added.

The **Use the pictures directory as observation id** button will set the directory name as the **observation id**.

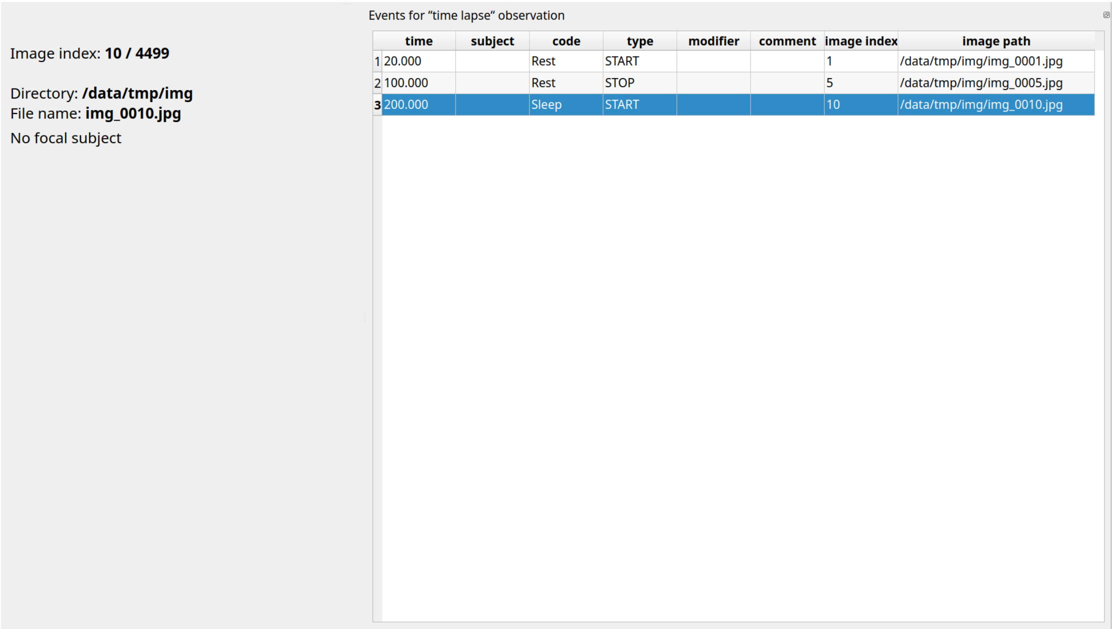
Time

You have three options for the coding time:

- No time: no time will be recorded. The image index (the position of the image in the directory) and the image file path will be recorded.
- Use the EXIF DateTimeOriginal tag: the time will be extracted from the EXIF tag of the picture file (if available).
- Time lapse: this option allows you to define the time interval between pictures.

Start the observation

Click the **Start** button to start coding. The **Observation** window will close, and you'll be transferred to the main **BORIS** window. If you do not want to start the observation, click the **Save** button. The observation will be saved in the observations list.



The main window during the coding of a picture directory

See the Media based coding section to start coding.

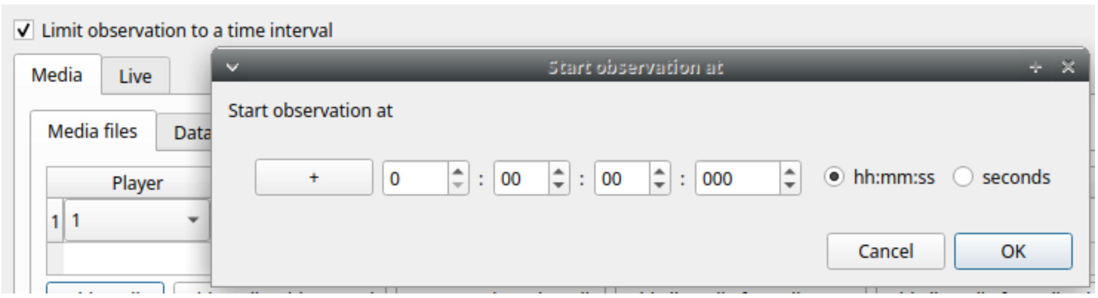
2.4.4 Various options

Scan sampling (Live and media observation)

You can select a time for **Scan sampling** observation. In this case, the timer will stop at every time offset you indicated, and all coded events will have the same time value.

Limit observation to a time interval (Live and media observation)

This option can be used to limit the observation to a time interval for live or media-based observations.



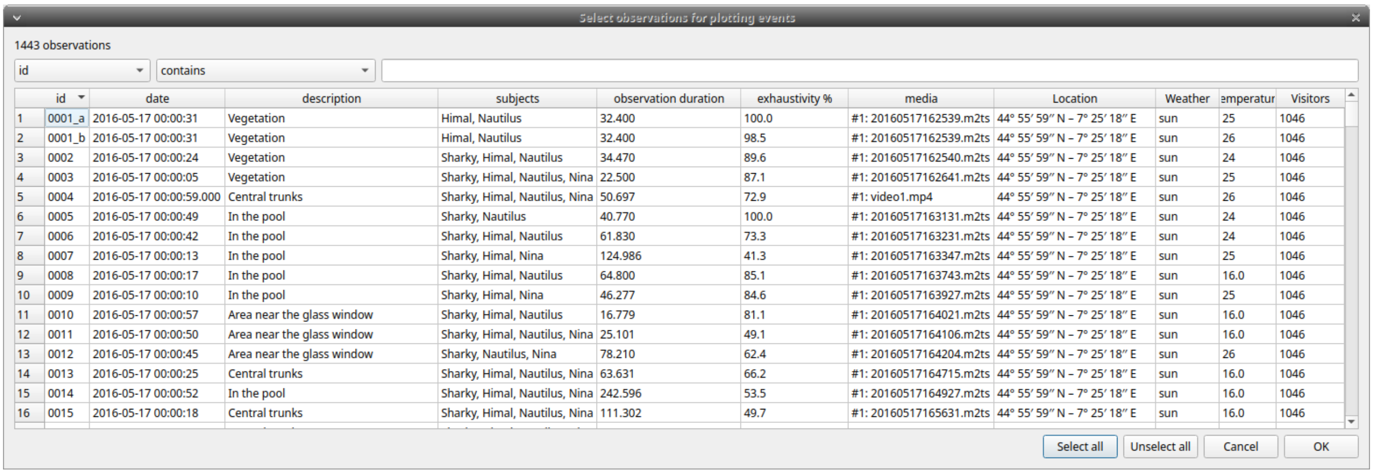
2.5 Managing observations

2.5.1 Observations list

Observations > **Observations list** shows all the observations contained in the current BORIS project.

The following values are displayed:

- The observation id (**id**)
- The **description** of the observation
- The coded subjects (**subjects**)
- The **observation duration** (the difference between the last recorded event and the first one)
- The percentage of **exhaustivity** of the coding (the sum of the length of the coded events divided by the observation duration)
- The **media** file path, **LIVE** for a live observation, or the pictures directory path for an observation based on pictures
- The values of the independent variables (if defined)

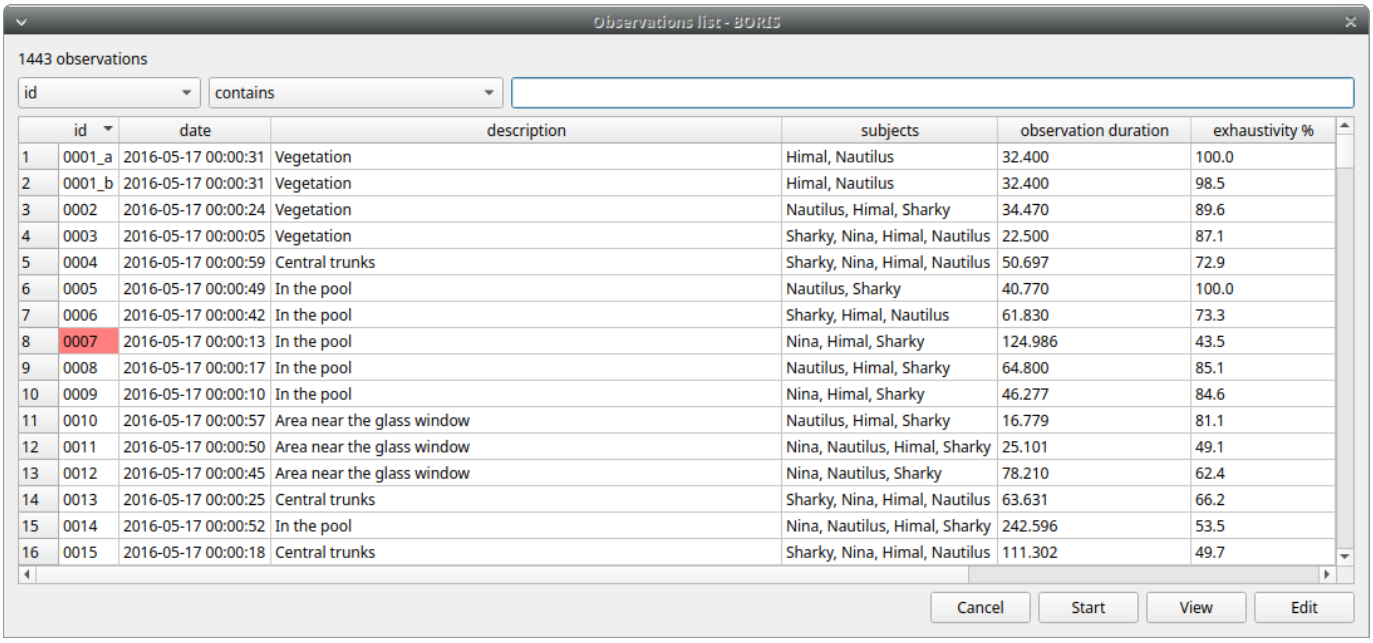


	id	date	description	subjects	observation duration	exhaustivity %	media	Location	Weather	emperatur	Visitors
1	0001_a	2016-05-17 00:00:31	Vegetation	Himal, Nautilus	32.400	100.0	#1: 20160517162539.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	25	1046
2	0001_b	2016-05-17 00:00:31	Vegetation	Himal, Nautilus	32.400	98.5	#1: 20160517162539.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
3	0002	2016-05-17 00:00:24	Vegetation	Sharky, Himal, Nautilus	34.470	89.6	#1: 20160517162540.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24	1046
4	0003	2016-05-17 00:00:05	Vegetation	Sharky, Himal, Nautilus, Nina	22.500	87.1	#1: 20160517162641.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	25	1046
5	0004	2016-05-17 00:00:59.000	Central trunks	Sharky, Himal, Nautilus, Nina	50.697	72.9	#1: video1.mp4	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
6	0005	2016-05-17 00:00:49	In the pool	Sharky, Nautilus	40.770	100.0	#1: 20160517163131.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24	1046
7	0006	2016-05-17 00:00:42	In the pool	Sharky, Himal, Nautilus	61.830	73.3	#1: 20160517163231.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24	1046
8	0007	2016-05-17 00:00:13	In the pool	Sharky, Himal, Nina	124.986	41.3	#1: 20160517163347.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	25	1046
9	0008	2016-05-17 00:00:17	In the pool	Sharky, Himal, Nautilus	64.800	85.1	#1: 20160517163743.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	16.0	1046
10	0009	2016-05-17 00:00:10	In the pool	Sharky, Himal, Nina	46.277	84.6	#1: 20160517163927.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	25	1046
11	0010	2016-05-17 00:00:57	Area near the glass window	Sharky, Himal, Nautilus	16.779	81.1	#1: 20160517164021.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	16.0	1046
12	0011	2016-05-17 00:00:50	Area near the glass window	Sharky, Himal, Nautilus, Nina	25.101	49.1	#1: 20160517164106.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	16.0	1046
13	0012	2016-05-17 00:00:45	Area near the glass window	Sharky, Nautilus, Nina	78.210	62.4	#1: 20160517164204.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
14	0013	2016-05-17 00:00:25	Central trunks	Sharky, Himal, Nautilus, Nina	63.631	66.2	#1: 20160517164715.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	16.0	1046
15	0014	2016-05-17 00:00:52	In the pool	Sharky, Himal, Nautilus, Nina	242.596	53.5	#1: 20160517164927.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	16.0	1046
16	0015	2016-05-17 00:00:18	Central trunks	Sharky, Himal, Nautilus, Nina	111.302	49.7	#1: 20160517165631.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	16.0	1046

The observations can be sorted by clicking the desired column header (alphabetical order, ascending or descending).

Checking the observations

The observation status is displayed in the first column (**id**). If the background of this column is **red**, the observation has one or more UNPAIRED state events. These UNPAIRED observations will not be analyzed. See Fix unpaired state events for details.



Observations list - BORIS

1443 observations

id | contains

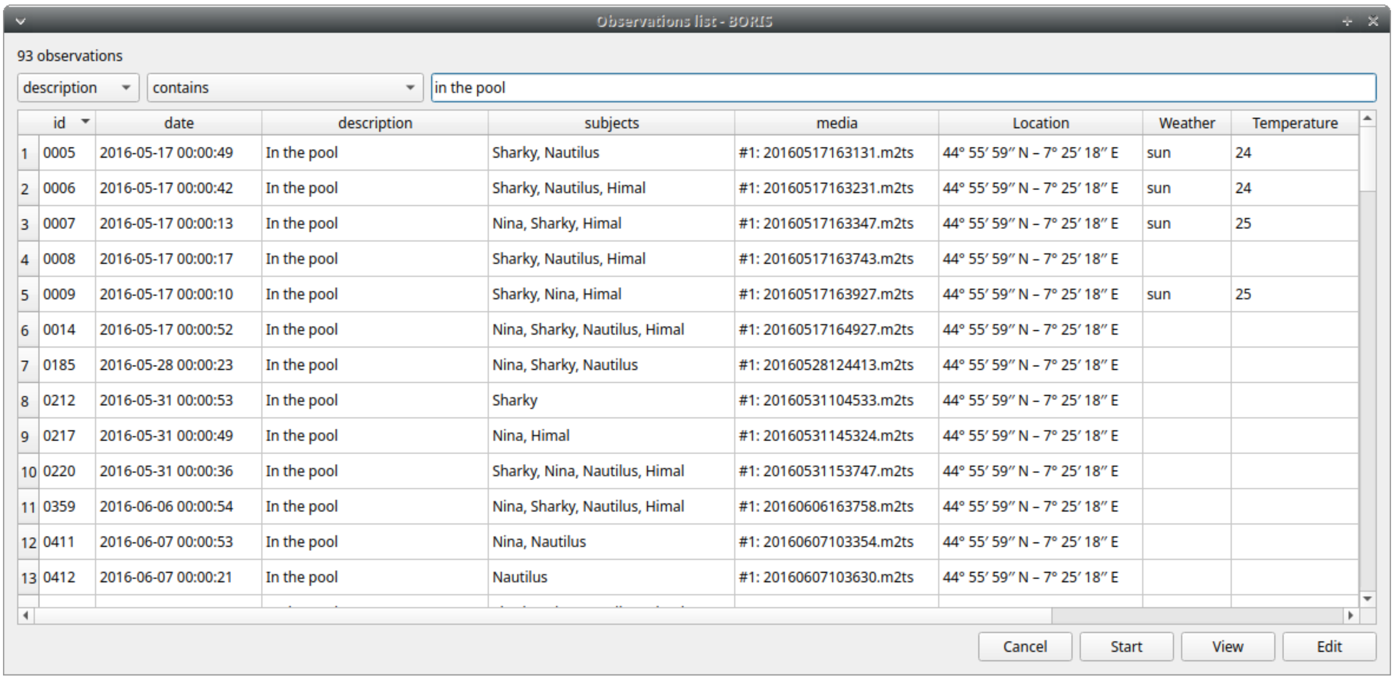
	id	date	description	subjects	observation duration	exhaustivity %
1	0001_a	2016-05-17 00:00:31	Vegetation	Himal, Nautilus	32.400	100.0
2	0001_b	2016-05-17 00:00:31	Vegetation	Himal, Nautilus	32.400	98.5
3	0002	2016-05-17 00:00:24	Vegetation	Nautilus, Himal, Sharky	34.470	89.6
4	0003	2016-05-17 00:00:05	Vegetation	Sharky, Nina, Himal, Nautilus	22.500	87.1
5	0004	2016-05-17 00:00:59	Central trunks	Sharky, Nina, Himal, Nautilus	50.697	72.9
6	0005	2016-05-17 00:00:49	In the pool	Nautilus, Sharky	40.770	100.0
7	0006	2016-05-17 00:00:42	In the pool	Sharky, Himal, Nautilus	61.830	73.3
8	0007	2016-05-17 00:00:13	In the pool	Nina, Himal, Sharky	124.986	43.5
9	0008	2016-05-17 00:00:17	In the pool	Nautilus, Himal, Sharky	64.800	85.1
10	0009	2016-05-17 00:00:10	In the pool	Nina, Himal, Sharky	46.277	84.6
11	0010	2016-05-17 00:00:57	Area near the glass window	Nautilus, Himal, Sharky	16.779	81.1
12	0011	2016-05-17 00:00:50	Area near the glass window	Nina, Nautilus, Himal, Sharky	25.101	49.1
13	0012	2016-05-17 00:00:45	Area near the glass window	Nina, Nautilus, Sharky	78.210	62.4
14	0013	2016-05-17 00:00:25	Central trunks	Sharky, Nina, Himal, Nautilus	63.631	66.2
15	0014	2016-05-17 00:00:52	In the pool	Nina, Nautilus, Himal, Sharky	242.596	53.5
16	0015	2016-05-17 00:00:18	Central trunks	Sharky, Nina, Himal, Nautilus	111.302	49.7

Cancel | Start | View | Edit

Filtering the observations

The observations list can be filtered by selecting a field and a condition from the drop-down lists.

In the following example, observations are filtered: only observations with **description** containing **In the pool** are shown:



Observations list - BORIS

93 observations

description | contains | in the pool

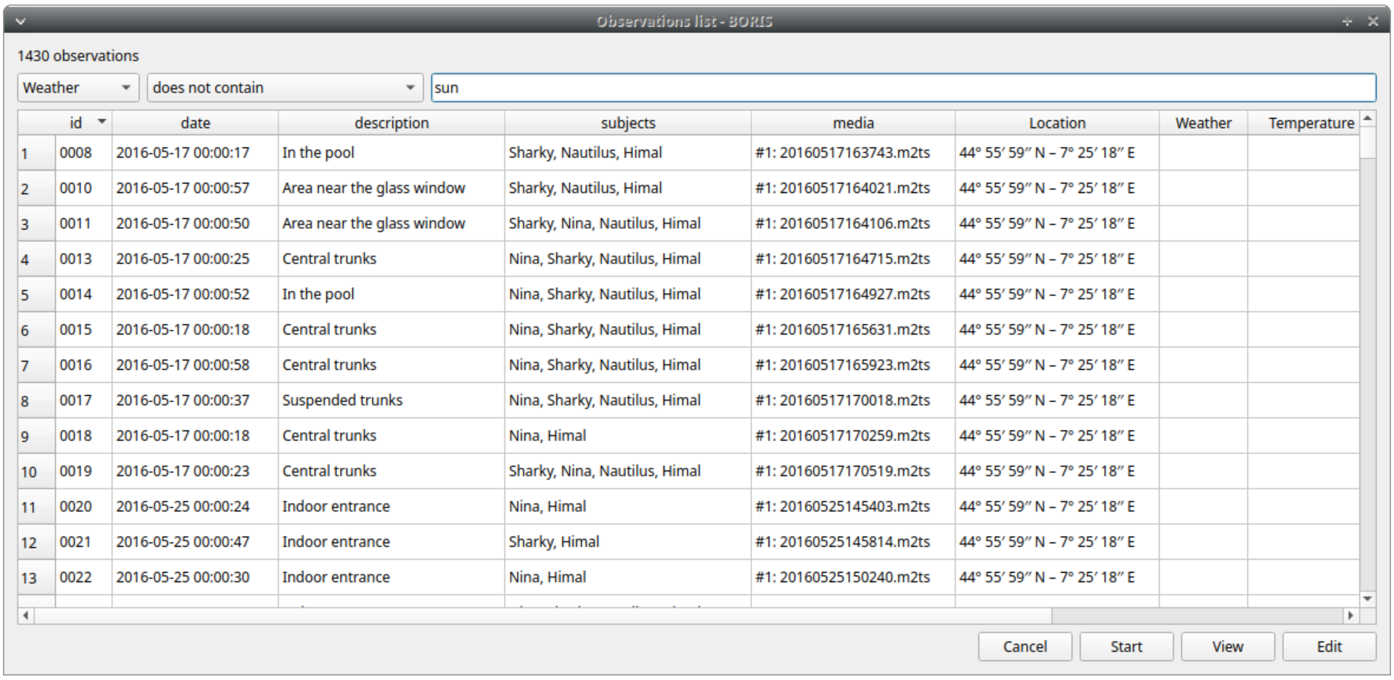
	id	date	description	subjects	media	Location	Weather	Temperature
1	0005	2016-05-17 00:00:49	In the pool	Sharky, Nautilus	#1: 20160517163131.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24
2	0006	2016-05-17 00:00:42	In the pool	Sharky, Nautilus, Himal	#1: 20160517163231.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24
3	0007	2016-05-17 00:00:13	In the pool	Nina, Sharky, Himal	#1: 20160517163347.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	25
4	0008	2016-05-17 00:00:17	In the pool	Sharky, Nautilus, Himal	#1: 20160517163743.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
5	0009	2016-05-17 00:00:10	In the pool	Sharky, Nina, Himal	#1: 20160517163927.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	25
6	0014	2016-05-17 00:00:52	In the pool	Nina, Sharky, Nautilus, Himal	#1: 20160517164927.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
7	0185	2016-05-28 00:00:23	In the pool	Nina, Sharky, Nautilus	#1: 20160528124413.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
8	0212	2016-05-31 00:00:53	In the pool	Sharky	#1: 20160531104533.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
9	0217	2016-05-31 00:00:49	In the pool	Nina, Himal	#1: 20160531145324.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
10	0220	2016-05-31 00:00:36	In the pool	Sharky, Nina, Nautilus, Himal	#1: 20160531153747.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
11	0359	2016-06-06 00:00:54	In the pool	Nina, Sharky, Nautilus, Himal	#1: 20160606163758.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
12	0411	2016-06-07 00:00:53	In the pool	Nina, Nautilus	#1: 20160607103354.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
13	0412	2016-06-07 00:00:21	In the pool	Nautilus	#1: 20160607103630.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		

Cancel | Start | View | Edit

Observations list

Observations can be filtered by **Independent variables** values.

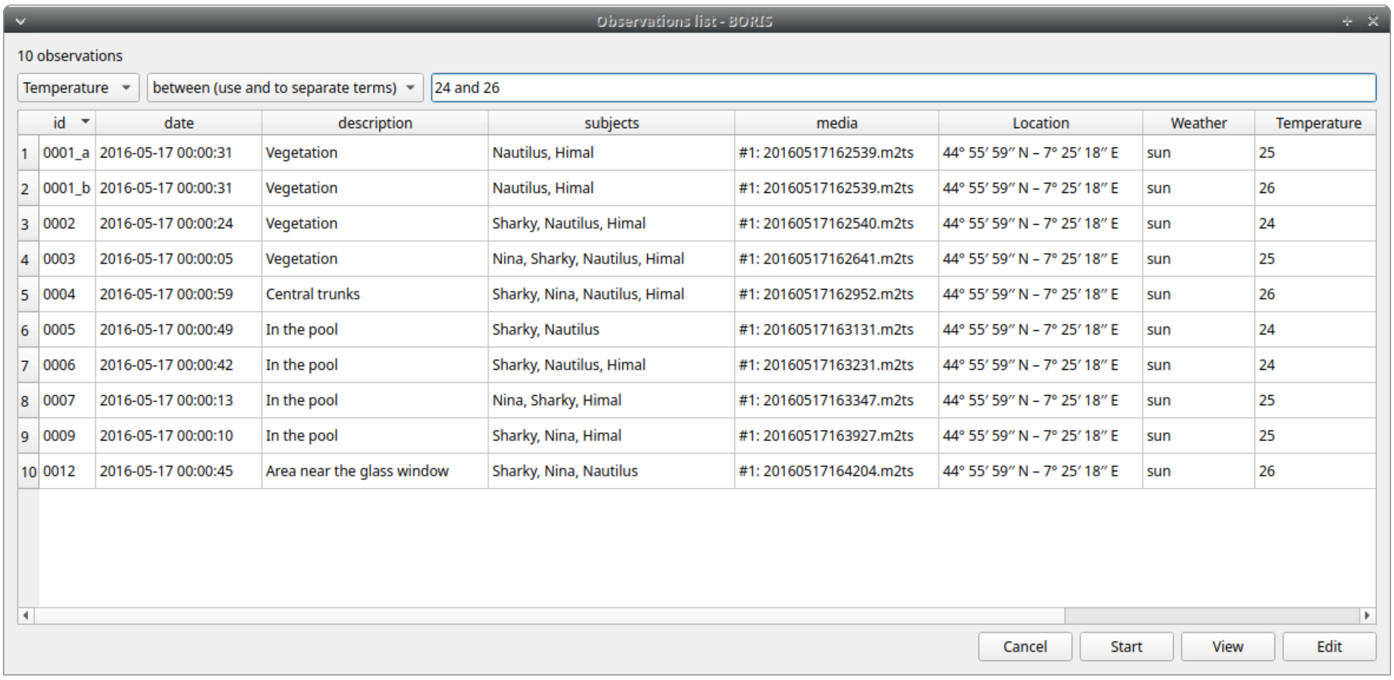
The following example displays only the observations that do not contain "Sunny" in the **Weather** independent variable:



	id	date	description	subjects	media	Location	Weather	Temperature
1	0008	2016-05-17 00:00:17	In the pool	Sharky, Nautilus, Himal	#1: 20160517163743.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
2	0010	2016-05-17 00:00:57	Area near the glass window	Sharky, Nautilus, Himal	#1: 20160517164021.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
3	0011	2016-05-17 00:00:50	Area near the glass window	Sharky, Nina, Nautilus, Himal	#1: 20160517164106.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
4	0013	2016-05-17 00:00:25	Central trunks	Nina, Sharky, Nautilus, Himal	#1: 20160517164715.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
5	0014	2016-05-17 00:00:52	In the pool	Nina, Sharky, Nautilus, Himal	#1: 20160517164927.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
6	0015	2016-05-17 00:00:18	Central trunks	Nina, Sharky, Nautilus, Himal	#1: 20160517165631.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
7	0016	2016-05-17 00:00:58	Central trunks	Nina, Sharky, Nautilus, Himal	#1: 20160517165923.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
8	0017	2016-05-17 00:00:37	Suspended trunks	Nina, Sharky, Nautilus, Himal	#1: 20160517170018.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
9	0018	2016-05-17 00:00:18	Central trunks	Nina, Himal	#1: 20160517170259.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
10	0019	2016-05-17 00:00:23	Central trunks	Sharky, Nina, Nautilus, Himal	#1: 20160517170519.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
11	0020	2016-05-25 00:00:24	Indoor entrance	Nina, Himal	#1: 20160525145403.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
12	0021	2016-05-25 00:00:47	Indoor entrance	Sharky, Himal	#1: 20160525145814.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		
13	0022	2016-05-25 00:00:30	Indoor entrance	Nina, Himal	#1: 20160525150240.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E		

Observations list

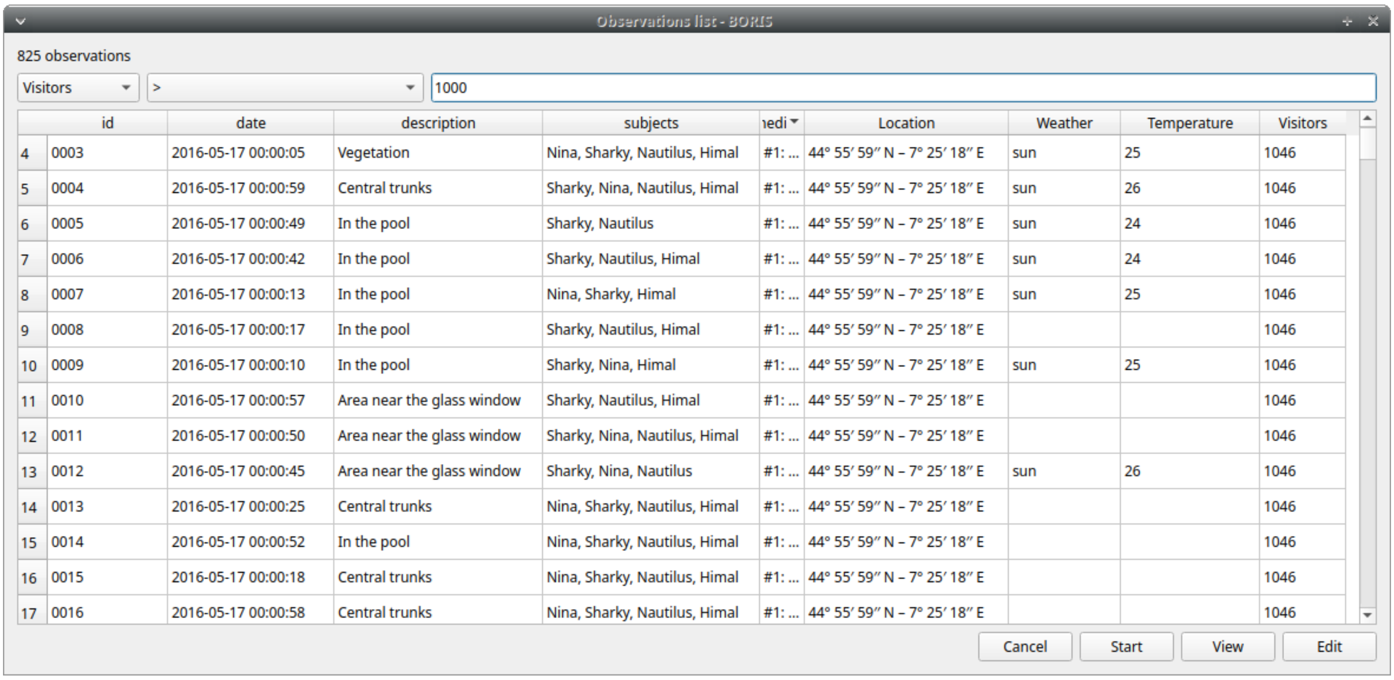
Observations with a **Temperature** independent variable value between 18 and 22:



	id	date	description	subjects	media	Location	Weather	Temperature
1	0001_a	2016-05-17 00:00:31	Vegetation	Nautilus, Himal	#1: 20160517162539.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	25
2	0001_b	2016-05-17 00:00:31	Vegetation	Nautilus, Himal	#1: 20160517162539.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26
3	0002	2016-05-17 00:00:24	Vegetation	Sharky, Nautilus, Himal	#1: 20160517162540.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24
4	0003	2016-05-17 00:00:05	Vegetation	Nina, Sharky, Nautilus, Himal	#1: 20160517162641.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	25
5	0004	2016-05-17 00:00:59	Central trunks	Sharky, Nina, Nautilus, Himal	#1: 20160517162952.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26
6	0005	2016-05-17 00:00:49	In the pool	Sharky, Nautilus	#1: 20160517163131.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24
7	0006	2016-05-17 00:00:42	In the pool	Sharky, Nautilus, Himal	#1: 20160517163231.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24
8	0007	2016-05-17 00:00:13	In the pool	Nina, Sharky, Himal	#1: 20160517163347.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	25
9	0009	2016-05-17 00:00:10	In the pool	Sharky, Nina, Himal	#1: 20160517163927.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	25
10	0012	2016-05-17 00:00:45	Area near the glass window	Sharky, Nina, Nautilus	#1: 20160517164204.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26

Observations list

Observations with a **Visitors** independent variable value greater than 1000:



Observations list - BORIS

825 observations

Visitors | > | 1000

	id	date	description	subjects	medi	Location	Weather	Temperature	Visitors
4	0003	2016-05-17 00:00:05	Vegetation	Nina, Sharky, Nautilus, Himal	#1: ...	44° 55′ 59″ N – 7° 25′ 18″ E	sun	25	1046
5	0004	2016-05-17 00:00:59	Central trunks	Sharky, Nina, Nautilus, Himal	#1: ...	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
6	0005	2016-05-17 00:00:49	In the pool	Sharky, Nautilus	#1: ...	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24	1046
7	0006	2016-05-17 00:00:42	In the pool	Sharky, Nautilus, Himal	#1: ...	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24	1046
8	0007	2016-05-17 00:00:13	In the pool	Nina, Sharky, Himal	#1: ...	44° 55′ 59″ N – 7° 25′ 18″ E	sun	25	1046
9	0008	2016-05-17 00:00:17	In the pool	Sharky, Nautilus, Himal	#1: ...	44° 55′ 59″ N – 7° 25′ 18″ E			1046
10	0009	2016-05-17 00:00:10	In the pool	Sharky, Nina, Himal	#1: ...	44° 55′ 59″ N – 7° 25′ 18″ E	sun	25	1046
11	0010	2016-05-17 00:00:57	Area near the glass window	Sharky, Nautilus, Himal	#1: ...	44° 55′ 59″ N – 7° 25′ 18″ E			1046
12	0011	2016-05-17 00:00:50	Area near the glass window	Sharky, Nina, Nautilus, Himal	#1: ...	44° 55′ 59″ N – 7° 25′ 18″ E			1046
13	0012	2016-05-17 00:00:45	Area near the glass window	Sharky, Nina, Nautilus	#1: ...	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
14	0013	2016-05-17 00:00:25	Central trunks	Nina, Sharky, Nautilus, Himal	#1: ...	44° 55′ 59″ N – 7° 25′ 18″ E			1046
15	0014	2016-05-17 00:00:52	In the pool	Nina, Sharky, Nautilus, Himal	#1: ...	44° 55′ 59″ N – 7° 25′ 18″ E			1046
16	0015	2016-05-17 00:00:18	Central trunks	Nina, Sharky, Nautilus, Himal	#1: ...	44° 55′ 59″ N – 7° 25′ 18″ E			1046
17	0016	2016-05-17 00:00:58	Central trunks	Nina, Sharky, Nautilus, Himal	#1: ...	44° 55′ 59″ N – 7° 25′ 18″ E			1046

Cancel | Start | View | Edit

Observations list

2.5.2 Delete observations from project

Observations can be deleted from the project using the following procedure:

Observations > **Remove observations**

Select the observations you want to delete.

Click the **OK** button and confirm the deletion.

Deletion is irreversible; deleted observations cannot be restored.

It is a good idea to back up your project before removing observations.

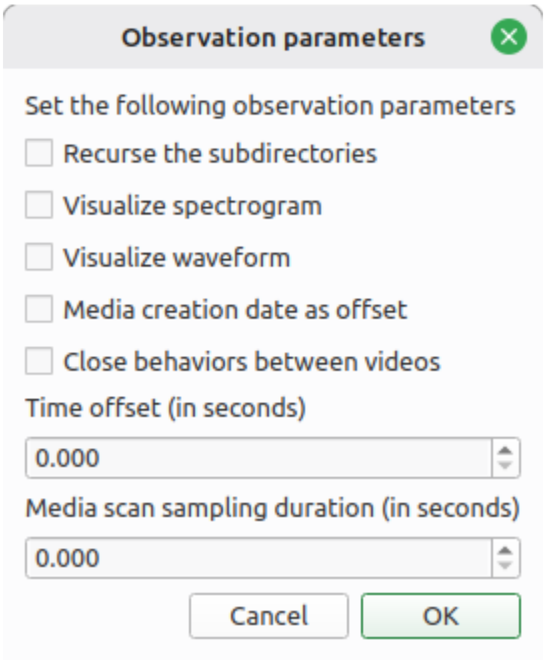
2.5.3 Create observations in bulk

Observations from media files can be created from a directory of media files:

Observations > **Create observations**

Choose the directory.

Select the parameters.



Parameters for creating observations

The ID of each created observation will be the path to the media file.

2.5.4 Import observations

The **Observations** > **Import observations** option allows you to import observations. Two formats are available for importing observations:

From a BORIS project file

Choose the BORIS project file and then the observations to import. BORIS will check whether observations with the same ID already exist in the current project. BORIS will also check if the behaviors and/or subjects used in the imported observations are defined in the current project.

From a spreadsheet file

Observations can be imported from:

- OpenDocument (ODS)
- Microsoft Excel (XLSX)

Choose the spreadsheet file.

2.5.5 Export a list of observations

This option allows you to export the selected observations as a list in various formats (CSV, TSV, ODS, XLSX, HTML):

Observations > **Export observations list**

The data frame will contain the following columns:

- Observation id
- Date
- Description
- Subjects
- Media files/Live observation
- Independent variables

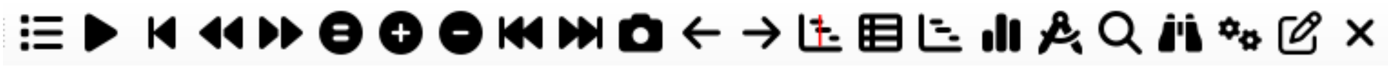
2.6 Coding

When looking at the BORIS main window, the window title bar shows the **Observation id** - **Project name** - **BORIS**. The media (the first in the queue) will be loaded in the media player and paused.

On macOS, the video is not embedded in the BORIS main window, so it is **VERY IMPORTANT to click on the BORIS main window** before coding.

2.6.1 Media based coding

The toolbar



The BORIS toolbar

List of observations

Play (becomes **Pause** when media is playing)

Rewind resets the media to the beginning

Fast backward jumps back by *n* seconds in the media (see preferences)

Fast forward jumps forward by *n* seconds in the media (see preferences)

Set the playback speed to 1x

Increase the playback speed (See preferences)

Decrease the playback speed (See preferences)

Jump to the previous media file

Jump to the next media file

Take a snapshot of current video or frame

Move one frame back

Move one frame forward

Real-time plot of events

Time budget of the current observation

Plot events of the current observation

Plot the time budget of the current observation

Geometric measurements

Find in events

Explore project

Preferences

Edit current observation

Close current observation

The toolbar can be resized in **Preferences** > **Interface**.

Media can also be controlled with dedicated keyboard shortcuts:

`⇞ Page Up` Switch to the next media file

`⇟ Page Down` Switch to the previous media file

`↑ Up` Jump forward in the current media

`↓ Down` Jump backward in the current media

`⤒ Home` Increase the playback speed (see Observations preferences to set the step value)

`⤓ End` Decrease the playback speed (see Observations preferences to set the step value)

`⌫ Backspace` Set the playback speed to 1x

`← Left` Go to the previous frame

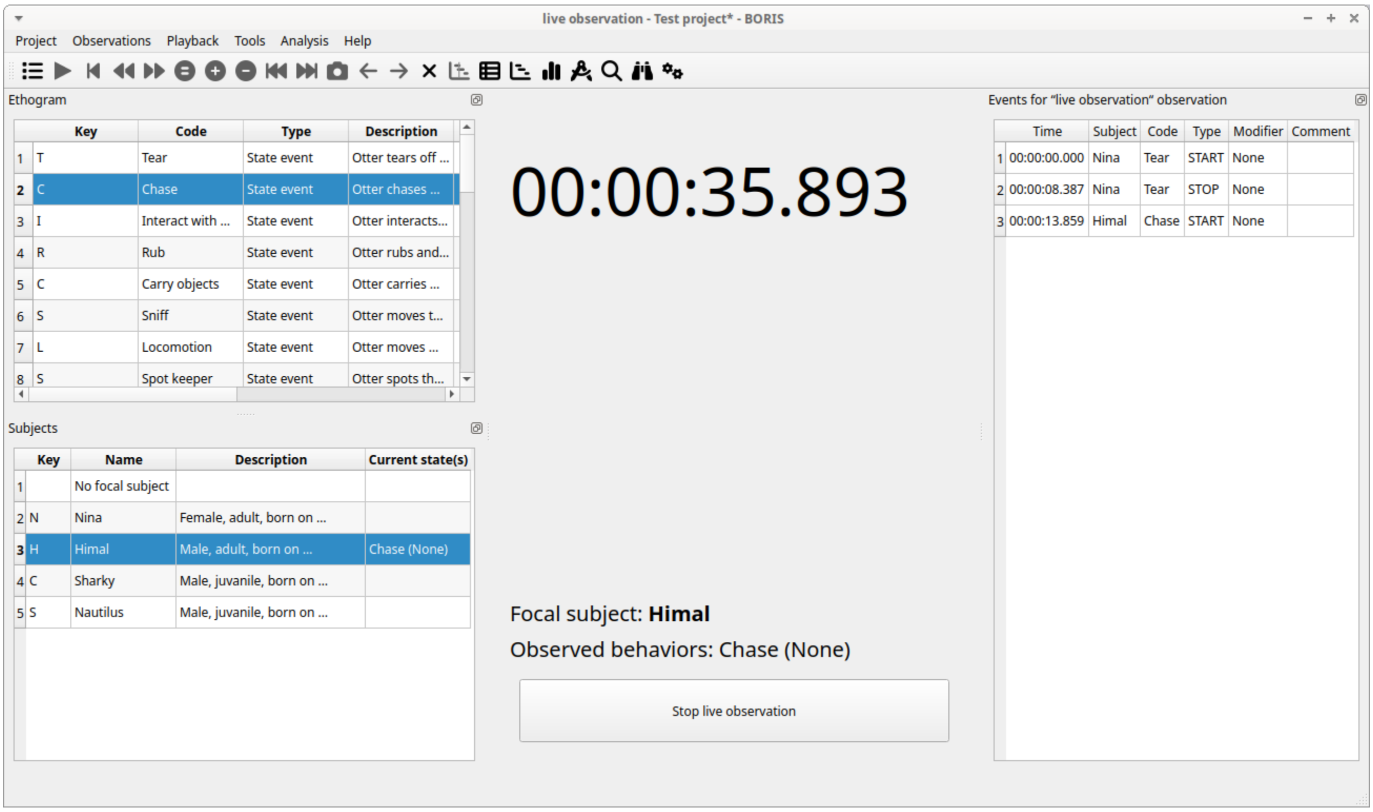
`→ Right` Go to the next frame

2.6.2 Live observations

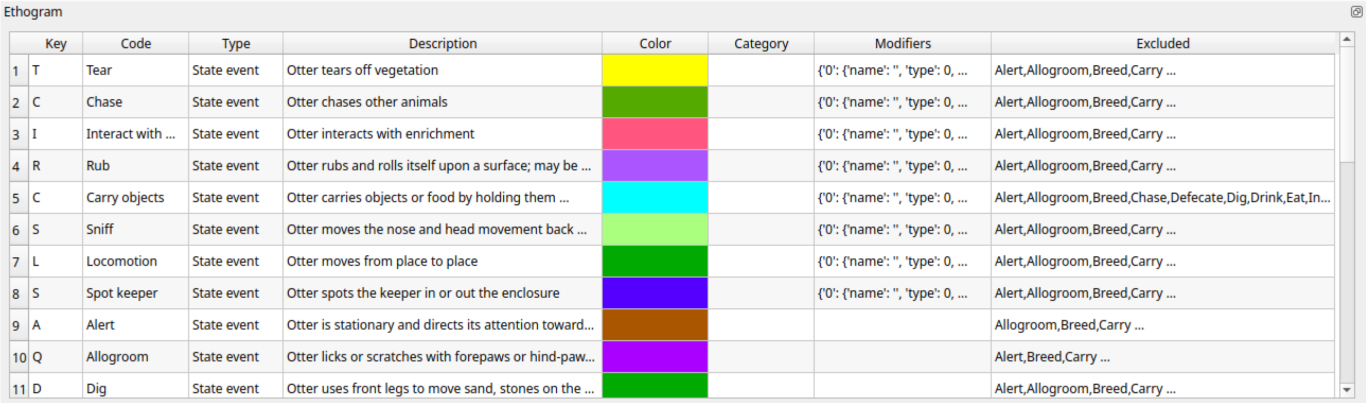
During a live observation, the media control toolbar is disabled.

Press the **Start live observation** button to start your observation. If some events have already been coded, BORIS will ask whether they should be deleted.

A timer will be displayed. Events will be recorded in the events widget.



2.6.3 **Ethogram** table in the main window



Ethogram

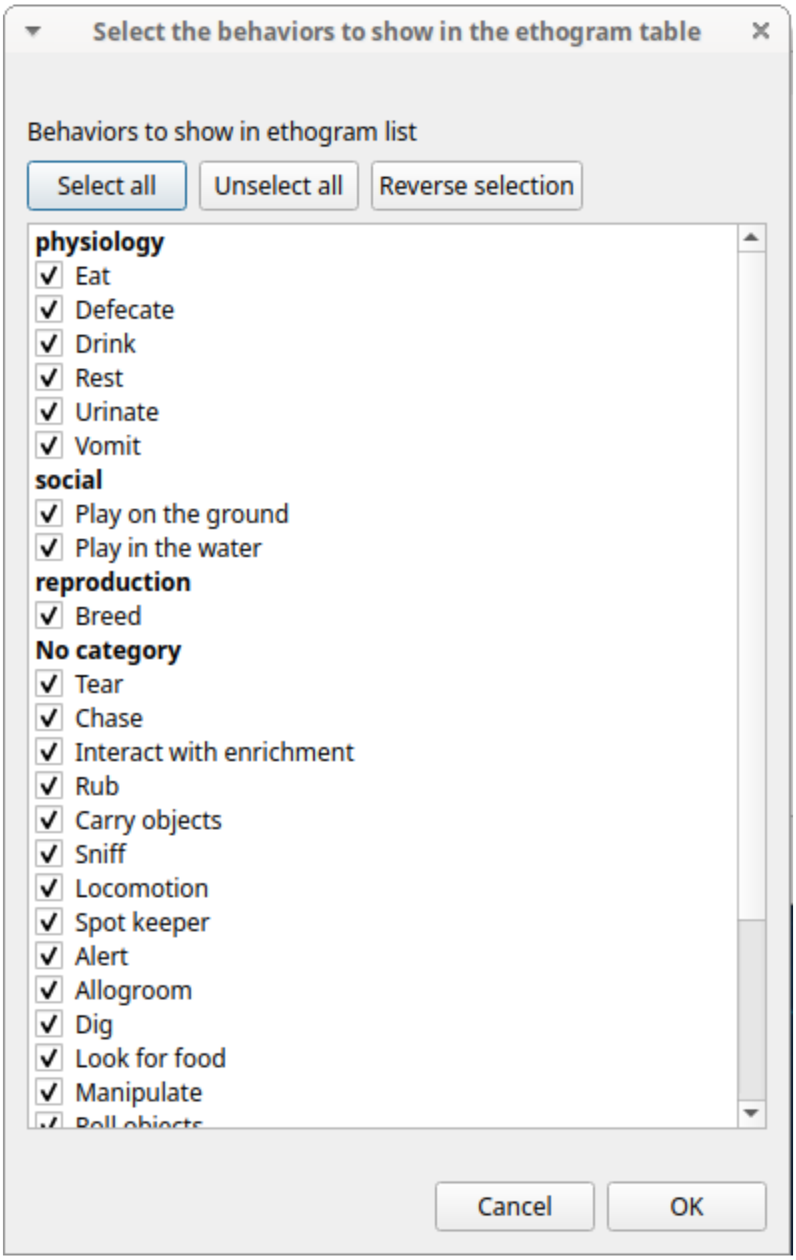
	Key	Code	Type	Description	Color	Category	Modifiers	Excluded
1	T	Tear	State event	Otter tears off vegetation			{'0': {'name': '', 'type': 0, ...	Alert,Allogroom,Breed,Carry ...
2	C	Chase	State event	Otter chases other animals			{'0': {'name': '', 'type': 0, ...	Alert,Allogroom,Breed,Carry ...
3	I	Interact with ...	State event	Otter interacts with enrichment			{'0': {'name': '', 'type': 0, ...	Alert,Allogroom,Breed,Carry ...
4	R	Rub	State event	Otter rubs and rolls itself upon a surface; may be ...			{'0': {'name': '', 'type': 0, ...	Alert,Allogroom,Breed,Carry ...
5	C	Carry objects	State event	Otter carries objects or food by holding them ...			{'0': {'name': '', 'type': 0, ...	Alert,Allogroom,Breed,Chase,Defecate,Dig,Drink,Eat,In...
6	S	Sniff	State event	Otter moves the nose and head movement back ...			{'0': {'name': '', 'type': 0, ...	Alert,Allogroom,Breed,Carry ...
7	L	Locomotion	State event	Otter moves from place to place			{'0': {'name': '', 'type': 0, ...	Alert,Allogroom,Breed,Carry ...
8	S	Spot keeper	State event	Otter spots the keeper in or out the enclosure			{'0': {'name': '', 'type': 0, ...	Alert,Allogroom,Breed,Carry ...
9	A	Alert	State event	Otter is stationary and directs its attention toward...				Allogroom,Breed,Carry ...
10	Q	Allogroom	State event	Otter licks or scratches with forepaws or hind-paw...				Alert,Breed,Carry ...
11	D	Dig	State event	Otter uses front legs to move sand, stones on the ...				Alert,Allogroom,Breed,Carry ...

The **Ethogram** table provides the list of behaviors defined in the **Ethogram**. It can be used to record an event by double-clicking the corresponding row. The **Key** column indicates the keyboard shortcut assigned to each behavior, if any. Pressing a key records the corresponding behavior, which then appears in the **Events** table.

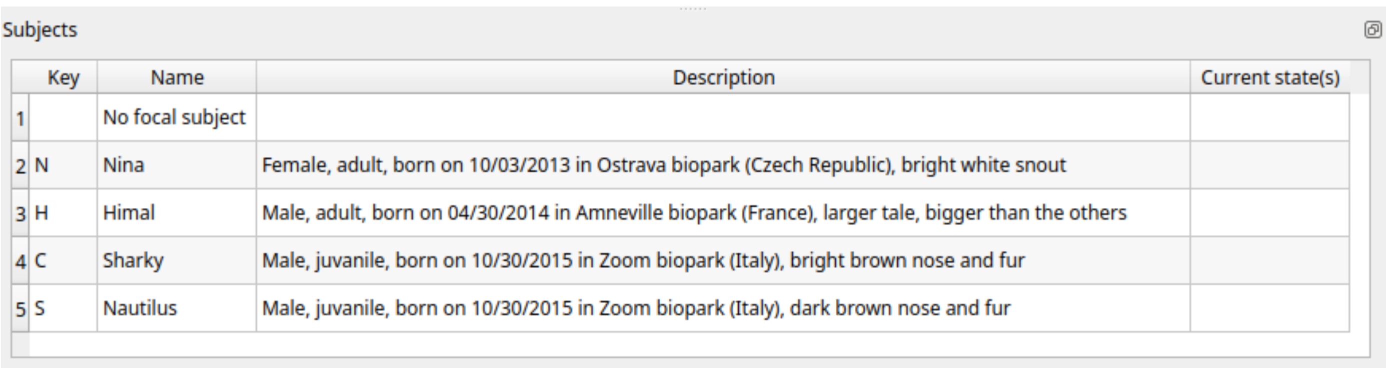
The behaviors shown in the Ethogram widget can be filtered:

Right-click on the ethogram widget > **Filter behaviors**

Check or uncheck individual behaviors, or double-click a behavioral category.



2.6.4 **Subjects** table in the main window


Subjects

	Key	Name	Description	Current state(s)
1		No focal subject		
2	N	Nina	Female, adult, born on 10/03/2013 in Ostrava biopark (Czech Republic), bright white snout	
3	H	Himal	Male, adult, born on 04/30/2014 in Amneville biopark (France), larger tale, bigger than the others	
4	C	Sharky	Male, juvanile, born on 10/30/2015 in Zoom biopark (Italy), bright brown nose and fur	
5	S	Nautilus	Male, juvanile, born on 10/30/2015 in Zoom biopark (Italy), dark brown nose and fur	

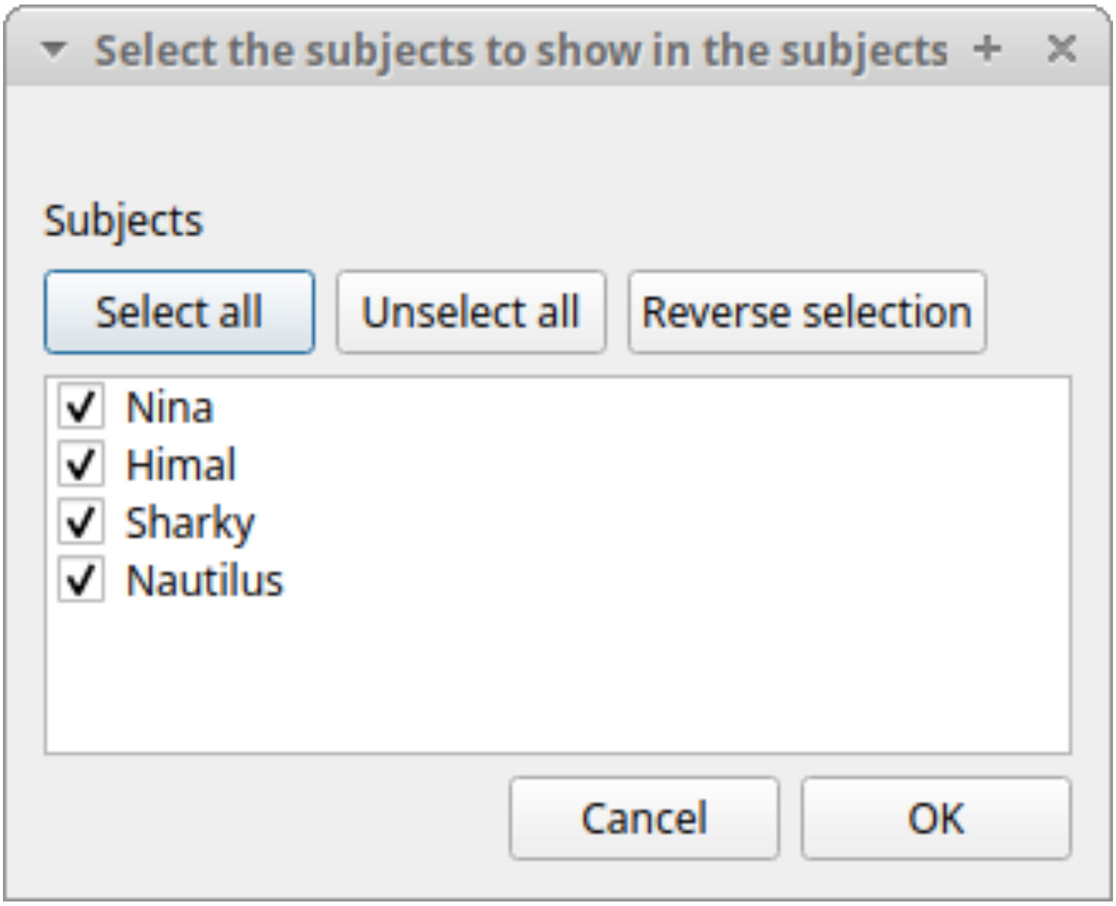
The subjects table in main window

The **Subjects** widget provides the list of subjects defined in the **Subject** tab of the **Project** window. It can be used to assign a focal subject to recorded behaviors by double-clicking the corresponding row. When a subject is selected, the name appears above the media player. The **Key** column indicates the keyboard shortcut assigned to each subject, if any.

The subjects shown in the Subjects widget can be filtered:

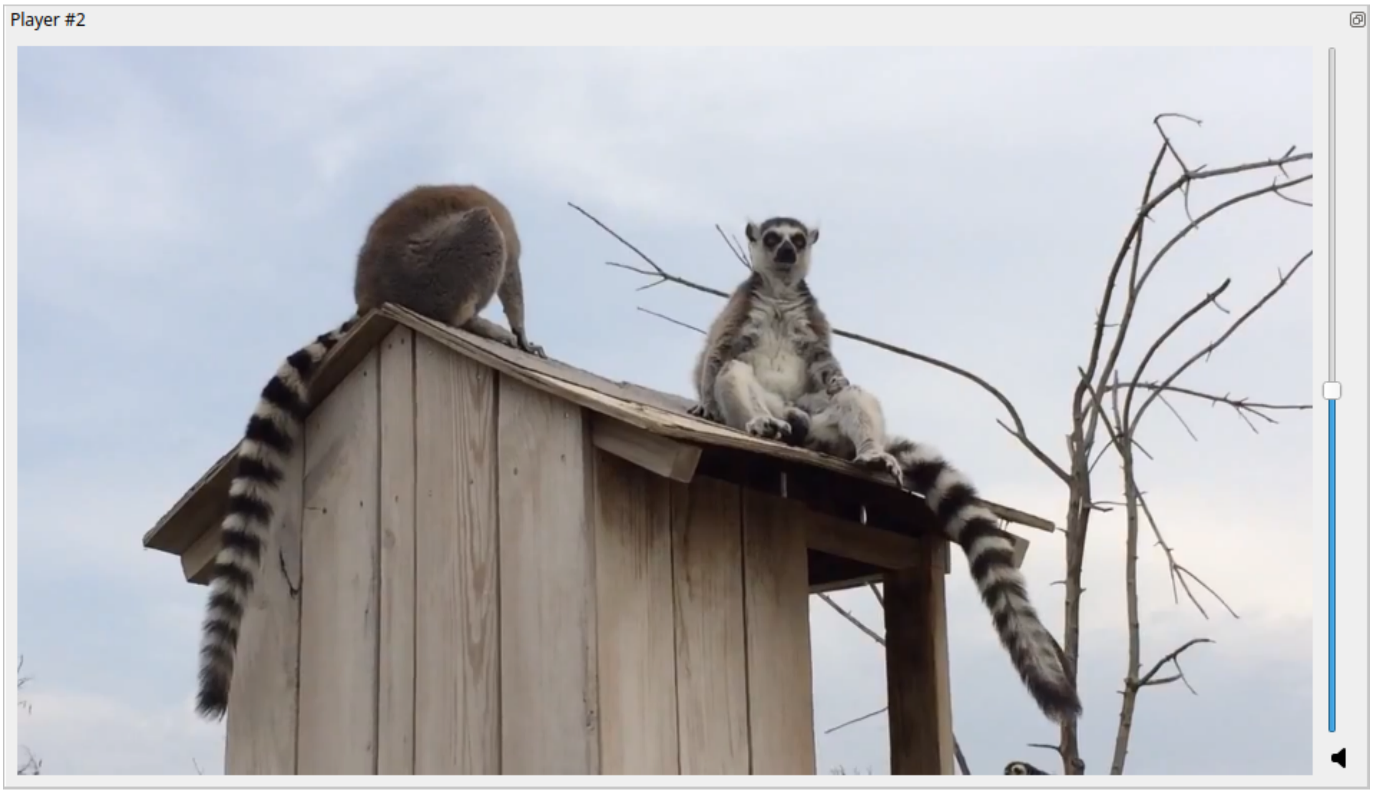
Right-click on the subjects widget > **Filter subjects**

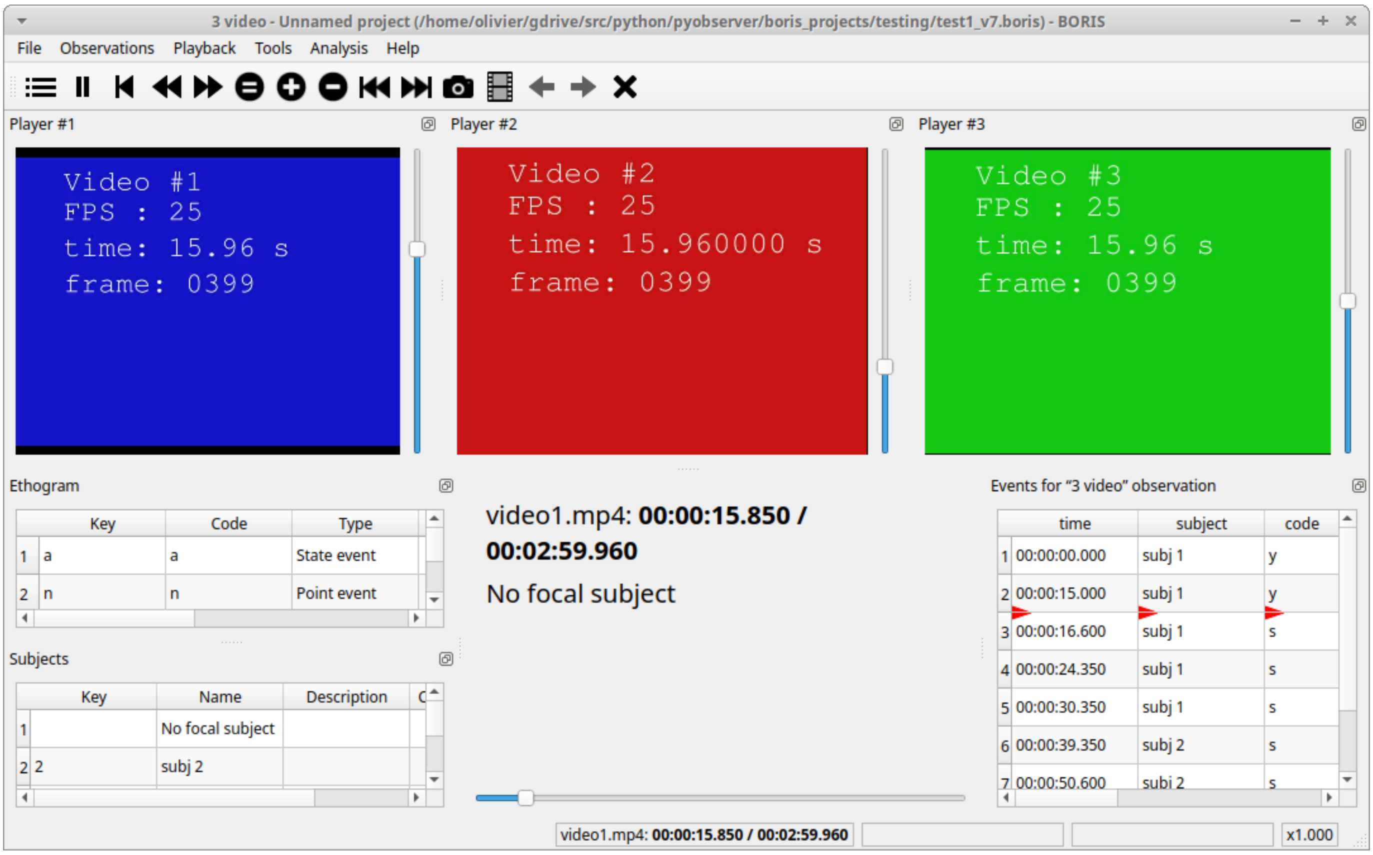
Check or uncheck subjects to show or hide them in the Subjects table.



Filter subjects in subjects table

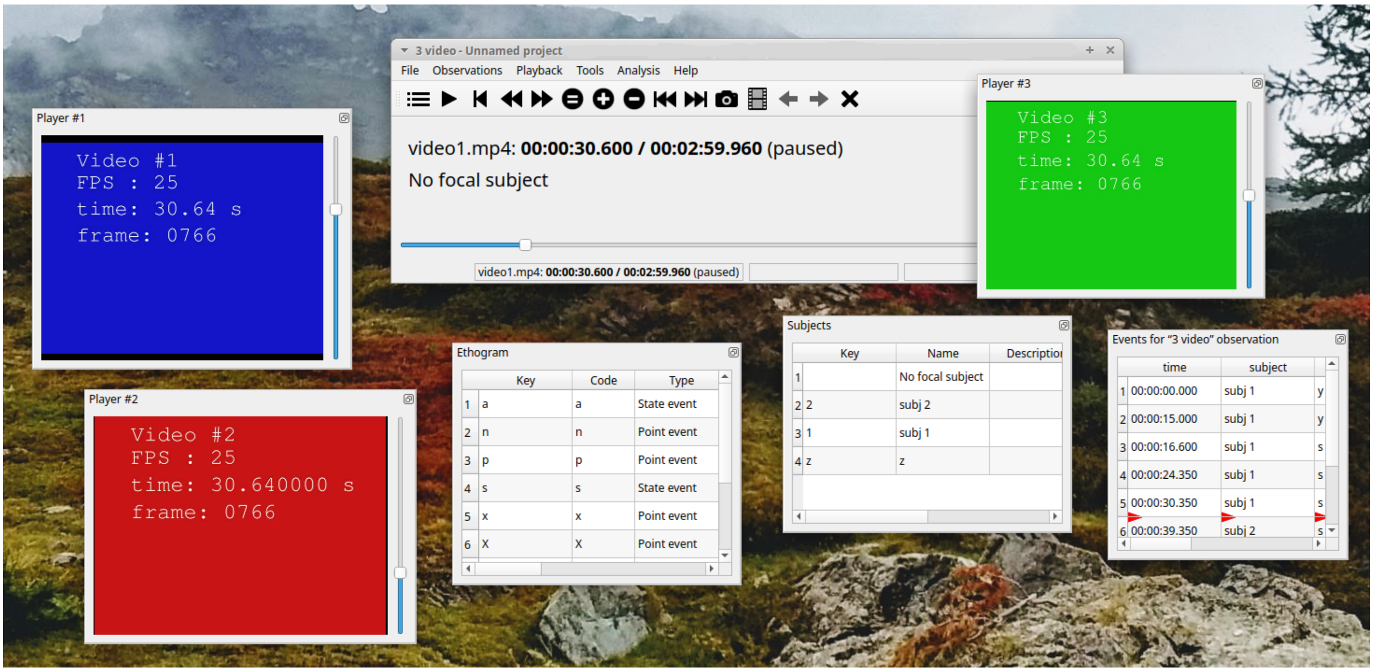
2.6.5 The **media player**





You can adjust the media position with the horizontal slider. Each media player is equipped with its own audio volume control, which can be adjusted using the vertical slider located on the right side of the player, and a mute/unmute button for easy sound management.

The arrangement of different widgets can be personalized to suit your preferences. In the screenshot provided, all widgets are detached from the main program window. This feature is particularly useful when using a multi-monitor setup.



2.6.6 The events table

The **events table** shows all the recorded behaviors (events).

The displayed parameters, organized in columns, depend on the observation type:

Observation from media file

Events for “0004” observation

	Time	Frame index	Subject	Code	Type	Modifier	Comment
1	00:00:08.253	207	Himal	Locomotion	START	Run	
2	00:00:08.329	209	Nautilus	Locomotion	START	Run	
3	00:00:10.400	260	Sharky	Swim	START		
4	00:00:11.779	295	Himal	Locomotion	STOP	Run	
5	00:00:12.778	320	Nina	Alert	START		
6	00:00:13.789	345	Nautilus	Locomotion	STOP	Run	
7	00:00:13.790	345	Nautilus	Locomotion	START	Walk	
8	00:00:14.348	359	Himal	Locomotion	START	Jump	
9	00:00:14.660	367	Nina	Alert	STOP		
10	00:00:14.865	372	Nautilus	Locomotion	STOP	Walk	
11	00:00:14.866	372	Nautilus	Locomotion	START	Jump	
12	00:00:15.001	375	Nina	Rest	START		
13	00:00:16.467	412	Himal	Locomotion	STOP	Jump	

The following parameters are displayed:

- **Time**, the time at which the event occurred;
- **Frame index**, the frame index corresponding to the event;
- **Subject**, the focal subject (if any);
- **Code**, the behavior code;
- **Type**, for a **state event**, indicates whether the time corresponds to the start or the stop. It is empty for a **point event**;
- **Modifier**, indicates the modifier(s) that was/were selected (if any);
- **Comment**, an open field where the user can add notes.

A tracking cursor (red triangle) marks the current event. This cursor can be positioned above the current event; see tracking cursor position in the **Preferences** window.

A double-click on a row will reposition the media player to the time of the corresponding event. See Time offset for media reposition in the **Preferences** window to customize the time offset for media repositioning.

Live observation

Events for “live observation” observation

	Time	Subject	Code	Type	Modifier	Comment
1	00:00:01.248	Nina	Alert	START		
2	00:00:15.664	Himal	Manipulate	START		vegetables
3	00:00:22.857	Himal	Manipulate	STOP		
4	00:00:24.841	Himal	Roll objects	START		
5	00:00:29.985	Himal	Roll objects	STOP		
6	00:00:38.009	Nina	Alert	STOP		
7	00:00:41.577	Sharky	Stomp	START		
8	00:00:49.105	Sharky	Stomp	STOP		

The following parameters are displayed:

- **Time**, the time at which the event occurred;
- **Subject**, the focal subject (if any);
- **Code**, the behavior code;
- **Type**, in case of a **state event**, indicates whether the time corresponds to the start or the stop. Empty for a **point event**;
- **Modifier**, indicates the modifier(s) that was/were selected (if any);
- **Comment**, an open field where the user can add notes.

Observation from pictures

Events for “cametra trap” observation

<table>
<tr><th></th><th>Time</th><th>Subject</th><th>Code</th><th>Type</th><th>Modifier</th><th>Comment</th><th>Image index</th><th>Image path</th></tr>
<tr><td>1</td><td>NA</td><td></td><td>Coleoptera</td><td></td><td>None|Bostrichoidea|Cerambycidae|Lagriinae|Anthaxia|None|Cetonia aurata|None|None|None</td><td></td><td>1</td><td>/data/[illegible]/08010001.JPG</td></tr>
<tr><td>2</td><td>NA</td><td></td><td>Diptera</td><td></td><td>Brachycera|None|Conopidae|Eristalinae|Chrysotoxum|None|Cheilosia impressa|None|None|None</td><td></td><td>2</td><td>/data/[illegible]/08010002.JPG</td></tr>
<tr><td>3</td><td>NA</td><td></td><td>Hymenoptera</td><td></td><td>None|Apoidea|Cephidae|Arginae|Anthidiellum|Bombus s.s.|Bombus distinguendus|None|None|None</td><td></td><td>3</td><td>/data/[illegible]/08010003.JPG</td></tr>
<tr><td>4</td><td>NA</td><td></td><td>Lepidoptera</td><td></td><td>Macroheterocera|None|Geometridae|Dismorphiinae|Albulina|None|Anarta melanopa|None|None|None</td><td></td><td>4</td><td>/data/[illegible]/08010004.JPG</td></tr>
<tr><td>5</td><td>NA</td><td></td><td>Others</td><td></td><td>flower_on</td><td></td><td>5</td><td>/data/[illegible]/08010005.JPG</td></tr>
<tr><td>6</td><td>NA</td><td></td><td>Hymenoptera</td><td></td><td>None|Cephoidea|Argidae|Cephinae|Andrena|Ashtonipsithyrus|Bombus bohemicus|None|None|None</td><td></td><td>6</td><td>/data/[illegible]/08010006.JPG</td></tr>
</table>

The following parameters are displayed:

- **Time**, the time at which the event occurred;
- **Subject**, the focal subject (if any);
- **Code**, the behavior code;
- **Type**, in case of a **state event**, indicates whether the time corresponds to the start or the stop. Empty for a **point event**;
- **Modifier**, indicates the modifier(s) that was/were selected (if any);
- **Comment**, an open field where the user can add notes;
- **Image index**, the image index (in the directory) corresponding to the event;
- **Image path**, the path of the image corresponding to the event (can be relative or absolute).

To simplify the **events table**, the relevant behaviors and subjects can be filtered; see Filter events.

2.6.7 Events

Recording an event

An event is a unique combination of a **time**, a **subject**, and a **behavior**. If the subject is not set, it will be **No focal subject**.

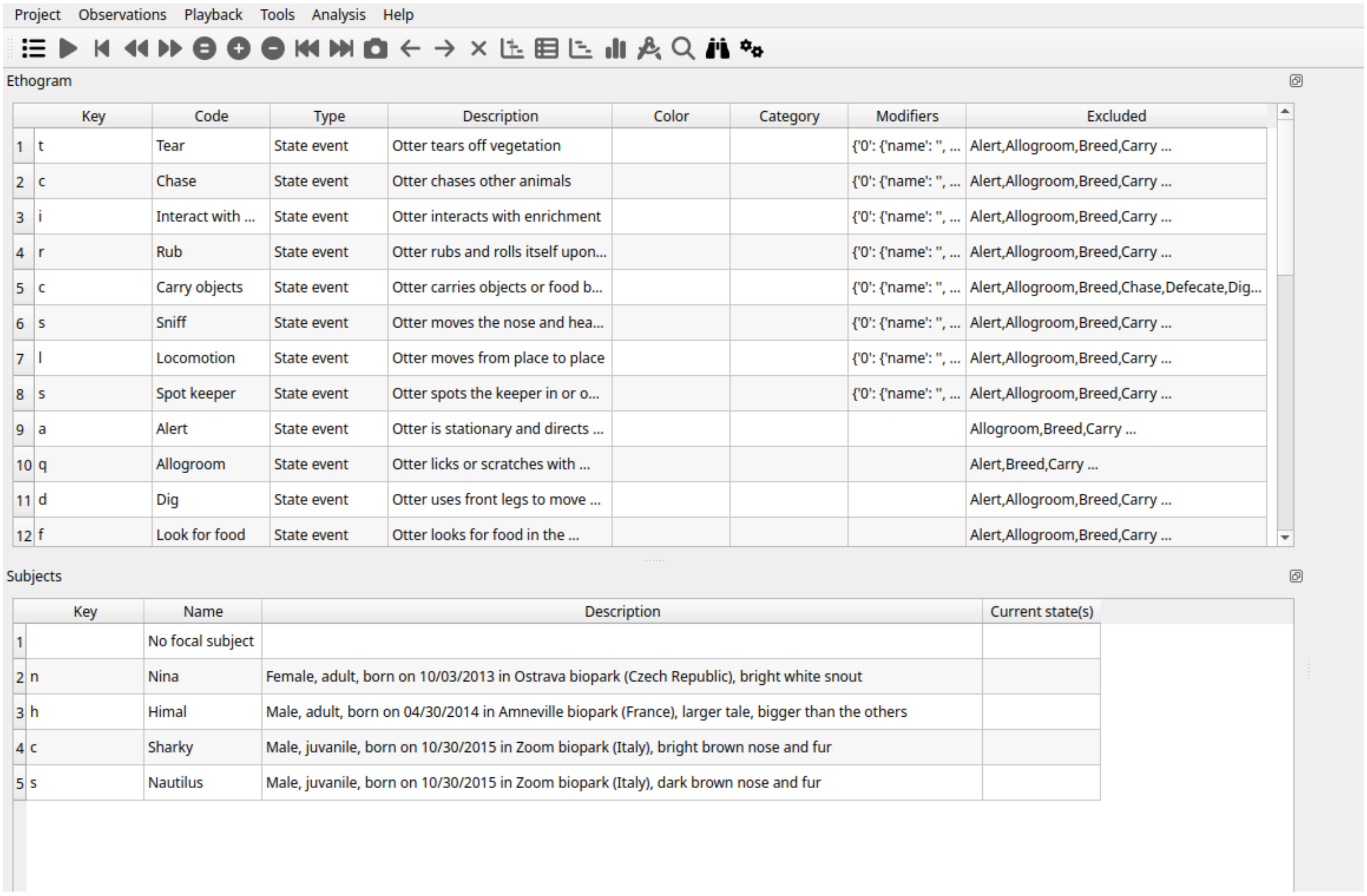
Once ready to begin your coding, you can start the media player using the **Play** ▶ button or the **Space bar**.

An **event** can be recorded by:

- Pressing the predefined **key** on the keyboard corresponding to the behavior to record.
- Double-clicking the corresponding row in the **Ethogram** table.
- Using the **Coding pad** (see coding pad).

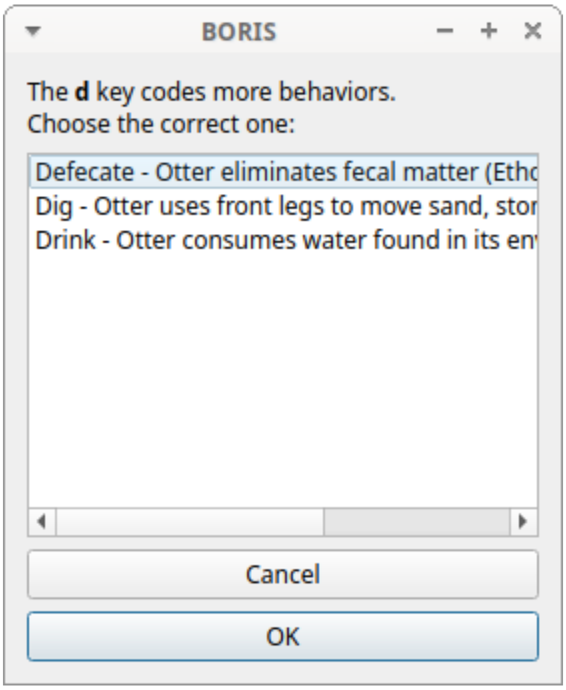
The **focal subject** can be selected by:

- Pressing the predefined **key** on the keyboard corresponding to the subject to select.
- Double-clicking the corresponding row in the **Subjects** table.
- Using the **Subject pad** (see subject pad).



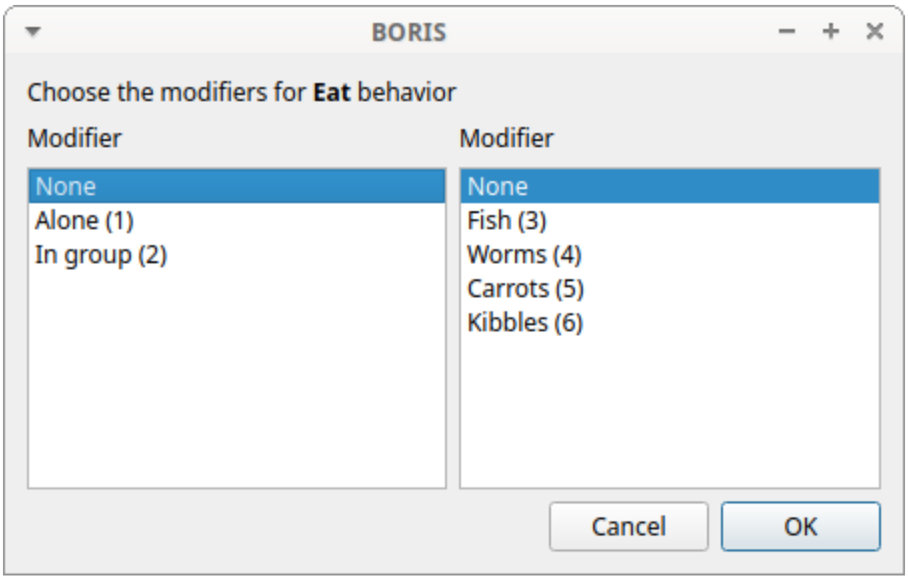
Ethogram and subjects widgets

If the pressed key defines a single event, the corresponding event will be recorded directly in the **Events** table. If you have specified the same key for two (or more) events (e.g., key **d** in the figure below), BORIS will prompt you for the desired behavior.



Ask for a behavior

If you have specified modifiers (one or more sets), BORIS will prompt you for the desired modifier(s), if any (e.g., **ball** or **opponent** in the figure below). You can select the modifiers using the mouse or the keyboard (1, 2, 3, 4, 5, or 6).



Ask for modifiers

If no keys are defined for modifier selection, you can type the first character of the modifier and use the **Up arrow** and **Down arrow** keyboard keys to select the correct modifier.

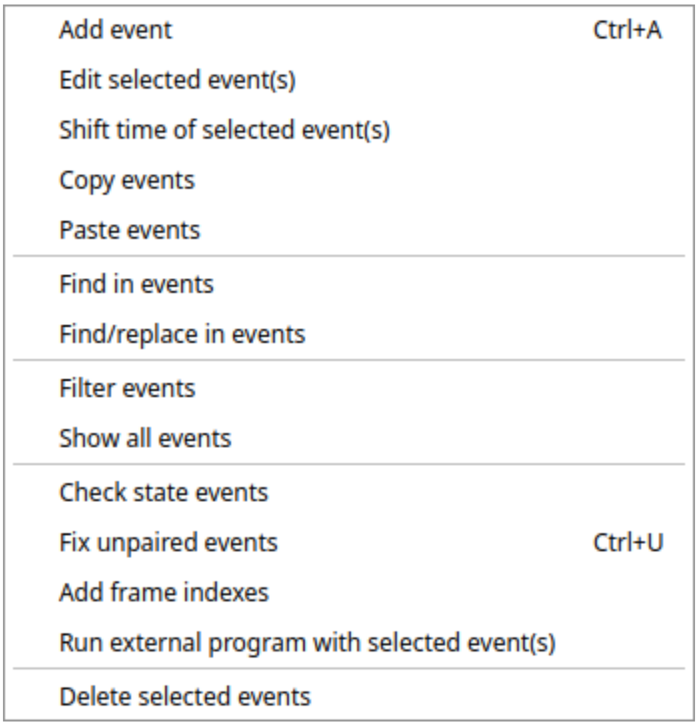
If your behavior type is a **Point event with coding map** or a **State event with coding map**, BORIS will show the **Coding map** window and allow you to select the desired area(s). If you click on a part of the map where two (or more) areas overlap, the corresponding codes will be recorded.

A recorded event can be edited (once selected) using the **Observations** > **Edit event** menu option. The resulting *Edit event parameters* window allows modifying every parameter (e.g., time, subject, code, modifiers, and comment).

The **Observations** > **Add event** menu option allows adding a new event by specifying its time and the other parameters.

The Events table context menu

Some functions are available in the Events table context menu. Right-click on the Events table to open the menu.



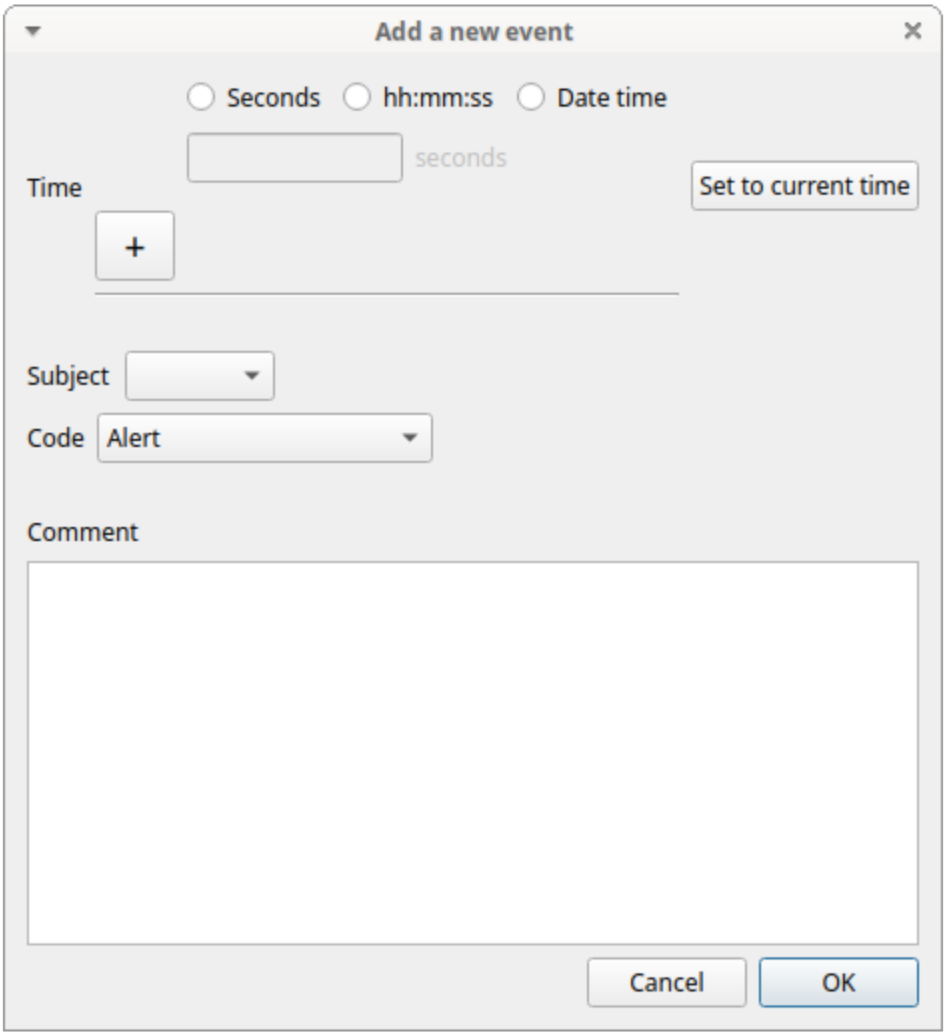
The various functions available in the menu are described below. The same functions and others are available in the **Observations** menu.

Undo an event recording

A wrong event can be removed from the events list using the **Undo** function (^ Ctrl + Z). You can undo up to 25 recorded events.

Add event

This option allows adding a new event by specifying its time and the other parameters.



Add a new event

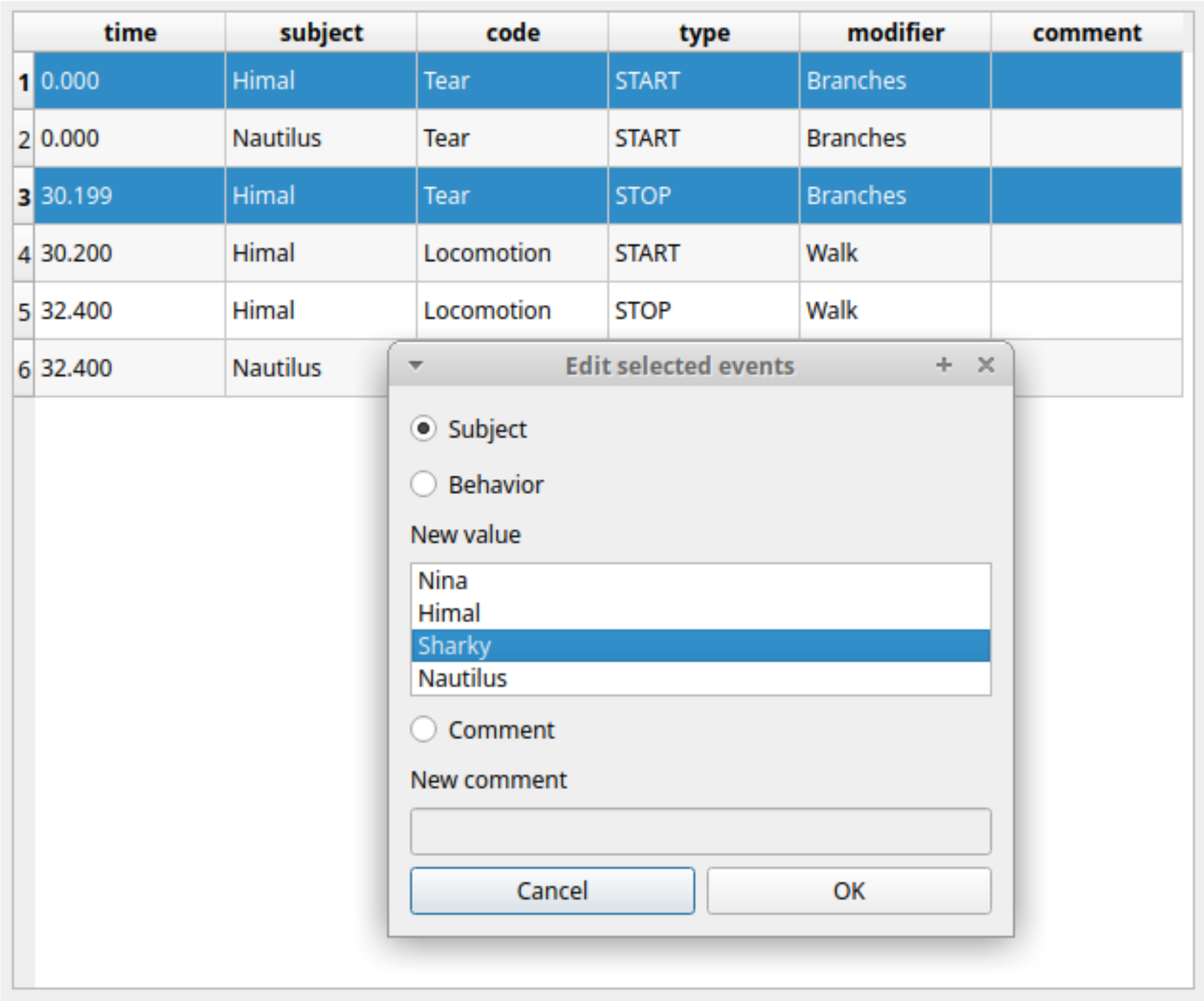
Select a time format and input the time value.

Select the **subject** from the drop-down menu or leave empty for **No focal subject**.

Select the **behavior** from the drop-down menu.

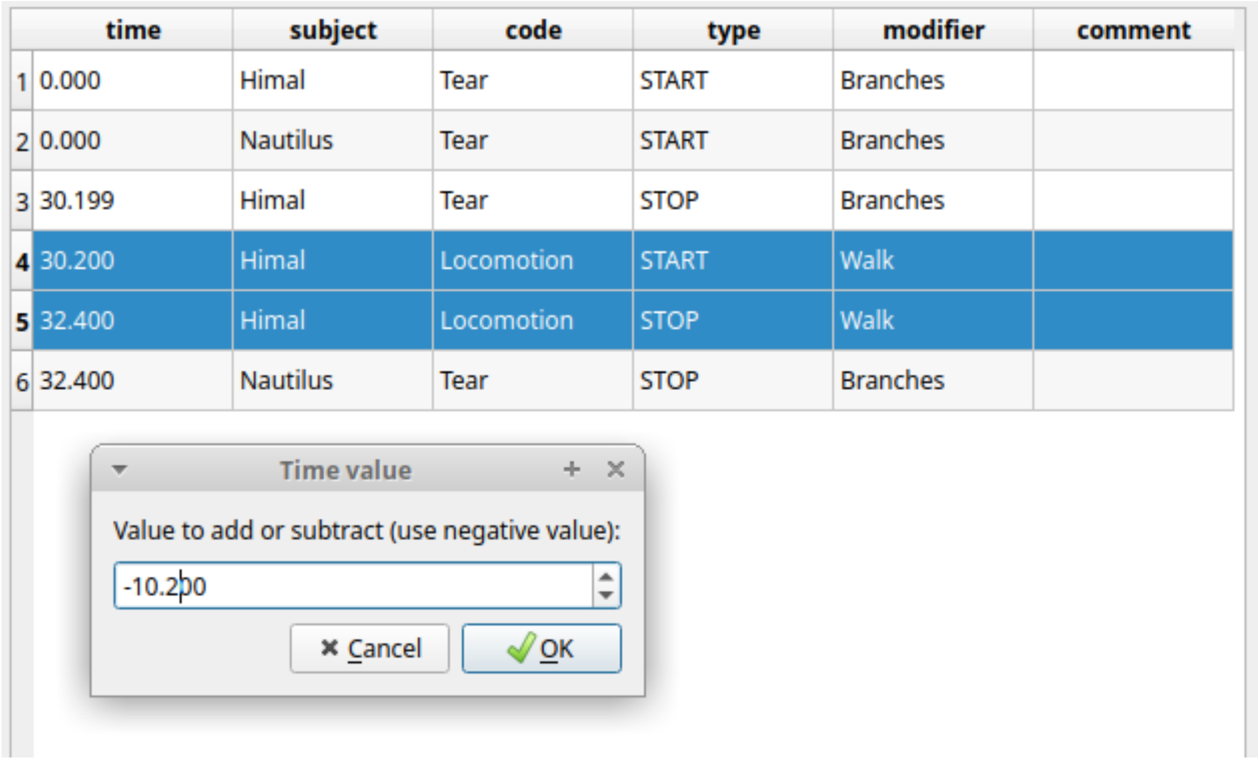
Edit selected event(s)

This option allows to edit the selected event(s). When many events are selected you have to choose the field to edit between **Subject**, **Behavior** and **Comment**. In this case the new value will apply to all selected events.



Edit time of selected event(s)

This option allows you to add or subtract a time value (in seconds) from all selected events. To subtract a value, use a negative number.



Copy events

This option allows you to copy the selected events to the clipboard. The clipboard will contain the values of the selected events (except the **type** field) separated by a character. The copied values are: **Time**, **Subject**, **Behavior**, **Modifier(s)**, **Frame index**.

Example of clipboard content:

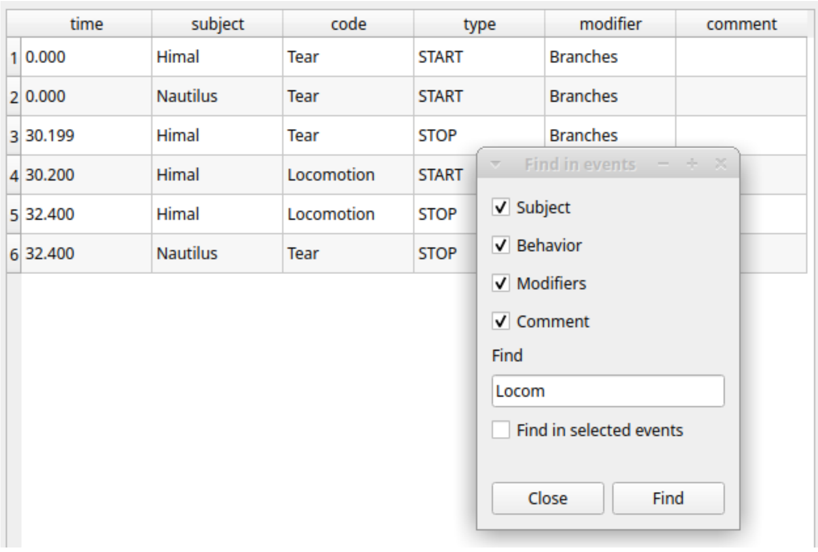
```
8.253	Himal	Locomotion	Run	207
8.329	Nautilus	Locomotion	Run	209
10.400	Sharky	Swim		260
11.778	Himal	Locomotion	Run	295
12.778	Nina	Alert		320
13.788	Nautilus	Locomotion	Run	345
13.789	Nautilus	Locomotion	Walk	345
14.348	Himal	Locomotion	Jump	359
14.660	Nina	Alert		367
14.865	Nautilus	Locomotion	Walk	372
14.865	Nautilus	Locomotion	Jump	372
15.000	Nina	Rest		375
16.466	Himal	Locomotion	Jump	412
16.467	Himal	Alert		412
23.600	Nautilus	Locomotion	Jump	590
23.600	Nautilus	Rest		590
24.228	Nautilus	Rest		606
24.407	Himal	Alert		611
24.917	Himal	Locomotion	Walk	623
39.682	Nautilus	Locomotion	Run	992
40.549	Nina	Rest		1014
42.313	Nautilus	Locomotion	Run	1058
42.314	Nautilus	Rest		1058
44.759	Himal	Locomotion	Walk	1119
44.761	Himal	Allogroom		1119
48.219	Nautilus	Rest		1206
48.363	Himal	Allogroom		1209
48.365	Himal	Locomotion	Walk	1209
49.274	Himal	Locomotion	Walk	1232
49.274	Himal	Drink		1232
50.408	Himal	Drink		1261
50.408	Himal	Swim		1261
58.851	Sharky	Swim		1472
58.950	Himal	Swim		1474
```

Paste events

This option allows you to paste the clipboard content into the events table. The clipboard must follow the format described in the previous section: 5 columns separated by a character.

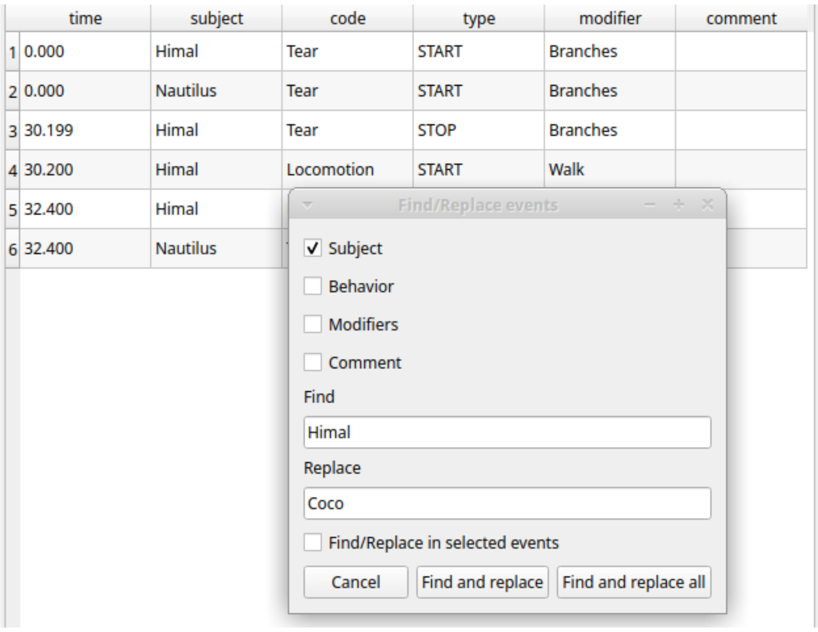
Find in events

This option allows you to search for a string in the various fields of events. Select the fields to be searched. The find/replace operation can be restricted to the selected events.



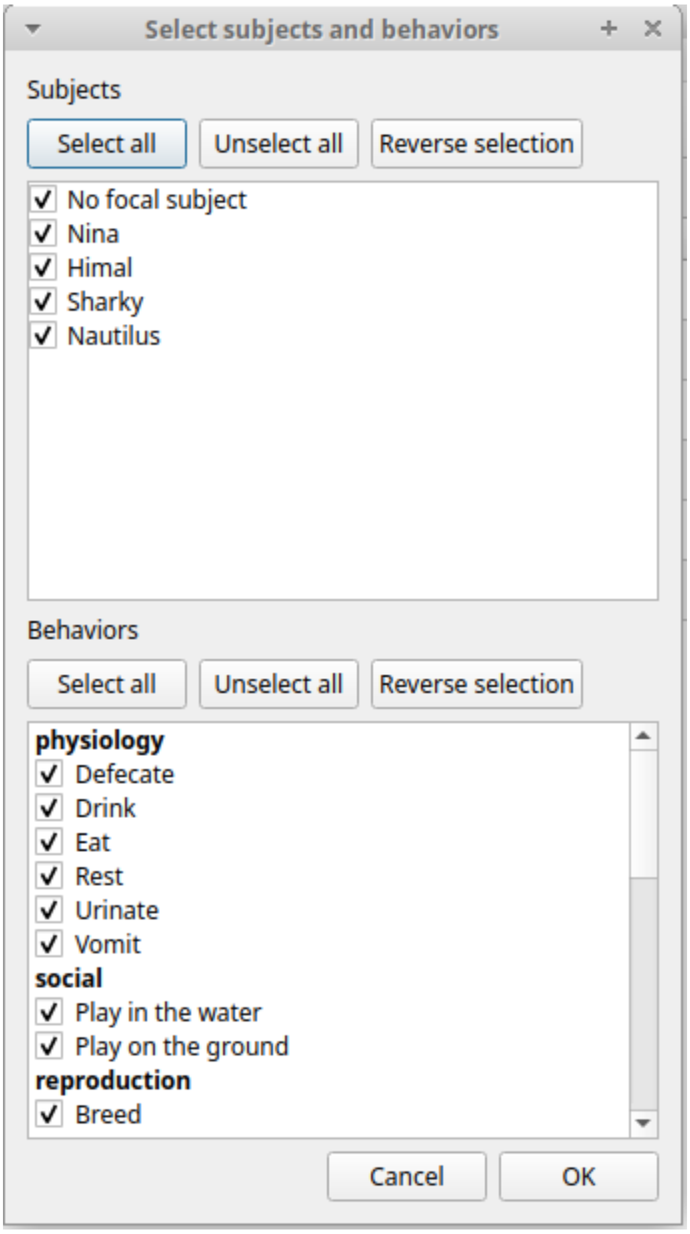
Find/Replace in events

This option allows you to search for a string and replace it with a new value in the various fields of events. Select the fields to be searched. The find operation can be restricted to the selected events.



Filter events

This option allows you to filter events by field value (Subject and Behavior).



Show all events

This option reverts the previous one and allows you to visualize all coded events.

Check state events

This option allows you to check if the **state events** are **PAIRED**, i.e., if they have both a **START** and a **STOP** occurrence.

Delete selected events

This option allows you to delete the selected events. This operation is irreversible!

Delete all events

This option is not present in the context menu but only in the main menu (**Observations** > **Delete all events**).

This option allows you to delete all events in the current observation. This operation is irreversible!

Fix unpaired state

You can use the **Fix unpaired events** function to fix **state events** without a STOP event.

Observations > **Fix unpaired events** (keyboard shortcut: ^ Ctrl + U)

The program will ask for a time at which to insert the STOP events for all unpaired **state events**.

This function can be run on a set of selected observations (when no observation is open). In this case, the STOP events will be inserted at the end of the observation.

Add frame indexes

This function can be used for observations from a video. The frame index corresponding to the coded events will be added to the events table.

Run external program with selected events

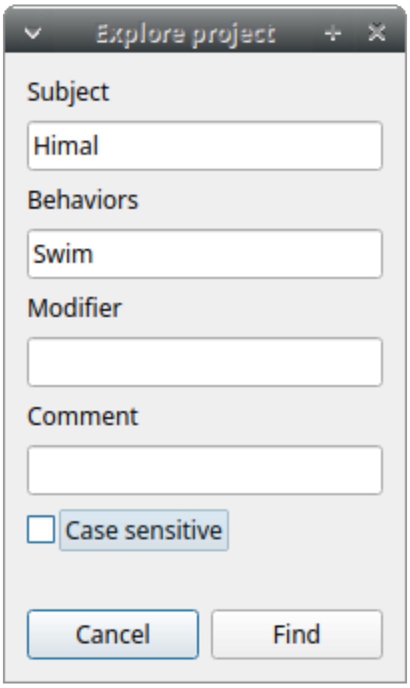
This function is not yet implemented.

Explore project

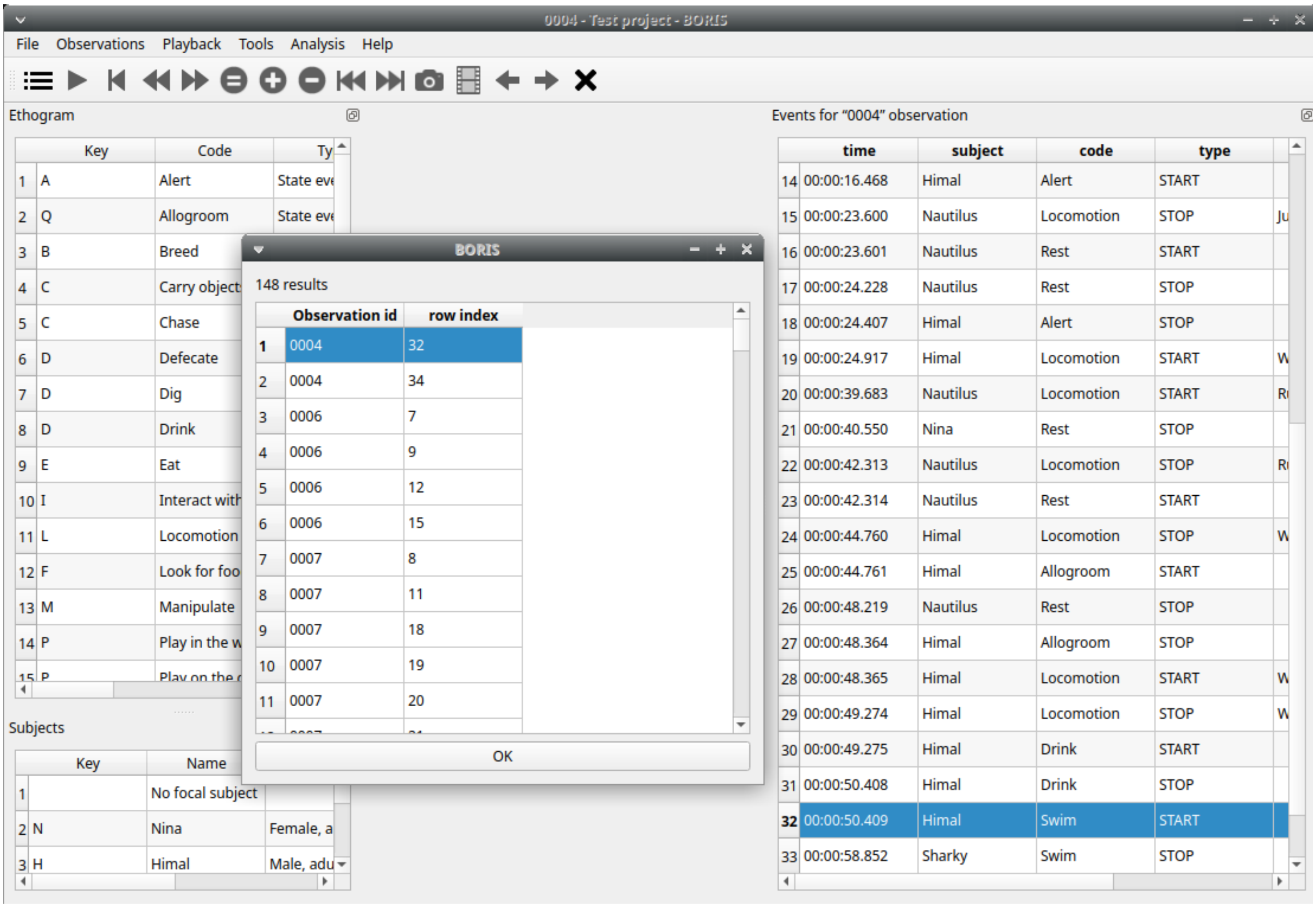
You can search for information in various fields in all observations in the current project (Observations > Explore project).

The searchable fields are: **subject**, **behavior**, **modifier**, and **comment**.

If more than one field is searched, a logical **AND** will apply.



The events that were found are listed in a table. By double-clicking on a row, the corresponding observation will be opened, and the visualization will scroll to the row corresponding to the event.



Frame-by-frame mode

You can switch between the media player and the frame-by-frame mode using the **arrow buttons** in the toolbar:

In frame-by-frame mode, the video will stop playing, and the user can view the video frame by frame.

> **Note**
>
> Some video files should be re-encoded to be used in frame-by-frame mode. Otherwise the extracted frames are not reliable or it will not be possible to move backward.

You can move between frames by using the arrow keys in the toolbar (on the right) or by using special keyboard keys:

`← Left` Go to the **previous frame**

`→ Right` Go to the **next frame**

`⇞ Page Up` Switch to the **next media**

`⇟ Page Down` Switch to the **previous media**

`↑ Up` **Jump forward** in the current media

`↓ Down` **Jump backward** in the current media

If you have a numeric keypad, you can use the following keys as an alternative:

- The key **/** will allow you to view the previous frame
- The key ***** will allow you to view the next frame

To return to the media player mode, press the **Play** ▶ button in the toolbar.

2.7 Export events

Coded events can be exported in various formats.

2.7.1 Export events in tabular format

Observations > **Export events** > **Tabular events**

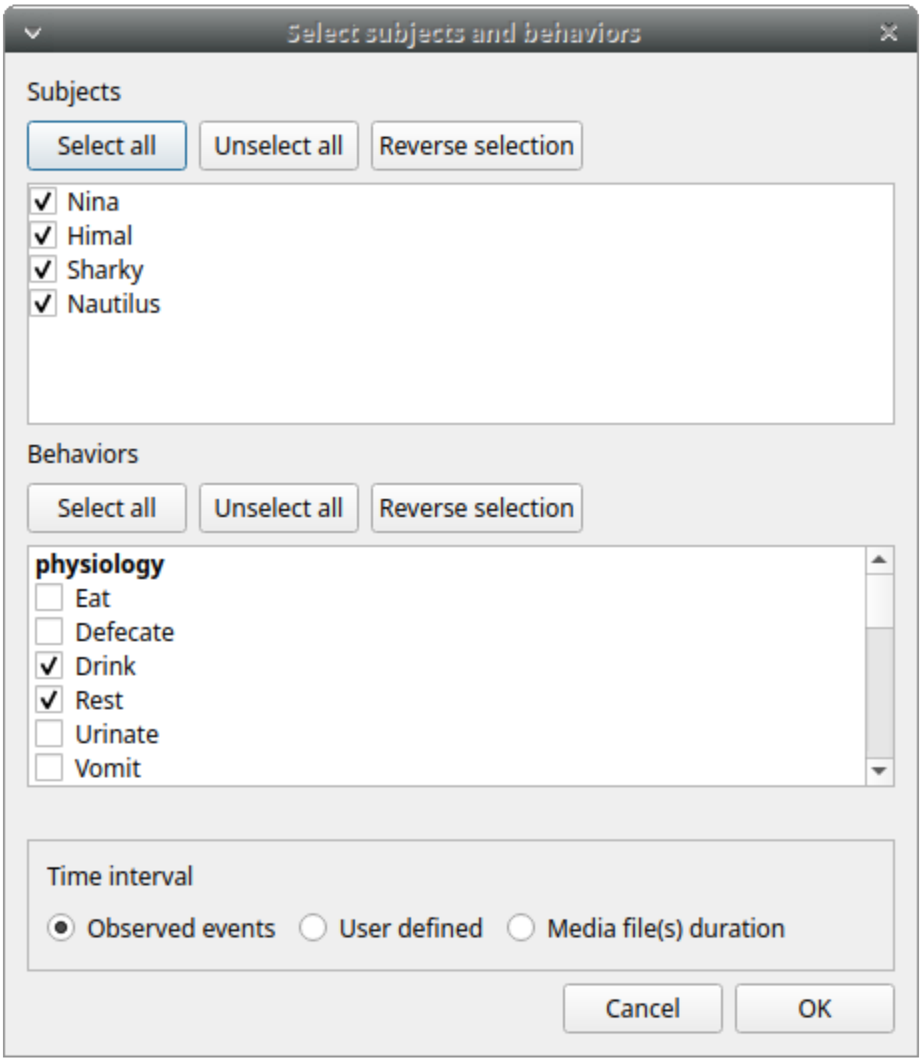
This function exports the events from selected observations to one or more files. Various formats are available:

- Plain text in tabular format
    - **Tab Separated Values** (TSV)
    - **Comma Separated Values** (CSV)
    - **Hyper Text Markup language** (HTML)
- Spreadsheet files
    - **OpenDocument** (ODS)
    - **Microsoft Excel** (XLSX, XLS)
- **Pandas dataframe** (to be loaded in Python with the pickle module)
- **R dataframe** (to be loaded in R with readRDS function)

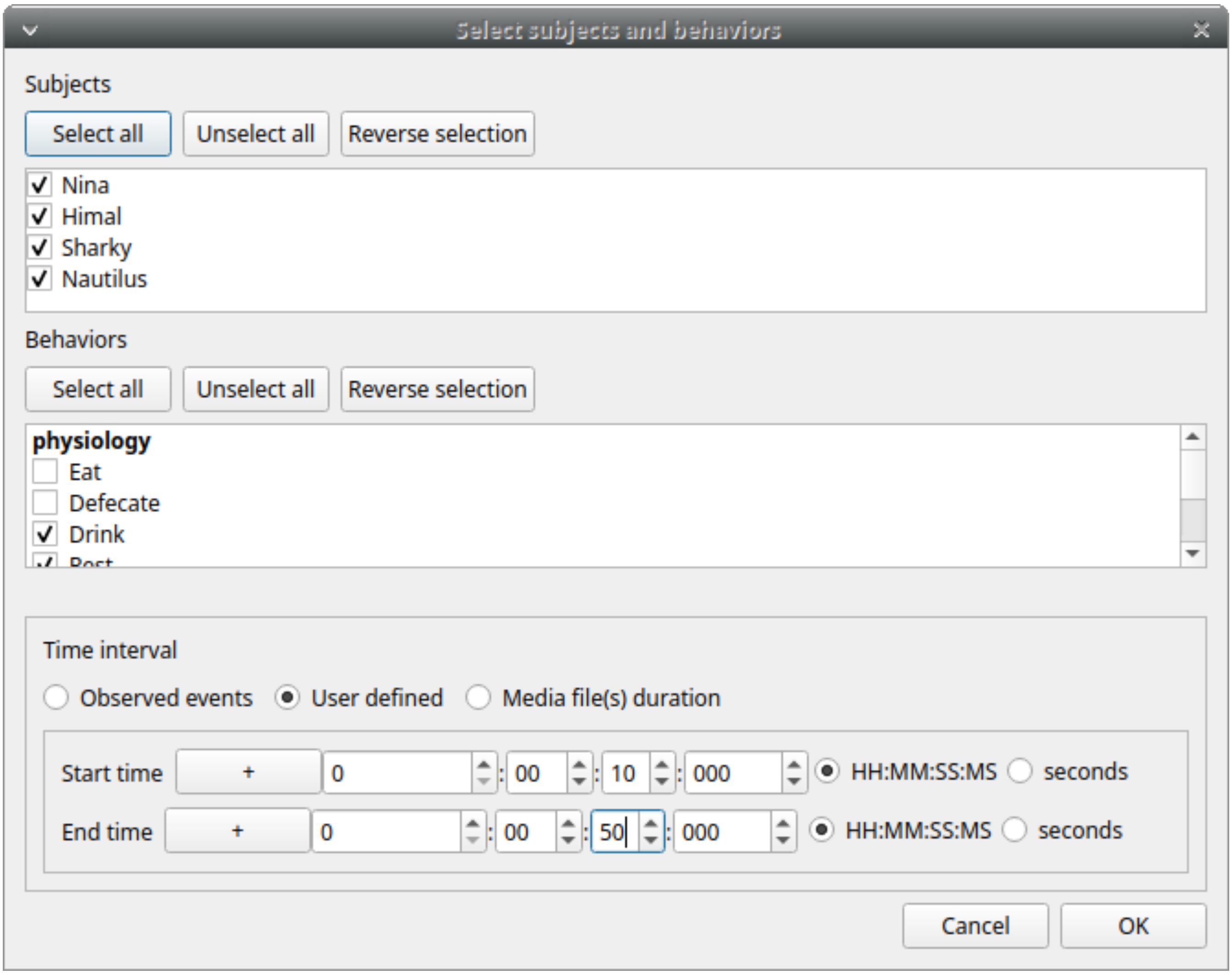
If many observations are selected, BORIS will ask you to choose a directory to save the files. For spreadsheet formats (XLSX and ODS), the events can be exported to multiple worksheets within a single workbook. All these formats are suitable for further analysis.

Select the subjects, behaviors, and time interval.

Set the time interval to **Observed events**.



Select a **User defined** time interval.



Example output of tabular events:

Observation id	Observation date	Description	Observation duration	Observation type	Source	Media duration (s)	FPS	Location	Weather	Temperature	Visitors
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55′ 59″ N – 7° 25′ 18″	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55′ 59″ N – 7° 25′ 18″	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55′ 59″ N – 7° 25′ 18″	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55′ 59″ N – 7° 25′ 18″	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55′ 59″ N – 7° 25′ 18″	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55′ 59″ N – 7° 25′ 18″	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55′ 59″ N – 7° 25′ 18″	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55′ 59″ N – 7° 25′ 18″	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55′ 59″ N – 7° 25′ 18″	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55′ 59″ N – 7° 25′ 18″	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55′ 59″ N – 7° 25′ 18″	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55′ 59″ N – 7° 25′ 18″	sun	24	1046

Subject	Behavior	Behavioral category	Modifier #1	Behavior type	Time	Media file name	Image index	Image file path	Comment
Himal	Tear		Branches	START	0.000	20160517162540.m2ts	NA	NA	
Sharky	Tear		Branches	START	0.000	20160517162540.m2ts	NA	NA	
Nautilus	Tear		Branches	START	1.359	20160517162540.m2ts	NA	NA	
Nautilus	Tear		Branches	STOP	25.776	20160517162540.m2ts	NA	NA	
Nautilus	Carry objects		Branches	START	25.777	20160517162540.m2ts	NA	NA	
Nautilus	Carry objects		Branches	STOP	27.732	20160517162540.m2ts	NA	NA	
Sharky	Tear		Branches	STOP	30.688	20160517162540.m2ts	NA	NA	
Sharky	Locomotion		Walk	START	30.689	20160517162540.m2ts	NA	NA	
Sharky	Locomotion		Walk	STOP	31.819	20160517162540.m2ts	NA	NA	
Himal	Tear		Branches	STOP	33.898	20160517162540.m2ts	NA	NA	
Himal	Locomotion			START	33.899	20160517162540.m2ts	NA	NA	
Himal	Locomotion			STOP	34.470	20160517162540.m2ts	NA	NA	

2.7.2 Export aggregated events

Observations > **Export events Aggregated events**

This function exports the events corresponding to the selected subjects and behaviors from the selected observations.

Various formats are available:

- Plain text in tabular format
    - **Tab Separated Values** (TSV)
    - **Comma Separated Values** (CSV)
    - **Hyper Text Markup language** (HTML)
- Spreadsheet files
    - **OpenDocument** (ODS)
    - **Microsoft Excel** (XLSX, XLS)
- **SQL format** for populating a SQL database
- **SDIS format** for analysis with the GSEQ program available at http://www2.gsu.edu/~psyrab/gseq
- **Pandas dataframe** (to be loaded in Python with the pickle module)
- **R dataframe** (to be loaded in R with readRDS function)

If two or more observations are selected, you can choose to group all results into one file. If you do not want to group the results, BORIS will ask you to choose a directory to save the files (the observation ID will be used as the file name).

State events are paired, and in this case, the event duration is available.

An arbitrary time interval can be selected by checking the **Limit to time interval** option. In this case, ongoing events will be started at the start time and stopped at the end time in the export file.

The following fields are available in the output:

- Observation id
- Observation date and time
- Observation description
- Observation type (Media file / Live / Pictures)
- Source (for media files and pictures)
- Total duration (in seconds, the duration of the observation based on the selected time interval)
- Media duration(s) (in seconds, for media file observations)
- FPS (frames/s, for video files, number of images per second)
- Independent variables (one column per variable)
- Subject name
- Observation duration by subject
- Behavior
- Behavioral category (if any)
- Behavior modifier(s) (one column per modifier)
- Behavior type (STATE / POINT)
- Start (seconds)
- Stop (seconds)
- Duration (seconds, duration of the event for STATE events)
- Media file name (for media file observations, the media file in which the event occurs)
- Image index start (for observations from pictures, the index of the image where the event starts)
- Image index stop (for observations from pictures, the index of the image where the event stops)
- Image file path start (for observations from pictures, the path of the image where the event starts)
- Image file path stop (for observations from pictures, the path of the image where the event stops)
- Comment start
- Comment stop

Example of aggregated event export (TSV, CSV, XLSX, ODS, HTML):

Observation id	Observation date	Description	Observation type	Source	Total duration	Media duration (s)	FPS (frame/s)	Location	Weather	Temperature	Visitors
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
0005	2016-05-17 00:00:49	In the pool	Media file	player #1:20160517163131.m2ts	40.770	40.800	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24	1046
0005	2016-05-17 00:00:49	In the pool	Media file	player #1:20160517163131.m2ts	40.770	40.800	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24	1046
0006	2016-05-17 00:00:42	In the pool	Media file	player #1:20160517163231.m2ts	61.830	61.920	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24	1046
0006	2016-05-17 00:00:42	In the pool	Media file	player #1:20160517163231.m2ts	61.830	61.920	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24	1046
0006	2016-05-17 00:00:42	In the pool	Media file	player #1:20160517163231.m2ts	61.830	61.920	25.000	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24	1046

Subject	Observation duration by subject by observation	Behavior	Behavioral category	Modifier #1	Behavior type	Start (s)	Stop (s)	Duration (s)	Media file name	Image index start	Image index stop	Image file path start	Image file path stop	Comment start	Comment stop
Himal	47.613	Locomotion	Not defined	Run	STATE	8.253	11.779	3.526	20160517162952.m2ts	NA	NA				
Nautilus	24.431	Locomotion	Not defined	Run	STATE	8.329	13.789	5.460	20160517162952.m2ts	NA	NA				
Sharky	48.452	Swim	Not defined		STATE	10.400	58.852	48.452	20160517162952.m2ts	NA	NA				
Nina	27.431	Alert	Not defined		STATE	12.778	14.660	1.882	20160517162952.m2ts	NA	NA				
Nautilus	24.431	Locomotion	Not defined	Walk	STATE	13.790	14.865	1.075	20160517162952.m2ts	NA	NA				
Himal	47.613	Locomotion	Not defined	Jump	STATE	14.348	16.467	2.119	20160517162952.m2ts	NA	NA				
Nautilus	24.431	Locomotion	Not defined	Jump	STATE	14.866	23.600	8.734	20160517162952.m2ts	NA	NA				
Nina	27.431	Rest	physiology		STATE	15.001	40.550	25.549	20160517162952.m2ts	NA	NA				
Himal	47.613	Alert	Not defined		STATE	16.468	24.407	7.939	20160517162952.m2ts	NA	NA				
Nautilus	24.431	Rest	physiology		STATE	23.601	24.228	0.627	20160517162952.m2ts	NA	NA				
Himal	47.613	Locomotion	Not defined	Walk	STATE	24.917	44.760	19.843	20160517162952.m2ts	NA	NA				
Nautilus	24.431	Locomotion	Not defined	Run	STATE	39.683	42.313	2.630	20160517162952.m2ts	NA	NA				
Nautilus	24.431	Rest	physiology		STATE	42.314	48.219	5.905	20160517162952.m2ts	NA	NA				
Himal	47.613	Allogroom	Not defined		STATE	44.761	48.364	3.603	20160517162952.m2ts	NA	NA				
Himal	47.613	Locomotion	Not defined	Walk	STATE	48.365	49.274	0.909	20160517162952.m2ts	NA	NA				
Himal	47.613	Drink	physiology		STATE	49.275	50.408	1.133	20160517162952.m2ts	NA	NA				
Himal	47.613	Swim	Not defined		STATE	50.409	58.950	8.541	20160517162952.m2ts	NA	NA				
Sharky	40.77		Manipulate	Not defined	STATE	0.000	40.770	40.770	20160517163131.m2ts	NA	NA				
Nautilus	40.77		Manipulate	Not defined	STATE	0.000	40.770	40.770	20160517163131.m2ts	NA	NA				
Himal	44.368	Manipulate	Not defined		STATE	0.000	40.760	40.760	20160517163231.m2ts	NA	NA				
Nautilus	45.037	Manipulate	Not defined		STATE	0.000	17.448	17.448	20160517163231.m2ts	NA	NA				
Sharky	46.537	Manipulate	Not defined		STATE	1.824	48.361	46.537	20160517163231.m2ts	NA	NA				

Example of SQL export of aggregated events:

```
CREATE TABLE aggregated_events (id INTEGER PRIMARY KEY ASC, observation TEXT, subject TEXT, behavior TEXT, type TEXT, modifiers TEXT, start FLOAT, stop FLOAT,
comment TEXT, comment_stop TEXT, image_index_start INTEGER,image_index_stop INTEGER,image_path_start TEXT,image_path_stop TEXT);
INSERT INTO "aggregated_events" VALUES(1,'0001_a','Himal','Tear','STATE','Branches',0.0,30.199,'','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(2,'0001_a','Himal','Locomotion','STATE','Walk',30.2,32.4,'','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(3,'0001_a','Nautilus','Tear','STATE','Branches',0.0,32.4,'','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(4,'0001_b','Himal','Tear','STATE','Branches',0.0,30.199,'','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(5,'0001_b','Himal','Locomotion','STATE','Walk',30.2,32.4,'','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(6,'0001_b','Nautilus','Tear','STATE','Branches',0.0,31.4,'','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(7,'0002','Himal','Tear','STATE','Branches',0.0,33.898,'','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(8,'0002','Himal','Locomotion','STATE','',33.899,34.47,'','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(9,'0002','Sharky','Tear','STATE','Branches',0.0,30.688,'','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(10,'0002','Sharky','Locomotion','STATE','Walk',30.689,31.819,'','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(11,'0002','Nautilus','Tear','STATE','Branches',1.359,25.776,'','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(12,'0002','Nautilus','Carry objects','STATE','Branches',25.777,27.732,'','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(13,'0003','Nina','Locomotion','STATE','Walk',21.626,22.5,'','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(14,'0003','Nina','Manipulate','STATE','',0.0,21.625,'','','NA','NA',NULL,NULL);
```

2.7.3 Export events as behavioral sequences

Observations > **Export events** > **as behavioral sequences**

Behavioral strings can be used with the **Behatrix** program: Behatrix

Example:

```
# observation id: demo#1
# observation description:
# Media file name: video1.mp4, video2.mp4


Subject #1:
eat|jump|eat|jump

Subject #2:
eat|rest|jump|eat|jump
```

2.7.4 Export events as Praat TextGrid

Observations > **Export events** > **as Praat TextGrid**

Example:

```
File type = "ooTextFile"
Object class = "TextGrid"

xmin = 4.3
xmax = 113.988
tiers? <exists>
size = 2
item []:
    item [1]:
        class = "IntervalTier"
        name = "Subject #1"
        xmin = 4.3
        xmax = 10.0
        intervals: size = 1
        intervals [1]:
            xmin = 4.3
            xmax = 10.0
            text = "eat"
    item [2]:
        class = "IntervalTier"
        name = "Subject #2"
        xmin = 26.6
        xmax = 113.988
```

```
        intervals: size = 1
        intervals [1]:
            xmin = 26.6
            xmax = 113.988
            text = "eat"
```

2.7.5 Export events for analysis with JWatcher

Observations > **Export events for analysis with JWatcher**

JWatcher is a powerful tool for the quantitative analysis of behavior.

Events coded with BORIS can be exported for analysis with JWatcher.

Click **Observations** > **Export events** > **for analysis with JWatcher** to export the coded events.

BORIS will ask you to select a directory. Then, for each combination of selected observation and selected subject, the following files will be created:

- the Focal Data File (.dat)
- the Focal Analysis Master File (.faf)
- the Focal Master File (.fmf)

These files can be used to analyze your observations with JWatcher.

2.7.6 Export events as Behaviors Binary Table

Observations > **Export events as Behaviors Binary Table**

A time interval will be requested (in seconds). The observation will be checked every n seconds, and the presence (1, absence: 0) of the selected behaviors will be exported in a table for each selected subject.

Example for a time interval of 1 second:

```
time    Alert     Drink    Locomotion    Swim
0.0     0         1        0             0
1.0     0         1        0             0
2.0     0         1        0             0
3.0     0         1        0             0
4.0     0         1        0             0
5.0     0         1        0             0
6.0     0         1        0             0
7.0     0         1        0             0
8.0     0         1        0             0
9.0     1         0        0             0
10.0    1         0        0             0
11.0    0         0        1             0
12.0    1         0        0             0
13.0    1         0        0             0
14.0    1         0        0             0
15.0    1         0        0             0
16.0    1         0        0             0
17.0    1         0        0             0
18.0    0         0        1             0
19.0    0         0        1             0
20.0    0         0        1             0
21.0    0         0        1             0
22.0    1         0        0             0
23.0    0         0        0             0
24.0    0         0        0             0
25.0    0         0        0             0
26.0    0         0        0             0
27.0    0         0        0             0
28.0    0         0        0             0
29.0    0         0        0             0
30.0    0         0        0             0
31.0    0         0        0             0
32.0    0         0        0             0
33.0    0         0        0             1
34.0    0         0        0             1
35.0    0         0        0             1
36.0    0         0        0             1
37.0    0         0        0             1
38.0    0         0        0             1
39.0    0         0        0             1
40.0    0         0        0             1
41.0    0         0        0             1
42.0    0         0        0             1
43.0    0         0        0             1
44.0    0         0        0             1
```

```
45.0    0        0       0           1
46.0    0        0       0           1
47.0    0        0       0           1
48.0    0        0       0           1
49.0    0        0       0           1
```

2.7.7 Extract clips from media files corresponding to coded events

Media sequences corresponding to coded events can be extracted from media files:

1. Click on **Observations** > **Extract clips from media files**.
2. Choose the observation(s).
3. Select the subjects and events to be extracted. Modifiers can be included.
4. Select the time interval around the events (in seconds; the default value is 0).
5. Select the tracks to include in clips:
   - Video and audio
   - Only video
   - Only audio
6. Select a destination directory to contain the extracted clips.

The time offset will be subtracted from the event start time and added to the stop time. All extracted clips will be saved in the selected directory following the file name format:

{observation id}_{player}_{subject}_{behavior}_{start time}-{stop time}

If modifiers are included:

{observation id}_{player}_{subject}_{behavior}_{start time}-{stop time}_{modifiers concatenated with +}

2.7.8 Extract frames corresponding to coded events

Frames corresponding to coded events can be extracted and saved as images.

1. Click on **Observations** > **Extract frames from media files**.
2. Choose the observation(s).
3. Select the subjects and events to be extracted. Modifiers can be included.
4. Select the time interval around the events (in seconds; the default value is 0).
5. Select the image format:
   - JPG - small size / low quality
   - PNG - large size / high quality
6. Select a destination directory to contain the extracted frames.

2.7.9 Export transitions matrix

Three transition matrix outputs are available: the matrix of frequencies of transitions, the matrix of frequencies of transitions after each behavior, and the matrix of the number of transitions.

Matrix of frequencies of transitions

This matrix contains the frequencies of total transitions. The sum of all frequencies must be 1.

Example of frequencies of transitions matrix:

```
eat   sleep     walk
eat        0.0   0.286    0.143
```

```
sleep    0.143     0.0    0.143
walk     0.286     0.0      0.0
```

In this matrix, you can see that the **eat** behavior precedes the **sleep** behavior with a frequency of **0.286** of the total number of transitions.

Matrix of frequencies of transitions after behavior

This matrix contains the frequencies of transitions after each behavior. The sum of each row must be 1.

Example:

```
eat    sleep     walk
eat     0.0    0.667    0.333
sleep   0.5      0.0      0.5
walk    1.0      0.0      0.0
```

In this example, you can see that **sleep** follows **eat** with a frequency of **0.667**, and **walk** follows with a frequency of **0.333**.

Matrix of number of transitions

This matrix contains the number of transitions after each behavior.

Example:

```
eat   sleep   walk
eat       0       2      1
sleep     1       0      1
walk      2       0      0
```

2.8 Playback menu

2.8.1 Jump

Jump forward

Allows you to jump forward in the current media file. See **File** > **Preferences** for setting the jump value.

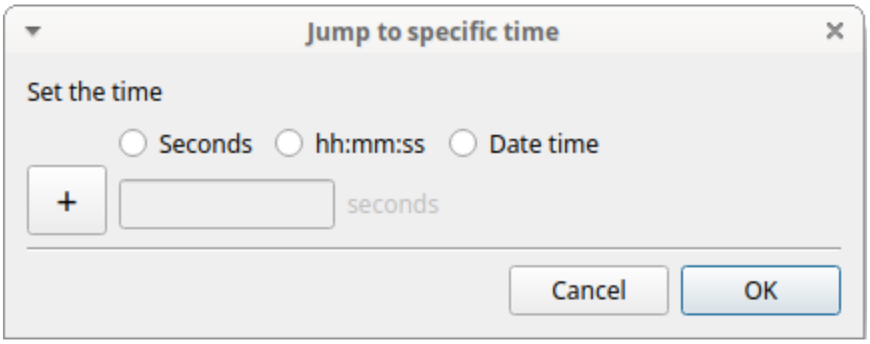
Jump backward

Allows you to jump backward in the current media file. See **File** > **Preferences** for setting the jump value.

Jump to specific time

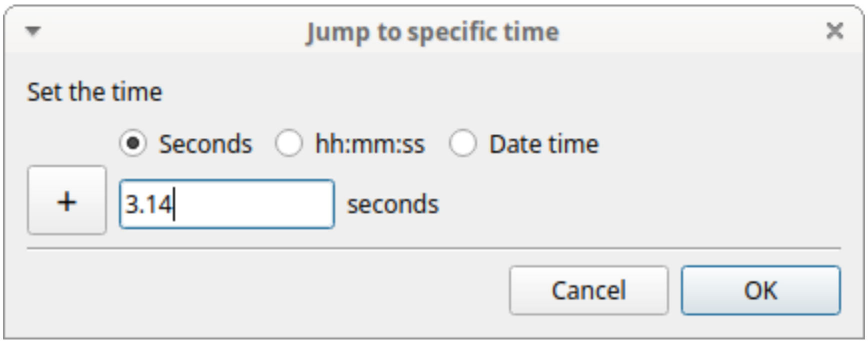
Allows you to go to a specific time in the current media file.

The time selection widget will pop up:

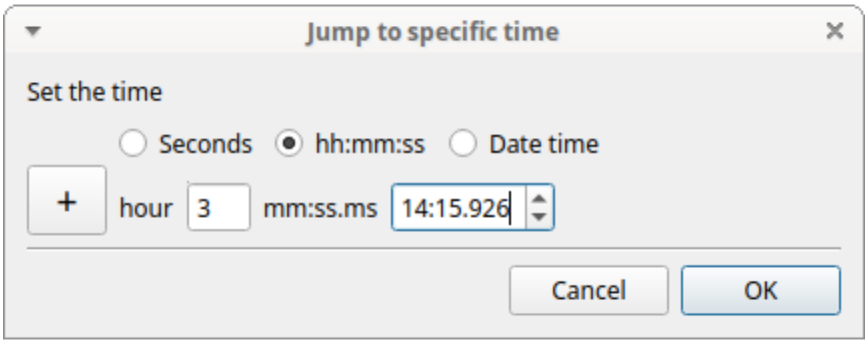


Three time formats are available:

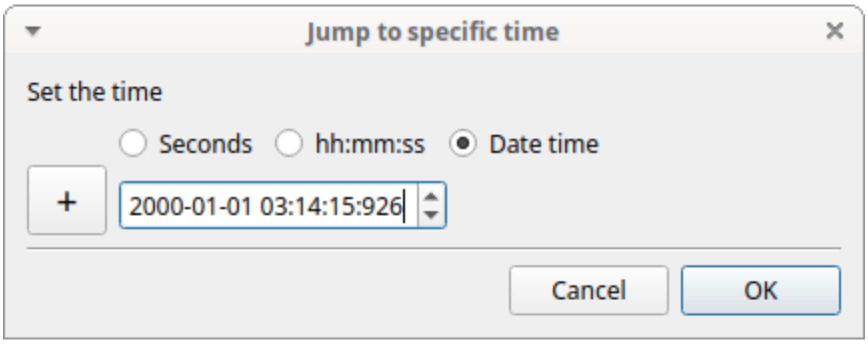
- Decimal seconds:



- HH:MM:SS:ZZZ format (ZZZ indicates the milliseconds):

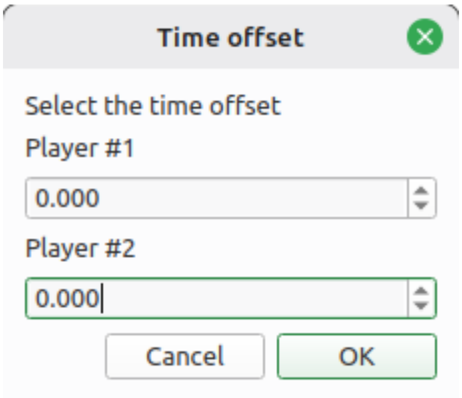


• A date-time format (YYYY-MM-DD hh:mm:ss.zzz):



2.8.2 Change time offset of players

Allows you to change the time offset for each player (in seconds)

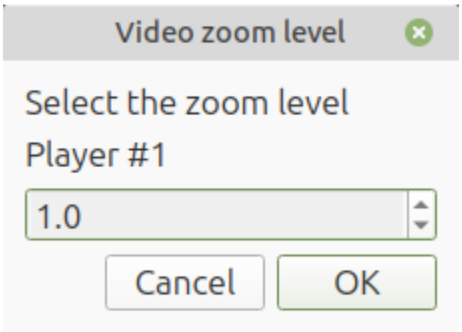


2.8.3 Deinterlace

Check this option if your video is interlaced

2.8.4 Zoom level

The zoom level can also be set using the menu **Playback** > **Zoom level**



Using the keyboard

Click on the media player for which you want to set the zoom level.

Zoom in `^ Ctrl` + `+` or `^ Ctrl` + `⇞ Page Up`

Zoom out `^ Ctrl` + `-` or `^ Ctrl` + `⇟ Page Down`

Reset zoom level `^ Ctrl` + `0`

> **Important**
>
> On Microsoft Windows, only the `+` and `-` keys on the numeric keypad can be used.

Using the mouse

Click on the media player for which you want to set the zoom level.

Zoom in `^ Ctrl` + **Mouse Wheel Up** or Double click on left mouse button

Zoom out `^ Ctrl` + **Mouse Wheel Down** or Double click on right mouse button

Reset zoom level Click the **right mouse button** on the video.

> **Important**
>
> On Microsoft Windows, the mouse cannot be used to zoom the video

2.8.5 Pan video

Click on the media player you want to pan.

Using the keyboard

Pan Up `^ Ctrl` + `↓ Down`

Pan Down `^ Ctrl` + `↑ Up`

Pan Left `^ Ctrl` + `← Left`

Pan Right `^ Ctrl` + `→ Right`

Using the mouse

Pan Up: **Mouse Wheel Up** (the video moves down)

Pan Down: **Mouse Wheel Down** (the video moves up)

Pan Left: `⇧ Shift` + **Mouse Wheel Up** (the video moves to the right)

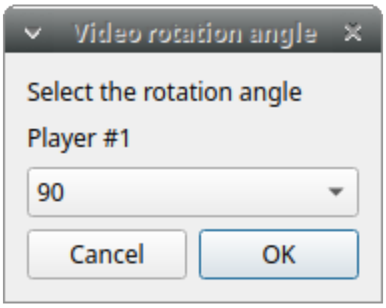
Pan Right: `⇧ Shift` + **Mouse Wheel Down** (the video moves to the left)

Reset Pan and zoom: `⇧ Shift` + **Left mouse button**

> **Important**
>
> On Microsoft Windows, the mouse cannot be used to pan the video

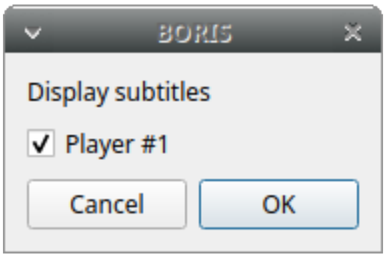
2.8.6 Rotate video

Select the video rotation angle for each player using the menu **Playback** > **Rotate video**. The available rotation angles are: 0, 90, 180 and 270.



2.8.7 Display subtitles

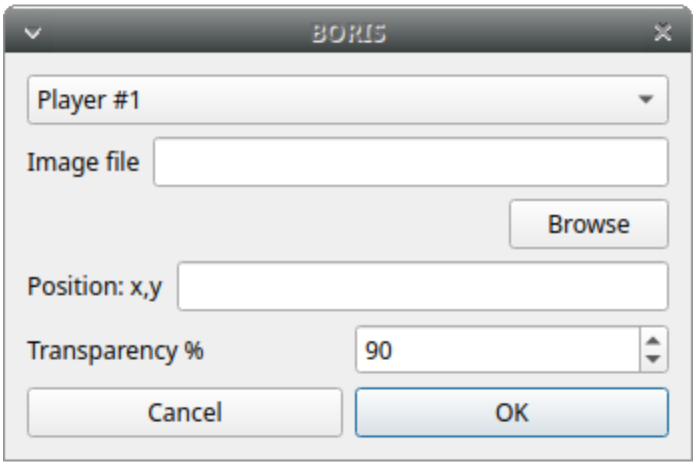
Select to display or hide subtitles using the menu **Playback** > **Display subtitles**. The subtitles file must have exactly the same name as the video file, except for the extension, and be placed in the same directory.



2.8.8 Image overlay on video

Select an image overlay to display on the video via **Playback** > **Image overlay on video** > **Add**. If the selected image does not have a transparent background, transparency can be set from 0 (full transparency) to 255 (no transparency).

The image must be in PNG format. If the image is smaller than the video resolution, its position can be set from the top-left corner (x: horizontally, y: vertically).

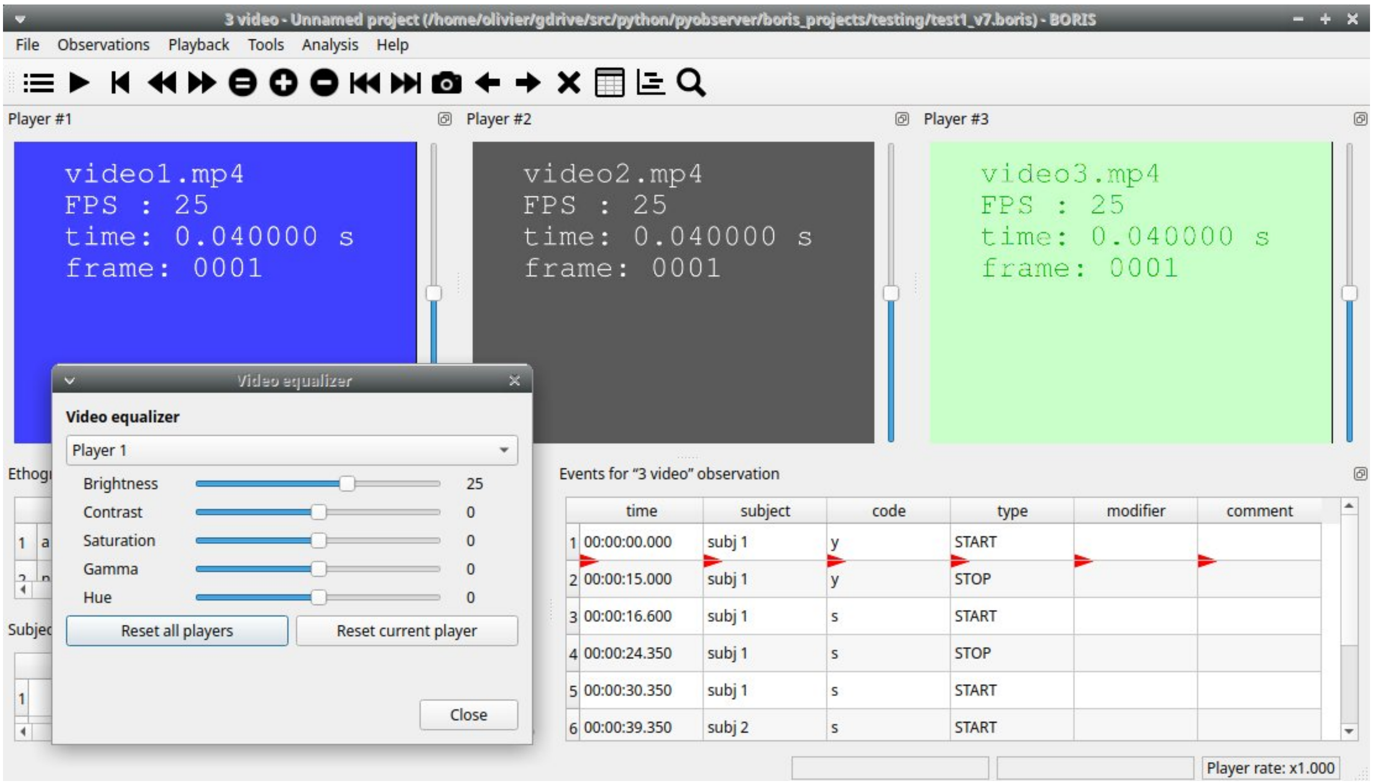


Select **> Playback > Image overlay on video > Remove** to remove the image overlay.

2.8.9 Video equalizer

Playback > **Video equalizer**

This function allows you to adjust the **brightness**, **contrast**, **saturation**, **gamma**, and **hue** for each player.



2.9 Tools

2.9.1 Media file

MEDIA FILE INFORMATION

Obtain information about your media file, such as duration, resolution, FPS, bitrate, etc.

RE-ENCODE / RESIZE VIDEO

BORIS can re-encode and resize your video files to reduce their size and ensure smooth coding (especially when two video files play simultaneously). The re-encoding and resizing operations are performed using the embedded ffmpeg program with high-quality parameters (bitrate 2000k).

Select the files you want to re-encode and resize, and select the horizontal resolution in pixels (the default is 1024). The aspect ratio will be maintained.

You can continue to use BORIS during the re-encoding/resizing operation.

The re-encoded/resized video files are renamed by adding the re-encoded.avi extension to the original file names.

ROTATE VIDEO

BORIS can rotate your video files to allow coding with the correct view. The rotation operation is performed using the embedded ffmpeg program with the same quality parameters as the original video.

Select the files you want to rotate and select the rotation angle: **Rotate 90 clockwise**, **Rotate 90 counter-clockwise**, or **Rotate 180**.

The aspect ratio will be maintained.

You can continue to use BORIS during the rotation operation.

The rotated video files are renamed by adding **rotated\<ANGLE>** to the original file name.

MERGE MEDIA FILES

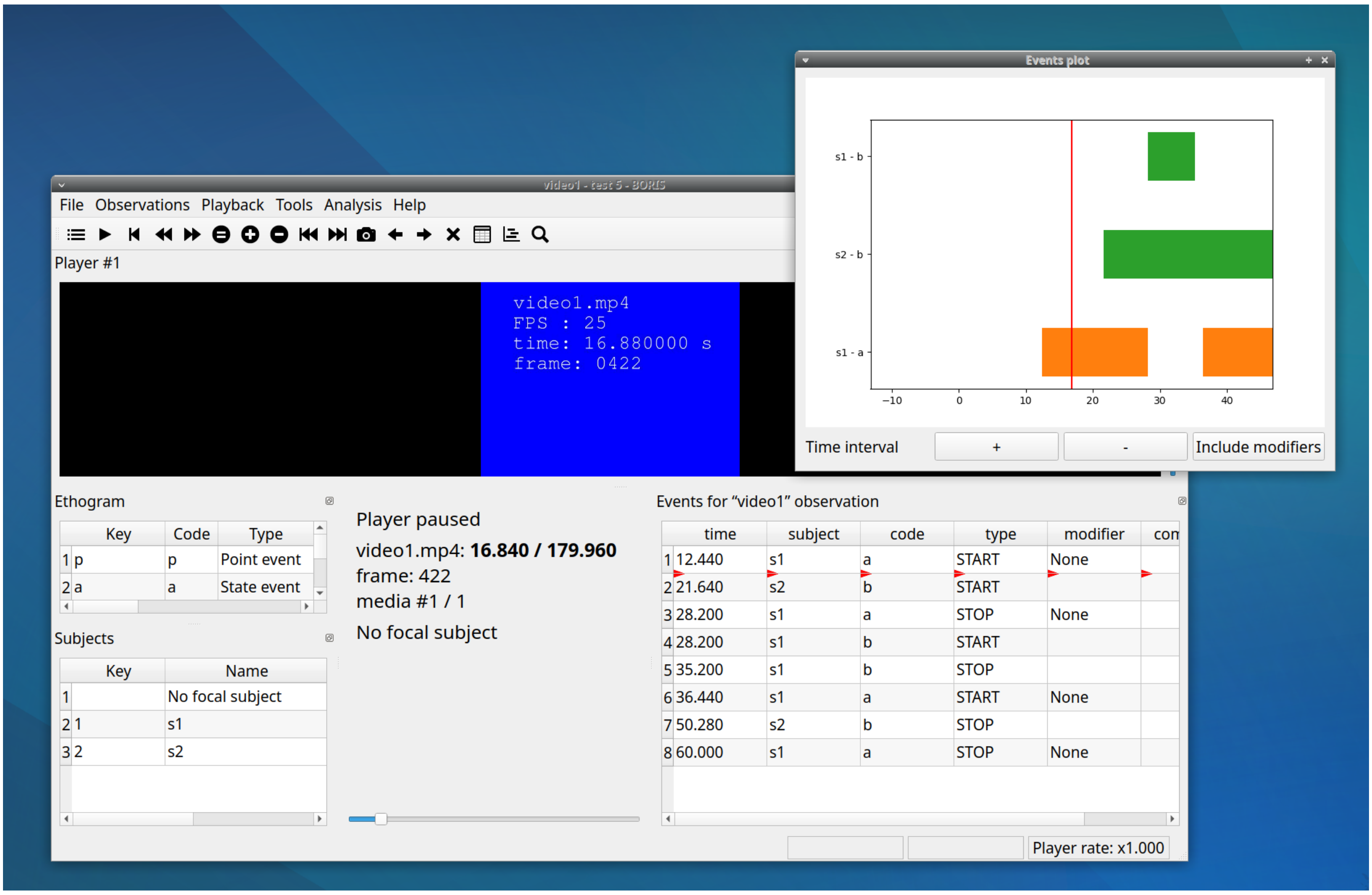
Use this function to concatenate multiple media files together.

CREATE VIDEO SPECTROGRAM

Create a video with the spectrogram of the audio track.

2.9.2 Plot events in real-time

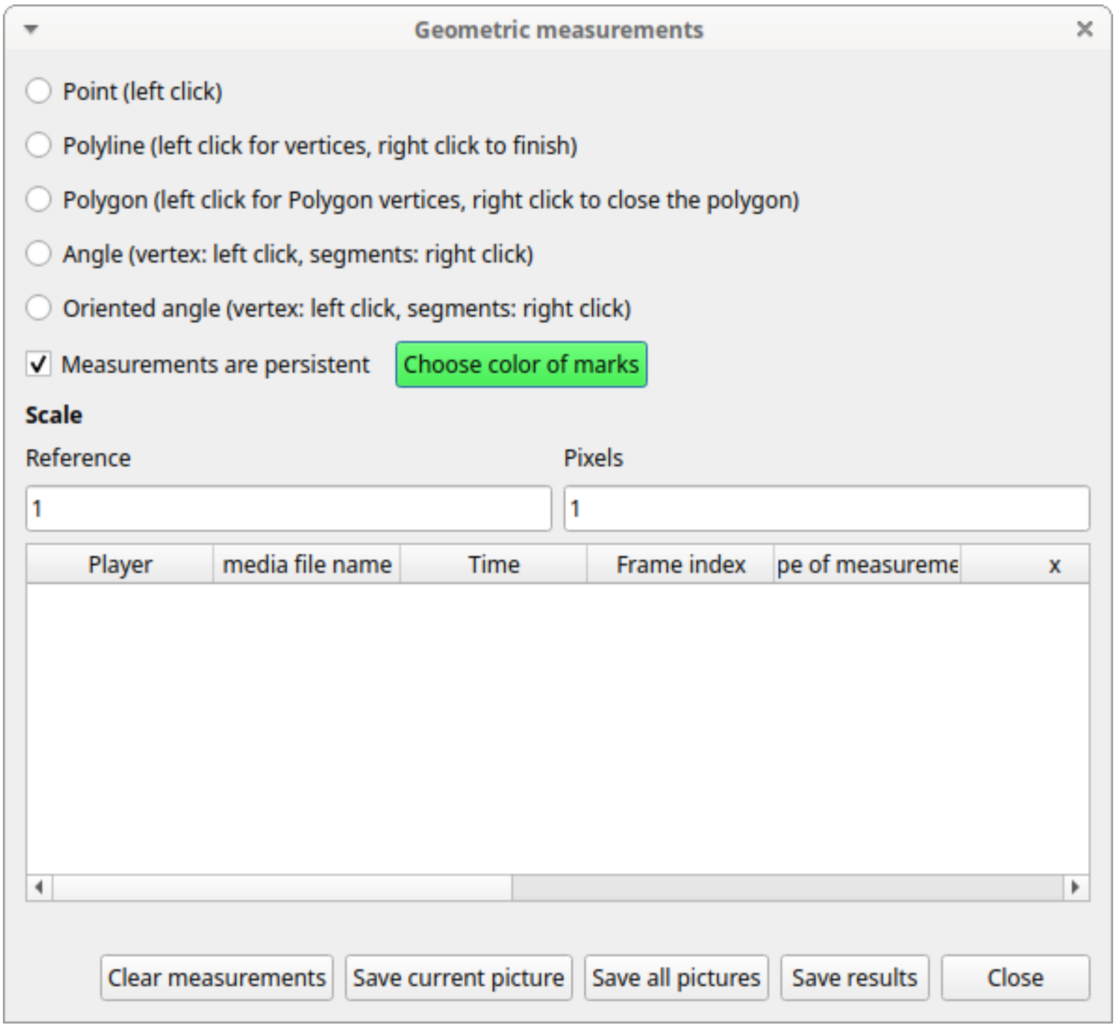
This function can be activated with **Tools** > **Plot event in real time**.



2.9.3 Geometric measurements

Several geometric measurements are available: **distances**, **areas**, and **angles** can be measured, and **point positions** can be recorded.

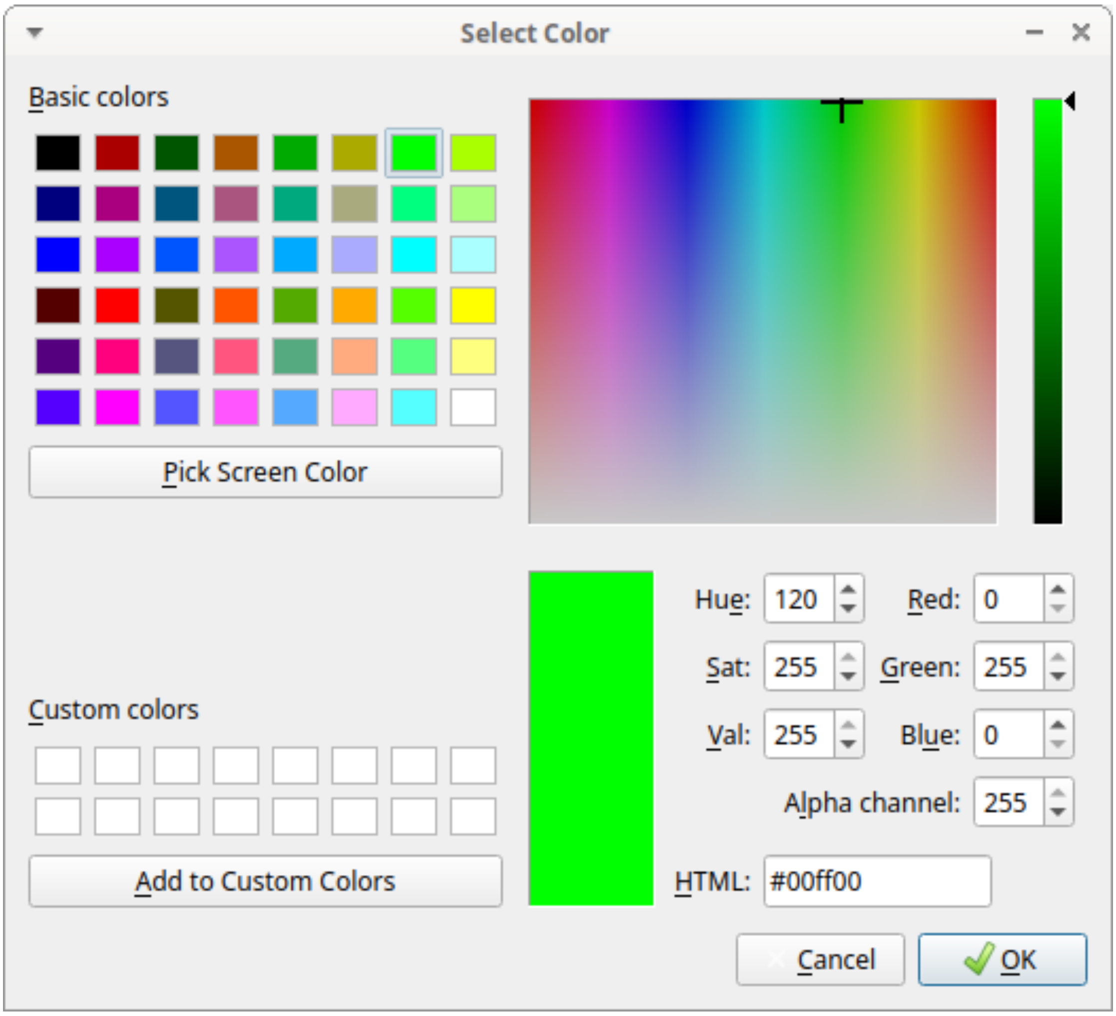
Click **Tools** > **Geometric measurements** to activate the measurement tools.



The geometric measurements window

Mark color

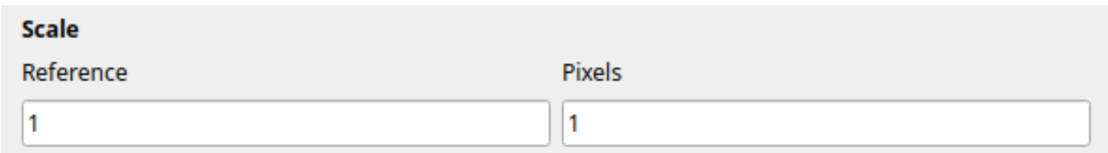
Use the **Choose color of marks** button to select a color. All marks will be drawn with the selected color. The color transparency can be set using the **Alpha channel** value (0 for 100% transparent, 255 for a solid color).



The color selection window

Setting the scale

For distance and area measurements, you can set a scale so that the results are expressed in real units such as centimeters or meters.

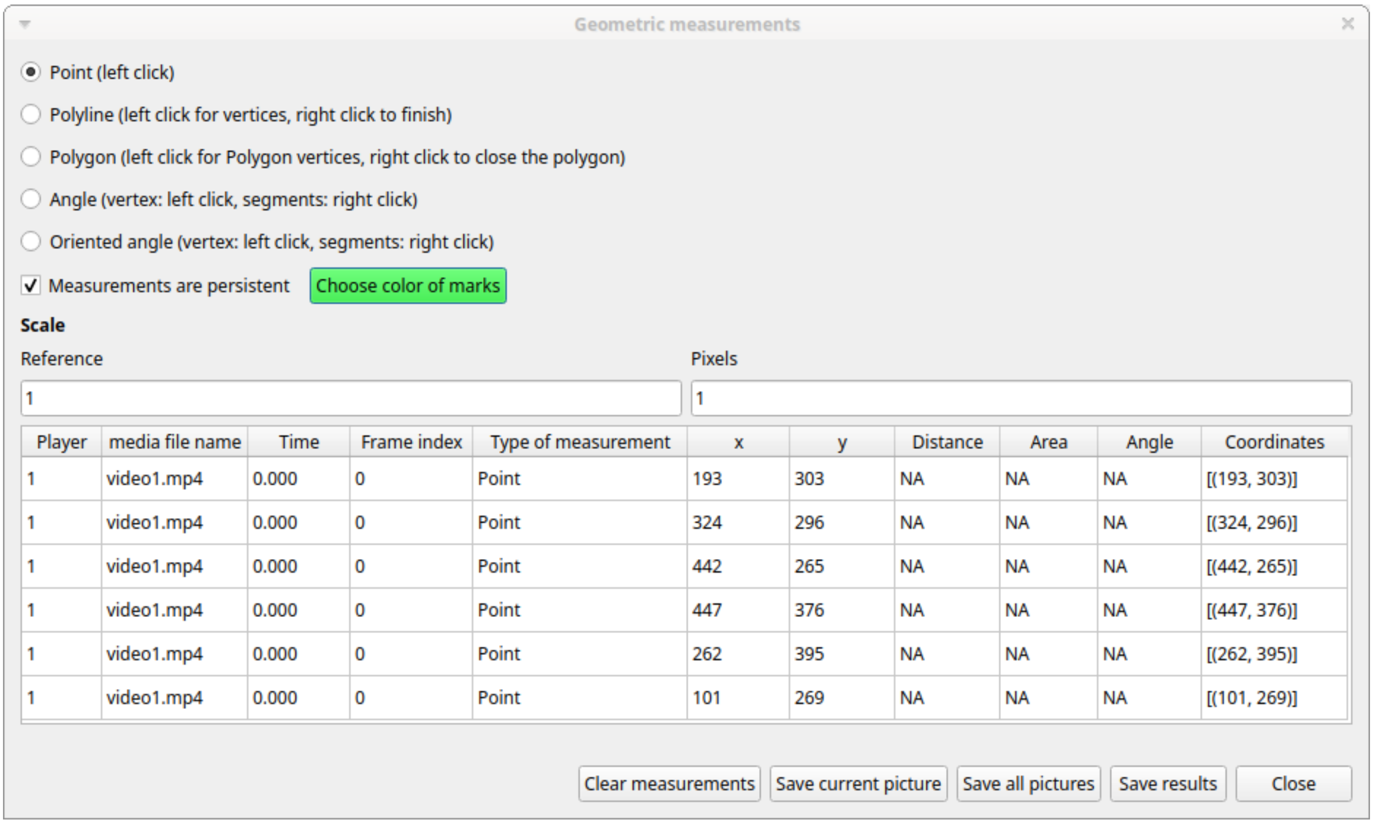


Setting the scale

1. Measure a reference object with a known size on the frame (with the distance tool; see the next section for details) and enter the pixel distance in the **Pixel** text box.
2. Enter the real size of the reference object in the **Reference** text box (must be a number without units).

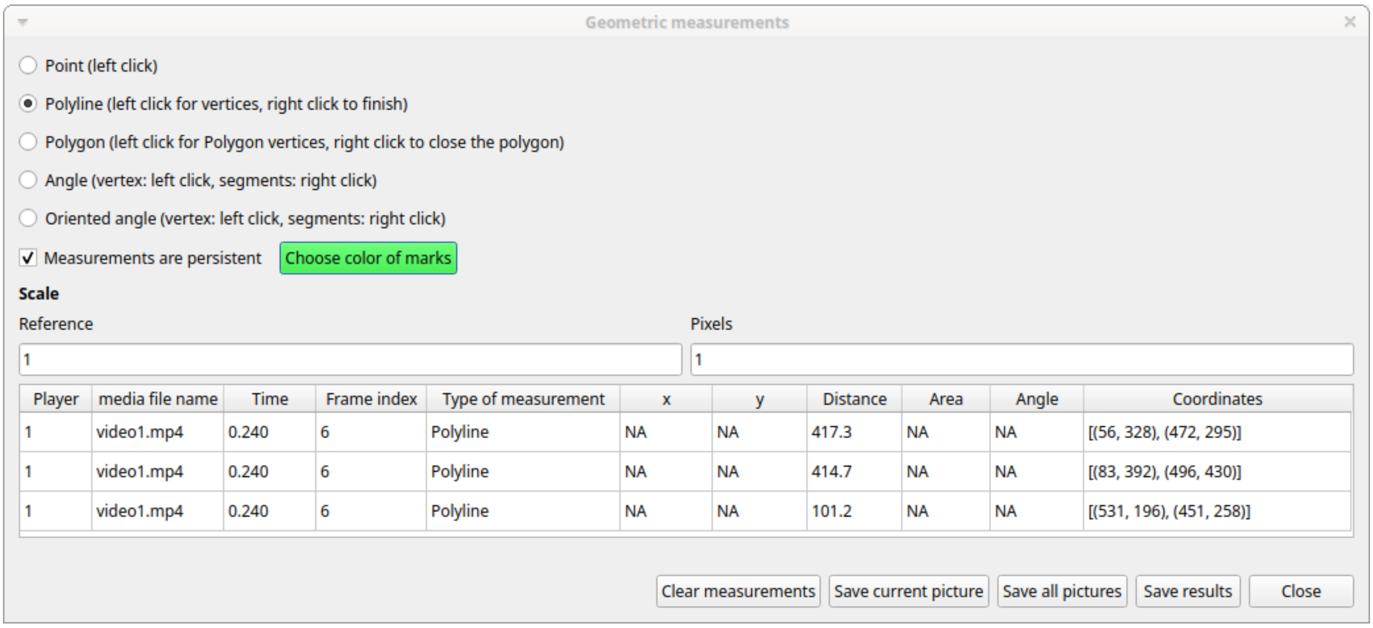
Point

Select the **Point** radio button. Click the left mouse button on the video/image to record the position of the clicked pixel.



Distance measurements

Select the **Distance** radio button. Click the left mouse button on the frame bitmap to set the start of the segment to be measured. A circle with a cross will be drawn. Click the right mouse button to set the end. A red circle with a cross will be drawn. The distance between the two selected points will be displayed in the text area of the **Measurements window**.



Area measurements

Select the **Area** radio button. Click the left mouse button on the frame bitmap to set the area vertices. Circles with a cross will be drawn. Click the right mouse button to close the area. The area of the drawn polygon will be displayed in the text area of the **Measurements window**.

Angle measurements

Select the **Angle** radio button. Click the left mouse button on the frame bitmap to set the angle vertex. A red circle with a cross will be drawn. Click the right mouse button to set the two segments. Circles with a cross will be drawn. The angle between the two drawn segments will be displayed in the text area of the **Measurements window**.

Persistent measurements

If the **Measurements are persistent** checkbox is selected, the measurement overlays will remain visible on all frames; otherwise, they will be cleared between frames.

The marks selected on other frames will be drawn in red.

2.9.4 Coding pad

During an observation, a coding pad with the available behaviors can be displayed (**Tools** > **Coding pad**). This **Coding pad** allows you to code using a touch screen or by clicking the buttons. When the **Coding pad** is displayed, you can still code using the keyboard or the ethogram.

The button size can be increased or decreased.

Button colors can be set per behavior, per behavioral category, or disabled entirely.

See the drop-down list in the upper-left corner of the Coding pad window.

2.9.5 Subjects pad

A pad with all defined subjects, or only filtered subjects, can be displayed during the observation (**Tools** > **Subjects pad**). This **Subjects pad** allows you to select the focal subject using a touch screen or by clicking the buttons. When the **Subjects pad** is displayed, you can continue to select the focal subject using the keyboard or the subjects list.

2.9.6 Transitions flow diagram

BORIS can generate DOT scripts and flow diagrams from the transition matrices (see Observations > Create transition matrix to obtain the transition matrices).

DOT script (Graphviz language)

Tools > Transitions flow diagram > Create transitions DOT script

Choose one or more transition matrix files, and BORIS will create the corresponding DOT script file(s).

The DOT script files can then be used with Graphviz (Graph Visualization Software) or WebGraphviz (Graphviz in the Browser) to generate transition flow diagrams.

See DOT (graph description language) for details.

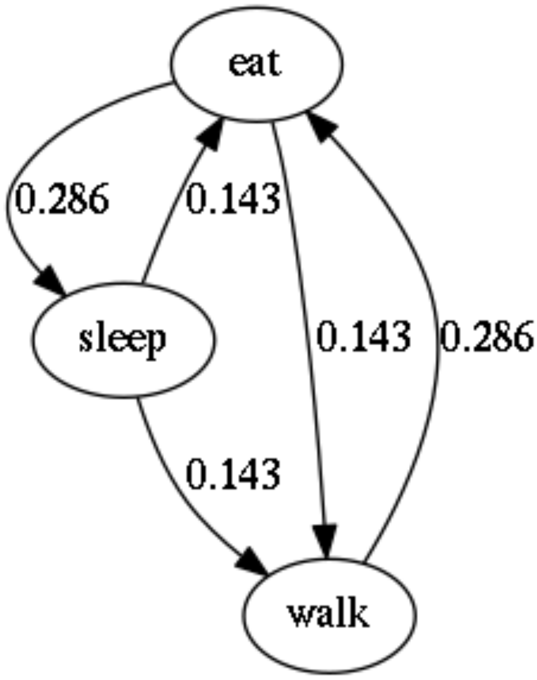
Flow diagram

If Graphviz (Graph Visualization Software) is installed on your system (and the **dot** program is available in the path), BORIS can generate a flow diagram (PNG format) from a transition matrix file.

Tools > Transitions flow diagram > Create transitions flow diagram

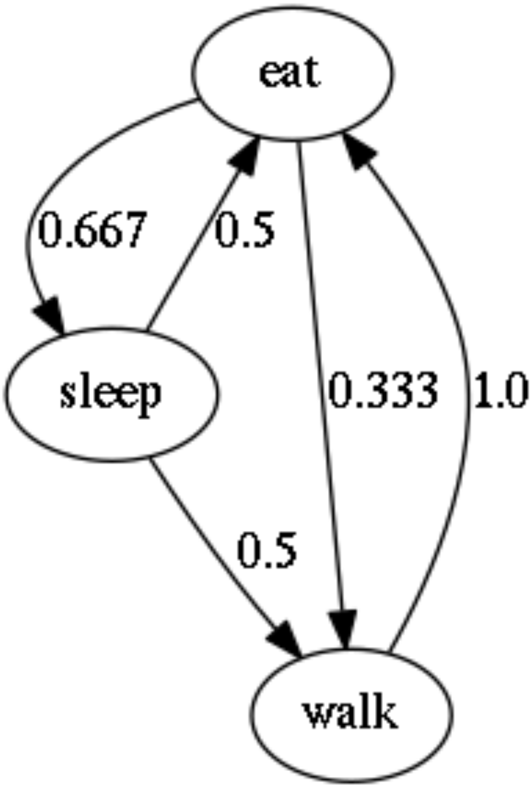
Choose one or more transition matrix files, and BORIS will create the corresponding flow diagrams.

Flow diagram of frequencies of transitions



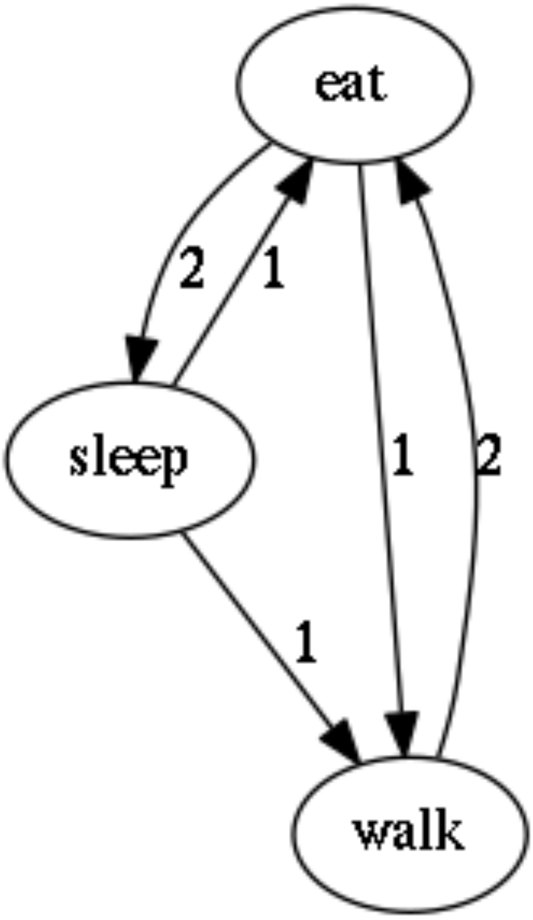
Frequencies of total transitions (the frequencies are plotted on the edges)

Flow diagram of frequencies of transitions after behavior



Frequencies of transitions after behavior (the frequencies are plotted on the edges)

Flow diagram of number of transitions



Number of transitions (the frequencies are plotted on the edges)

2.10 Coding map

A coding map is a bitmap image with user-defined clickable areas that help code behaviors or modifiers for a behavior.

2 types of coding maps are available:

- Behaviors coding map
- Modifiers coding map

2.10.1 The Behaviors coding map

BORIS allows you to create a **Behaviors coding map** using the **Map creator** tool (**Tools** > **Create a coding map** > **for behaviors**).

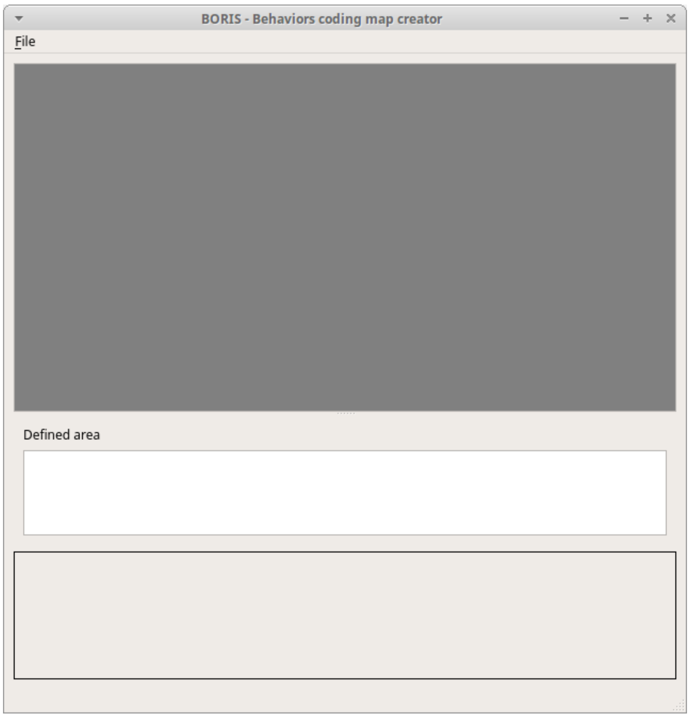
A **Behaviors coding map** can be created only if you have defined behaviors in your ethogram.

Creating a Behaviors coding map

To create a new **Behaviors coding map**, launch the **Behaviors coding map creator**:

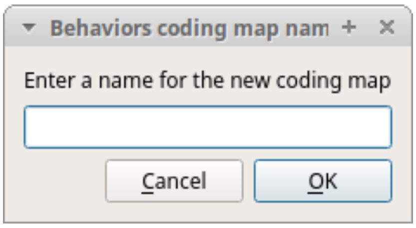
Tools > **Create a coding map** > **for behaviors**.

A new window will open



File > **New behaviors coding map**

Enter a name for the new **Behaviors coding map**

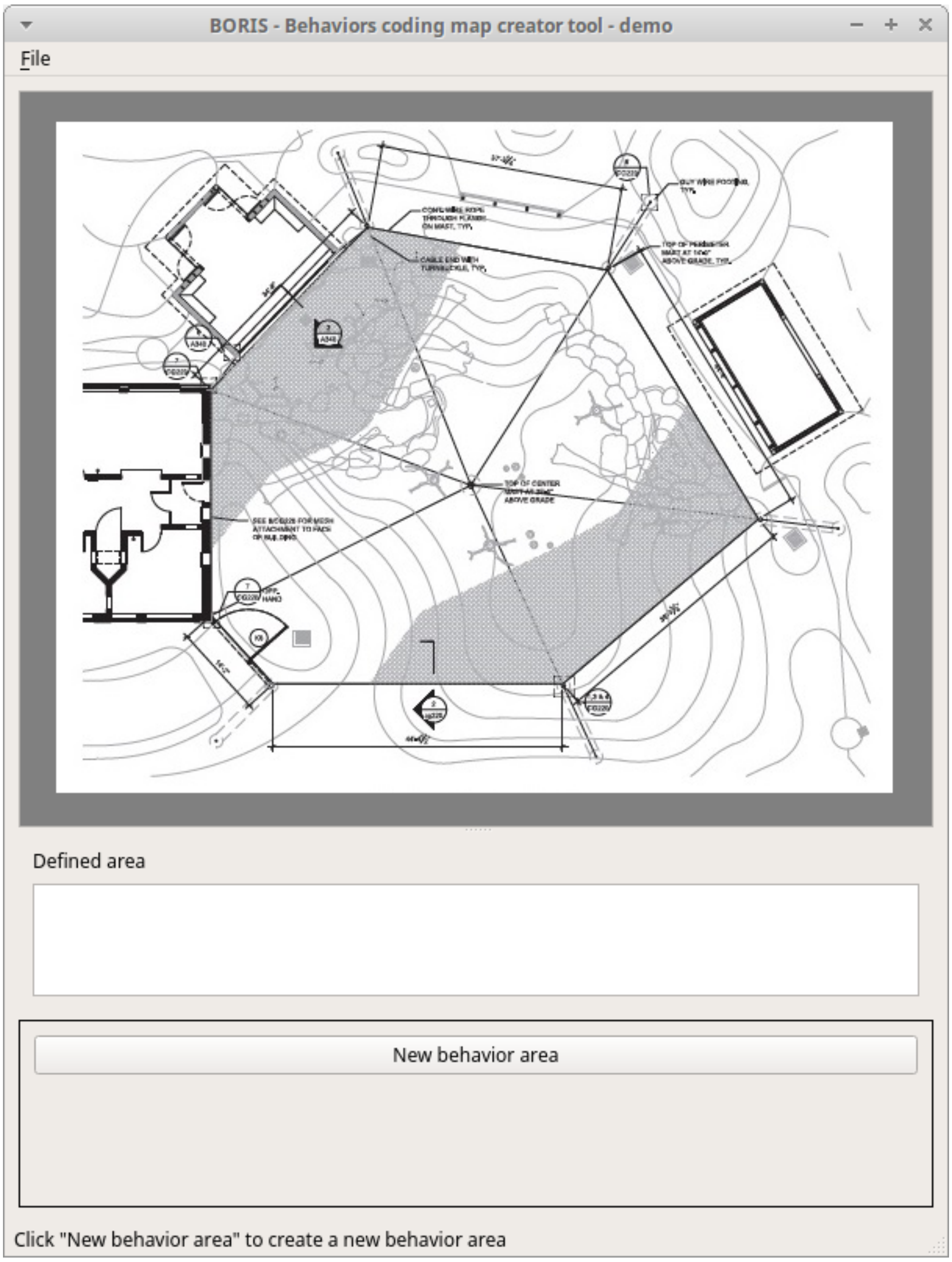


Loading a bitmap for a behaviors coding map

Click the **Load bitmap** button at the bottom of the window and select a bitmap image (PNG and JPEG formats are accepted).

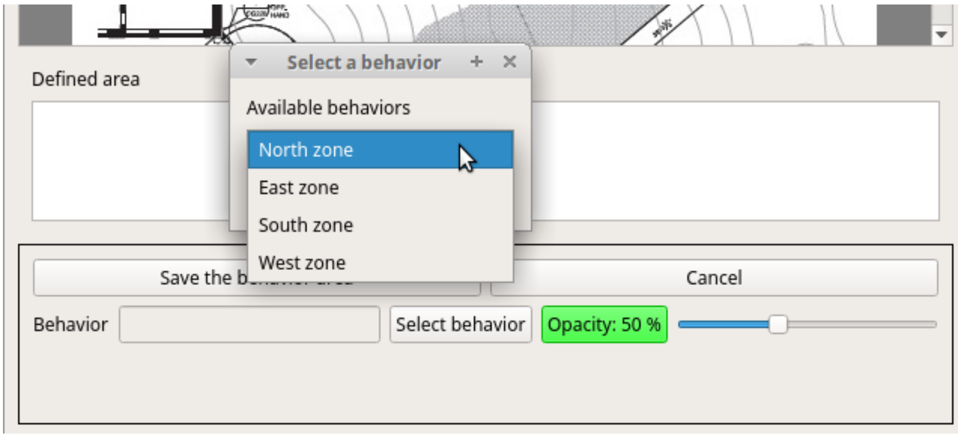
If your bitmap image is larger than 640 x 640 pixels, BORIS will resize it to 640 x 640 pixels, keeping the aspect ratio, and store the resized version in the coding map file.

The bitmap will be displayed



Adding areas corresponding to the behaviors

Click the **New behavior area** button at the bottom of the window and select a behavior by clicking on the **Select behavior** button.



The available behaviors are taken from the ethogram of the current project.

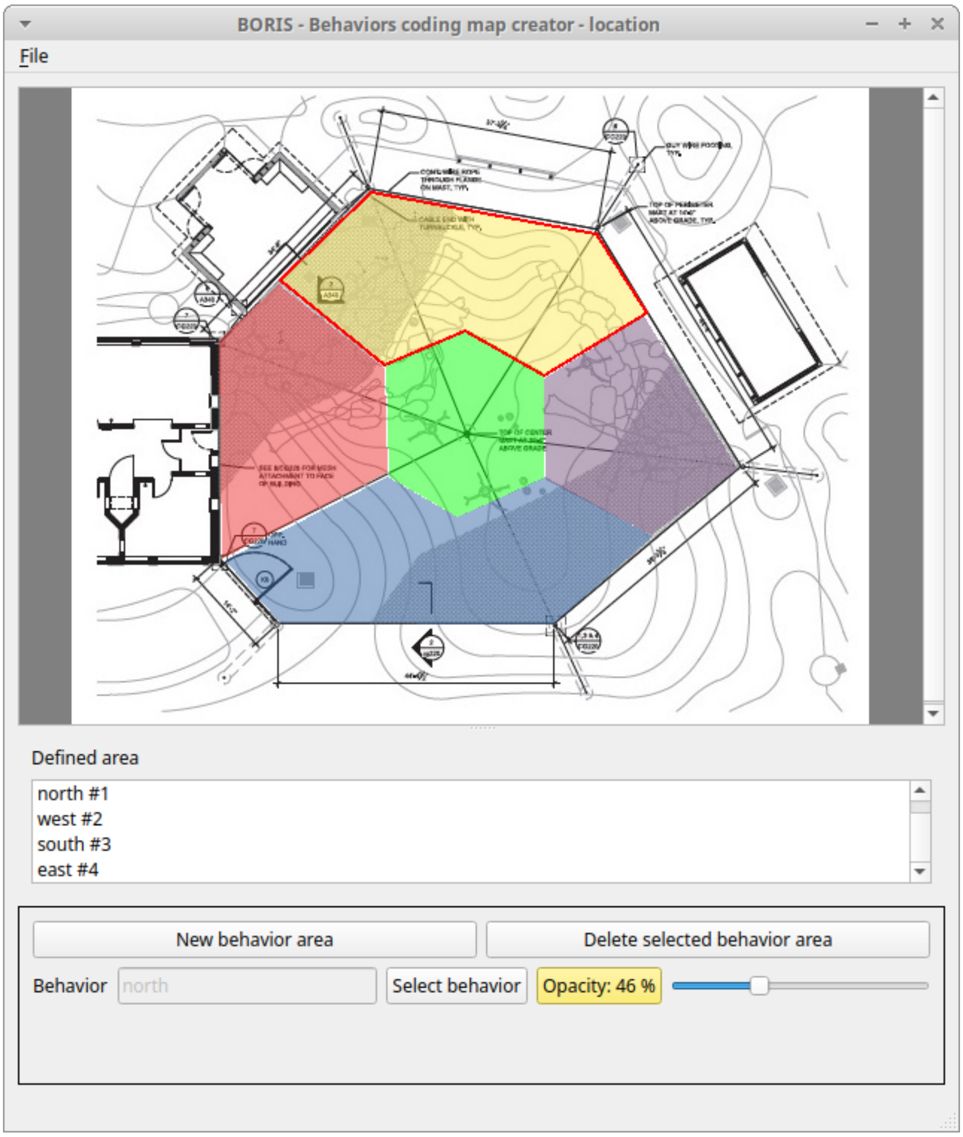
Click on the bitmap to define the vertices of the area that will code the selected behavior. Close the area by clicking again on the first point.

The color of the new area can be changed using the **Opacity** button. The opacity can be adjusted (from 0 to 100%) using the slider.

Save the behavior area by clicking the **Save the behavior area** button.

The area will be added to the **Defined area** list.

You can add more areas and also add more than one area for the same behavior. Two or more areas can overlap. In this case, all corresponding behaviors will be triggered.



Add the Behaviors coding map to the current project

File > **Add coding map to project**

The coding map will be added to the current project.

You can add a **Behaviors coding map** to the current project from a file containing the coding map:

(**File** > **Edit project** > **Behaviors coding map** > **Add a behaviors coding map**)

Saving the Behaviors coding map

Saving the **Behaviors coding map** creates a file containing the **Behaviors coding map**, including the bitmap image.

File > **Save the current Behaviors coding map**

The file containing the **Behaviors coding map** can then be reloaded in the **Behaviors coding map creator** or added to a BORIS project (**File** > **Edit project** > **Behaviors coding map** > **Add a behaviors coding map**)

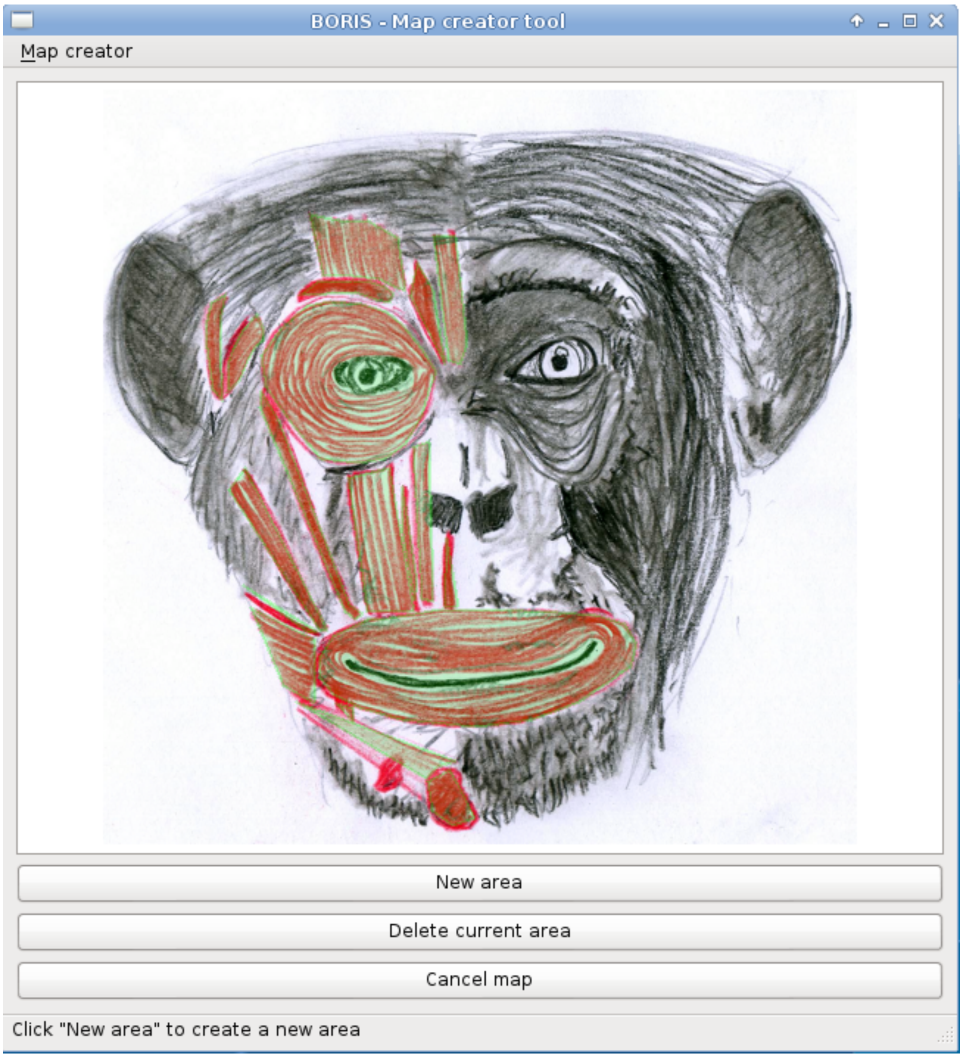
2.10.2 The Modifiers coding map

BORIS allows you to create a modifiers coding map using the **Modifiers Map creator** tool (**Tools** > **Create a coding map** > **for modifiers**). Clickable areas may correspond to specific modifiers that can be meaningful for behavioral coding. Facial expression is the primary use case we had in mind when developing this function.

Creating a modifiers coding map

Loading a bitmap for a modifiers coding map

To create a new **Modifiers coding map**, launch the **Modifiers Map creator** tool (**Tools** > **Create a coding map** > **for modifiers**). The BORIS main window will be replaced by the **Modifiers Map creator** window. Click on **Modifiers Map creator** > **New Modifiers map** and enter a name for the new map in the edit box. Then load a bitmap image (JPEG or PNG) using the **Load bitmap** button. The loaded image will be displayed.



If your bitmap image is larger than 640 x 640 pixels, BORIS will resize it to 640 x 640 pixels, keeping the aspect ratio, and store the resized version in the coding map file.

Adding areas corresponding to the modifiers

To create clickable areas on a coding map, click the **New area** button and enter an **Area code** in the edit box. The new area can then be defined by clicking on the image. The drawing tool allows you to define an irregular polygon (a plane shape with straight sides, which does not have all sides equal and all angles equal) by clicking to determine subsequent vertices. It can be convex or concave. Straight sides must not cross each other. Once selected, an area can be deleted using the **Delete area** button. When an area is closed and its name has been defined in the **Area code** field, it can be saved using the **Save area** button. The areas can partially overlap each other. See the **Using a Coding map** section for more details. Once all areas are added, the entire map can be saved using the **Save map** option menu (**Map creator** > **Save map**). The map is saved in its own file (.boris_map), which is NOT part of the BORIS project. A map can be edited at any time by opening the map file from the **Open map** menu option (**Map creator** > **Open map**).

Adding a modifiers coding map to your project

Creating a Coding map does not automatically add the map to your project. The Coding map has to be added to your project by selecting the corresponding **Behavior type** (**Point event with coding map**, **State event with coding map**). BORIS will ask you to select the file name containing the coding map (.boris_map) and load the coding map into the project. The coding map name will appear in the **Coding map** column and will be saved in the BORIS project file.

> **Important**
>
> If you later modify your coding map you must reload the new version in your BORIS project.

2.11 Analysis and plot

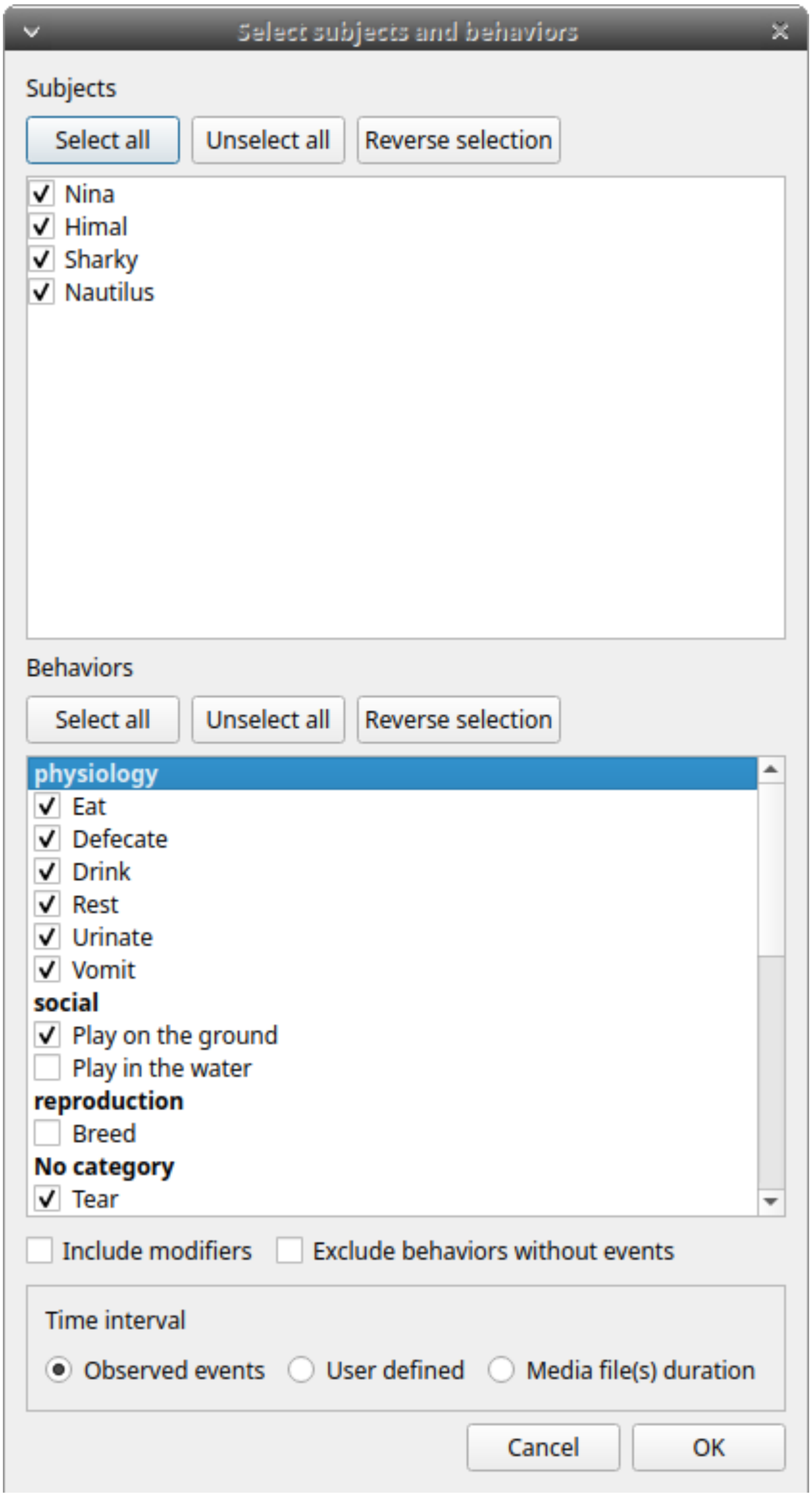
2.11.1 Time budget analysis

The **Time budget analysis** can be performed by behavior (including or excluding modifiers) or by behavioral category. Select the option from the **Analysis** menu.

The **Time budget analysis** can be performed on one or more observations. If you select more than one observation, you can choose either a global time budget analysis that includes all selected observations or a separate time budget analysis for each observation.

Choose **Yes** to group all observations into a single time budget analysis.

The **Analysis** > **Time budget** menu option shows the time budget for the events of the selected observations. Select the subjects and behaviors you want to include in the time budget analysis:



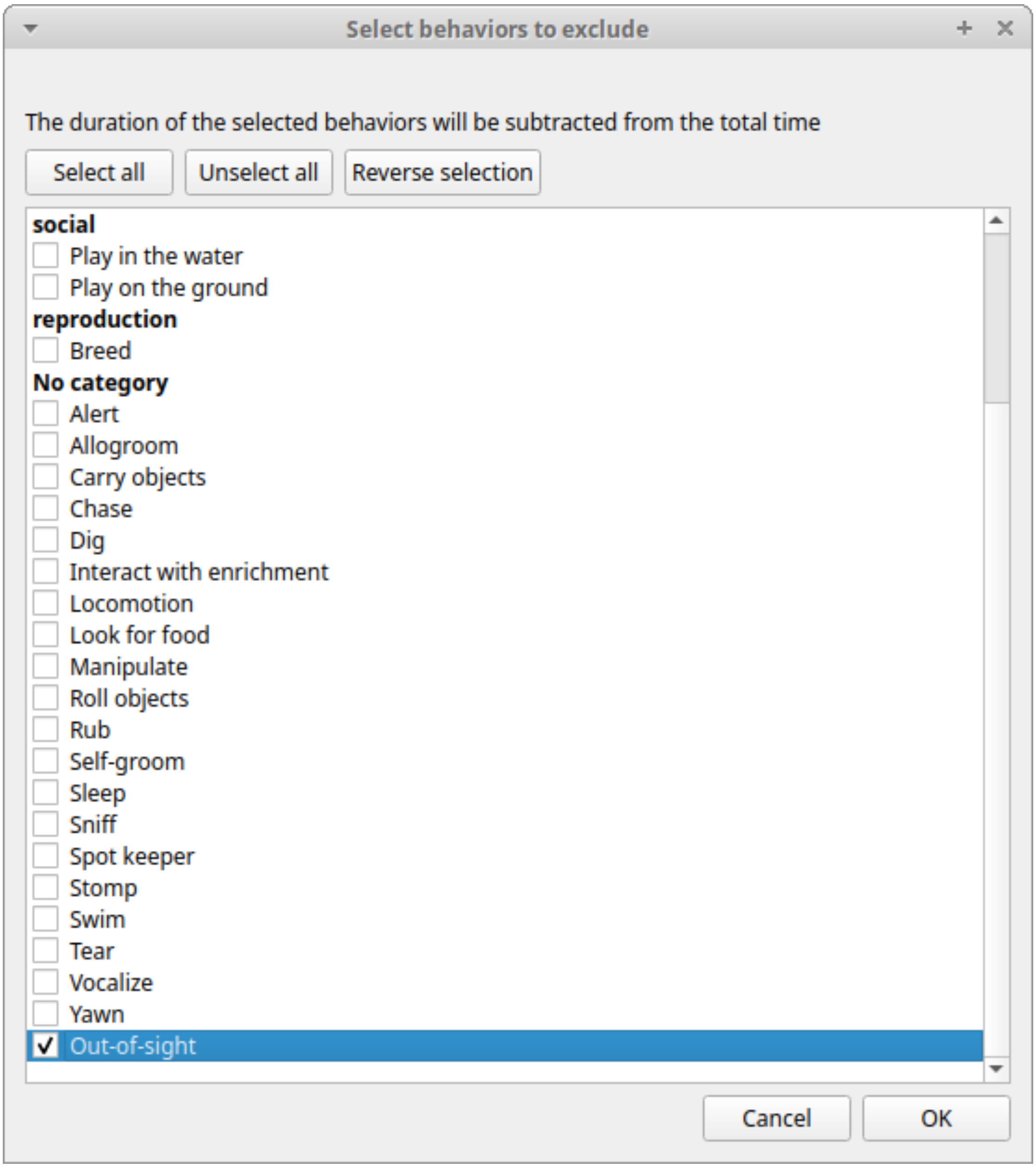
All behaviors can be selected or deselected by clicking on the category name (in bold).

You can choose whether to include behavior modifiers in the time budget analysis and to exclude behaviors without coded events.

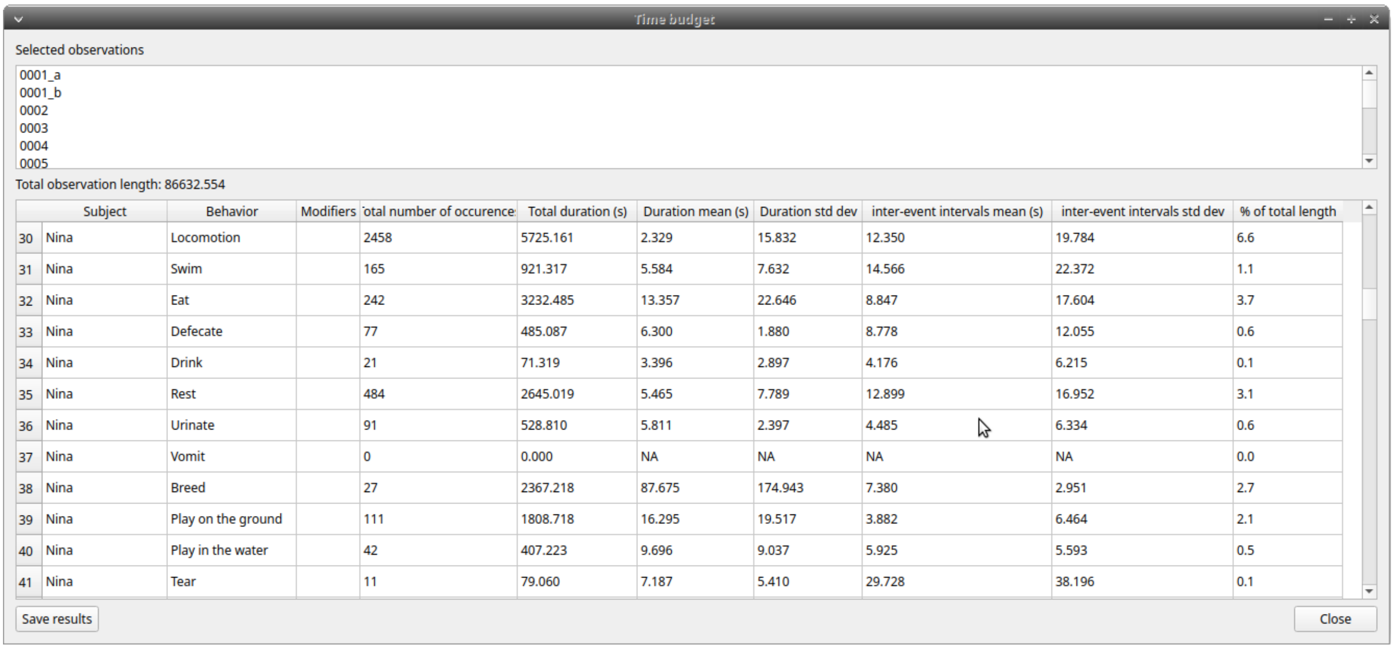
The time budget analysis can be restricted to part of the observation:

- Full observation(s): the analysis will be made on the full observation length.
- Limit to time interval: use the **Start time** and **End time** boxes to select starting time and ending time.
- Limit to observed events: the analysis will be made from the first observed event to the last observed event.

The last dialog window allows you to subtract the duration of one or more behaviors from the total observation duration. This can be useful, for example, if you have defined an "out-of-sight" code in your ethogram:



For each subject and behavior, the results include the **total number of occurrences**, the **total duration** (for behaviors defined as state events), the **mean duration** (for behaviors defined as state events), the **standard deviation of duration**, the **mean inter-event interval**, the **standard deviation of the inter-event interval**, and the **percentage of the total observation duration**. All durations are expressed in seconds (s).



	Subject	Behavior	Modifiers	otal number of occurence	Total duration (s)	Duration mean (s)	Duration std dev	inter-event intervals mean (s)	inter-event intervals std dev	% of total length
30	Nina	Locomotion		2458	5725.161	2.329	15.832	12.350	19.784	6.6
31	Nina	Swim		165	921.317	5.584	7.632	14.566	22.372	1.1
32	Nina	Eat		242	3232.485	13.357	22.646	8.847	17.604	3.7
33	Nina	Defecate		77	485.087	6.300	1.880	8.778	12.055	0.6
34	Nina	Drink		21	71.319	3.396	2.897	4.176	6.215	0.1
35	Nina	Rest		484	2645.019	5.465	7.789	12.899	16.952	3.1
36	Nina	Urinate		91	528.810	5.811	2.397	4.485	6.334	0.6
37	Nina	Vomit		0	0.000	NA	NA	NA	NA	0.0
38	Nina	Breed		27	2367.218	87.675	174.943	7.380	2.951	2.7
39	Nina	Play on the ground		111	1808.718	16.295	19.517	3.882	6.464	2.1
40	Nina	Play in the water		42	407.223	9.696	9.037	5.925	5.593	0.5
41	Nina	Tear		11	79.060	7.187	5.410	29.728	38.196	0.1

Results of the time budget analysis

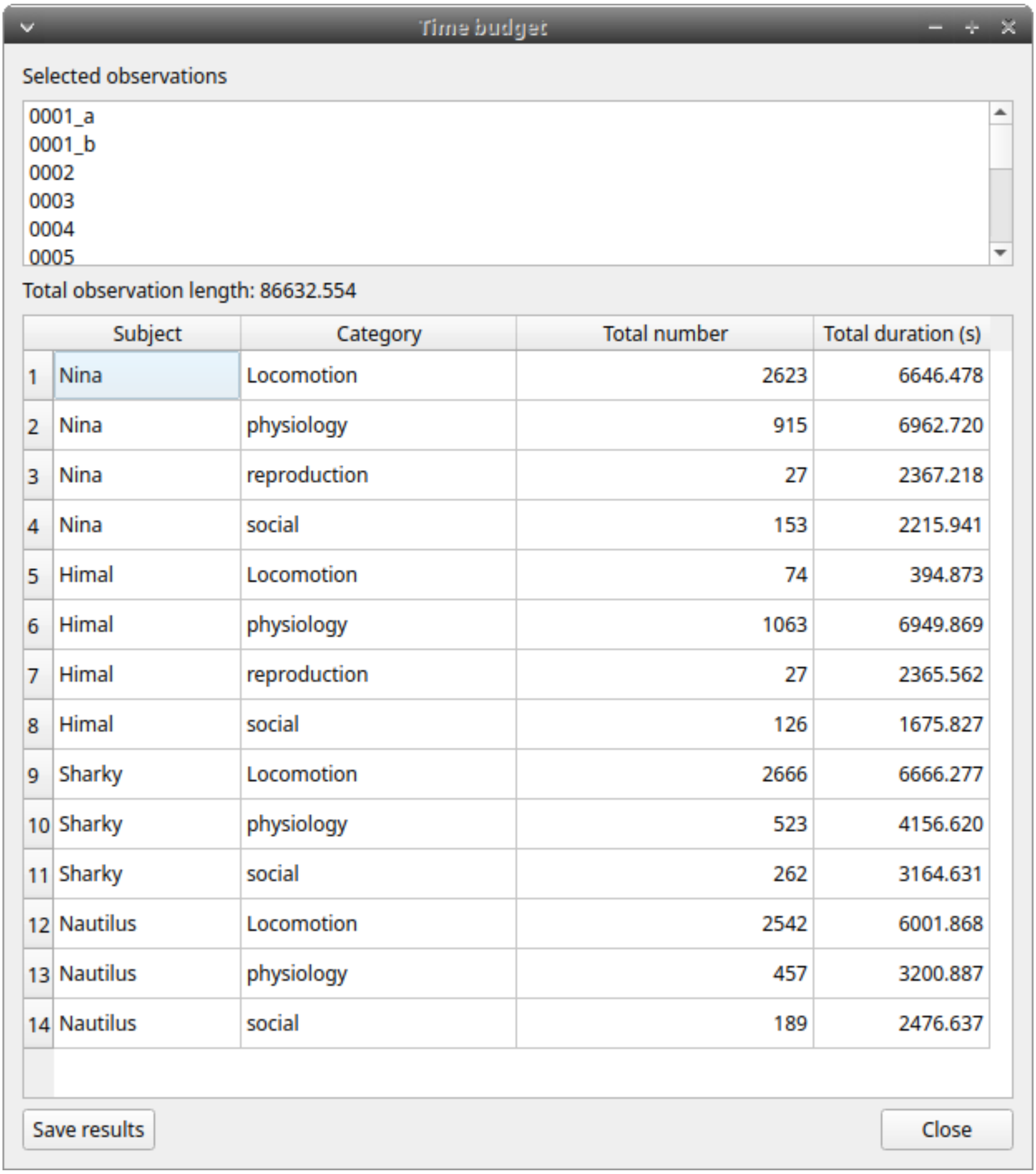
The time budget results can be saved in various formats for further analysis:

- Plain text in tabular format
  - **Tab Separated Values** (TSV)
  - **Comma Separated Values** (CSV)
  - **Hyper Text Markup language** (HTML)
- Spreadsheet files
  - **OpenDocument** (ODS)
  - **Microsoft Excel** (XLSX, XLS)
- **Pandas dataframe** (to be loaded in Python with the pickle module)
- **R dataframe** (to be loaded in R with readRDS function)

> ⚠ **Important**
>
> If a STATE behavior has an odd number of coded events, BORIS will report **UNPAIRED** instead of results.

2.11.2 Time budget by behavioral category

The **Time budget by behavioral category** is similar to the **Time budget analysis** except that the behaviors are grouped into **behavioral categories**.



	Subject	Category	Total number	Total duration (s)
1	Nina	Locomotion	2623	6646.478
2	Nina	physiology	915	6962.720
3	Nina	reproduction	27	2367.218
4	Nina	social	153	2215.941
5	Himal	Locomotion	74	394.873
6	Himal	physiology	1063	6949.869
7	Himal	reproduction	27	2365.562
8	Himal	social	126	1675.827
9	Sharky	Locomotion	2666	6666.277
10	Sharky	physiology	523	4156.620
11	Sharky	social	262	3164.631
12	Nautilus	Locomotion	2542	6001.868
13	Nautilus	physiology	457	3200.887
14	Nautilus	social	189	2476.637

Results of a time budget by behavioral category analysis

2.11.3 Synthetic time budget

The synthetic time budget is similar to the time budget analysis but with fewer parameters and a different organization of results. Results from all selected observations are organized in columns on a single page. Two parameters are currently available: **number of occurrences** and **total duration** (for behaviors defined as state events).

	A	B	C	D	E	F	G	H	I	J	
1			Nina	Nina	Nina	Nina	Nina	Nina	Nina	Nina	Nina
2			Alert	Alert	Allogroom	Allogroom	Breed	Breed	Carry objects	Carry objects	Chas
3		Total length (s)	Total duration	Number of occurrences	Total duration	Number of occurrenc	Total duration	Number of oc	Total duration	Number of oc	Total
40	0037	32.160									
41	0038	86.880	1.675	1							
42	0039	96.960	13.471	2							
43	0040	335.520	86.615	12	15.84	2					
44	0041	130.560	9.069	2	5.85	1					
45	0042	36.960	24.298	3							
46	0043	107.040	42.928	5							
47	0044	109.920	11.769	4	12.471	2					
48	0045	102.240	72.857	6							
49	0046	84.960	44.459	6							
50	0047	77.280	68.929	2							
51	0048	23.520	3.1	1	10.797	1					
52	0049	34.080	9.897	2	8.431	1					
53	0050	23.040									
54	0051	44.160	1.079	1							
55	0052	18.240									
56	0053	39.840									
57	0054	35.040									
58	0055	256.800	61.321	15	19.73	1					
59	0056	26.400	15.241	5							
60	0057	45.120	6.887	3	10.984	2					
61	0058	45.120	2.647	1							
62	0059	41.760									
63	0060	292.800	1.543	2							
64	0061	25.920									

All duration times are expressed in seconds (s).

The time budget results can be saved in various formats for further analysis:

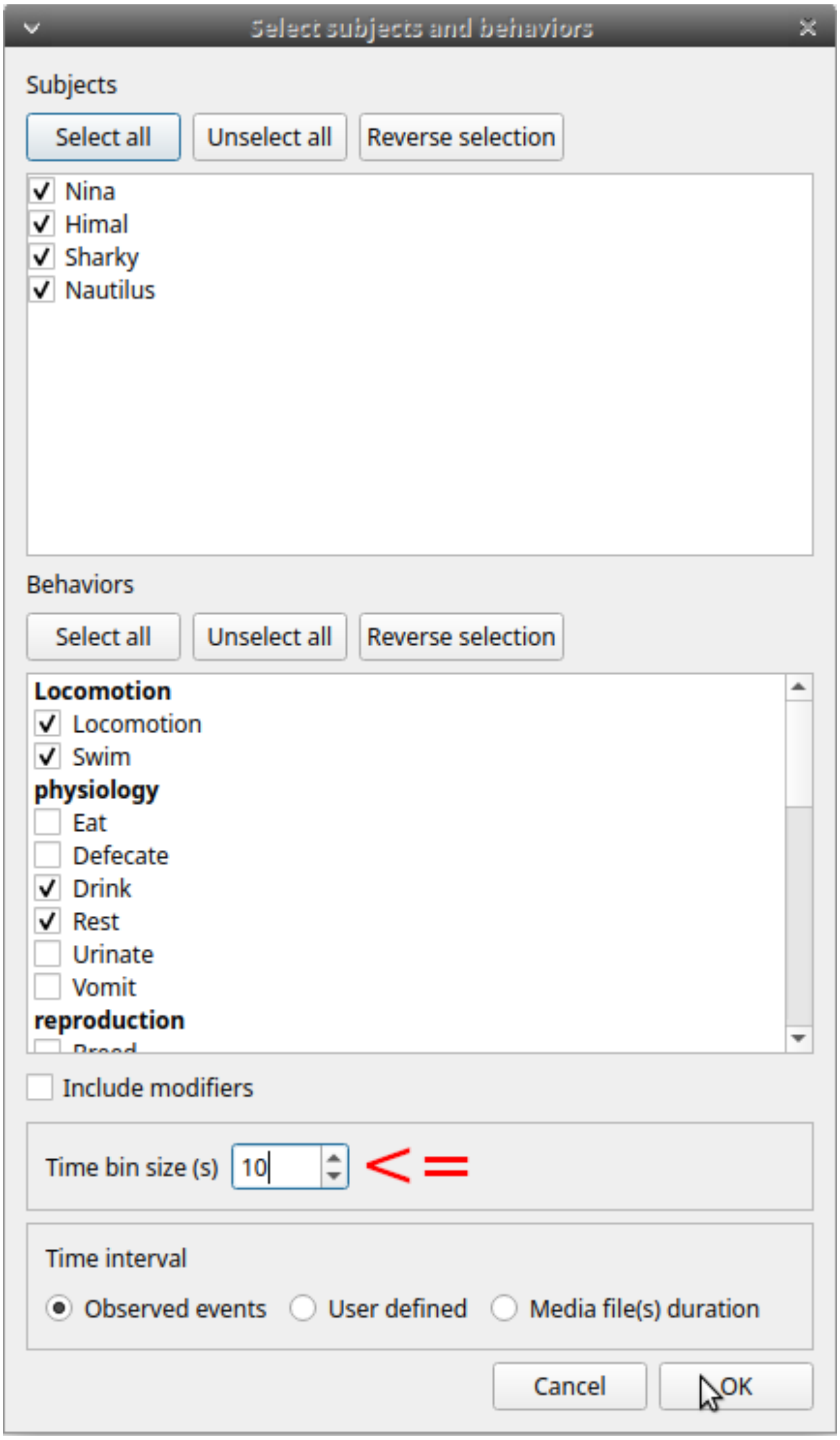
- Plain text in tabular format
  - **Tab Separated Values** (TSV)
  - **Comma Separated Values** (CSV)
  - **Hyper Text Markup language** (HTML)
- Spreadsheet files
  - **OpenDocument** (ODS)
  - **Microsoft Excel** (XLSX, XLS)

2.11.4 Synthetic time budget with time bin

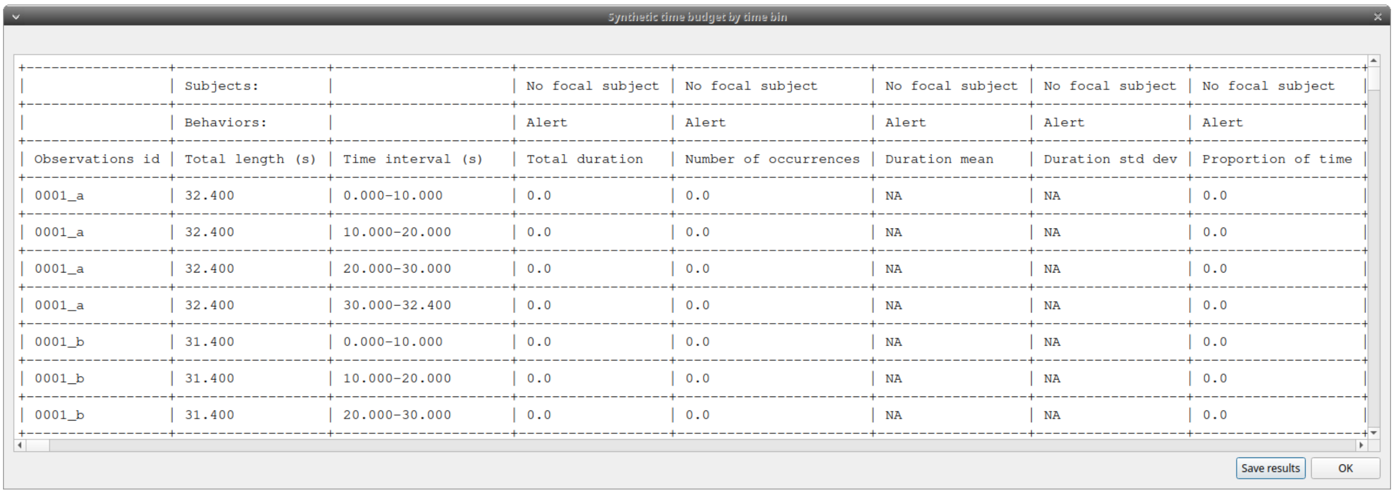
The **synthetic time budget with time bin** is similar to the **Synthetic time budget**, but the results are divided into time bins.

Analysis > **Synthetic time budget with time bin**

Choose a time bin size (in seconds)



Time bin size of 10 seconds



	Subjects:		No focal subject	No focal subject	No focal subject	No focal subject	No focal subject
	Behaviors:		Alert	Alert	Alert	Alert	Alert
Observations id	Total length (s)	Time interval (s)	Total duration	Number of occurrences	Duration mean	Duration std dev	Proportion of time
0001_a	32.400	0.000-10.000	0.0	0.0	NA	NA	0.0
0001_a	32.400	10.000-20.000	0.0	0.0	NA	NA	0.0
0001_a	32.400	20.000-30.000	0.0	0.0	NA	NA	0.0
0001_a	32.400	30.000-32.400	0.0	0.0	NA	NA	0.0
0001_b	31.400	0.000-10.000	0.0	0.0	NA	NA	0.0
0001_b	31.400	10.000-20.000	0.0	0.0	NA	NA	0.0
0001_b	31.400	20.000-30.000	0.0	0.0	NA	NA	0.0

Results of a Synthetic time budget with time bin of 10 seconds

The **time budget with time bin** results can be saved in various formats for further analysis:

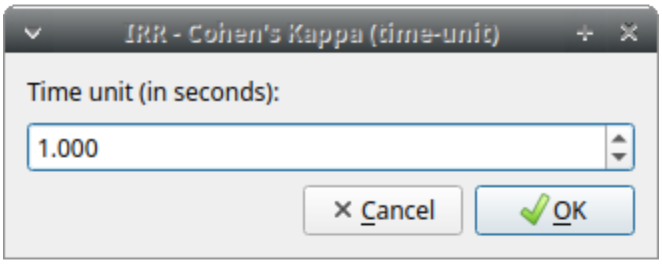
- Plain text in tabular format
    - **Tab Separated Values** (TSV)
    - **Comma Separated Values** (CSV)
    - **Hyper Text Markup language** (HTML)
- Spreadsheet files
    - **OpenDocument** (ODS)
    - **Microsoft Excel** (XLSX, XLS)

2.11.5 Inter-rater reliability

The Cohen's kappa coefficient can be calculated (**Analysis** > **Inter-rater reliability** > **Cohen's kappa**).

Cohen's kappa on Wikipedia

After selecting two observations and a time window (in seconds) for the analysis (the default value is 10 seconds), Cohen's kappa will be displayed in the results window.



Implementation of IRR Cohen's Kappa in BORIS

If a time window of n seconds is set, the two selected observations will be checked every n seconds for agreement or disagreement, from the first event to the last event of the two observations. In the case of a point event, the presence of a corresponding event in the other observation will be verified using a time window of n seconds centered on the point event.

An IRR Cohen's Kappa analysis is available in the GSEQ program (http://www2.gsu.edu/~psyrab/gseq). For this purpose, the coded events can be exported as aggregated events in SDIS format. See export aggregated events.

2.11.6 Similarities

Needleman-Wunsch similarity

2.11.7 Co-occurrence

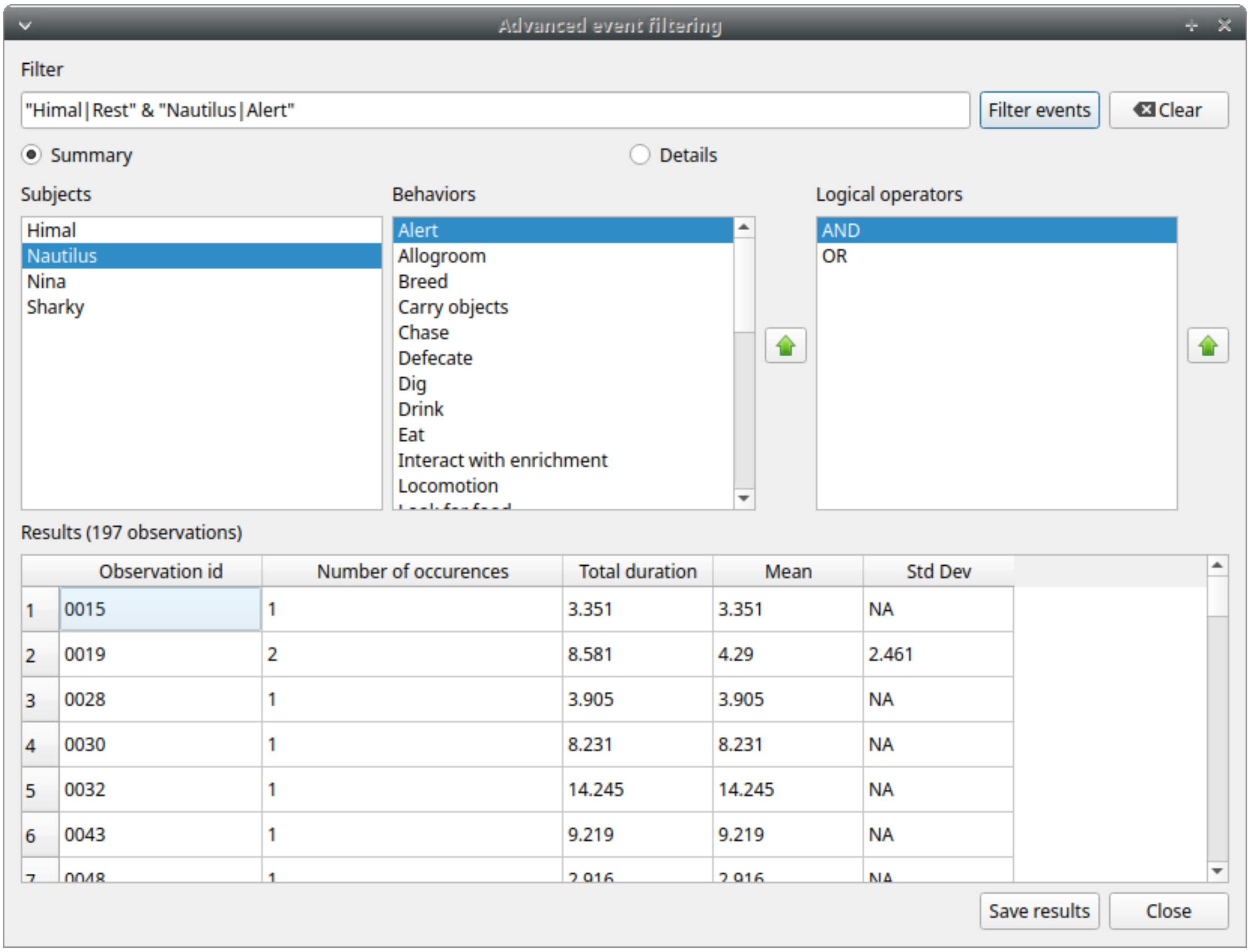
This function allows you to determine the co-occurrence of 2 behaviors.

2.11.8 Advanced event filtering

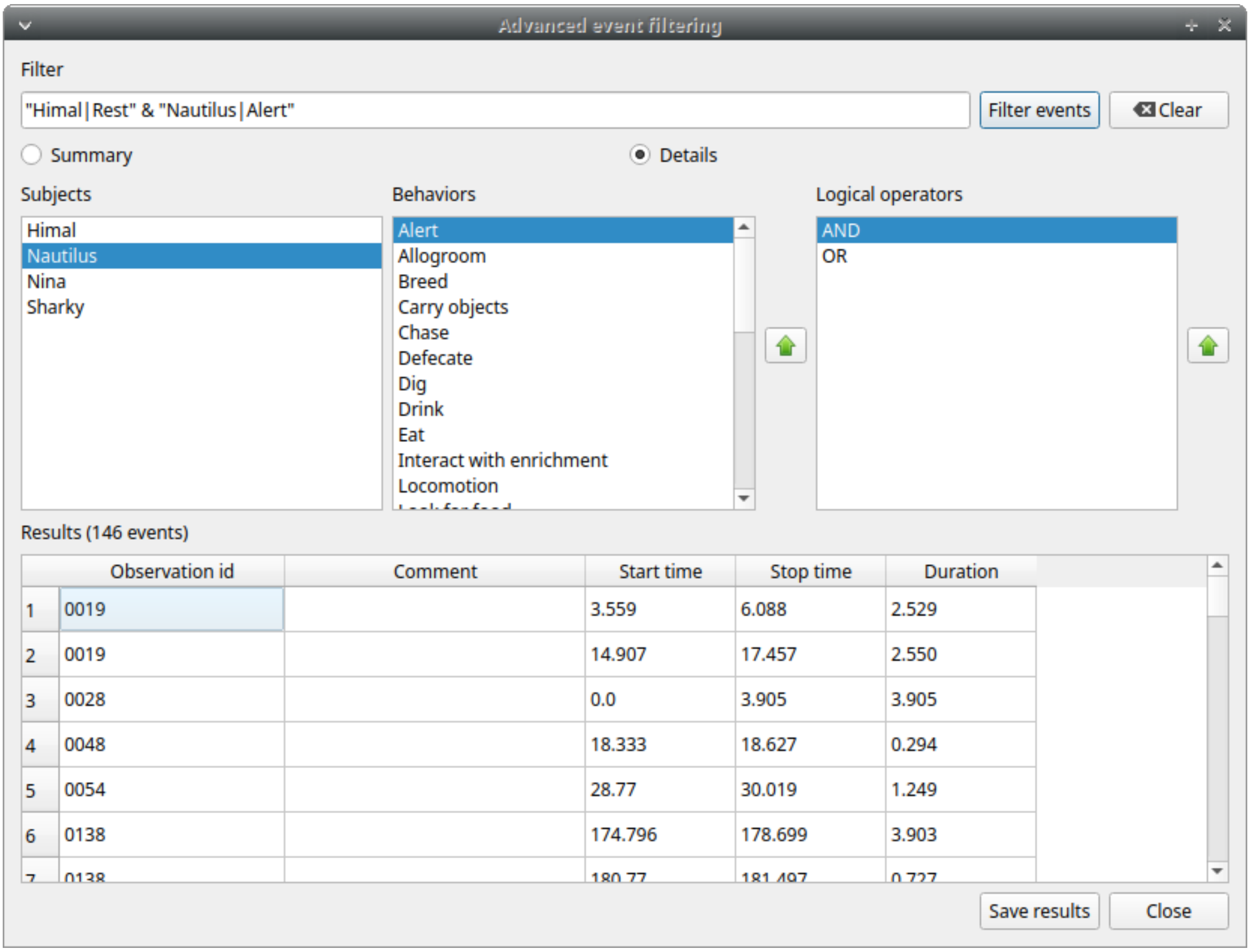
This function allows filtering events from one or more observations by selecting subjects, behaviors, and logical operators.

To use the filter, select a subject, select a behavior, and click on the button with the green arrow next to the behaviors list. The tuple subject/behavior will be added to the **filter text edit**. A complex filter query can be constructed by adding parentheses and logical operators **&** (AND) or | (OR) to combine various subjects and behaviors.

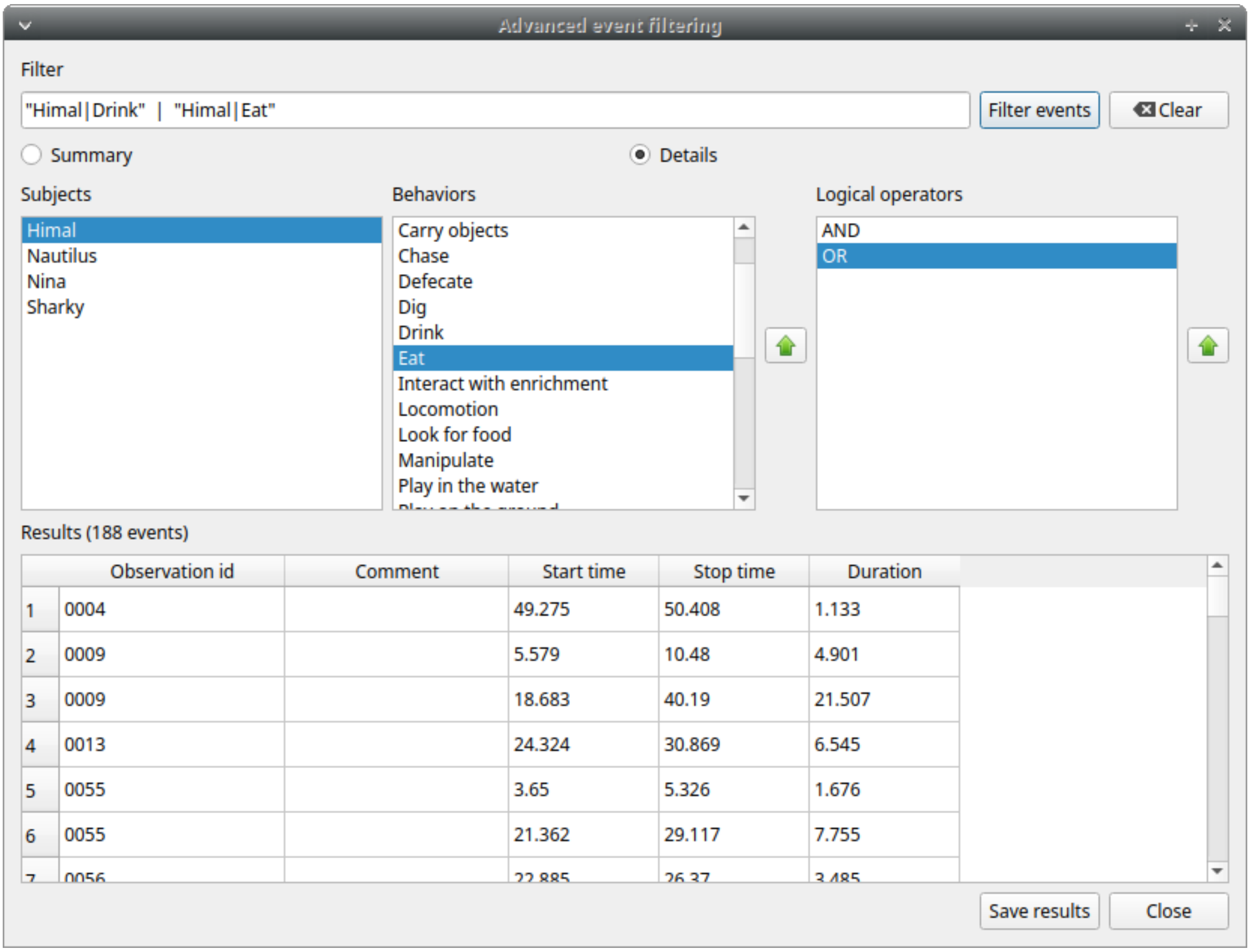
Example of a summarized output showing the occurrences of Himal resting while Nautilus in alert:



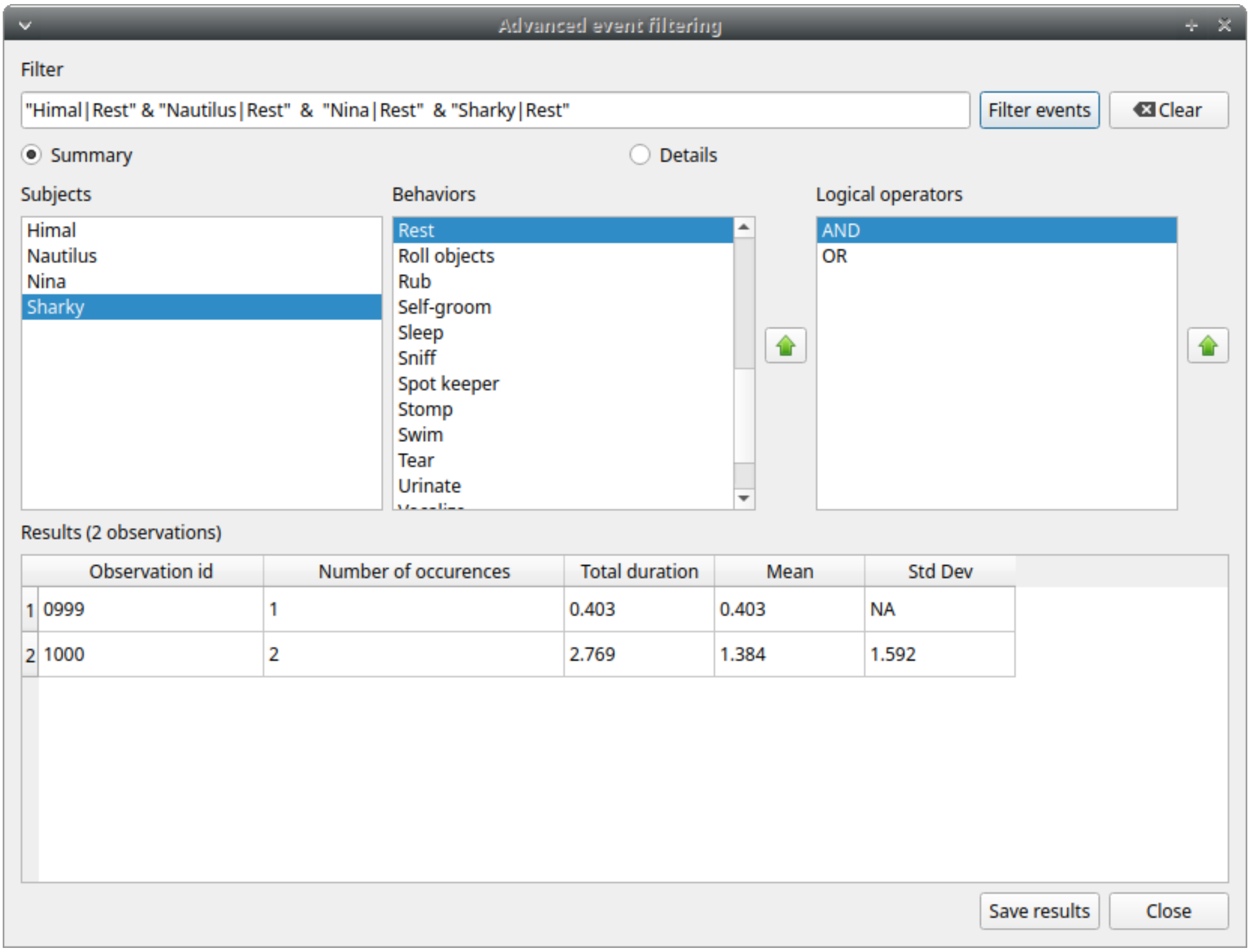
Example of a detailed output showing the overlapping intervals while Himal rests and Nautilus is in alert:



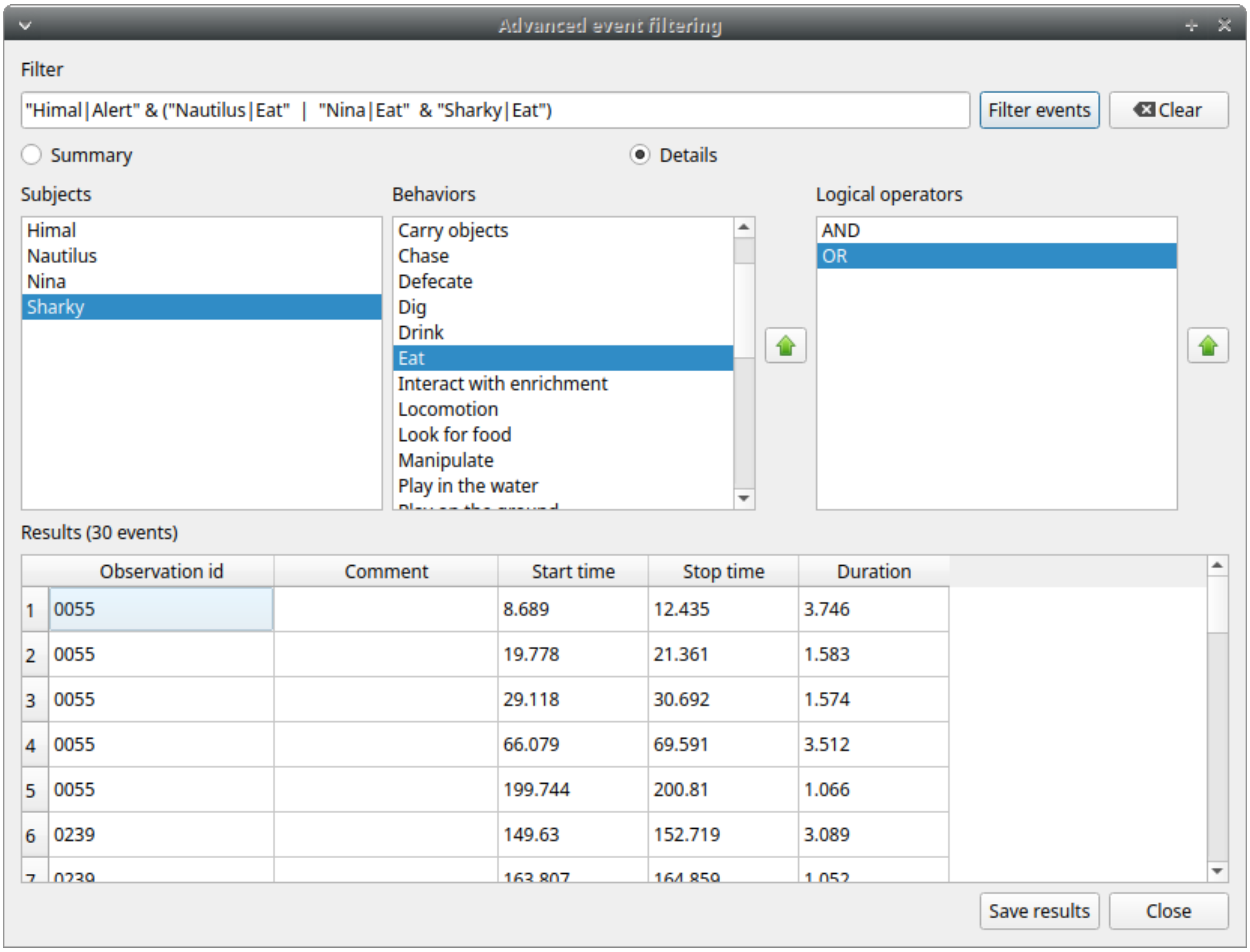
The same subject can be used multiple times in the query with OR or AND (in the case of non-exclusive behaviors):



An unlimited number of conditions can be used:



Parentheses can be used to group logical conditions into blocks:



The results can be saved in a Tab Separated Values (TSV) file using the **Save results** button. Other formats may be added in the future.

2.11.9 Latency

The latency analysis calculates the time between one or more markers (arbitrary behaviors) and other behaviors.

2.11.10 Plugins

BORIS can load Python and R analysis plugins. See Analysis plugins for plugin configuration, loading order, and plugin authoring details.

2.12 Analysis plugins

Starting from version 9, BORIS can load analysis plugins to process coded events. Plugins can be written in **Python** or **R**.

> ⚠ **Important**
>
> The plugin interface is intended for custom analyses and may change between BORIS versions.

2.12.1 Plugin sources

BORIS loads plugins from two sources.

Official BORIS plugins

Official plugins are loaded first. BORIS looks in the directory configured in **Preferences** > **Analysis plugins** > **Official BORIS plugins**.

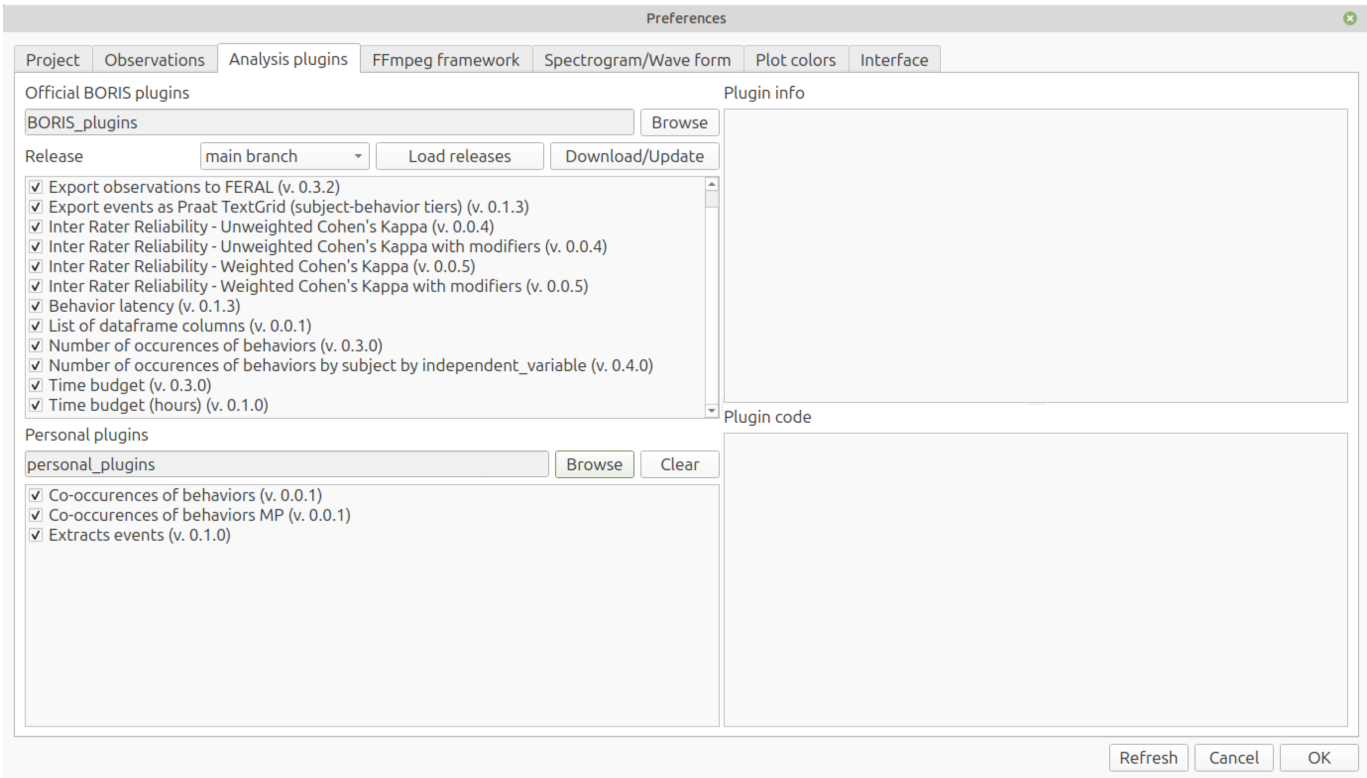
Personal plugins

Personal plugins are loaded from the directory selected in **Preferences** > **Analysis plugins** > **Personal plugins**. BORIS loads Python files (`.py`) and R files (`.R`) found directly in that directory. Files whose names start with `_` are ignored.

If an official plugin and a personal plugin use the same plugin name, the official plugin is kept, and the duplicate personal plugin is not loaded.

2.12.2 Managing plugins in Preferences

Open **File** > **Preferences** > **Analysis plugins**.



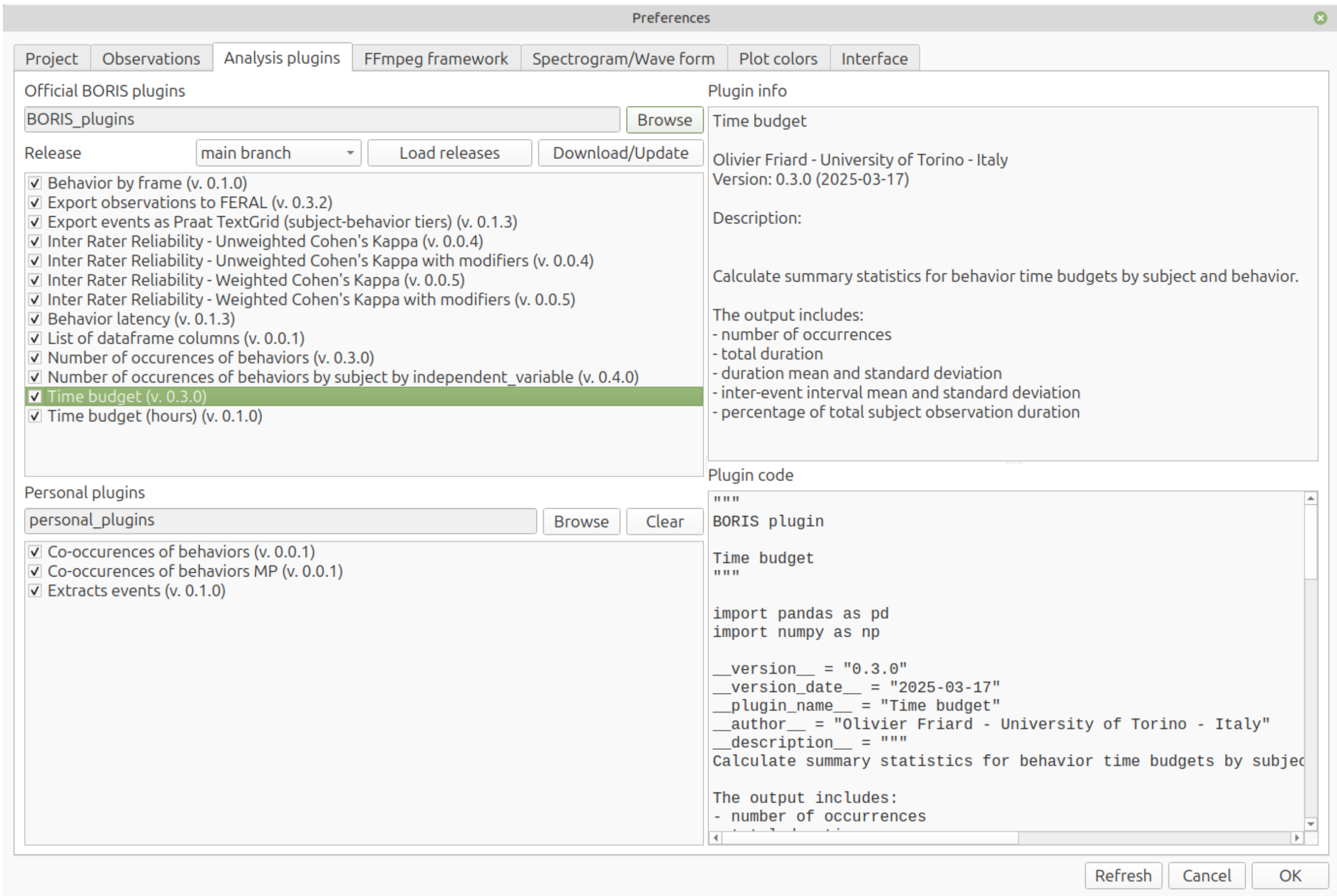
The **Official BORIS plugins** section contains:

- The directory currently used for the official plugin repository;
- A **Browse** button to select another official plugin repository directory;
- A **Release** drop-down list to select the release of the BORIS plugins;
- A **Load releases** button to update the available releases;
- A **Download/Update** button to download or update the official BORIS plugins directory from the BORIS plugins GitHub repository https://github.com/olivierfriard/BORIS_plugins;
- A list of official plugins with check boxes.

The **Personal plugins** section contains:

- The directory currently used for personal plugins;
- A **Browse** button to select a personal plugins directory;
- A **Clear** button to remove the saved personal plugin path without deleting the directory or its files;
- A list of personal plugins with check boxes.

Plugin lists show the plugin version next to the plugin name when a version is available. Click a plugin in the list to display its metadata and source code.



Changes are saved when you click **OK** in the Preferences window.

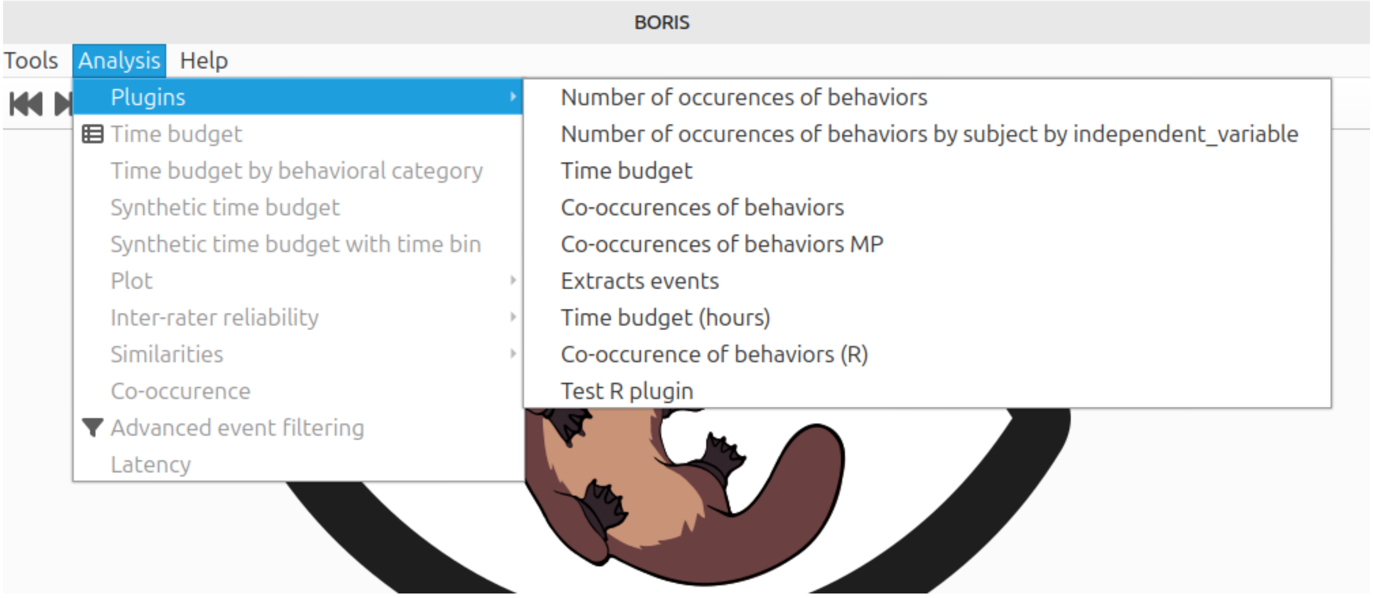
2.12.3 Downloading plugins from the official BORIS plugins GitHub repository

To select the BORIS plugin release you want to use, click **Load releases**, choose the release number from the **Release** drop-down list, and then click **Download/Update**.

To support reproducible analyses, all BORIS plugin releases will remain available.

2.12.4 Running a plugin

To run a plugin, select **Analysis** > **Plugins**, then choose the plugin.

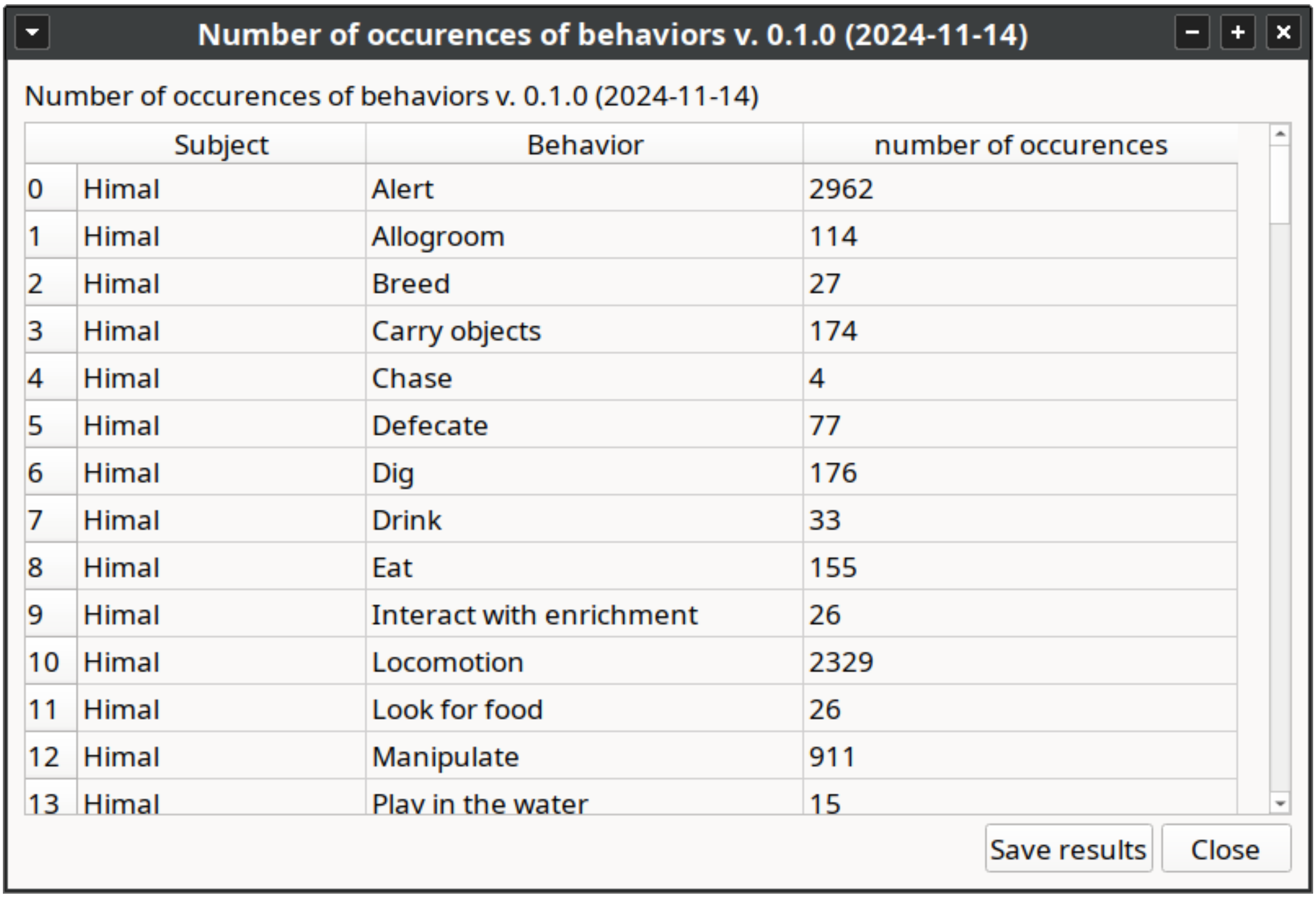


BORIS lists official plugins first. Personal plugins are listed after a separator.

When a plugin is started, BORIS asks you to select:

- The observations to analyze;
- The subjects and behaviors to include;
- The time interval or observation interval to use.

The plugin result is displayed in a result window.



	Subject	Behavior	number of occurences
0	Himal	Alert	2962
1	Himal	Allogroom	114
2	Himal	Breed	27
3	Himal	Carry objects	174
4	Himal	Chase	4
5	Himal	Defecate	77
6	Himal	Dig	176
7	Himal	Drink	33
8	Himal	Eat	155
9	Himal	Interact with enrichment	26
10	Himal	Locomotion	2329
11	Himal	Look for food	26
12	Himal	Manipulate	911
13	Himal	Play in the water	15

Plugin results can be saved from the result window in the available export formats.

2.12.5 Data passed to plugins

For Python plugins, BORIS inspects the arguments of the plugin `run` function and passes only the requested objects.

The following arguments are supported:

`df`

> A Pandas DataFrame containing the selected and filtered events. The parameter must be annotated as `pd.DataFrame` or as a union that includes `pd.DataFrame`, for example `pd.DataFrame | None`.

`project`

> A copy of the BORIS project dictionary containing only the selected observations. The parameter must be annotated as `dict` or as a union that includes `dict`, for example `dict | None`.

`parameters`

> A dictionary with the analysis selections made by the user, including selected subjects, selected behaviors, and time filtering options. The parameter must be annotated as `dict` or as a union that includes `dict`, for example `dict | None`.

String annotations are also accepted. This means that plugins using `from __future__ import annotations` can use the same signatures.

Example:

```
import pandas as pd

def run(
```

```
    df: pd.DataFrame | None = None,
    project: dict | None = None,
    parameters: dict | None = None,
) -> pd.DataFrame:
    if df is None:
        return pd.DataFrame()

    return df.groupby(["Subject", "Behavior"]).size().reset_index(name="count")
```

The DataFrame passed to the plugin includes columns such as:

```
Observation id
Observation date
Description
Observation type
Observation interval start
Observation interval stop
Source
Media duration (s)
FPS (frame/s)
independent variable #1
independent variable #2
...
Subject
Observation duration by subject by observation
Behavior
Behavioral category
(BEHAVIOR1, Modifiers set #1)
(BEHAVIOR1, Modifiers set #2)
(BEHAVIOR2, Modifiers set #1)
...
Behavior type
Start (s)
Stop (s)
Duration (s)
Comment start
Comment stop
```

The DataFrame also includes one column for each independent variable and one column for each behavior modifier set defined in the project.

2.12.6 Python plugin structure

A Python plugin is a `.py` file that defines a `run` function and plugin metadata.

The plugin metadata are:

```
__plugin_name__ = "PLUGIN NAME"
__version__ = "x.y.z"
__version_date__ = "YYYY-MM-DD"
__author__ = "AUTHOR - INSTITUTION"
__description__ = "Short plugin description"
```

You can optionally define the `__require_boris_version__` constant. This ensures that the plugin runs only with specific BORIS versions:

```
__require_boris_version__ = ">= 9.12"
```

The `run` function can return:

- A string;
- A Pandas DataFrame;
- A tuple of results.

Each result can optionally be returned as `(title, payload)`, where `payload` is a string or a Pandas DataFrame. For a single titled result, return it inside a tuple of results:

```
return (("Occurrences", result_df),)
```

For multiple results:

```
return (
    ("Summary", summary_text),
    ("Table", result_df),
)
```

Example of a Python plugin:

```
"""
BORIS plugin

Number of occurrences of behaviors.
"""

import pandas as pd

__version__ = "0.3.0"
__version_date__ = "2025-03-17"
__plugin_name__ = "Number of occurrences of behaviors"
__author__ = "Olivier Friard - University of Torino - Italy"
__description__ = "Count behavior occurrences by subject."


def run(df: pd.DataFrame) -> pd.DataFrame:
    return df.groupby(["Subject", "Behavior"]).size().reset_index(name="number of occurrences")
```

If you want a plugin to be usable outside BORIS, avoid importing BORIS internal modules such as `boris.config`. Use the DataFrame columns, the `project` dictionary, and the `parameters` dictionary passed to `run`.

2.12.7 R plugin structure

An R plugin is an `.R` file that defines a `run` function and plugin metadata. BORIS passes the selected and filtered events as an R data frame.

The plugin metadata are:

```
plugin_name <- "PLUGIN NAME"
version <- "x.y.z"
version_date <- "YYYY-MM-DD"
author <- "AUTHOR - INSTITUTION"
description <- "Short plugin description"
```

Example:

```
plugin_name <- "Number of occurrences of behaviors (R)"
version <- "0.0.1"
version_date <- "2026-05-29"
author <- "Olivier Friard - University of Torino - Italy"
description <- "Count behavior occurrences by subject."

run <- function(df) {
    aggregate(df$Behavior, by = list(Subject = df$Subject, Behavior = df$Behavior), FUN = length)
}
```

R plugins require the `rpy2` Python module.

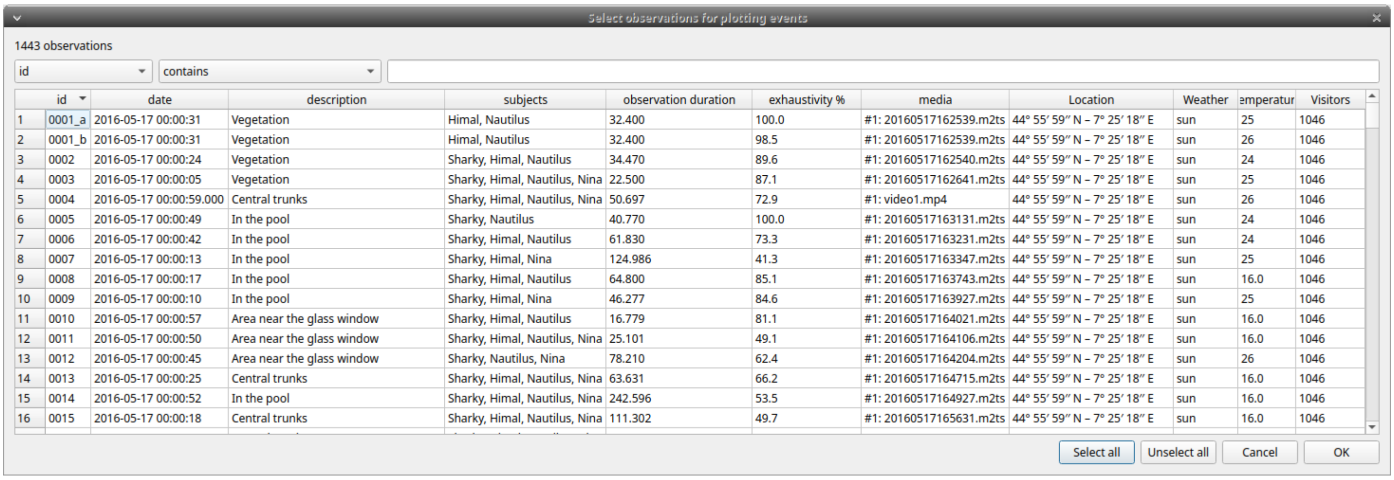
2.13 Plot

2.13.1 Plot events

Recorded events can be plotted along a time axis.

Analysis > **Plot** > **Plot events**

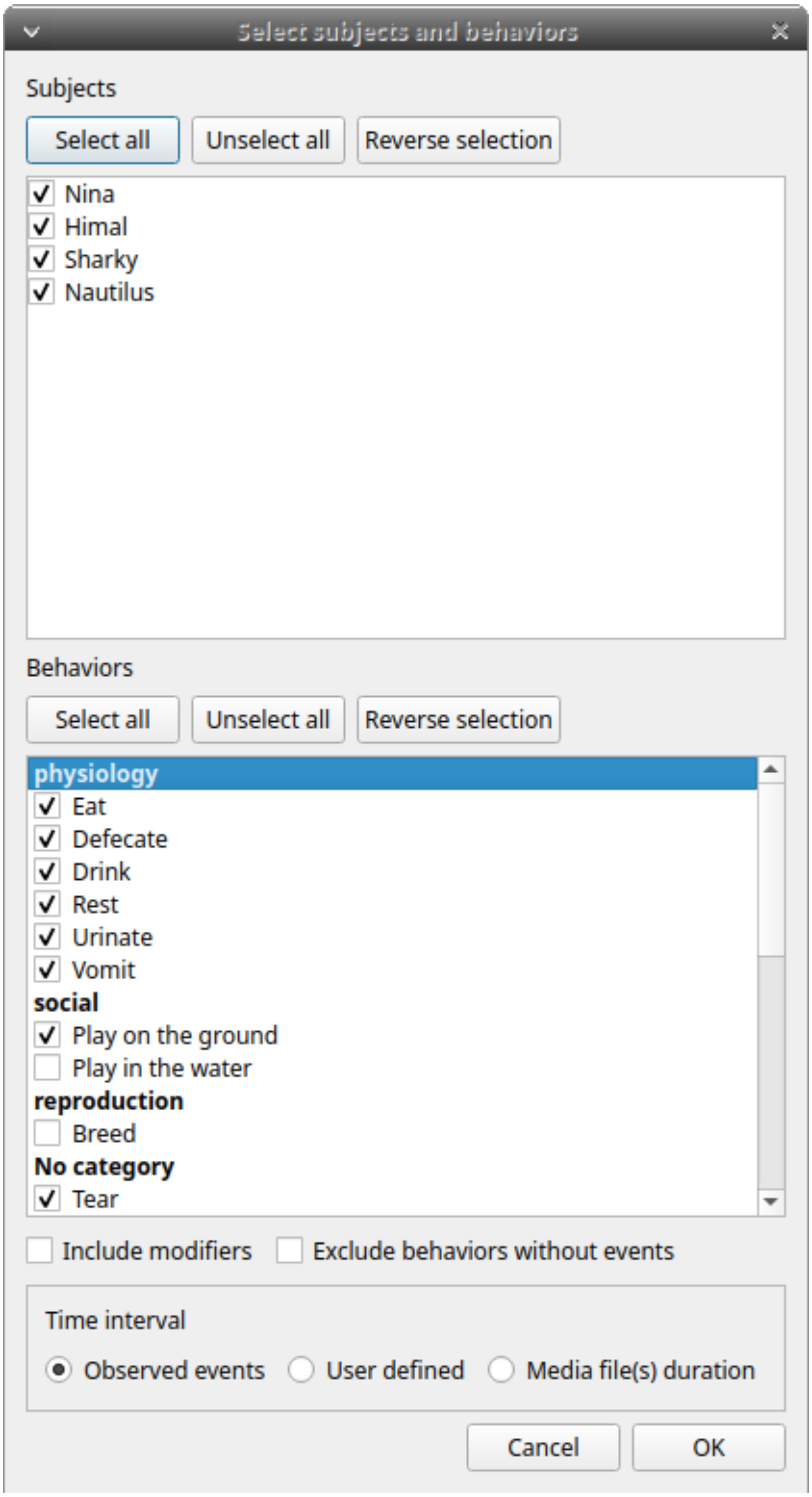
Select the **observations** you want to plot. If more than one observation is selected, BORIS will ask you to choose a directory to save the plots.



	id	date	description	subjects	observation duration	exhaustivity %	media	Location	Weather	emperatur	Visitors
1	0001_a	2016-05-17 00:00:31	Vegetation	Himal, Nautilus	32.400	100.0	#1: 20160517162539.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	25	1046
2	0001_b	2016-05-17 00:00:31	Vegetation	Himal, Nautilus	32.400	98.5	#1: 20160517162539.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
3	0002	2016-05-17 00:00:24	Vegetation	Sharky, Himal, Nautilus	34.470	89.6	#1: 20160517162540.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24	1046
4	0003	2016-05-17 00:00:05	Vegetation	Sharky, Himal, Nautilus, Nina	22.500	87.1	#1: 20160517162641.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	25	1046
5	0004	2016-05-17 00:00:59.000	Central trunks	Sharky, Himal, Nautilus, Nina	50.697	72.9	#1: video1.mp4	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
6	0005	2016-05-17 00:00:49	In the pool	Sharky, Nautilus	40.770	100.0	#1: 20160517163131.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24	1046
7	0006	2016-05-17 00:00:42	In the pool	Sharky, Himal, Nautilus	61.830	73.3	#1: 20160517163231.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	24	1046
8	0007	2016-05-17 00:00:13	In the pool	Sharky, Himal, Nina	124.986	41.3	#1: 20160517163347.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	25	1046
9	0008	2016-05-17 00:00:17	In the pool	Sharky, Himal, Nautilus	64.800	85.1	#1: 20160517163743.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	16.0	1046
10	0009	2016-05-17 00:00:10	In the pool	Sharky, Himal, Nina	46.277	84.6	#1: 20160517163927.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	25	1046
11	0010	2016-05-17 00:00:57	Area near the glass window	Sharky, Himal, Nautilus	16.779	81.1	#1: 20160517164021.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	16.0	1046
12	0011	2016-05-17 00:00:50	Area near the glass window	Sharky, Himal, Nautilus, Nina	25.101	49.1	#1: 20160517164106.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	16.0	1046
13	0012	2016-05-17 00:00:45	Area near the glass window	Sharky, Nautilus, Nina	78.210	62.4	#1: 20160517164204.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	26	1046
14	0013	2016-05-17 00:00:25	Central trunks	Sharky, Himal, Nautilus, Nina	63.631	66.2	#1: 20160517164715.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	16.0	1046
15	0014	2016-05-17 00:00:52	In the pool	Sharky, Himal, Nautilus, Nina	242.596	53.5	#1: 20160517164927.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	16.0	1046
16	0015	2016-05-17 00:00:18	Central trunks	Sharky, Himal, Nautilus, Nina	111.302	49.7	#1: 20160517165631.m2ts	44° 55′ 59″ N – 7° 25′ 18″ E	sun	16.0	1046

Select the observations to plot

The **subjects** and **behaviors** to include in the plot can be selected in the following window:



You can choose whether to include behavior modifiers (if any) and to exclude behaviors without coded events.

The time interval can be selected (see time budget).

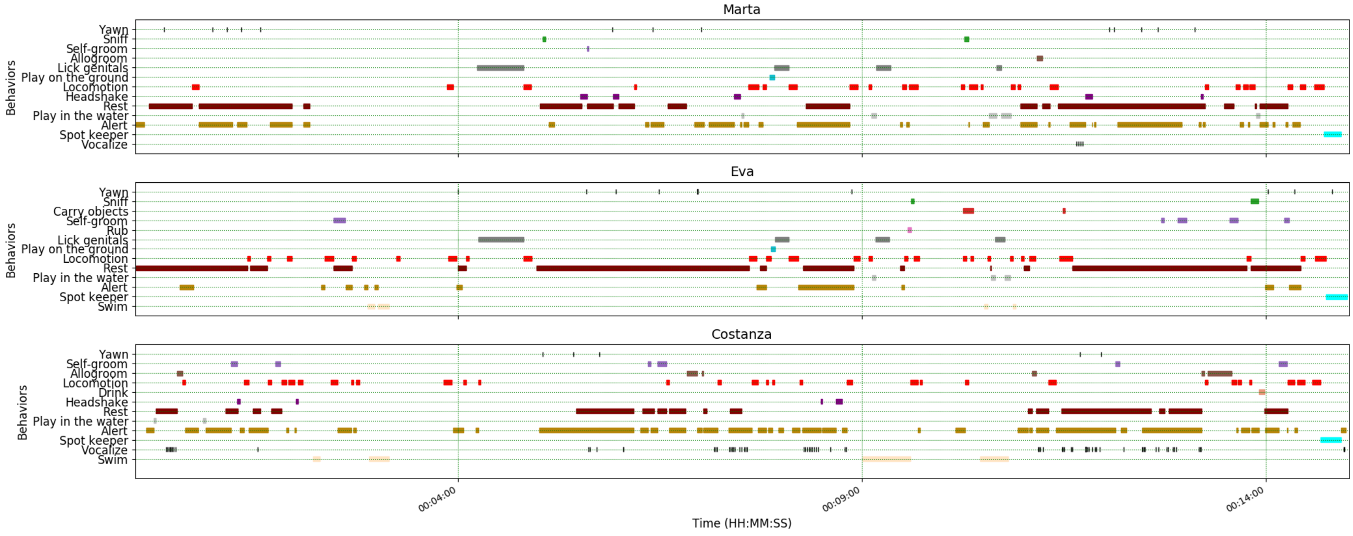
The plot can be exported in various bitmap formats (PNG, JPG, TIFF) or vector graphics formats (SVG, PDF, EPS, PS). The SVG format can be further edited with the Inkscape vector graphics editor.

Important

If a STATE behavior has an odd number of coded events, you will see this error message: "The STATE behavior XXX is not paired"

This function creates one plot per subject on a single figure.

Behavior colors can be customized. See Plot colors.

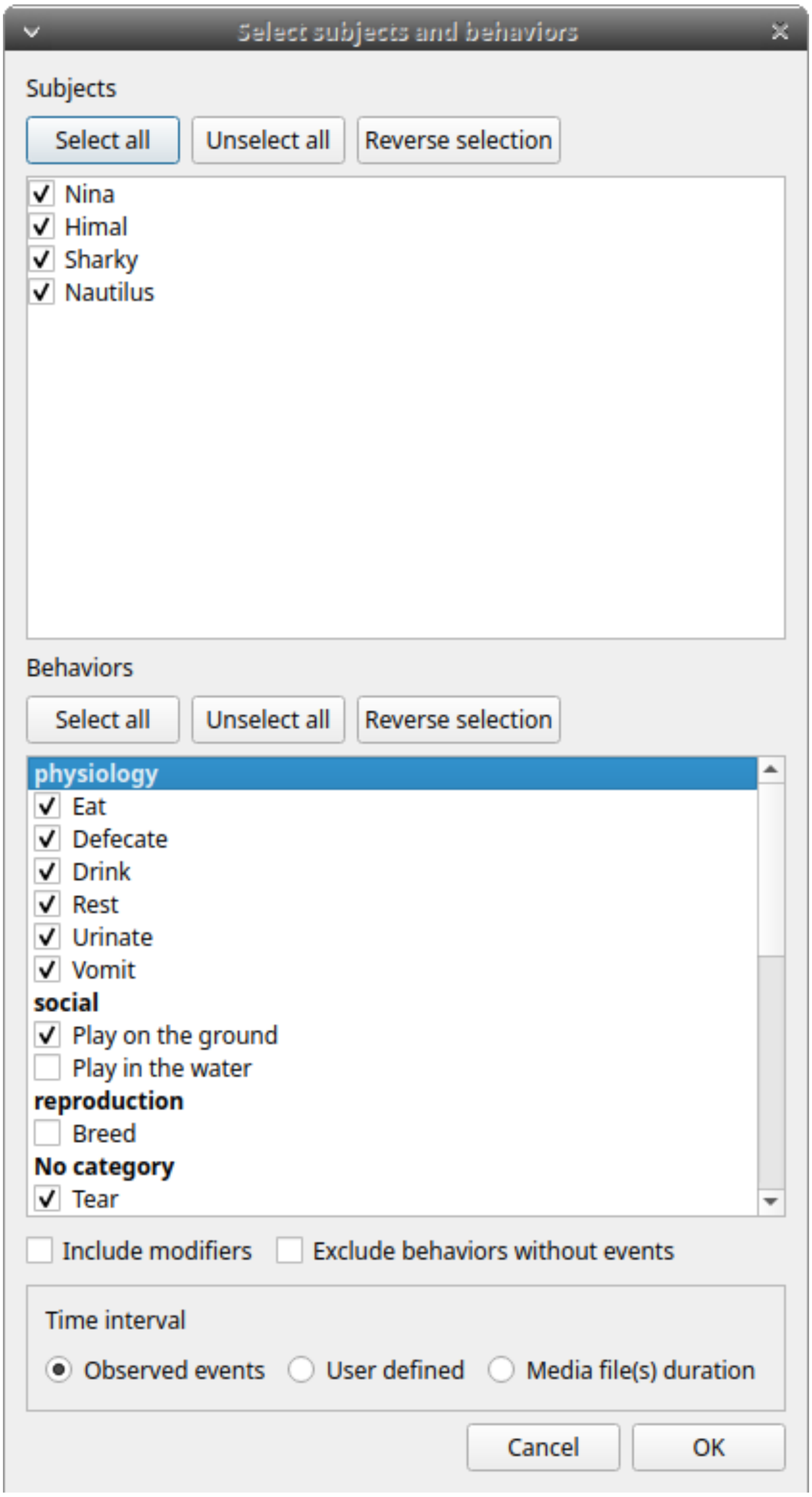


2.13.2 Plot time budget

The duration and number of occurrences can be plotted for each subject and behavior.

Analysis > **Plot** > **Plot time budget**

The subjects and behaviors to include in the plot can be selected in the following window:



Behavior modifiers cannot currently be included in the plot.

The time interval can be selected (see time budget).

The plot can be exported in various bitmap formats (PNG, JPG, TIFF) or vector graphics formats (SVG, PDF, EPS, PS). The SVG format can be further edited with the Inkscape vector graphics editor.

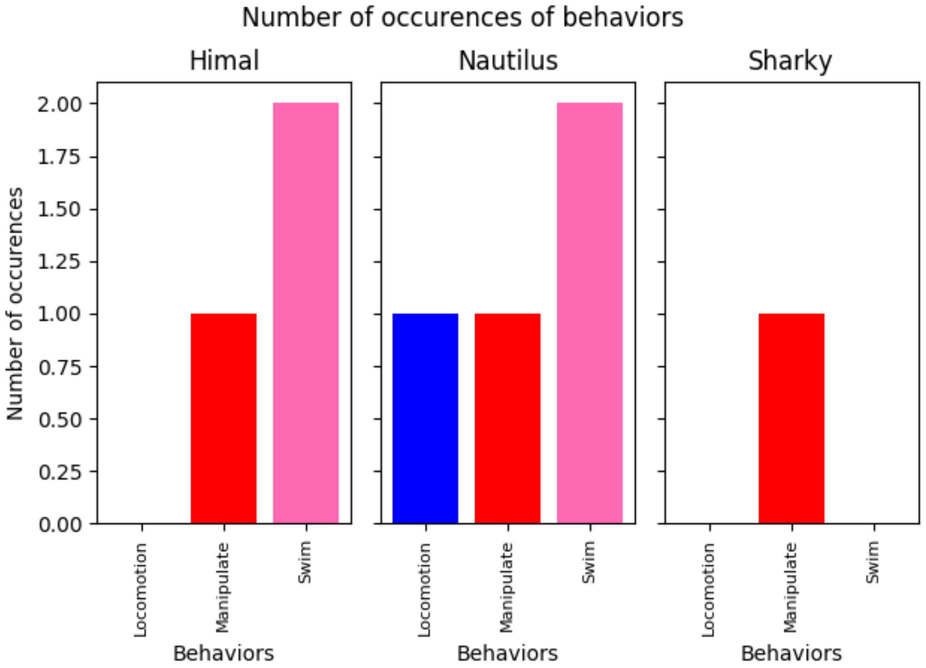
Important

If a STATE behavior has an odd number of coded events, you will see this error message: "The STATE behavior XXX is not paired"

This function creates two plots including all subjects for each observation:

- A plot of the behavior durations for behaviors defined as STATE events.
- A plot of the number of occurrences for all behaviors.

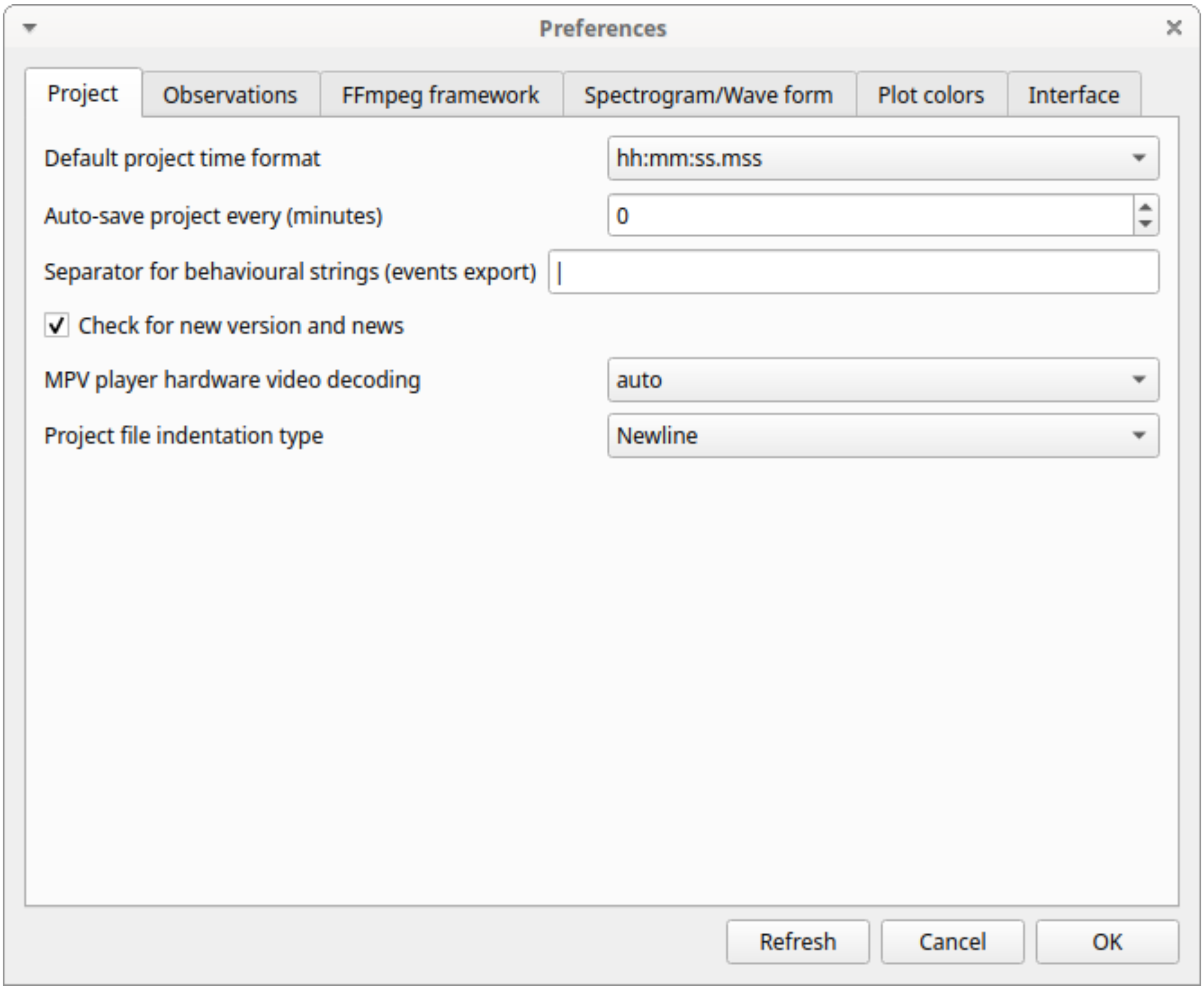
Behavior colors can be customized. See Plot colors.



2.14 Preferences

You can customize BORIS in the Preferences window (**File** > **Preferences** or the **Preferences** button in the toolbar).

2.14.1 Project preferences



Default project time format

This option allows you to choose the format used to display time in the project. Time is always stored internally in seconds with a precision of 3 decimal places.

Auto-save project every (minutes)

If set, BORIS will save your project automatically every n minutes. A value of 0 disables automatic backup. The project will be saved only if it has already been saved and an observation is open.

Separator for behavioral strings

Character or string used to separate behaviors when exporting events as behavioral strings. See also Behatrix.

Check for new version

Checks for a new version on the BORIS website every 15 days (internet access required).

MPV hardware video decoding

If you experience problems with the embedded MPV player, try changing this value.

Project file indentation type

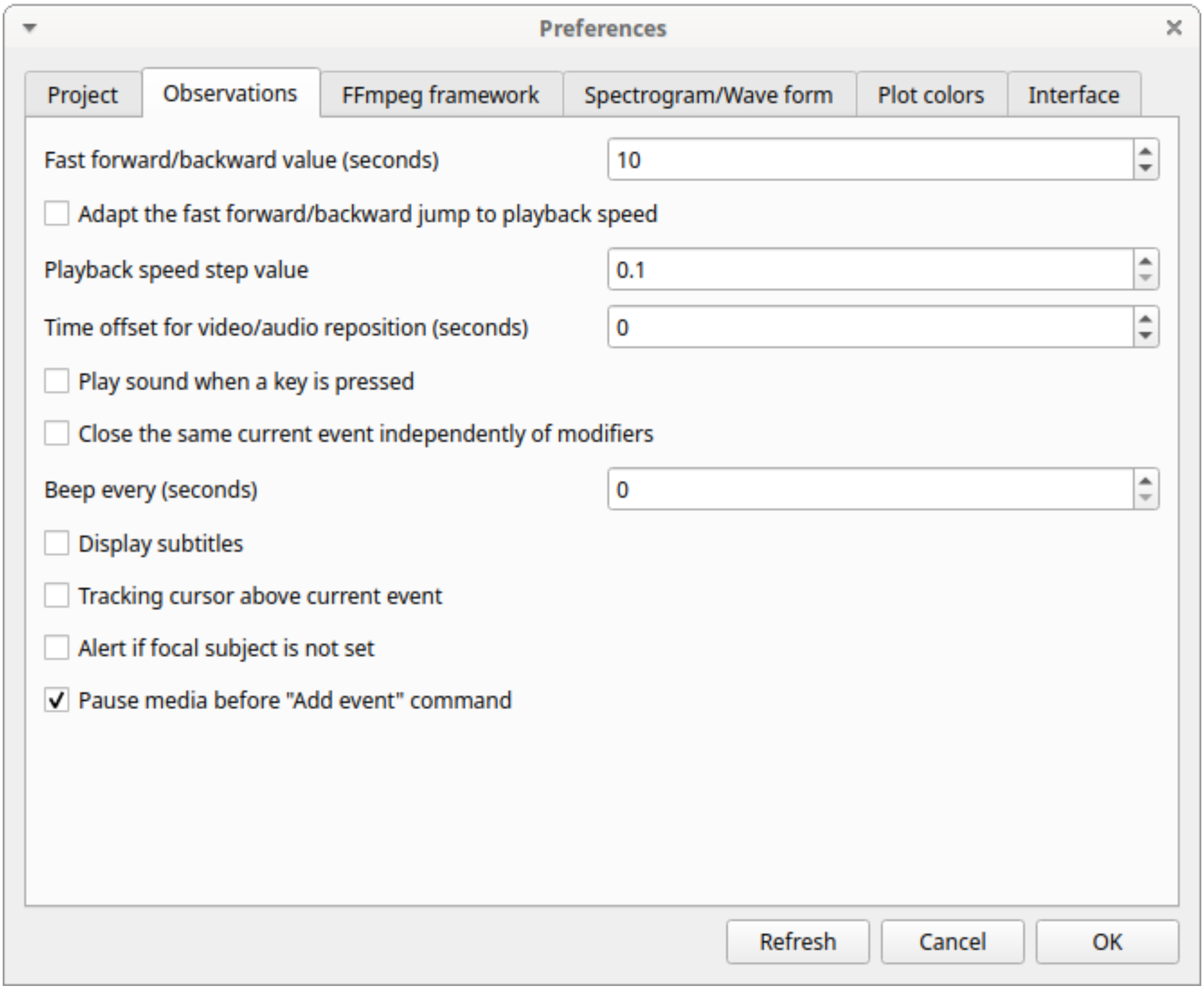
The BORIS project file is encoded in JSON format. Choose the indentation style for the project file between:

- None
- Newline
- Tab
- 2 spaces
- 4 spaces

Refresh button

Resets the configuration to the default values. BORIS will then close.

2.14.2 Observations



Fast forward/backward value (seconds)

This option allows you to customize the jump amount (in seconds) when moving forward or backward in media.

Adapt the fast forward/backward jump to playback speed

The jump value will be adjusted to the playback speed.

Playback speed step value

This value indicates how much the speed will increase or decrease after pressing the *change playback speed* buttons.

Time offset for media reposition (seconds)

This value indicates the time offset used when repositioning the media after double-clicking an event row in the *Events* table. For example, `-4` means that after a double-click, the media will be repositioned 4 seconds before the recorded event.

Play sound when a key is pressed

Plays a sound after each keypress.

Close the same current event independently of modifiers

Stops the current behavior regardless of modifiers.

Display subtitles

Displays or hides subtitles. If subtitles are stored in a separate file, the subtitle file must have the same base name as the video file and use the `.srt` extension.

Tracking cursor above current event

Positions the tracking cursor above the current event in the events list table.

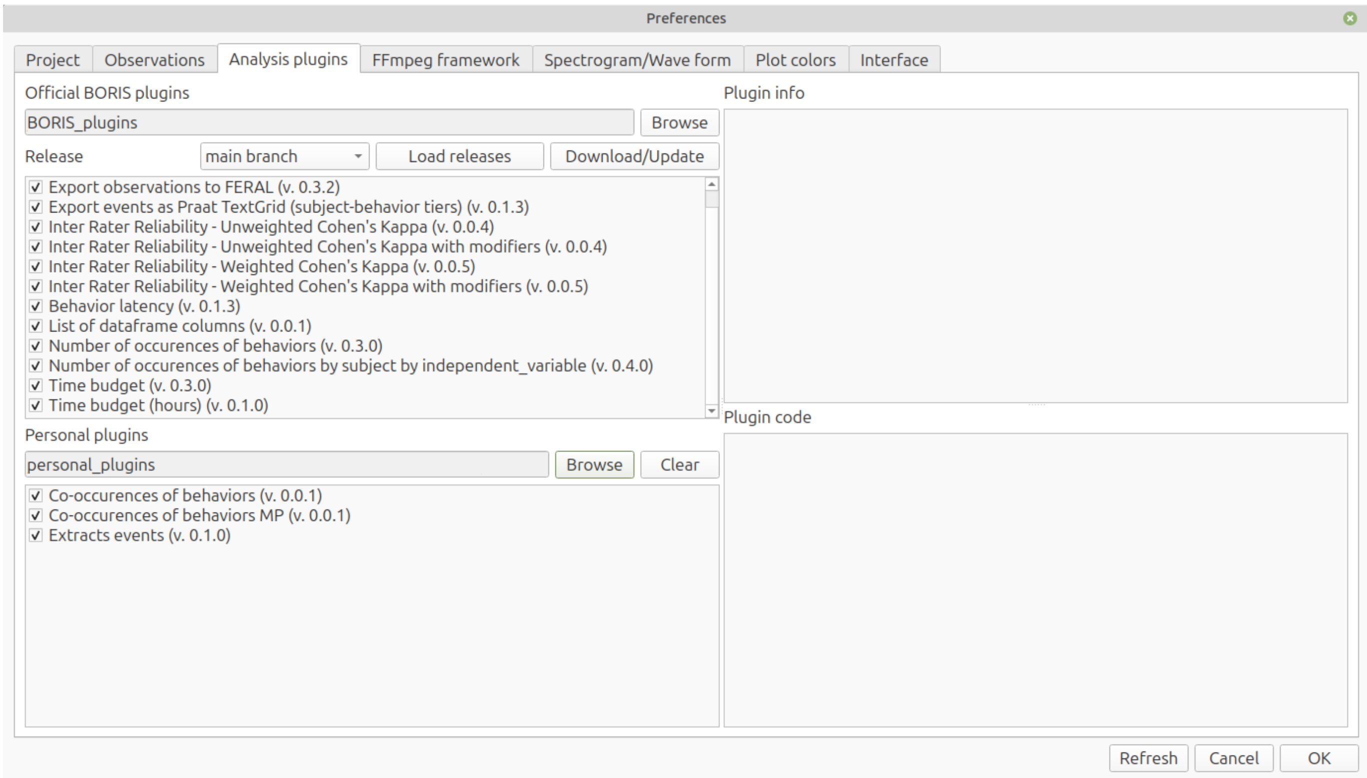
Alert if focal subject is not set

If this option is enabled, BORIS will display an alert if no focal subject is selected.

Pause media before "Add event" command

Pauses the media before manually adding an event.

2.14.3 Analysis plugins



The **Analysis plugins** tab lets you configure official BORIS plugins and personal plugins. The plugin lists show the plugin version when available.

Official BORIS plugins

Select, download, update, enable, or disable official BORIS plugins.

Personal plugins directory

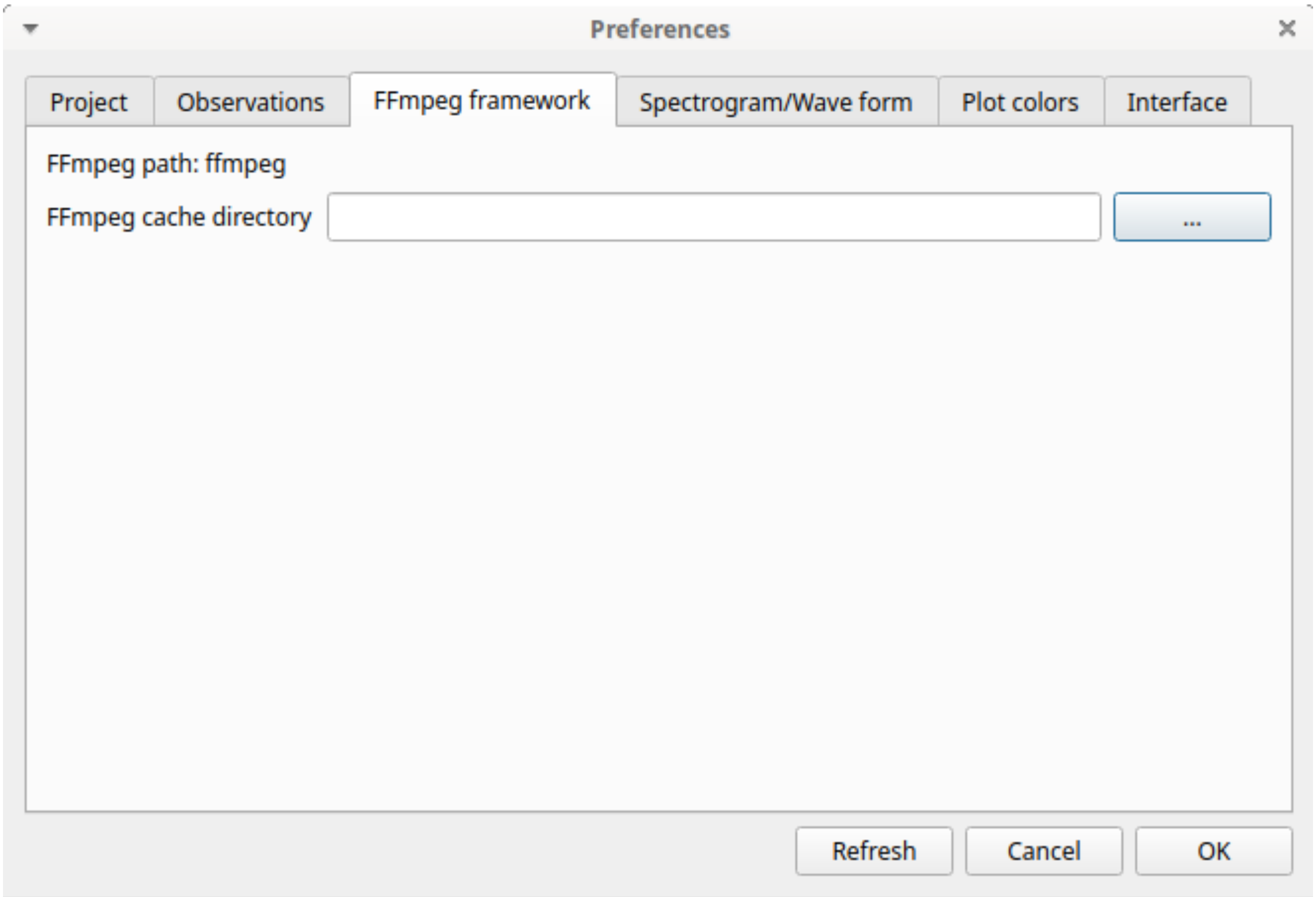
Select the directory that contains your personal plugins.

Clear

Remove the saved personal plugins path without deleting the directory or any plugin file.

See Analysis plugins for details about plugin loading, writing, and running.

2.14.4 FFmpeg framework

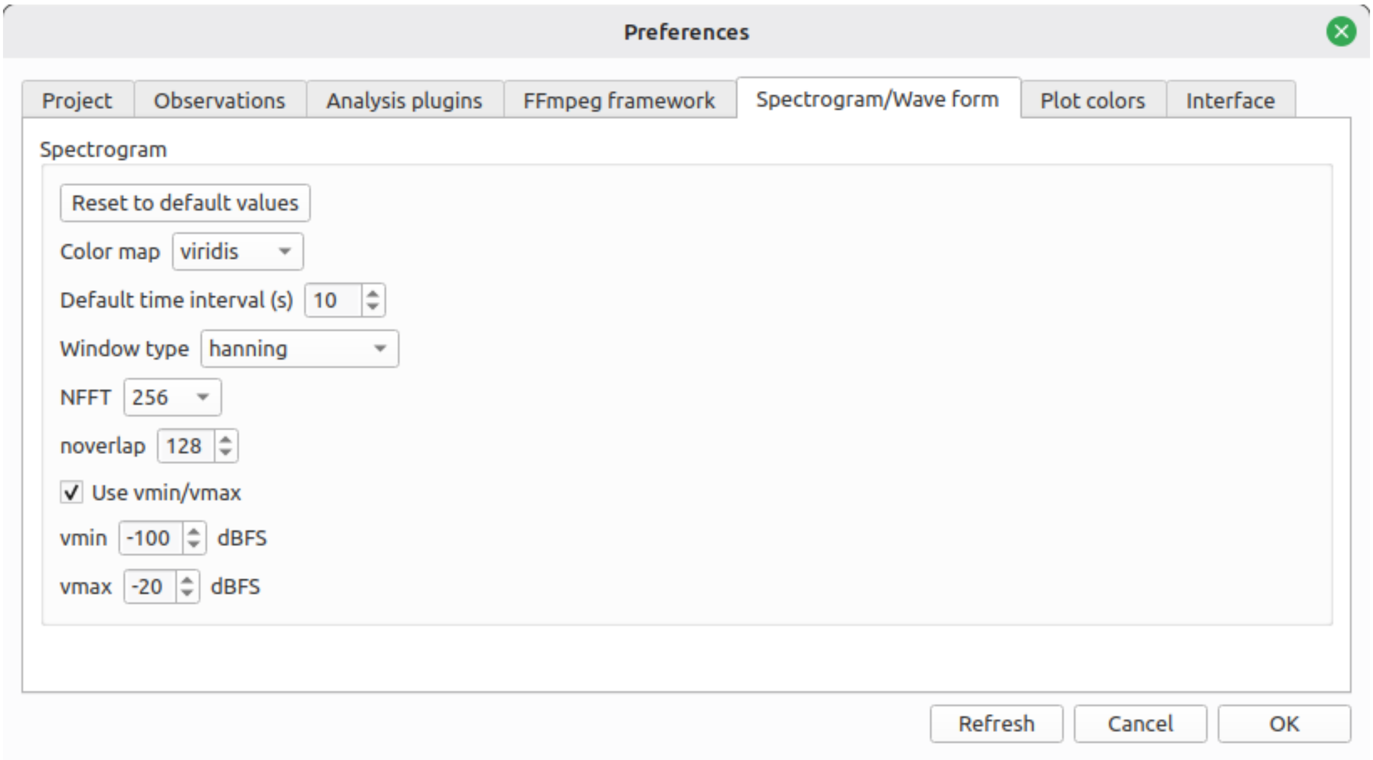


The path to the `ffmpeg` executable is displayed. The FFmpeg executable is included with BORIS for Windows. The FFmpeg framework is required to run BORIS.

FFmpeg cache directory

Specifies the directory used as the image cache for frame-by-frame mode and spectrogram visualization. If you do not specify a path, BORIS will use your system's default temporary directory.

2.14.5 Spectrogram / wave form



Color map

Selects the color map used to display the generated spectrogram. See Matplotlib colormaps for details.

Default time interval

Selects the time interval (in seconds) used to display the spectrogram and waveform.

Window type

Selects the window type: Hanning, Hamming, or Blackmanharris.

NFFT size

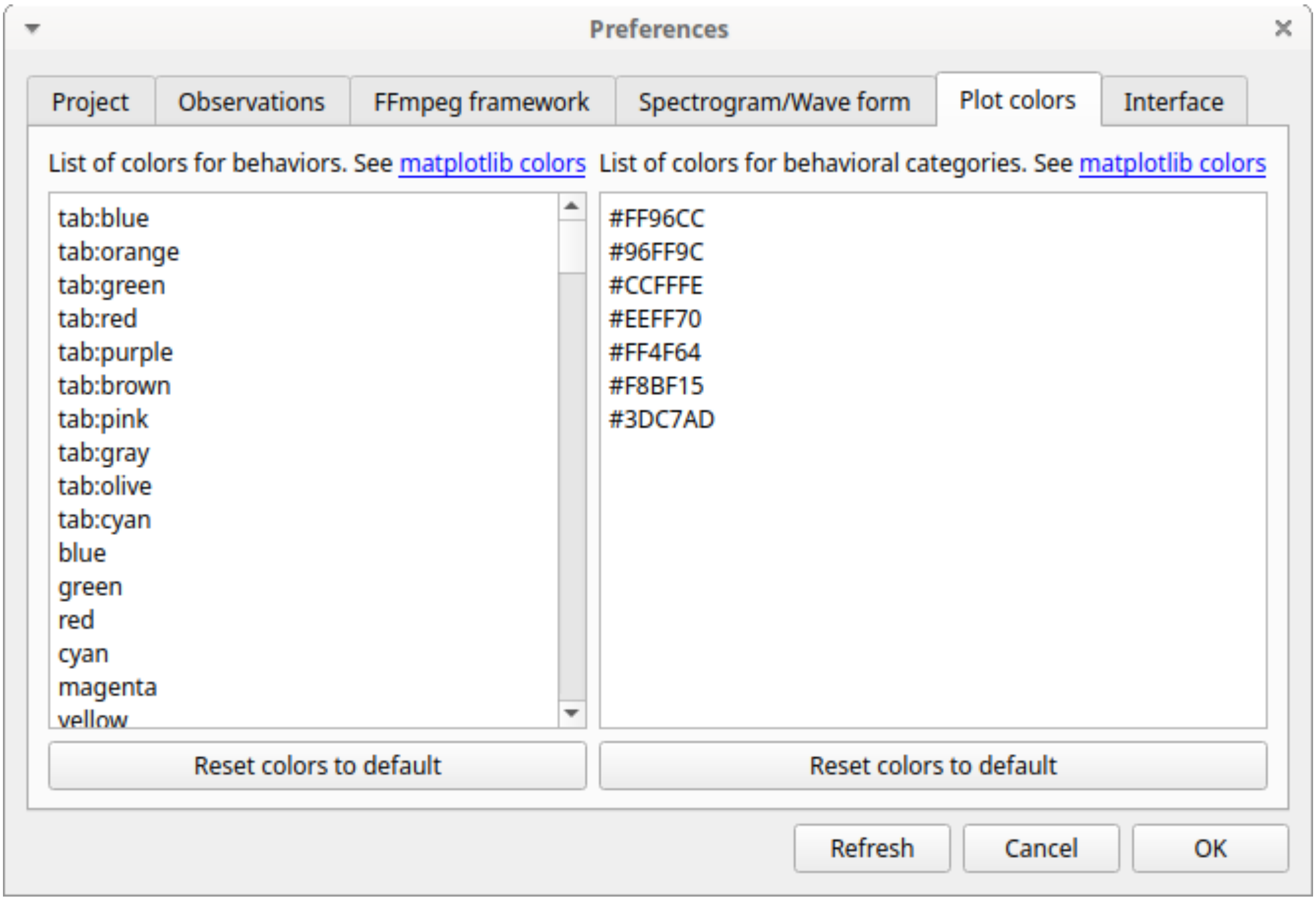
noverlap

Use vmin/vmax

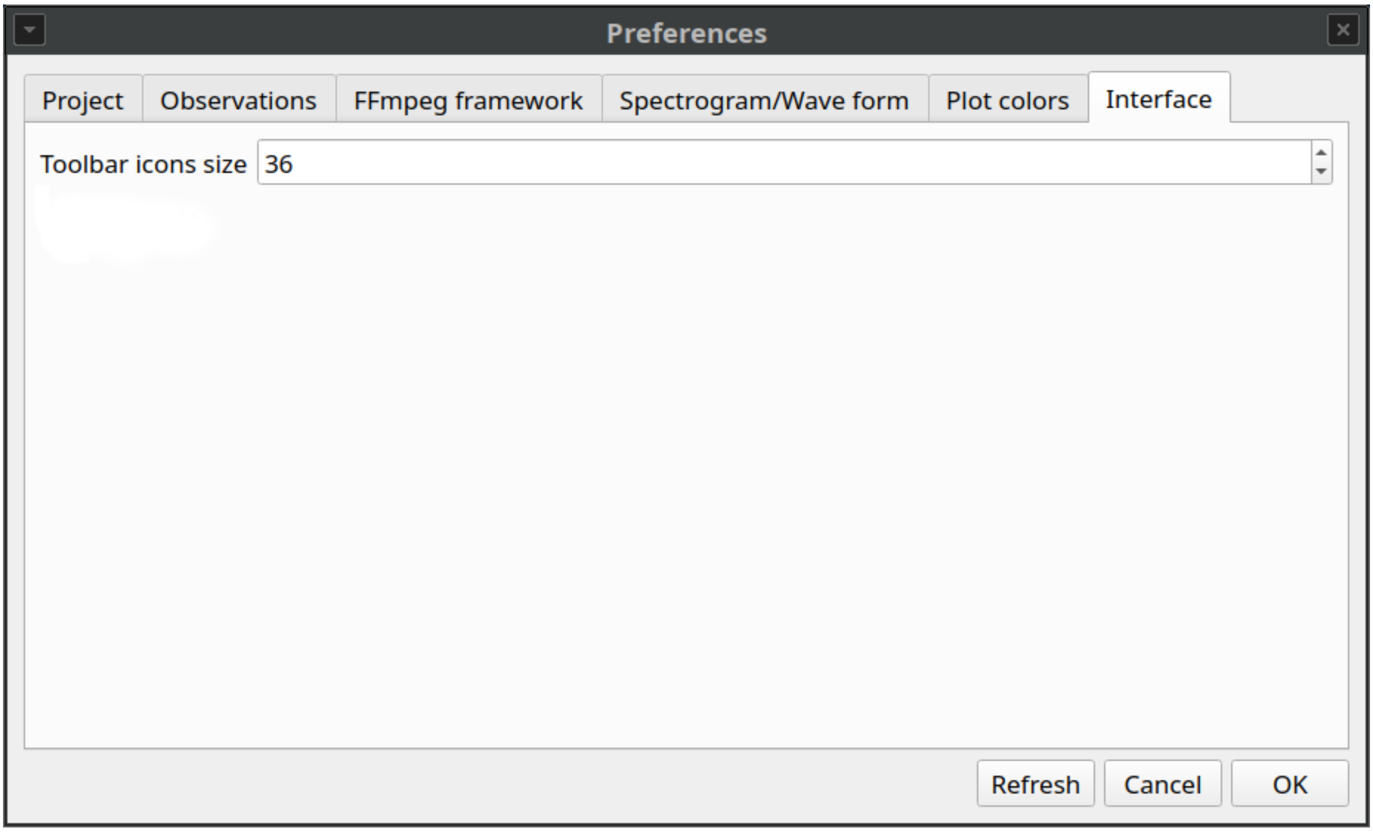
2.14.6 Plot colors

Behavior colors used in the plotting functions can be customized. The first color will be associated with the first behavior in your ethogram, the second color with the second behavior, and so on. Various color formats can be used to specify a color: **named color** or **hex RGB** (like #0F0F0F). See https://matplotlib.org/api/colors_api.html and https://matplotlib.org/examples/color/named_colors.html for details.

The **reset colors to default** button will reload the default colors.



2.14.7 Interface



Toolbar icons size

Sets the size of the toolbar icons in pixels.

2.15 Various

2.15.1 Removing path of media files

With BORIS, you can choose to store the full path of media/data files in the project file (for example: `/home/user/Video/video_n1.mp4` or `c:\Users\user\Documents\video1.avi`).

If you want to move your project to a different computer, or if you want to move your media/data files, you may choose not to store the full path. To do this, you can add media/data files using relative paths (see the **Add media files** section). You can also remove the full path of your media/data files from all observations in the current project (**File** > **Remove path from media files**). Please note that this operation is irreversible. After removal, the full path of your media will be lost and cannot be recovered.

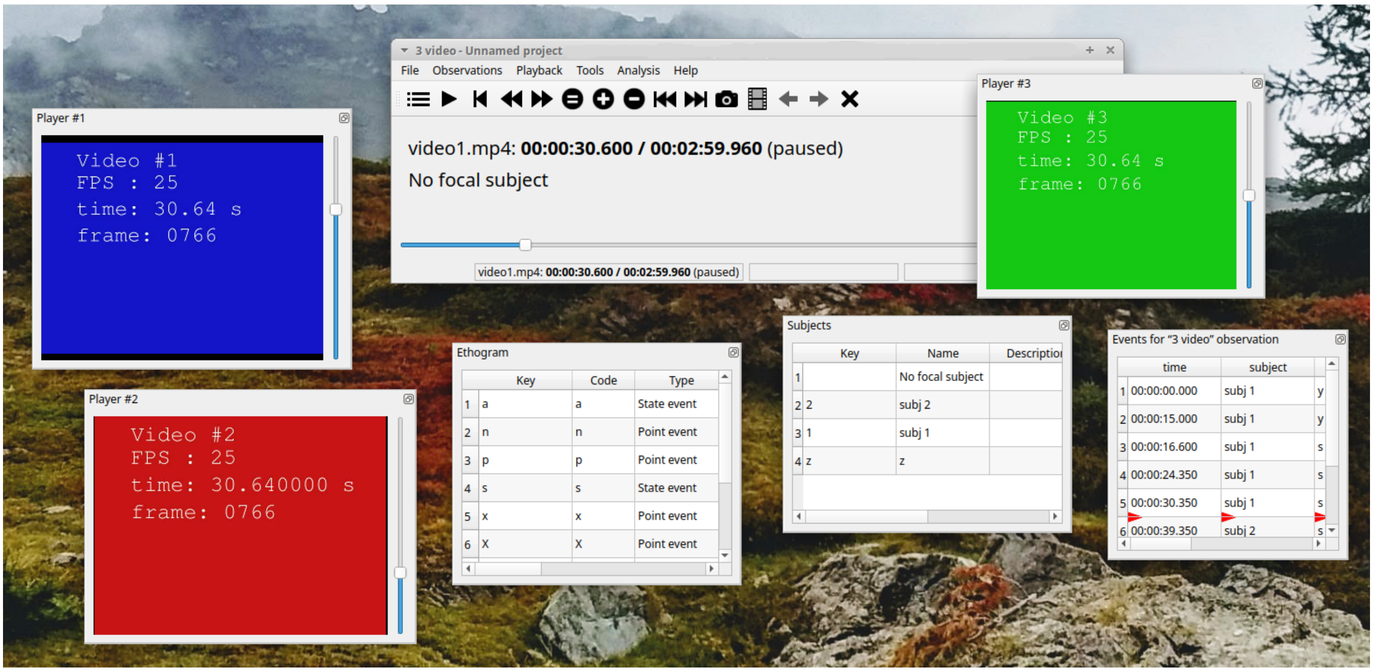
If you choose not to store the full path of media/data files, the media/data files must be located within the directory tree of your BORIS project file.

Example: If your BORIS project file is saved in `/home/user/projects/test.project`, your media/data files can be saved in the `/home/user/projects/videos` directory but **NOT** in the `/home/user/videos` directory.

2.15.2 Docking / undocking graphical elements

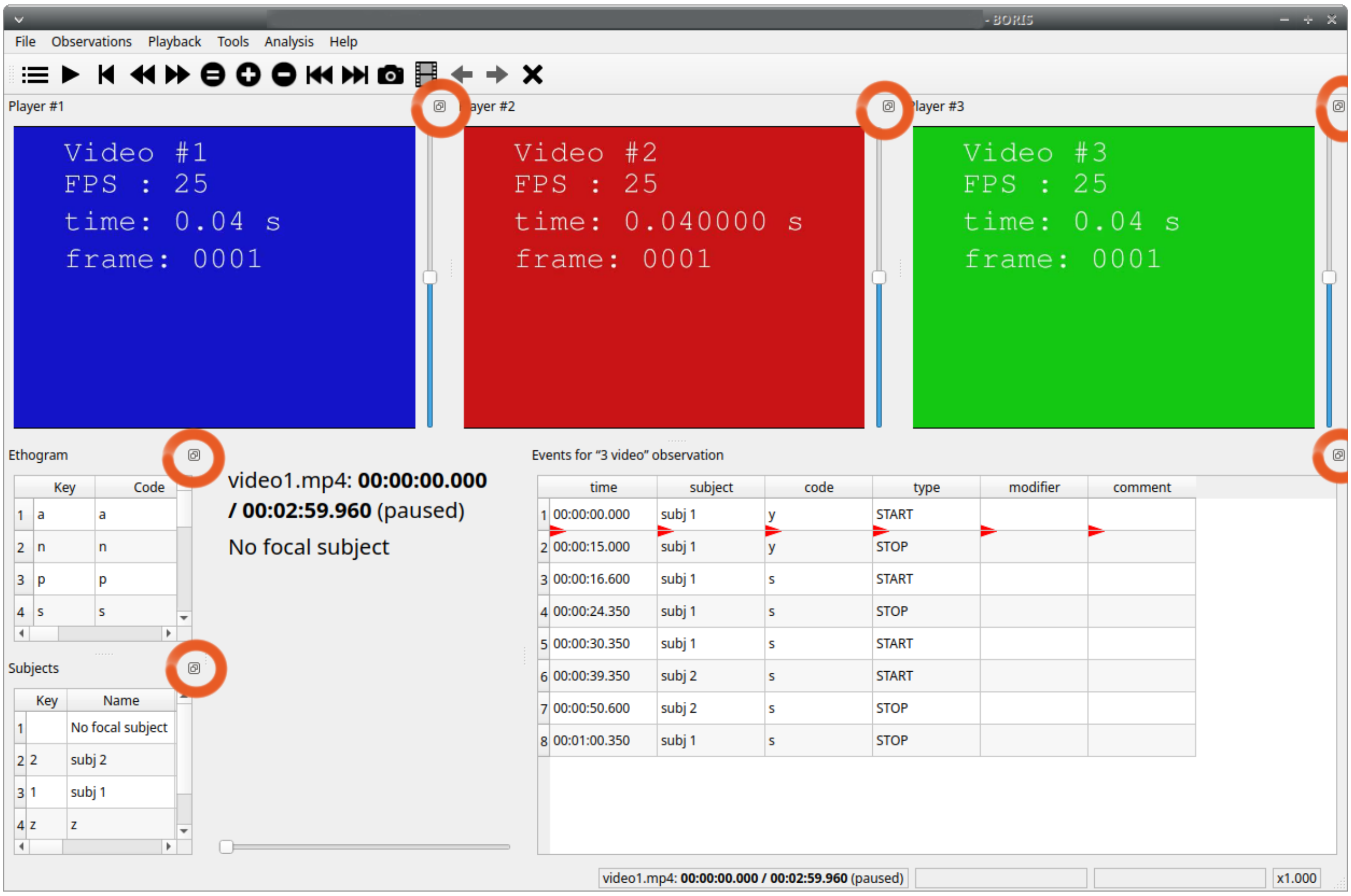
All elements, including all media players, can be undocked from the main window and positioned wherever you prefer (e.g., they can be on the same desktop across one or more screens).

The position of the various widgets is saved in the configuration file at the end of the work session.

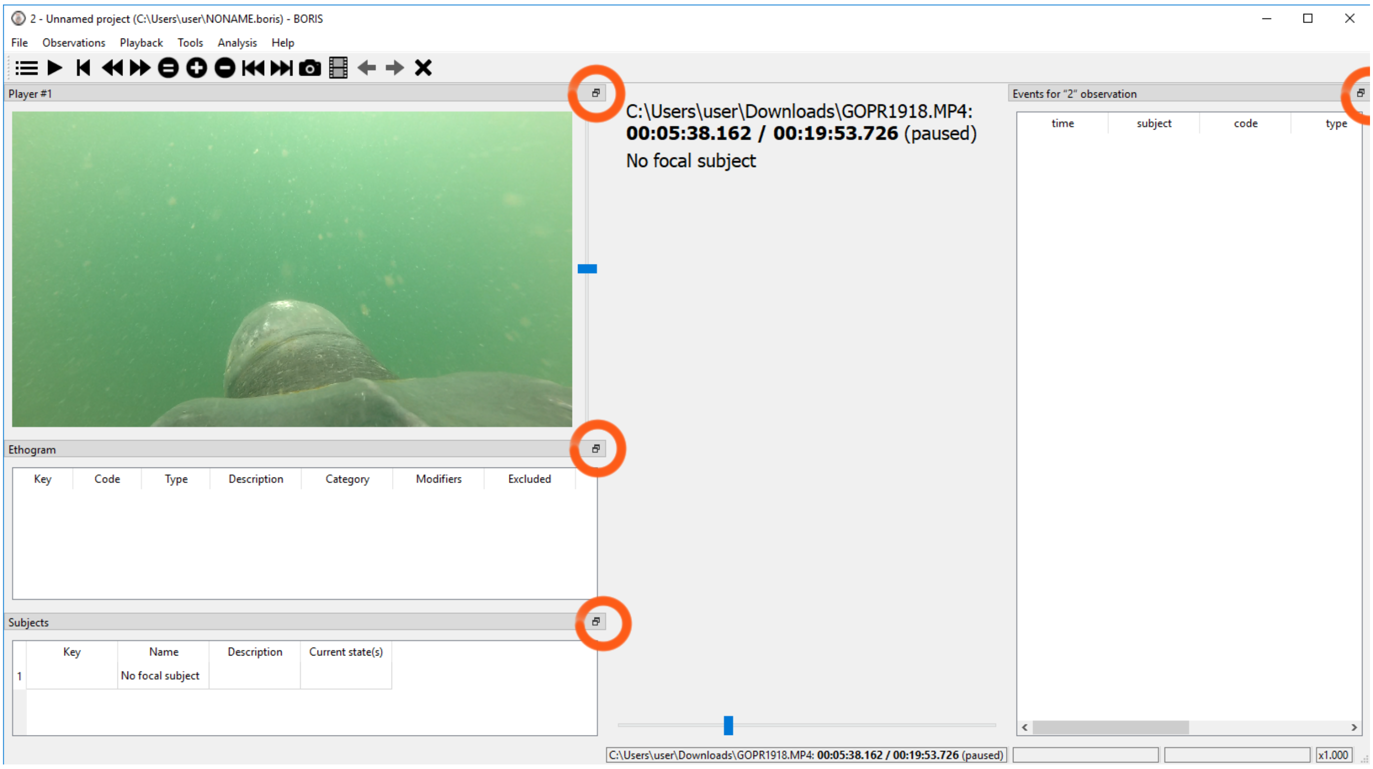


Clicking the icon in the top-right corner of the widget (on macOS, the icon is located in the top-left corner) undocks the widget so that it can be repositioned in another docking area or moved outside the main window. Double-clicking the title bar of the widget docks it back into the main window.

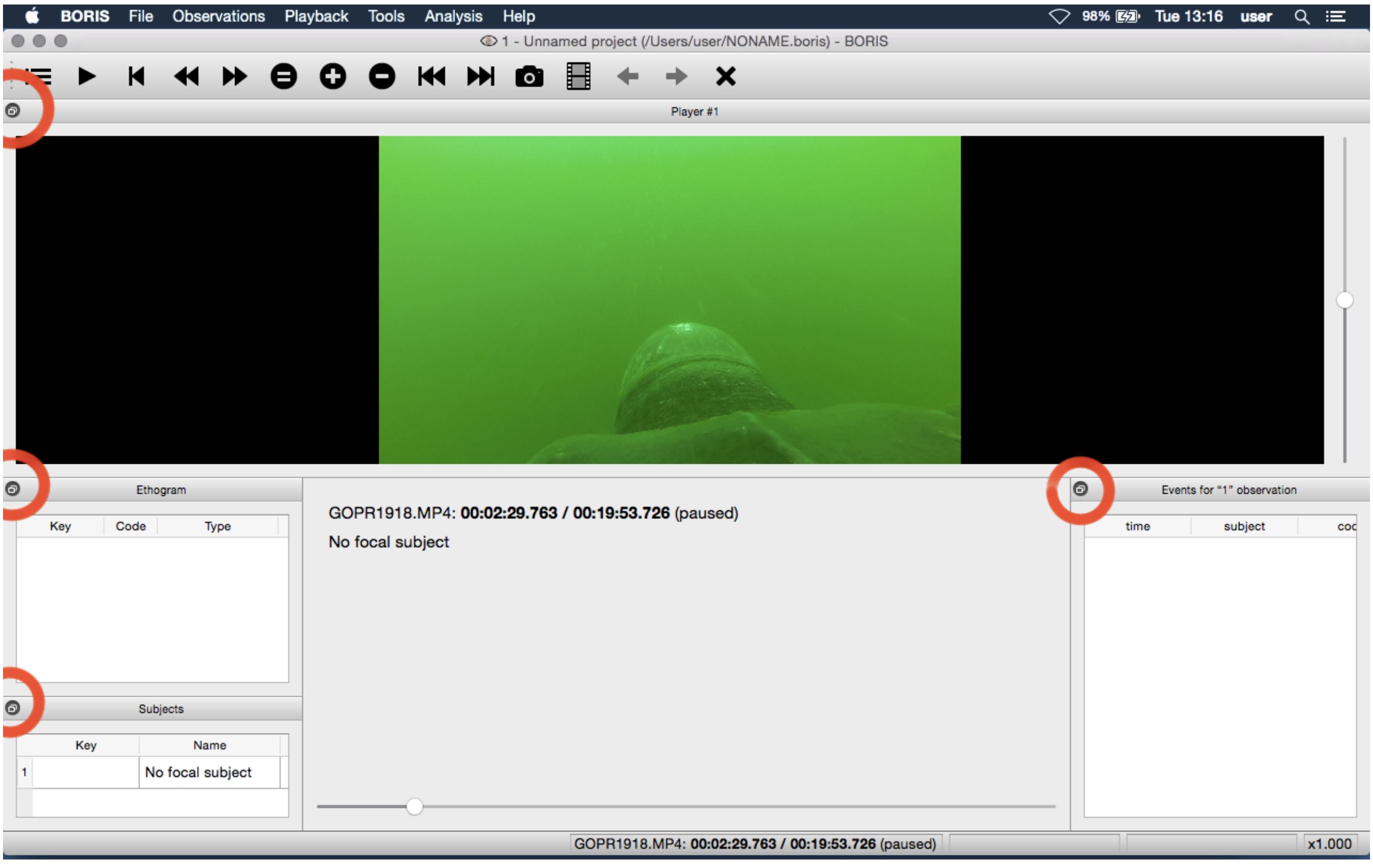
For Linux:



For Microsoft-Windows:



For MacOS:



If you prefer not to move dock widgets, you can lock them in the main window by enabling **Lock dockwidgets** (see **Tools** > **Lock dockwidgets**). All dock widgets will be docked in the main window and locked, except for the player dockwidgets.

2.15.3 Configuration files

BORIS saves the configuration (user preferences, window positions, widget positions) in a configuration file named **.boris**, stored in the home directory of the current user:

```
For Linux:
/home/USERNAME/.boris

For Microsoft-Windows:
C:\Users\USERNAME\.boris

For MacOS:
/Users/USERNAME/.boris
```

If you have trouble using BORIS, try closing the program, deleting this file, and relaunching BORIS.

The **recent projects list** is saved in the **.boris_recent_projects** file in the home directory of the current user.

2.15.4 Lock the dockwidgets

The dockwidgets (except the player dockwidgets) can be locked to the main window (see **Tools** > **Lock dockwidgets**).

2.15.5 Valid keys for triggering behavior

BORIS distinguishes between lowercase and uppercase characters.

- Keys from a to z
- Keys from A to Z
- Keys from 0 to 9
- Function keys from F1 to F12
- à é è ù ì ç
- ! " £ $ % & / ( ) = ? ^ [ ] { } @ | § ° #

3. Community

3.1 Acknowledgement

The authors would like to thank all users who report bugs and request new features for their valuable contributions.

3.2 Citing BORIS

If you use BORIS in a publication, please cite:

```
Friard, O. and Gamba, M.,
BORIS: a free, versatile open-source event-logging software for video/audio coding and live observations.
(2016) Methods in Ecology and Evolution, 7: 1325-1330.
```

DOI: 10.1111/2041-210X.12584

You can also send us a nice postcard.

Please consider starring the BORIS GitHub repository.

3.3 Bug Reports and Feature Requests

Please report any bugs you find in the latest BORIS version through the GitHub repository.

Before reporting a bug, please:

- Check the Frequently Asked Questions (FAQ) section.
- Check whether the issue has already been reported on GitHub.
- Delete the configuration file and try again (see configuration file).

Remember to include:

- Your operating system
- Your operating system version
- Your computer specifications (model, RAM, etc.)
- The BORIS version you are using
- Information about the media file you are coding, if applicable. See `Tools > Media file information`

Provide all the information needed to reproduce the bug, such as a detailed procedure or a screen recording.

In case of a crash, please send the `boris_error.log` file generated in your home directory immediately after the crash and before restarting BORIS:

```
Linux:
/home/YOUR_PROFILE_NAME/boris_error.log

Microsoft-Windows:
c:\Users\YOUR_PROFILE_NAME\boris_error.log
```

> **Note**
>
> If the bug you reported has been fixed, please remember to close the issue.